
Deutsche Sammlung von Mikroorganismen und Zellkulturen
Abt. Menschliche und Tierische Zellkulturen

German Collection of Microorganisms and Cell Cultures
Dept. Human and Animal Cell Cultures

Mascheroder Weg 1b, D-38124 Braunschweig, Germany

Tel. +49-531-2616.161, FAX +49-531-2616.150

E-Mail: mutz@dsmz.de

<http://www.dsmz.de>

INFORMATION SHEET FOR CELL LINES

Cell Line:	1G1
Cell Type:	mouse hybridoma (anti-chicken dorsal retina cytoplasmic membranes, E12)
DSMZ No:	ACC 424
Origin:	cell line was established by fusion of mouse NS-1 cells with spleen from a BALB/c mouse immunized with enriched retinal cell membranes; described as secreting a mouse monoclonal antibody (IgM) reacting with the chicken dorsal retina cytoplasmic membranes (E12) and binding outer photoreceptor elements of adult vertebrate retina (inclusive chick, mouse, rat)
References:	Schlosshauer et al., Eur. J. Cell Biol. 73: 150-157 (1997)
Depositor:	Dr. B. Schlosshauer, Natural and Medical Science Institute (NMI), Reutlingen, Germany
DSMZ Cell Culture Data:	
Morphology:	spheroidal cells growing singly in suspension, or occasionally loosely adherent
Medium:	90% RPMI 1640 + 10% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement" for the first 1-2 weeks post-thawing)
Subculture:	maintain at ca. $0.1-0.3 \times 10^6$ cells/ml; split saturated culture 1:3 to 1:10 every 2-5 days (some cells attach to the surface of the culture flask, but can be detached by tapping the flask); seed out at ca. 0.3×10^6 cells/ml; a maximal density of 1.0×10^6 cells/ml should not be exceeded
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgM monoclonal antibody
Species:	confirmed as mouse with IEF of AST, LDH, NP
Cytogenetics:	murine flat-modulated hypertetraploid karyotype; 94-102<4n>; no centric fusion markers present
Viruses:	ELISA: reverse transcriptase negative

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/muthti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	22RV1
Cell Type:	human prostate carcinoma
DSMZ No:	ACC 438
Origin:	cell line was derived from a human prostate carcinoma xenograft (CWR22R) that was serially propagated in nude mice after castration-induced regression and relapse of the parental, androgen-dependent CWR22 xenograft
References:	Wainstein et al., Cancer Res. 54: 6049-6052 (1994); Sramkoski et al., In Vitro Cell Dev. Biol. 35: 403-409 (1999)
Depositor:	Dr. J. W. Jacobberger, Case Western Reserve University, Cleveland, Ohio, USA
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently in monolayers
Medium:	40% RPMI 1640 + 40% Dulbecco's MEM + 20% FBS
Subculture:	split confluent culture 1:3 to 1:6 every 3-6 days using trypsin/EDTA; seed out at ca. 2.5×10^4 cells/cm ² ; maximum density of ca. 5×10^5 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-60 hours
Harvest:	cell harvest of ca. 4×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype with 8% polyploidy; 50(48-52)<2n>XY, +1, +7, +8, +12, i(1q), der(2)t(2;4)(p1?4;q3?3)del(2)(q11q3?1), der(4)t(2;4)(p1?4;q3?3), add(6)(q15), add(14)(q24); sideline with del(1)(q21), +3, +18; matches published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	23132/87
Cell Type:	human gastric adenocarcinoma
DSMZ No:	ACC 201
Origin:	established from the primary tumor of a 72-year-old man with gastric adenocarcinoma in 1987
References:	Vollmers et al., Virchows Archiv B Cell Pathol. 63: 335-343 (1993)
Depositor:	Dr. H. P. Vollmers, Institute for Pathology, Würzburg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial cells growing as confluent monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:8 every 5-6 days using trypsin/EDTA; seed out with $1-2 \times 10^6/25 \text{ cm}^2$ flask; initially after thawing, cells may grow slowly and loosely adherent;
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 50 hours
Harvest:	cell harvest of about 0.3×10^6 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, cytokeratin-7+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-; (initial data published by the originators showed negativity, we found the cells to express cytokeratin-7)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypertetraploid karyotype with 12% polyploidy; 47(45-52)<2n>XY, +20, t(1;15)(p11;p11), t(6;12) (p21;q21), i(13q), t(13;14)(p10;q10); closely resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	293
Cell Type:	human embryonal kidney
DSMZ No:	ACC 305
Origin:	established from a human primary embryonal kidney transformed by adenovirus type 5 (Ad 5); classified as risk category 1 according to the German Central Commission for Biological Safety (ZKBS); cell line also known as HEK-293 (human embryonic kidney-293)
References:	Graham et al., J. Gen. Virol. 36: 59-72 (1977)
Depositor:	Dr. W. Dirks, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblastoid cells growing as monolayer
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:5 to 1:6 every 2-3 days (without using trypsin/EDTA, cells can be detached by tapping the flask); seed out at about $2.5\text{-}3.0 \times 10^5$ cells/25 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of about 20-24 hours
Harvest:	cell harvest of about 40×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2\text{-}3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18(+), desmin-, endothel-, GFAP-, neurofilament+, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH
Cytogenetics:	human near-triploid karyotype with 6% polyploidy; 69(64-69)XX?X, +1, +6, -10, -13, -16, -18, -20, +21, +22, +mar, add(?X)(p22), inv(1)(p35q43), del(1)(p35), i(5p), del(6)(q11), add(8)(q24), add(13)(p13), add(17)(qter), der(?)t(5;?)(q13;?)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	2A1
Cell Type:	mouse hybridoma (anti-chicken retina)
DSMZ No:	ACC 281
Origin:	established by fusion of mouse NS-1 cells with spleen from a BALB/c ByJ mouse immunized with White Leghorn chicken embryonic retinal and tectal membranes; secretes 2A1 monoclonal antibody which binds epitopes involved in chicken retinal morphogenesis
References:	Schlosshauer et al., Development 109: 395-409 (1990)
Depositor:	Prof. B. Schlosshauer, Natural and Medical Science Institute (NMI), Reutlingen, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly in suspension or loosely adherent
Medium:	80% RPMI 1640 + 20% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement") for the first 1-2 weeks post-thawing
Subculture:	cells are difficult to culture! maintain at ca. $0.05\text{--}0.5 \times 10^6$ cells/ml; split saturated culture 1:4 to 1:8 every 2-3 days; seed out at ca. 0.2×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 25 hours
Harvest:	maximal density at ca. 0.7×10^6 cells/ml (which should not be exceeded)
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgM monoclonal antibody
Species:	confirmed as mouse with IEF of LDH, NP
Cytogenetics:	murine hypertetraploid karyotype
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	2E10-H2
Cell Type:	mouse hybridoma (anti-canine adenovirus type 1)
DSMZ No:	ACC 178
Origin:	established by fusion of SP2/0-AG14 myeloma cells with BALB/c mouse spleen cells immunized with purified canine adenovirus type 1 (CAV-1); the secreted antibody binds to and neutralizes CAV-1 and is type-specific
References:	Intervirology 23: 116-120 (1985); Vet. Microbiol. 16: 1-8 (1988)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single cells in suspension
Medium:	92.5% Dulbecco's MEM (4.5 g/L glucose) + 5% horse serum + 2.5% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; optimal split ratio of 1:2 every 2-3 days
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximum density at about $1-2 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 10% horse serum, 5% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, NP
Cytogenetics:	92(88-101)<4n> with 1.7% polyploidy; predominantly mouse chromosome morphology with 3 submetacentric marker chromosomes
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	2HX-2
Cell Type:	mouse hybridoma (anti-adenovirus 2)
DSMZ No:	ACC 183
Origin:	established by fusion of SP2/0-AG14 myeloma cells with spleen cells from BALB/c mice immunized with HELA cells that were infected with adenovirus 2; cells secrete an IgG monoclonal antibody reacting with an adenovirus group specific antigen; cell line is patented
References:	Virology 110: 385-401 (1981)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single cells growing in suspension
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS
Subculture:	optimal split ratio of 1:2 every 2-3 days
Incubation:	at 37 °C with 5% CO ₂
Harvest:	cell harvest of about 1-2 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, NP, PEP B
Cytogenetics:	murine karyotype with centric fusion and 16% polyploidy; 66(58-74)<3n>
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	2M6
Cell Type:	mouse hybridoma (anti-chicken retina)
DSMZ No:	ACC 205
Origin:	cell line was established by fusion of mouse NS-1 cells with spleen from a BALB/c mouse immunized with Triton X114-purified chicken retina E10 proteins; described as secreting a mouse monoclonal antibody (IgG) reacting with chicken retina (Müller glia cell-specific antigen)
References:	Schlosshauer et al., Development 111: 789-799 (1991); Schlosshauer & Wild, Develop. Brain Res. 59: 197-208 (1991)
Depositor:	Prof. B. Schlosshauer, Natural and Medical Science Institute (NMI), Reutlingen, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly or in small clumps in suspension, some cells are slightly adherent
Medium:	80-85% RPMI 1640 + 10-15% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement") for the first 1-2 weeks post-thawing
Subculture:	cells are difficult to culture! maintain at ca. $0.2-0.8 \times 10^6$ cells/ml; split saturated culture 1:2 to 1:4 every 2-3 days; seed out at ca. 0.2×10^6 cells/ml; after thawing cultivate cells for the first 1-2 weeks in 24- or 12-well-plates using HFCS
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	maximal density at ca. 0.8×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of IgG ₁ monoclonal antibody
Species:	confirmed as mouse with IEF of LDH, NP
Cytogenetics:	murine hypertetraploid karyotype
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	32D
Cell Type:	mouse bone marrow
DSMZ No:	ACC 411
Origin:	established from long-term bone marrow cultures of C3H/HeJ mice infected with the Friend murine leukemia virus; cells are constitutively growth factor-dependent
References:	Greenberger et al., Proc. Natl. Acad. Sci. USA 80: 2931-2935 (1983); Valtieri et al., J. Immunol. 138: 3829-3835 (1987); Migliaccio et al., J. Cell Biol. 109: 833-841 (1989)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	mostly round cells growing singly or in small clusters in suspension, some cells are loosely adherent
Medium:	90% RPMI 1640 + 10% FBS + 10-20% vol conditioned medium of cell line WEHI-3B (DSM ACC 26)
Subculture:	maintain at $0.02\text{--}1.5 \times 10^6$ cells/ml; split 1:10 to 1:50 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml. CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 15 hours
Harvest:	maximum density at ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of LDH, NP
Cytogenetics:	murine hypodiploid karyotype; 35(34-35)<2n>; carries 5 centric fusion markers
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	380
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 39
Origin:	established from the peripheral blood of a 15-year-old boy with acute lymphoblastic leukemia (ALL FAB L2) in relapse in 1983; cells were described to be EBNA-negative and to carry (8;14) and (14;18) translocations
References:	Pegoraro et al., Proc. Natl. Acad. Sci. USA 81: 7166-7170 (1984)
Depositor:	Dr. L. Pegoraro, University of Torino, Turin, Italy
DSMZ Cell Culture Data:	
Morphology:	round, small, single cells in suspension
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	cells are very difficult to culture and grow very slowly! cells appear to grow better in 24-well-plates than in flasks; optimal density at about $1-3 \times 10^6$ cells/ml; split carefully
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 80 hours
Harvest:	maximal density at about $4-5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a-, cyCD79a+, CD80-, CD138(+), HLA-DR+, sm/cyIgM-, sm/cyIgG-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human near-diploid karyotype with 1.8% polyploidy; 46(43-47)<2n>XY, -14, t(8;14;18)(q24;q32;q21), +der(14)t(8;14;18)(q24;q32;q21); corresponds to published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	3T3
Cell Type:	Swiss albino mouse fibroblasts
DSMZ No:	ACC 173
Origin:	established from disaggregated Swiss albino mouse embryos in 1962; cells are used for transfection studies with DNA viruses and as assay system for transformation studies; cells have a high sensitivity to contact inhibition (corresponds to ATCC CCL 92)
References:	Todaro & Green, J. Cell Biol. 17: 299-313 (1963); Todaro et al., Virology 27: 179-185 (1965)
Depositor:	Dr. J. Wehland, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	fibroblasts growing adherently as monolayer with contact inhibition
Medium:	90-95% Dulbecco's MEM + 5-10% FBS
Subculture:	split confluent culture at a ratio of up to 1:8 every 2 to 4 days using trypsin/EDTA; after trypsination cells might remain in aggregates that are difficult to disperse; upon reaching confluence cells stop to grow; cells grow fast at low titer; seed out at ca. $1-2 \times 10^6$ cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	cell harvest of about $5-8 \times 10^6$ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, LDH, PEP B
Cytogenetics:	murine hypertriploid (near tetraploid sideline present) karyotype with 13% polyploidy; 68(58-73)<3n>
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	3T6
Cell Type:	Swiss albino mouse fibroblasts
DSMZ No:	ACC 202
Origin:	established from disaggregated Swiss albino mouse embryos in 1963; cells were described to secrete collagen and hyaluronic acid (= ATCC CCL 96)
References:	Todaro & Green, J. Cell Biol. 17: 299-313 (1963); Todaro et al., Virology 27: 179-185 (1965)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	fibroblasts growing adherently as monolayer
Medium:	85-90% Dulbecco's MEM + 10-15% FBS
Subculture:	split confluent culture every 2-3 days using trypsin/EDTA, seed out about 2×10^6 cells/175 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 15 hours
Harvest:	cell harvest of about 15×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, G6PD, MDH, NP
Cytogenetics:	murine flat-modulated hypertriploid/hypotetraploid karyotype; 67(64-73)<3n>
Viruses:	ELISA: reverse transcriptase positive

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	42-MG-BA
Cell Type:	human glioma
DSMZ No:	ACC 431
Origin:	established from the temporal lobe brain biopsy from a 63-year-old man with glioblastoma multiforme in 1990 (prior to treatment)
References:	Perzelova et al., Neoplasma 45: 25-29 (1998)
Depositor:	Dr. A. Perzelova, Comenius University, Bratislava, Slovakia
DSMZ Cell Culture Data:	
Morphology:	flat epithelial-like polygonal cells growing adherently as monolayer or, if confluent, as multilayers with processes
Medium:	80% mixture of RPMI 1640 + MEM (with Earle's salts) (at 1:1) + 20% FBS
Subculture:	split confluent culture 1:2 to 1:4 every 3-6 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ² ; saturation density at ca. 0.6×10^5 /cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	cell harvest of maximal $0.3\text{--}0.5 \times 10^5$ /cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, MDH, NP
Cytogenetics:	human flat-moded hypertetraploid karyotype with 8% polyploidy; 88-95<4n>XX, -Y, -Y, +4, -5, -5, -7, -8, -9, -9, +11, +11, -13, -13, +14, +14, +15, -16, -17, -17, -18, -18, -19, -21, -22, +14mar, add(1)(p1?), del(1)(q21.2), del(2)(q13)x2, dup(2)(p12p2?4)add(2)(p2?4), del(3)(q13.3), add(4)(p11), add(5)(p1?5), add(6)(q2?)x2, add(7)(q11)x2, add(7)(q3?), del(9)(p21p22)x2, del(11)(p11)x2, inv(12)(q12q21)x2, der(12)del(p11)t(7;12)(q21;q1?4)x2, add(13)(p11)x2, add(14)(q32)x2, add(21)(p11)x2
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	4H1-A7
Cell Type:	mouse hybridoma (anti-canine adenovirus 2)
DSMZ No:	ACC 28
Origin:	established by fusion of SP2/0-AG14 myeloma cells with BALB/c mouse spleen cells immunized with purified canine adenovirus type 2 (CAV-2, Manhattan strain); the secreted antibody (IgG _{2a}) binds to and neutralizes CAV-2 and is type-specific
References:	Intervirology 23: 116-120 (1985); Vet. Microbiol. 16: 1-8 (1988)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single cells in suspension
Medium:	92.5% Dulbecco's MEM (4.5 g/L glucose) + 5% horse serum + 2.5% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; optimal split ratio of 1:2 every 2-3 days
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximum density at $1.0-1.5 \times 10^6$ cells/ml
Storage:	frozen with 75% medium, 10% horse serum, 5% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of MDH, NP, PEP B
Cytogenetics:	murine hypertetraploid/pentaploid karyotype with 5% polyploidy; $91-98 < 4n >$; 3 centric fusion markers and 1-3 dmin present in most cells
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	5637
Cell Type:	human urinary bladder carcinoma
DSMZ No:	ACC 35
Origin:	established from the primary bladder carcinoma of a 68-year-old man in 1974; cells were described to produce several growth factors (e.g. SCF, IL-1, IL-6, G-CSF, GM-CSF, etc.)
References:	Welte et al., Proc. Natl. Acad. Sci. USA 82: 1526-1530 (1985); Fogh, Natl. Cancer Inst. Monograph 49: 5 (1978); Kaashoek et al., Lymphokine Cytokine Res. 10: 231-235 (1991); Quentmeier et al., Leukemia Res. 21: 343-350 (1997)
Depositor:	Prof. K. Welte, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent cultures 1:4 to 1:5 using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	about $8-10 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypotriploid karyotype with 17% polyploidy; 62(47-64)<3n>XX, -Y, -4, -5, -10, -13, -13, -15, -18, -19, -21, -21, +4mar, der(1)t(1;?)(p21;?), der(3)t(3;?)(p22;?), del(6)(q23), der(9)t(9;?)(p12;?), der(11)t(5;11)(q14;p12), ?der(12)t(12;?)(q14;?), i(13q), i(14q), der(17)t(17;?)(p11/12;?), del(22)(q13); carries large submetacentric marker with q arm resembling tandem dup(7); 9p12 is a recurrent breakpoint in bladder carcinoma
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	639-V
Cell Type:	human ureter transitional cell carcinoma
DSMZ No:	ACC 413
Origin:	established from the transitional cell carcinoma (small, grade III) removed from the left ureter of a 69-year-old white man in 1973
References:	Elliott et al., Cancer Res. 36: 365-369 (1976); Elliott et al., Cancer Res. 37: 1279-1289 (1977)
Depositor:	Dr. W. A. Schulz, Urologische Klinik, Düsseldorf, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing adherently in monolayers
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 3-4 days using trypsin/EDTA; seed out at ca. 0.5×10^6 cells/25 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 6×10^6 cells/25 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hyperdiploid karyotype with 24% polyploidy; 55(50-59)<2n>XX, +1, +3, +5, +6, +8, +11, +12, +14, -15, +20, -21, add(1)(p?21), i(5p), del(6)(q13-14), i(7p), add(8)(p2?), add(11)(p11), del(12)(q21q24), add(18)(q23); additional copies of 6q-, i(7p), 8p+, der(18) markers present in sidelines together with del(7)(p13-14)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	647-V
Cell Type:	human urothelial bladder carcinoma
DSMZ No:	ACC 414
Origin:	established from a male patient with a primary urothelial bladder carcinoma (malignant, grade 2)
References:	Elliot et al., Cancer Res. 36: 365-369 (1976)
Depositor:	Dr. W. A. Schulz, Urologische Klinik, Düsseldorf, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing in monolayers
Medium:	85% Dulbecco's MEM + 15% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:5 to 1:6 every 3-4 days using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ² in 10 ml medium
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. $10\text{--}15 \times 10^6$ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1.5\text{--}2.0 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human flat-moded hyperdiploid/hypotriploid karyotype with 12% polyploidy; 52-60<3n>XX, -X/Y, -2, -4, -6, -9, -10, -12, -12, -14, -15, -16, -17, -18, +5mar, del(1)(p32), del(2)(q23)/add(2)(q2?), add(3)(q27), add(3)(q1?), i(5p), i(6p), i(7p)x2, add(8)(q24), del(11)(q22q24), del(13)(q?11q?14), der(19)t(12;19)(q13;q13), add(21)(p11); i(5p) associated with bladder cancer
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	697
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 42
Origin:	established from the bone marrow of a 12-year-old boy with acute lymphoblastic leukemia (cALL) at relapse in 1979; cells express bcl-2, bcl-3 and c-myc mRNA; recently termed "EU-3" in the literature
References:	Findley et al., Blood 60: 1305-1309 (1982)
Depositor:	Dr. M. Cooper, University of Alabama, Birmingham, Alabama, USA
DSMZ Cell Culture Data:	
Morphology:	round, single cells and clusters of cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; split saturated culture at 1:3 every 2-3 days; seed out at ca. $0.5-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30-40 hours
Harvest:	maximal density at about $2-3 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20(+), CD34-, CD37-, CD79a-, cyCD79a+, CD80-, CD138+, HLA-DR+, cyIgM+, cyIgG-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human near diploid karyotype; 46(45-48)<2n>XY, t(1;19)(q23;p13), del(6)(q21)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	7-TD-1
Cell Type:	mouse B cell hybridoma
DSMZ No:	ACC 23
Origin:	murine myeloma cells fused with C57Bl/6 spleen cells; indicator cell line for IL-6
References:	Van Snick et al., Proc. Natl. Acad. Sci. USA 83: 9679-9683 (1986)
Depositor:	Dr. E. Schmitt, Institute of Immunology, University of Mainz, Mainz, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells, growing single or clustered in suspension
Medium:	90% RPMI 1640 + 5% FBS + 5% ConA spleen cell supernatant (or 10 ng/ml IL-6) + 2.5 x 10 ⁻⁵ M mercaptoethanol + 2 mM L-glutamine
Subculture:	maintain at 0.1-0.3 x 10 ⁶ cells/ml, optimal split ratio (daily) of 1:3, maximal split ratio of 1:5; for an IL-6 bioassay, we recommend to initially use 1.25-2.5 x 10 ⁴ cells/ml and 100 : 1 in a 96-well-plate (it is advisable to determine the optimal cell density in various growth curves)
Incubation:	at 37 °C with 7.5% CO ₂
Doubling Time:	doubling time of ca. 24-25 hours
Harvest:	maximum density of 0.5-1.0 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁵ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	bioassay for IL-6 (low cell densities are essential for reliable assays; try doubling cell dilutions beginning with cell densities of 2.5 x 10 ⁴ cells/ml; duration of assay is commonly 2-3 days)
Species:	confirmed as mouse with IEF of MDH, NP, PEP B
Cytogenetics:	murine karyotype
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	72A1
Cell Type:	mouse hybridoma (anti-EBV)
DSMZ No:	ACC 89
Origin:	established by fusion of the P3X62AG8.653 mouse myeloma cell line with spleen cells of a BALB/c mouse immunized with Epstein-Barr virus (EBV) particles; cells produce a monoclonal antibody (IgG ₁) that reacts with a viral envelope glycoprotein (350/220 kD) and that neutralizes EBV
References:	Hoffman et al., Proc. Natl. Acad. Sci. USA 77: 2979-2983 (1980)
Depositor:	obtained from ATCC (HB 168), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	single cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at 0.1-1.0 x 10 ⁶ cells/ml; split every 2-3 days at 1:3 to 1:5
Incubation:	at 37 °C with 5% CO ₂
Harvest:	1.0-1.5 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, MDH, PEP B
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	8-MG-BA
Cell Type:	human glioma
DSMZ No:	ACC 432
Origin:	established from the frontal brain lobe of a 54-year-old woman with glioblastoma multiforme in 1989 (pre-treatment)
References:	Perzelova et al., Neoplasma 45: 25-29 (1998)
Depositor:	Dr. A. Perzelova, Comenius University, Bratislava, Slovakia
DSMZ Cell Culture Data:	
Morphology:	epithelial-like, adherent cells growing as monolayer
Medium:	80-85% MEM (with Earle's salts) + 15-20% FBS
Subculture:	split confluent culture 1:3 to 1:4 once a week using trypsin/EDTA; seed out at ca. 1×10^6 cells/25 cm ² ; optimal density at ca. $5-10 \times 10^6$ /175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 100 hours
Harvest:	cell harvest of maximal 15×10^6 /175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin(+), cytokeratin-7-, cytokeratin-8(+), cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH
Cytogenetics:	human flat-moded hyperdiploid karyotype with 15% polyploidy; 52(47-52)<2n>X, -X, +1, +1, +3, +5, +6, +7, +8, -19, +2-4mar, add(1)(p34-36;hsr), add(1)(p34-36), der(1)t(1;17)(q11-12;q12-21), der(3;5)(q10;q10), i(5p), del(6)(q24-25), del(7)(p14), i(7p), i(8q), add(8)(p21)x1-2, del(9)(q32q34)/add(9)(q32), add(11)(?q13), del(12)(q14), del(18)(q22.2), add(19)(p13.1); sideline with del(3)(p11), der(6;9)(p10;q10); 2-4 acf present in 31% cells
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	8305C
Cell Type:	human thyroid carcinoma
DSMZ No:	ACC 133
Origin:	established from the primary tumor of a 67-year-old woman with primary thyroid carcinoma (undifferentiated carcinoma, composed of spindle, polygonal and giant cells)
References:	Ito et al., Int. J. Oncol. 4: 583-586 (1994)
Depositor:	Dr. M. Akiyama, Radiation Effects Research Foundation, Hiroshima, Japan
DSMZ Cell Culture Data:	
Morphology:	adherent, large spindle-shaped cells growing as monolayer
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	split confluent culture 1:5 once a week using trypsin/EDTA; seed out at about 1×10^6 cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 54 hours
Harvest:	about $5-6 \times 10^6$ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $0.7-0.9 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human flat-moded hypertriploid karyotype with 22% polyploidy; 74(67-76)<3n>XXX, -2, +9, -10, -12, -13, +14, +15, +15, +18, -19, +20, +4mar, 1-20 dm in most cells, der(6)t(6;7)(q22;q36q11.23), add(15)(p11)x2, del(17)(p12)x2; subclonal rearrangements, e.g. del(2)(p16), del(2)(q22), del(3)(q22), i(4p), t(6p;7q), del(9)(p23)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	8505C
Cell Type:	human thyroid carcinoma
DSMZ No:	ACC 219
Origin:	established from the primary tumor of a 78-year-old woman with thyroid carcinoma (undifferentiated carcinoma, histologically a largely papillary adenocarcinoma with some spindle, polygonal and giant cells)
References:	Ito et al., Int. J. Oncol. 4: 583-586 (1994)
Depositor:	Dr. M. Akiyama, Radiation Effects Research Foundation, Hiroshima, Japan
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells growing as monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:5 every 2-3 days or 1:10 every 4-5 days using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30-40 hours
Harvest:	cell harvest of about 20 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hypotriploid karyotype with 16% polyploidy; 65(57-68)<3n>XX, -X, -2, +3, -7, -10, +11, -13, -16, -18, -19, +2mar, del(1)(q13), del(3)(q22), ?del(4)(q21), i(8q)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	A-10
Cell Type:	rat smooth muscle
DSMZ No:	ACC 132
Origin:	established from the thoracic aorta of a DB1X embryonic rat in 1976; cells were described to display properties characteristic of smooth muscle cells
References:	Kimes & Brandt, Exp. Cell Res. 98: 349-366 (1976)
Depositor:	obtained from ATCC (CRL 1476), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	large, epithelial-like, some fibroblast-like, adherent cells growing in monolayers
Medium:	80% Dulbecco's MEM + 20% FBS
Subculture:	cell growth is very slow! split confluent culture 1:2 to 1:3 once a week using trypsin/EDTA; initially seed out cells at best in a 25 cm ² flask; after thawing, cells may contain a significant amount of granulae; confluent cultures look more transparent
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 70-100 hours
Harvest:	about 1-2 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1-3 x 10 ⁵ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Fingerprinting:	unique Hinf I-(gtg) ₅ DNA profile
Species:	confirmed as rat with IEF of AST, MDH, NP
Cytogenetics:	rat karyotype
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	A-2
Cell Type:	fish Poeciliid (<i>Xiphophorus xiphidium</i>)
DSMZ No:	ACC 276
Origin:	derived from the sweetwater poeciliid fish (<i>xiphophorus xiphidium</i>); origin: Rio Purificacion; strain: Y ^F], embryonal
References:	Kuhn et al., In Vitro 15: 537-544 (1979); Mäueler et al., Oncogene 3: 113-122 (1988)
Depositor:	Prof. M. Scharl, University of Würzburg, Würzburg, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing as monolayer
Medium:	80-85% mixture of Ham's F12 + Dulbecco's MEM (at 1:1) + 15-20% FBS
Subculture:	split culture 1:3 about 2-3 times per week before cells reach confluency; cells grow loosely adherent and can be detached from the plastic surface by shaking or pipetting; old cultures tend to form foci and lose adherence, but the cells remain attached to each other and are difficult to disperse
Incubation:	at 26-28 °C with 5% CO ₂
Doubling Time:	doubling time of about 2-3 days
Harvest:	maximal density of 6-10 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 0.5-3.0 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	A-204
Cell Type:	human rhabdomyosarcoma
DSMZ No:	ACC 250
Origin:	established from the rhabdomyosarcoma of a 1-year-old girl in 1973; reported to form small malignant tumors in nude mice consistent with embryonic rhabdomyosarcoma
References:	Giard et al., J. Natl. Cancer Inst. 51: 1417-1423 (1973)
Depositor:	obtained from DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells growing in monolayers
Medium:	85-90% McCoy's 5A + 10-15% FBS
Subculture:	split confluent culture 1 x per week at up to 1:5 using trypsin/EDTA; seed out directly in a 80 cm ² culture flask, cells will reach confluency within 5-8 days; cells may clump after trypsinization and may be difficult to disperse
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 36 hours
Harvest:	cell harvest of about 5-10 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1-2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin(+), desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human near diploid karyotype with tetraploid sideline (31%) and 3.5% polyploidy; 46(41-46)<2n>XX; no consistent chromosome abnormality detected
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	A-427
Cell Type:	human lung carcinoma
DSMZ No:	ACC 234
Origin:	established from the lung carcinoma of a 52-year-old Caucasian man
References:	Giard et al., J. Natl. Cancer Inst. 51: 1417-1423 (1973); Pulciani et al., Nature 300: 539-542 (1982)
Depositor:	obtained from DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells growing in monolayers
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	split confluent culture 1:3 every 4-5 days using trypsin/EDTA; seed out at ca. $2-3 \times 10^6$ cells/80 cm ² ; after thawing, recovery may be slow and adherence may be reduced
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 60-70 hours
Harvest:	cell harvest of about 25×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2.5-3.0 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human near triploid karyotype with 2% polyploidy; 67(59-72)<3n>XXY, +3, +7, -13, -14, -15, -18, -19, -20, -22, +4mar, add(1)(p11), del(1)(q11), del(3)(p11)x2, der(3)t(1;3)(q21;q11), add(4)(q31), del(6)(q21), del(7)(p11), i(10q), add(11)(q24), add(14)(p11), add(15)(p13), add(18)(q23); 3p- deletion is characteristic of lung carcinoma; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	A-431
Cell Type:	human epidermoid carcinoma
DSMZ No:	ACC 91
Origin:	established from the solid tumor of an 85-year-old woman; cells were reported to have large numbers of EGF binding sites; cell line was used as indicator cell line for anti-TGF binding
References:	Giard et al., J. Natl. Cancer Inst. 51: 1417-1423 (1973)
Depositor:	Dr. K. Welte, Medical School of Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like, adherent cell line growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent cultures 1:4 to 1:6 every 3-5 days using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 80-100 hours
Harvest:	about $7-12 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-1.5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, PEP B
Cytogenetics:	human hypotriploid karyotype with 24% polyploidy; 64(58-64)<3n>XX/XXX, -2, +3, +7, -8, +10, -11, -13, -14, -15, -16, -18, -22, +2mar; del(3)(p11)x1-2, add(4)(p15), add(6)(p21.3), del(7)(p15), der(7)t(7;11)(p11/13;q11/13)add(11)(q23), add(8)(p21), add(11)(q23), der(14)t(13;14)(q11;p11), add(16)(p13); carries t(7;11) with breakpoint at EGFR (which it overexpresses); corresponds to published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	A-498
Cell Type:	human kidney carcinoma
DSMZ No:	ACC 55
Origin:	established from the kidney carcinoma of a 52-year-old man in 1973
References:	Giard et al., J. Natl. Cancer Inst. 51: 1417-1423 (1973)
Depositor:	Dr. H. Kirchner, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like, adherent cells growing as monolayers
Medium:	90% MEM (with Earle's salts) + 10% FBS
Subculture:	split confluent cultures 1:3 to 1:5 every 3-5 days using trypsin/EDTA; seed out at ca. 1.0×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of > 60 hours
Harvest:	about $2-3 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypertriploid karyotype with 7% polyploidy; 72(68-72)<3n>XX, -X, +2, -3, -6, +7, +7, +9, -11, -14, +17, -18, -20, +21, +3mar, dm/r (31%); der(1)t(1;1)(p32;q12), der(2)t(2;3)(q37;q13), i(8q); sideline with der(3)t(3;3)(q27;p14)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	A-549
Cell Type:	human lung carcinoma
DSMZ No:	ACC 107
Origin:	established from an explanted lung tumor which was removed from a 58-year-old Caucasian man in 1972; cells were described to induce tumors in athymic mice and to synthesize lecithin
References:	Giard et al., J. Natl. Cancer Inst. 51: 1417-1423 (1973); Lieber et al., Int. J. Cancer 17: 62-70 (1976)
Depositor:	Dr. J. Wehland, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial cells, growing adherently as monolayer
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent cultures 1:5 to 1:10 every 4-7 days using trypsin/EDTA; cells grow easily; seed out at ca. $1-2 \times 10^6$ cells/80 cm ² + 10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	about 20×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human hypotriploid karyotype with 8% polyploidy; 65(59-66)<3n>XXY, -1, -3, -6, +12, -13, -15, -18, -19, -21, -22, +4mar, der(6)t(1;6)(q11;q27), del(11)(q22-23); presence of large distinctive der(6) marker confirms identity of this cell line
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	A-S-30D
Cell Type:	rat hepatoma
DSMZ No:	ACC 208
Origin:	established from the hepatoma of a 16-month-old SD-female rat in 1988
References:	Holstege et al., Verh. Dt. Ges. Inn. Med. 85: 1296-1300 (1979); Church et al., Biochem. Cell Biol. 66: 1-12 (1988)
Depositor:	obtained from DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	grows as single cells in suspension
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	optimal split ratio of 1:2 to 1:3 twice per week
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 29 hours
Harvest:	cell harvest of about 0.6 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Fingerprinting:	unique DNA profile with (gtg) ₅ multilocus probe
Species:	confirmed as rat with IEF of MDH, NP
Cytogenetics:	hypodiploid rat karyotype with 12% tetraploidy; 38(35-41)<2n>; 2 large submetacentric and 1 large metacentric centric fusion markers; sideline with 30-32/60-62 present at 14%
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	A4-1025
Cell Type:	mouse hybridoma (anti-fiber/myosin)
DSMZ No:	ACC 296
Origin:	described as secreting a mouse monoclonal antibody (IgG ₂ a) detecting all fibers in skeletal muscle and myosin heavy chains on Western blots and detecting mammalian, chicken, zebrafish, axolotl, xenopus skeletal muscle myosin heavy chains; the cell line was established in 1985 by fusion of X63AG8.653 cells with spleen cells from a BALB/c mouse that had been immunized with human skeletal muscle myosin
References:	Dan-Goor et al., J. Muscle Res. Cell Motility 11: 216-226 (1990)
Depositor:	Dr. S. M. Hughes, King's College, London, England
DSMZ Cell Culture Data:	
Morphology:	round single cells in suspension, partly in clusters, lightly adherent
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + 2 mM L-glutamine + 1 mM sodium pyruvate + 1x non-essential amino acids
Subculture:	split saturated culture 1:5 to 1:10 every 2-3 days; seed out at about 1.0-1.5 x 10 ⁵ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 40-50 hours
Harvest:	cell harvest of about 0.5-1.0 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgG ₂ a monoclonal antibody
Species:	confirmed as mouse with IEF of AST, NP
Cytogenetics:	murine hypertetraploid karyotype with 5% polyploidy; 84(79-84); one large submetacentric and up to 5 centric fusion markers present
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	A4-1077
Cell Type:	mouse hybridoma (anti-fiber/myosin)
DSMZ No:	ACC 297
Origin:	described as secreting a mouse monoclonal antibody (IgG ₁) reacting with human nascent secondary fibers and adult fast fibers and myosin heavy chains in Western blots (does not detect rat); the cell line was established in 1985 by fusion of X63AG8.653 cells with spleen cells from a BALB/c mouse that had been immunized with human skeletal muscle myosin
References:	Cho et al., J. Cell Sci. 107: 2361-2371 (1994)
Depositor:	Dr. S. M. Hughes, King's College, London, England
DSMZ Cell Culture Data:	
Morphology:	single round cells, lightly adherent
Medium:	85% Dulbecco's MEM (4.5 g/L glucose) + 15% FBS + 2 mM L-glutamine + 1 mM sodium pyruvate + 1x non-essential amino acids
Subculture:	split saturated culture 1:2 to 1:3 every 2 days; optimal density of about 0.5 x 10 ⁶ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 20-40 hours
Harvest:	maximal density of about 0.6-1.0 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgG ₁ monoclonal antibody
Species:	confirmed as mouse with IEF of AST, MDH
Cytogenetics:	murine flat-modulated hypertetraploid/pentaploid karyotype with 10% polyploidy; 90(81-96)<4n>; 2-3 centric fusion markers and 1-2 dmin present in most cells
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	A4-840
Cell Type:	mouse hybridoma (anti-fiber/myosin)
DSMZ No:	ACC 213
Origin:	described as secreting a mouse monoclonal antibody (IgM) reacting with slow muscle fibers in rodent, chicken and human and with slow myosin heavy chain on Western blots; the cell line was established in 1985 by fusion of X63AG8.653 cells with spleen cells from a BALB/c mouse that had been immunized with human skeletal muscle myosin
References:	Hughes et al., Develop. Biol. 158: 183-199 (1993)
Depositor:	Dr. S. M. Hughes, King's College, London, England
DSMZ Cell Culture Data:	
Morphology:	single round cells, lightly adherent
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + 2 mM L-glutamine + 1 mM sodium pyruvate + 1x non-essential amino acids
Subculture:	split saturated culture 1:5 every 3 days; seed out at about 0.3×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 25 hours
Harvest:	maximal density of about 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgM monoclonal antibody
Species:	confirmed as mouse with IEF of AST, NP
Cytogenetics:	murine karyotype
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	A4-951
Cell Type:	mouse hybridoma (anti-fiber/myosin)
DSMZ No:	ACC 214
Origin:	described as secreting a mouse monoclonal antibody (IgG ₁) reacting with adult slow fibers in rat and human and with slow myosin heavy chain on Western blots; the cell line was established in 1985 by fusion of X63AG8.653 cells with spleen cells from a BALB/c mouse that had been immunized with human skeletal muscle myosin
References:	Hughes et al., Dev. Biol. 158: 183-199 (1993)
Depositor:	Dr. S. M. Hughes, King's College, London, England
DSMZ Cell Culture Data:	
Morphology:	round cells, lightly adherent, growing singly and in clumps
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + 2 mM L-glutamine + 1 mM sodium pyruvate + 1x non-essential amino acids
Subculture:	split saturated culture 1:2 to 1:3 every 2-3 days; seed out at ca. 0.2 x 10 ⁶ cells/ml; optimal density of about 0.2 x 10 ⁶ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 24 hours
Harvest:	maximal density of about 0.8 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3-5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgG ₁ monoclonal antibody
Species:	confirmed as mouse with IEF of AST, NP
Cytogenetics:	murine flat-modulated hypertetraploid karyotype; 81(42-92)<4n>; 1-2 dmin and 1-2 centric fusion markers present in most cells
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	AC-1M32
Cell Type:	human choriocarcinoma-trophoblast hybrid
DSMZ No:	ACC 442
Origin:	one of five phenotypically and cytogenetically distinct subclones of a hybrid (ACH1P, DSM ACC 481) formed by fusing an HGPRT-defective mutant (AC1-1) of JEG-3 human choriocarcinoma (DSM ACC 463) with primary human trophoblast (chorion laeve) cells
References:	Frank et al., Cancer Genet. Cytogenet. 116: 16-22 (2000); King et al., Placenta 21, Suppl. A, Trophoblast Res. 14: 113-119 (2000)
Depositor:	Dr. H. G. Frank, Technical University of Aachen, Aachen, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently in monolayers
Medium:	90% Ham's F12 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:6 every 3-5 days using trypsin/EDTA; seed out at ca. $0.1-0.2 \times 10^5$ cells/cm ² ; maintain at ca. $0.1-1.3 \times 10^5$ cells/cm ² ; reaches max. density of ca. 1.8×10^5 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 20-30 hours
Harvest:	cell harvest of ca. $1.0-1.5 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP+, neurofilament-, vimentin-
Fingerprinting:	near identical DNA profiles shared by the AC-sister cell lines, partly identical with JEG-3 and BEWO
Cytogenetics:	human flat-moded hypertetraploid karyotype with 4% polyploidy; 80-104<4n>XXXY, +1, +2, +3, -4, +7, +9, -13, -13, -15, +16, +16, -18, -20, +21, +22, +5-7mar, del(X)(p11p22), der(X)(Xpter->Xq27.8::4p1?->4p1?::11q1?1->11qter), del(1)(p11), der(7)t(7;7)(p22;q11.22)add(7)(q32), add(7)(q32), add(9)(p24), add(11)(p15), i(13q), add(15)(p13), add(19)(p13); sideline with: idem, +7, +16, -18, del(1)(p11), +del(1)(p22); resembles published karyotype and those of sisters with minor, though consistent, differences (mirroring DNA profiles)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	AC-1M46
Cell Type:	human choriocarcinoma-trophoblast hybrid
DSMZ No:	ACC 454
Origin:	subclone of ACH1P; ACH1P was established by somatic hybridization of primary male trophoblast cells and the HGPRT-defective choriocarcinoma cell line AC1-1 which was originally derived from cell line JEG-3 (DSM ACC 463)
References:	Frank et al., Cancer Genet. Cytogenet. 116: 16-22 (2000); King et al., Placenta 21, Suppl. A, Trophoblast Res. 14: 113-119 (2000)
Depositor:	Dr. H. G. Frank, Technical University of Aachen, Aachen, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing as monolayer and carrying many intracellular dark granula
Medium:	90% Ham's F12 + 10% FBS
Subculture:	split confluent culture 1:3 every 3-4 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ² ; maintain at ca. 3×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	cell harvest of ca. 10×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	near identical DNA profiles shared by the AC-sister cell lines, partly identical with JEG-3 and BEWO
Cytogenetics:	human flat-modulated hypertetraploid karyotype with 4% polyploidy; 84-107<4n>XXXY, -4, +5, +7, -8, +9, -15, -13, -13, -15, +16, -18, -20, +21, +22, +4-6mar, del(X)(p11p22), der(X)(Xpter->Xq2?8::4p1? ->4p1?::11q1?1->11qter), der(7)t(7;7)(p22;q11.22)add(7)(q32), add(7)(q32), add(9)(p24), add(11)(p15), i(13q), add(15)(p13), add(19)(p13), der(19)t(11;19)(q11;q13); sideline with: idem, add(8)(q22), +der(7)t(7;7)(p22;q11.22)add(7)(q32); resembles published karyotype and those of sisters with minor, though consistent, differences (mirroring DNA profiles); the DNA fingerprint finding of 3 alleles at loci D2S44 and D17S5 suggests additional human chromosomes
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	AC-1M59
Cell Type:	human choriocarcinoma-trophoblast hybrid
DSMZ No:	ACC 455
Origin:	subclone of ACH1P; ACH1P was established by somatic hybridization of primary male trophoblast cells and the HGPRT-defective choriocarcinoma cell line AC1-1 which was originally derived from cell line JEG-3 (DSM ACC 463); in contrast to the respective sister cell lines, AC-1M59 was described as expressing ERK5, cytokeratin I, TNF-R2, VEGFR-2 and IL-2
References:	Frank et al., Cancer Genet. Cytogenet. 116: 16-22 (2000)
Depositor:	Dr. H. G. Frank, Technical University of Aachen, Aachen, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing as monolayer and carrying many intracellular dark granula
Medium:	90% Ham's F12 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:4 every 3 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ² ; maintain at ca. 3×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of ca. 5×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	near identical DNA profiles shared by the AC-sister cell lines, partly identical with JEG-3 and BEWO
Cytogenetics:	human flat-modulated hypertetraploid karyotype with 8% polyploidy; 86-97<4n>XXXY, -4, +7, -8, +9, -13, -13, -18, +19, +2-5mar, del(X)(p11p22), der(X)(Xpter->Xq2?8::4p1?->4p1?::11q1?1->11qter), der(7)t(7;7)(p22;q11.22)add(7)(q32), add(7)(q32), add(9)(p24), add(11)(p15), i(13q), add(15)(p13), add(19)(p13)x1-2; resembles published karyotype and those of sisters with minor, though consistent, differences (mirroring DNA profiles); the DNA fingerprint finding of 3 alleles at loci D2S44 and D17S5 suggests additional human chromosomes
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	AC-1M81
Cell Type:	human choriocarcinoma-trophoblast hybrid
DSMZ No:	ACC 456
Origin:	subclone of ACH1P which was established by fusion of the JEG-3-derived HGPRT-defective mutant cell line AC1-1 with primary male trophoblast cells isolated from term human chorion laeve; described as possessing both, a distinct karyotype and a distinct expression pattern of tumor-related genes
References:	Frank et al., Cancer Genet. Cytogenet. 116: 16-22 (2000)
Depositor:	Dr. H. G. Frank, Technical University of Aachen, Aachen, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells growing in monolayers
Medium:	90% Ham's F12 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:5 twice a week using trypsin/EDTA; seed out at ca. $2-3 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	cell harvest of ca. 0.4×10^6 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with sparflloxacin, then negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	near identical DNA profiles shared by the AC-sister cell lines, partly identical with JEG-3 and BEWO
Cytogenetics:	human flat-moded hypertetraploid karyotype with 4% polyploidy; 80-106<4n>XX, -Y, -3, -4, +7, +9, -13, -18, +22, +4-6mar, del(X)(p11p22), der(X)(Xpter->Xq2?8::4p1?->4p1?::11q1?1->11qter), der(7)t(7;7)(p22;q11.22)add(7)(q32), add(7)(q32)x2, i(13q), add(15)(p13), add(19)(p13), der(19)t(11;19)(q11;q13); resembles published karyotype and those of sisters with minor, though consistent, differences (mirroring DNA profiles)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	AC-1M88
Cell Type:	human choriocarcinoma-trophoblast hybrid
DSMZ No:	ACC 457
Origin:	one of five phenotypically and cytogenetically distinct subclones of a hybrid (ACH1P, DSM ACC 481) formed by fusing an HGPRT-defective mutant (AC1-1) of JEG-3 human choriocarcinoma (DSM ACC 463) with primary human trophoblast (chorion laeve) cells
References:	Frank et al., Cancer Genet. Cytogenet. 116: 16-22 (2000); King et al., Placenta 21, Suppl. A, Trophoblast Res. 14: 113-119 (2000)
Depositor:	Dr. H. G. Frank, Technical University of Aachen, Aachen, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently in monolayers
Medium:	90% Ham's F12 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:6 every 3-5 days using trypsin/EDTA; seed out at ca. $0.1-0.2 \times 10^5$ cells/cm ² ; maintain at ca. $0.1-0.5 \times 10^5$ cells/cm ² ; reaches max. density of ca. 1.3×10^5 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-50 hours
Harvest:	cell harvest of ca. 0.9×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	near identical DNA profiles shared by the AC-sister cell lines, partly identical with JEG-3 and BEWO
Cytogenetics:	human flat-moded hypertetraploid karyotype with 11% polyploidy; 80-99<4n>XXX, -Y, -3, +7, +7, -8, +9, +9, +12, -13, +3-5mar, del(X)(p11p22), der(7)t(7;7)(p22;q11.22)add(7)(q32)x2, add(7)(q32), add(9)(p24), i(13q), add(19)(p13), der(19)t(11;19)(q11;q13); sideline with idem, +1, +7, +12, der(4)t(4;11)(p14-16;q1?3); resembles published karyotype and those of sisters with minor, though consistent, differences (mirroring DNA profiles)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	ACH1P
Cell Type:	human choriocarcinoma-trophoblast hybrid
DSMZ No:	ACC 481
Origin:	established by somatic hybridization of primary male trophoblast cells and the HGPRT-defective choriocarcinoma cell line AC1-1 which was originally derived from cell line JEG-3 (DSM ACC 463)
References:	Frank et al., Cancer Genet. Cytogenet. 116: 16-22 (2000)
Depositor:	Dr. H. G. Frank, Technical University of Aachen, Aachen, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing as monolayers, cells carry a large number of dark granula
Medium:	90% Ham's F12 + 10% FBS
Subculture:	split confluent culture 1:4 every 4-5 days week using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	cell harvest of ca. 8×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with ciprofloxacin, then negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Cytogenetics:	human flat-modulated hypertetraploid karyotype with 10% polyploidy; 85-104<4n>XXY/XXYY/XXXY, +2, +5, +7, -10, -11, +12, -13, +16, -18, +21, +1-4 mar, add(7)(q32), i(13q), add(15)(p11) add(19)(p13)x1-2; sideline with del(X)(p11p22), der(X)(Xpter->Xq278::4p1? ->4p1?::11q1?1->11qter), der(4)t(4;11)(p14-16;q1?3), der(7)t(7;7)(p22;q11.22), add(11)(p15); karyotype closely resembles AC-1M32/46/59/81/88 (daughters) and JEG-3 (parent)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	ALL-SIL
Cell Type:	human T cell leukemia
DSMZ No:	ACC 511
Origin:	established from the peripheral blood of a 17-year-old man with T-ALL (T cell acute lymphoblastic leukemia) at relapse; also found in the literature as SIL-ALL
References:	Drexler, The Leukemia-Lymphoma Cell Line FactsBook, Academic Press, 2000
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	mostly round cells growing singly or in small clusters in suspension
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. $0.5-1.5 \times 10^6$ cells/ml; maximal density of ca. 1.5×10^6 cells/ml; it may be of advantage to first culture the cells after thawing in a 24-well-plate with 20% FBS
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of about 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with ciprofloxacin (Ciprobay), then negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3-, cyCD3+, CD4+, CD5+, CD6+, CD7+, CD8+, CD13-, CD19-, CD34-, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypertetraploid karyotype; 90-95<4n>XX/XXYY, +6, +8, +8, t(1;13)(p32;q32)x2, del(6)(q25)x2, del(9)(p23p24)x2, t(10;14)(q24;q11.2)x2, add(17)(p11)x2; sideline with idem, -20, -20; carries t(10;14) effecting HOX11-TCRD juxtaposition; matches published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	AM-C6SC8
Cell Type:	porcine normal kidney cells
DSMZ No:	ACC 152
Origin:	derived as a subclone from the cell line AM II which was established from a normal porcine kidney in 1983; cells have been used in assays to test toxins and mutagens
References:	Milczewski, Mycotoxin Res. 3: 69-76 (1987)
Depositor:	Dr. K. E. von Milczewski, Bundesanstalt für Milchforschung, Institut für Mikrobiologie, Kiel, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing in monolayers
Medium:	80-90% Medium 199 (Hank's salts) + 10-20% FBS
Subculture:	cells are difficult to culture and grow slowly to very slowly! split confluent culture 1:5 to 1:10 about once a week using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Harvest:	cell harvest of about 8 x 10 ⁶ cells/80 cm ² or 1 x 10 ⁷ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Viruses:	ELISA: reverse transcriptase positive

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	AML-193
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 549
Origin:	established from a 13-year-old girl with acute myeloid leukemia (AML FAB M5) at relapse
References:	Lange et al., Blood 70: 192-199 (1987); Drexler et al., Leukemia 11: 701-708 (1997)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round to polymorph cells growing singly or in small clusters in suspension
Medium:	70% Iscove's MDM + 10% FBS + 20% vol conditioned medium of cell line 5637 (DSM ACC 35)
Subculture:	split saturated culture 1:2 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. $0.2-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	cell harvest of ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD4+, CD13-, CD14+, CD15+, CD19-, CD33+, CD34-, HLA-DR-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype with 4% polyploidy; 49(46-52)<2n>XX, +3, +6, +8, +13, i(17q); matches published karyotype
Viruses:	PCR: EBV-, HBV, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	AMO-1
Cell Type:	human plasmacytoma
DSMZ No:	ACC 538
Origin:	established from the ascitic fluid of a 64-year-old woman with plasmacytoma (IgAkappa) of the duodenum two months following resection of the tumor mass (additional therapy not specified) in 1984; plasmacytoma is related to multiple myeloma
References:	Shimizu et al., Leukemia 7: 274-280 (1993); Drexler & Matsuo, Leukemia Res. 24: 681-703 (2000)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2-4 days; seed out at ca. 0.8×10^6 cells/ml; maintain at ca. $0.5-1.2 \times 10^6$ cells/ml; minimum density at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-90 hours
Harvest:	maximum cell density of ca. 1.8×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $6-8 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in microbiological culture, PCR assays
Immunology:	CD3-, CD4+, CD10-, CD13-, CD19-, CD20-, CD34-, CD37-, CD38+, CD79a-, cyCD79a+, CD80-, CD138+, HLA-DR+, sm/cyIgG-, sm/cyIgM-, smkappa-, cykappa+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	AN3-CA
Cell Type:	human endometrial adenocarcinoma
DSMZ No:	ACC 267
Origin:	established in 1962 from the lymph node metastasis resected from a 55-year-old white woman with an endometrial adenocarcinoma (epithelial undifferentiated small cell type) presenting with acanthosis nigricans; forms heterotransplantable tumors
References:	Dawe et al., J. Natl. Cancer Inst. 33: 441-456 (1964)
Depositor:	obtained from DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial cells growing as adherent monolayer; prominent cytoplasmic vacuoles
Medium:	90% MEM (with Earle's salts) + 10% FBS + 1 mM sodium pyruvate and non-essential amino acids
Subculture:	split confluent culture 1:3 to 1:8 once or twice weekly using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 55 hours
Harvest:	cell harvest of about 5 x 10 ⁴ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1.0-1.5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hyperdiploid karyotype with 14% polyploidy; 49(45-50)<2n>XX, +5, +6, +7, der(3)t(1;3)(q24;p22)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	ARH-77
Cell Type:	human plasma cell leukemia
DSMZ No:	ACC 512
Origin:	established from the peripheral blood of a 33-year-old woman with IgG plasma cell leukemia; cell line is widely used as model for multiple myeloma/plasma cell leukemia, but cells are clearly EBV+ indicating that ARH-77 might be rather a "normal" B-lymphoblastoid cell line than a tumor cell line
References:	Drewinko et al., Cancer 54: 1883-1892 + 1893-1903 (1984); Pellat-Deceunynk et al., Blood 86: 4001-4002 (1995); Drexler & Matsuo, Leukemia Res. 24: 681-703 (2000)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round single cells or clustered growing in suspension, some cells loosely adherent
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:2 to 1:5 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml; maintain at $0.2-1.5 \times 10^6$ cells/ml; adherent cells can be detached by tapping the flask
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	saturation density at ca. $1.5-2.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with enrofloxacin (Baytril), then negative in microbiological culture, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD34-, CD37+, CD71+, CD79a+, cyCD79a+, CD80+, CD138-, HLA-DR+, sm/cyIgG+, sm/cyIgM-, sm/cykappa+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human near diploid karyotype with 8% polyploidy; 46(44-48)<2n>XX, +9, del(1)(q23), add(2)(q21), add(3)(p11), der(3)t(2;3)(q23;q26), del(6)(p21), der(9)t(9;17)(q10;q10); sideline with der(X)t(X;1)(q23;p32), del(16)(p13.2) and unresolved der(3) and der(9)hsr markers; no IGH translocations detected
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	B-16V
Cell Type:	mouse melanoma
DSMZ No:	ACC 370
Origin:	established from a B16 melanoma tumor grown in syngeneic C57BL/6 mice
References:	Supino et al., Brit.J. Cancer 54: 33-42 (1986)
Depositor:	Dr. R. Supino, Istituto Nazionale Tumori, Milano, Italia
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial cells growing in monolayers, but not reaching complete confluence (ca 80%)
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split semi-confluent culture 1:7 about every 3 days using trypsin/EDTA; seed out at ca. $1.5\text{-}2.0 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-50 hours
Harvest:	cell harvest of ca. 35×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1\text{-}2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, LDH, NP
Cytogenetics:	murine karyotype
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	B-CPAP
Cell Type:	human thyroid carcinoma
DSMZ No:	ACC 273
Origin:	established from the tumor tissue of a 76-year-old woman with metastasizing papillary thyroid carcinoma in 1992
References:	Fabien et al., Cancer 73: 2206-2212 (1994)
Depositor:	Dr. C. Paulin, Medical School, Lyon, France
DSMZ Cell Culture Data:	
Morphology:	spindle- or round-shaped cells growing adherently as monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:4 to 1:6 every 3-4 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30 hours
Harvest:	about 10×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human near-diploid karyotype with hexaploid sideline; 70/71(69-72)<3n>XX/XXX, +5, -7, -8, +11, +12, -13, +15, +19, -22, +mar; i(1p), 1qh+, t(5;5)(p13;p15), add(10)(p11), add(12)(p11), del(17)(p11), del(17)(p11)dup(17)(q11q23), add(19)(q13); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	B9
Cell Type:	mouse hybridoma
DSMZ No:	ACC 211
Origin:	cells are a subclone derived from B13.29 (first described as clone 9.9) by fusion of BALB/c spleen cells with SP2/0-AG14 myeloma cells; this hybridoma is a non-producer; cells are IL6-dependent
References:	Aarden et al., Eur. J. Immunol. 17: 1411-1416 (1987)
Depositor:	Dr. L. Aarden, Netherlands Red Cross Blood Transfusion Service, Amsterdam, The Netherlands
DSMZ Cell Culture Data:	
Morphology:	round cells growing in suspension
Medium:	90% RPMI 1640 + 10% FBS + 50 : M mercaptoethanol + 10-100 pg/ml human IL-6
Subculture:	split saturated culture 1:3 to 1:4 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml, best in 24- or 6-well plate. CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	maximal density at about 0.8×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	bioassay for IL-6 (low cell densities are essential for reliable assays; try doubling cell dilutions beginning with cell densities of 2.5×10^4 cells/ml; duration of assay is commonly 2-3 days)
Species:	confirmed as mouse with IEF of AST, PEP B
Cytogenetics:	apparently murine chromosome morphology with extensive centric fusions
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	B95-8
Cell Type:	marmoset monkey lymphocytes
DSMZ No:	ACC 100
Origin:	established from peripheral blood lymphocytes of a marmoset monkey (<i>saguinus oedipus</i>); cells were described to release high titers of Epstein-Barr virus (EBV) providing a source for transformation of B cells and establishment of continuous cell lines; this cell line has been assigned by the ZKBS (German Central Commission for Biological Safety) to risk group 2
References:	Miller & Lipman, Proc. Natl. Acad. Sci. USA 70: 190-194 (1973)
Depositor:	Prof. Dr. M. Hadam, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round to oval cells or clustered in small clumps in suspension and partially adherent; ca. 10% adhere, but trypsin is not necessary to detach the cells
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5\text{--}1.0 \times 10^6$ cells/ml; split 1:2 to 1:5 every 72 hours; minimal density at 0.2×10^6 cells/ml; adherent cells might be detached by tapping the flask; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 41 hours
Harvest:	maximal density at about 3×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Fingerprinting:	unique Hinf I-(gtg) ₅ DNA profile
Species:	IEF analysis with LDH and NP confirmed the cells as non-human and non-murine
Cytogenetics:	near-diploid simian karyotype with 4% polyploidy; 44(42-47)<2n>; chromosome rearrangements present
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BA-D5
Cell Type:	mouse hybridoma (anti-rat fiber/myosin)
DSMZ No:	ACC 75
Origin:	this hybridoma cell line was described to secrete a mouse monoclonal antibody (IgG) that reacts with rat type 1 fibers in immunofluorescence and with rat type 1 MHCs (myosin heavy chain) in Western blotting; the cell line was established in 1986 by fusion of NSO myeloma cells with spleen cells from a BALB/c mouse that had been immunized with purified myosin from bovine atrium
References:	Schiaffino et al., J. Muscle Res. Cell Motil. 10: 197-205 (1989)
Depositor:	Prof. S. Schiaffino, University of Padova, Padova, Italy
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension (some cells adhere partly, but trypsin is not necessary to detach these cells)
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS (or horse serum) + 2 mM L-glutamine
Subculture:	split saturated culture at 1:5 every 2-3 days; optimal density at about 0.5×10^6 cells/ml; after thawing, culture first with 20% FBS, viability may drop to as low as 50-60%
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 25 hours
Harvest:	cell harvest of about 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgG ₂ b monoclonal antibody
Species:	confirmed as mouse with IEF of AST, NP, PEP B
Cytogenetics:	murine hypertetraploid/pentaploid karyotype with 10% polyploidy; 90(88-100)<4n>; 1-3 centric fusion markers and 2 dmin present in most cells
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BA-F8
Cell Type:	mouse hybridoma (anti-rat fiber/myosin)
DSMZ No:	ACC 154
Origin:	this hybridoma cell line was described to secrete a mouse monoclonal antibody (IgG ₂ b) that reacts with rat type 1 fibers in immunofluorescence and with rat type 1 MHCs (myosin heavy chain) in Western blotting; the cell line was established in 1986 by fusion of NSO myeloma cells with spleen cells from a BALB/c mouse that had been immunized with purified myosin from bovine atrium
References:	Borrione et al., Differentiation 38: 49-59 (1988)
Depositor:	Prof. S. Schiaffino, University of Padova, Padova, Italy
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension (some cells adhere partly, but trypsin is not necessary to detach these cells)
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% horse serum + 2 mM L-glutamine
Subculture:	split saturated culture at 1:3 to 1:4 every 2-3 days
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	maximal density at about 0.8 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3-5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgG ₂ monoclonal antibody
Species:	confirmed as mouse with IEF of AST, NP
Cytogenetics:	murine hypertetraploid/pentaploid karyotype; 92(89-99)<4n>; 1-3 centric fusion markers and 1-3 dmin present in most cells
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BA/F3
Cell Type:	mouse pro B cells
DSMZ No:	ACC 300
Origin:	IL-3 dependent murine pro B cell line established from peripheral blood; apparently derived from BALB/c mouse
References:	direct publication is not available; related publications: Palacios et al., Nature 309: 126-131 (1984); Palacios et al., Cell 41: 727-734 (1985)
Depositor:	obtained from RIKEN Cell Bank (RCB0805), Tsubuka Science City, Ibaraki, Japan
DSMZ Cell Culture Data:	
Morphology:	mostly single, round (some polymorph) cells in suspension (or occasionally in clumps)
Medium:	90% RPMI 1640 + 10% FBS + 10 ng/ml IL-3 or 10% vol conditioned medium of WEHI-3B cell line (DSM ACC 26)
Subculture:	split saturated culture 1:10 every 3 days; seed out at about $1-3 \times 10^5$ cells/ml; at high cell density ($>2 \times 10^6$ cells/ml), a cytokine-independent subclone may grow out relatively quickly. CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 20 hours
Harvest:	maximal density of about 2.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, MDH, NP, PEP B
Cytogenetics:	murine near diploid karyotype with 33% polyploidy; 40(36-42)<2n>
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BAG-12G2
Cell Type:	mouse hybridoma (CDw78)
DSMZ No:	ACC 470
Origin:	described as secreting a mouse monoclonal antibody (IgG ₂ akappa) CDw78; the cell line was established in 1988 by fusion of PX63AG8.653 cells with spleen cells of a mouse immunized with human peripheral blood lymphocytes from a patient with high numbers of gamma-delta T cells
References:	Hadam, Leucocyte Typing VI: 178-179 (1997)
Depositor:	Dr. M. R. Hadam, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	single round cells growing in suspension (some cells adhere partly)
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement")
Subculture:	split saturated cultures 1:2 to 1:4 every 1-3 days; seed out at 0.25 x 10 ⁶ cell/ml; maintain at 0.1-0.8 x 10 ⁶ cells/ml; cells may grow slowly after initial recovery from N ₂
Incubation:	at 37/C with 5-10% CO ₂
Doubling Time:	doubling time of about 24 hours
Harvest:	maximal density of about 0.8 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgG ₂ a monoclonal antibody
Viruses:	ELISA: reverse transcriptase negative

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BAG-85D10
Cell Type:	mouse hybridoma (CD31)
DSMZ No:	ACC 471
Origin:	described as secreting a mouse monoclonal antibody (IgG ₁) CD31; the cell line was established in 1988 by fusion of PX63AG8.653 cells with spleen cells of a mouse immunized with human peripheral blood lymphocytes from a patient with high numbers of gamma-delta T cells
References:	not published
Depositor:	Dr. M. R. Hadam, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells, growing singly and in small clumps
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement" for the first 1-2 weeks post-thawing)
Subculture:	split saturated culture 1:2 every 3 days; seed out at 0.2×10^6 cells/ml; maintain at $0.2\text{--}0.8 \times 10^6$ cells/ml
Incubation:	at 37°C with 5-10% CO ₂
Doubling Time:	doubling time of about 72 hrs
Harvest:	maximal density at about 0.8×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgG ₁ monoclonal antibody
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BC-3C
Cell Type:	human urinary bladder transitional cell carcinoma
DSMZ No:	ACC 450
Origin:	derived from the invasive solid transitional cell carcinoma of the bladder of a 82-year-old Caucasian woman (grade IV, stage C); cell growth described as being inhibited by EGF
References:	Pratsinis et al., In Vitro Cell Dev. Biol. 34: 722-728 (1998)
Depositor:	Prof. P. Davaris, Medical School of Athens, Athens, Greece
DSMZ Cell Culture Data:	
Morphology:	epitheloid adherent cells growing adherently in monolayers and in multilayer foci
Medium:	90% McCoy's 5A + 10% FBS
Subculture:	split confluent culture 1:10 every 3-7 days using trypsin/EDTA; seed out at ca. $0.25-0.4 \times 10^5$ cells/cm ² ; recover and initially culture cells in 20% FBS; maximum density of ca. 3.5×10^5 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 25-30 hours
Harvest:	cell harvest of ca. $2-3 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Cytogenetics:	human flat-moded hypertriploid karyotype with 10% polyploidy; 65-72<3n>XX, -X, -2, -4, +9, +9, -11, -13, +14, +16, -17, -17, -18, +19, +20, +20, +2-3mar, add(2)(p1?), del(3)(q2?2), der(5;7)(q10;q10), del(9)(p2?2p2?3), add(12)(q24); sideline with del(2)(q2?), i(3q), add(7)(p22), add(8)(p12), add(11)(q22-24), i(?17p), i(17q) and extra numerical changes; matches published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BC3H1
Cell Type:	mouse brain tumor
DSMZ No:	ACC 171
Origin:	established from a mouse tumor induced by nitrosoethylurea with properties characteristic of muscle; cells were described to synthesize adenylate and creatine phosphokinases
References:	Schubert et al., J. Cell Biol. 61: 398-413 (1974)
Depositor:	Dr. J. Wehland, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, flat cells forming a confluent monolayer
Medium:	80-90% Dulbecco's MEM + 10-20 % FBS
Subculture:	split confluent culture 1:5 every week or 1:10 every two weeks (cells grow slowly) using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Harvest:	about 1-2 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 0.5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, PEP B
Cytogenetics:	murine near-tetraploid karyotype; 74-82<4n>; 1-3 centric fusion markers and 1-2 dm present in most cells; does not correspond with published karyotype
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BD-215
Cell Type:	human B lymphoblastoid cells (from patient with Fanconi anemia)
DSMZ No:	ACC 398
Origin:	B lymphoblastoid cell line (B-LCL) established from the peripheral blood of a Scottish man with Fanconi anemia (complementation group C); homozygote with unrelated non-consanguineous parents; homozygous for nonsense mutation in exon 6 of the FACC coding sequences
References:	Gibson et al., Hum. Mol. Genet. 2: 797-799 (1993)
Depositor:	Dr. N. Morgan, Guy's Hospital, London, UK
DSMZ Cell Culture Data:	
Morphology:	lymphoblastoid-like cells growing as single cells or in clumps in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:2 to 1:5 every 2-4 days; maintain at $0.4-1.0 \times 10^6$ cells/ml; seed out at ca. 0.8×10^6 cells/ml; after recovery from the frozen stock, first culture with 20% FBS
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	saturation density at ca. $2.0-2.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD34-, CD37+, CD138+, HLA-DR+, sm/cyIgM-, smIgG-, cyIgG+, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human diploid karyotype with 12% polyploidy; 46, XY, 1qh+; no consistent abnormality detected
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BE-13
Cell Type:	human T cell leukemia
DSMZ No:	ACC 396
Origin:	established from bone marrow cells of an 11-year-old girl with T cell acute lymphoblastic leukemia (T-ALL) in relapse; cell lines BE-13 and PEER (DSM ACC 6) show identical DNA fingerprints and hence share common origin, presumably due to cross-contamination; however, their diploid and tetraploid karyotypes indicative of earlier and later passage numbers, respectively, suggest that BE-13 is derived from PEER, rather than vice-versa.
References:	Galili et al., Human Lymphocyte Differentiation 1: 123-130 (1981)
Depositor:	Prof. H. Ben-Bassat, Hadassah University Hospital, Jerusalem, Israel
DSMZ Cell Culture Data:	
Morphology:	single round cells growing in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	maintain at $0.5\text{--}2.0 \times 10^6$ cells/ml; split 1:2 to 1:3 every 2-4 days; seed out at about 1×10^6 cells/ml in a 24- or 6-well plate (cells grow clearly better in 24- or 6-well plates than in flasks)
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-60 hours
Harvest:	maximal density at ca. 2.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3+, CD4+, CD5+, CD6+, CD7+, CD8-, CD13-, CD19-, CD34-, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile, but identical to DNA profile of PEER
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human flat-moded hypotetraploid karyotype; 81-89<4n>XXXX, +1, +4, -9, -9, -10, -16, -21, del(1)(q13.2), add(1)(p11-21), del(4)(q11q31.2)/i(4p)x2, i(4q)del(4q31.2)/i(4q)add(4)(q31)x2, del(5)(q14q21)x2, del(6)(q22)x2, del(9)(p22)x2, der(17)t(17;?21)(p11;q11)x2; 5q- and apparent 9p22-pter nullisomy
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BEN
Cell Type:	human lung carcinoma
DSMZ No:	ACC 254
Origin:	established from the supraclavicular tumor cells-containing lymph node of a 71-year-old man with poorly differentiated carcinoma of the lung
References:	Ellison et al., Brit. J. Cancer 32: 373-379 (1975)
Depositor:	Dr. R. Rizzoli, Hopital Cantonal Universitaire, Geneve, Switzerland
DSMZ Cell Culture Data:	
Morphology:	adherent, epitheloid cells growing in monolayers
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS
Subculture:	split confluent culture 1:3 every 3-4 days using trypsin/EDTA
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of about 40-50 hours
Harvest:	cell harvest of about 20 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human highly rearranged flat-moded near-tetraploid karyotype; 82-92<4n>X/XY, -X, -Y, -Y, -2, -2, -3, -3, -4, -4, -5, +6, +6, +7, -8, -8, -9, -10, -13, -13, -14, -16, -16, -17, -18, -18, -19, -21, -21, -21, -22, +18mar; del(1)(q22)x2, add(1)(p11), i(2q), i(4q), add(5)(p11), del(6)(q13)x3, del(6)(p21.2), der(7)(t(7;12)(p11;q11), del(7)(q21)x2, i(7p), del(8)(p21), del(9)(p12), del(10)(q24), del(11)(p11)x2, add(12)(q24), i(12q), del(?12)(q14), add(14)(p11), add(18)(q21/23), add(19)(p13)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BETA-TC-3
Cell Type:	mouse pancreas insulinoma (β -tumor)
DSMZ No:	ACC 324
Origin:	established from pancreatic insulinoma excised from a transgenic mouse carrying hybrid insulin-SV40
References:	large T genes in 1988; described as producing pro-insulin I and II processed into mature insulin (inducible 30-fold by glucose) Efrat et al., Proc. Natl. Acad. Sci. USA 85: 9037-9041 (1988); Shing et al., Science 259: 1604-1607 (1993)
Depositor:	Dr. H. Weich, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent cells growing as multilayers
Medium:	82.5% Dulbecco's MEM (4.5 g/L glucose) + 15% horse serum + 2.5% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 2-5 days by mechanical disruption (trypsin/EDTA can be used for a very short time); seed out at ca. $1-4 \times 10^6$ cells/175 cm ²
Incubation:	at 37 °C with 7.5-10% CO ₂
Doubling Time:	doubling time of ca. 34 hours
Harvest:	cell harvest of ca. $2-3 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, NP
Cytogenetics:	murine hypotetraploid karyotype; 74(70-78)<4n>; 2-4 centric fusion markers and 1-3 dmin present
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BEWO
Cell Type:	human choriocarcinoma
DSMZ No:	ACC 458
Origin:	a human choriocarcinoma sample was taken at autopsy from a cerebral metastasis; cells were initially transplanted to the cheek pouch of a hamster and maintained by serial passage for 8 years prior to growth in culture vessels
References:	Pattillo et al., Cancer Res. 28: 1231-1236 (1968); Pattillo et al., Science 159: 1467-1469 (1968); Hertz et al., Proc. Soc. Exptl. Biol. Med. 102: 77-80 (1959)
Depositor:	Dr. B. Ugele, University Hospital, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing as monolayer
Medium:	85-90% Ham's F12 + 10-15% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:6 every 3-4 days using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ² ; maintain at about 8×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 20×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	same DNA profile as JEG-3 using multiplex PCR of minisatellite markers
Cytogenetics:	human hypertriploid karyotype with 24% polyploidy; 78(77-80)<3n>XXYY, +1, +2, +5, +5, +6, +8, +12, -13, +14, -18, +19, -20, +21, +4-6mar, +0-2 dmin, add(X)(p21-22), del(X)(p11), add(1)(p36), der(1)t(1;1)(p3?6;q2?1), i(13q), add(18)(q2?), der(19)t(9;19)(q13;p13), del(22)(q12); sideline with del(3)(p12p23); resembles published karyotype and (marginally) that of JEG-3 (DSM ACC 463)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BF-32
Cell Type:	mouse hybridoma (anti-rat fiber/myosin)
DSMZ No:	ACC 272
Origin:	described as secreting a mouse monoclonal antibody (IgM) reacting with type I and IIA fibers in immunofluorescence and with type 1 and 2A myosin heavy chains (MHCs) in Western blotting; this cell line was produced by fusing NSO myeloma cells with spleen cells from a BALB/c mouse immunized with purified myosin from bovine fetal muscles
References:	Schiaffino et al., J. Muscle Res. Cell Motil. 10: 197 (1989)
Depositor:	Prof. S. Schiaffino, University of Padova, Padova, Italy
DSMZ Cell Culture Data:	
Morphology:	grows as single cells or in clumps in suspension; loosely adherent at times
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% horse serum + 2 mM L-glutamine
Subculture:	split saturated culture at 1:2 to 1:3 every 2-4 days
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of about 34 hours
Harvest:	maximal density at about 1.0 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% horse serum, 10% DMSO at about 4 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgM monoclonal antibody
Species:	confirmed as mouse with IEF of AST, LDH, MDH, NP
Cytogenetics:	murine hypotetraploid karyotype with 4.5% polyploidy; 72-80<4n>; 2-4 dms present in most cells
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BF-34
Cell Type:	mouse hybridoma (anti-rat fiber/myosin)
DSMZ No:	ACC 156
Origin:	this hybridoma cell line was described as secreting a mouse monoclonal antibody (IgG ₁) that reacts with rat neonatal fibers and, to a lower extent, with all rat adult type 2 fibers in immunofluorescence; it reacts specifically with neonatal MHCs (myosin heavy chain) in Western blotting; established in 1984 by fusion of NSO myeloma cells with spleen cells from a BALB/c mouse immunized with purified myosin from fetal bovine muscles
References:	Borriane et al., Differentiation 38: 49-59 (1988)
Depositor:	Prof. S. Schiaffino, University of Padova, Padova, Italy
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension (some cells adhere partly, but trypsin is not necessary to detach these cells)
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% horse serum + 2 mM L-glutamine
Subculture:	split saturated culture at 1:2 to 1:3 every 2-3 days
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	maximal density at about 1.5-2.0 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% horse serum, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgG ₁ monoclonal antibody
Species:	confirmed as mouse with IEF of AST, MDH, PEP B
Cytogenetics:	murine chromosome morphology
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BF-45
Cell Type:	mouse hybridoma (anti-rat fiber/myosin)
DSMZ No:	ACC 221
Origin:	described as secreting a mouse monoclonal antibody (IgG ₁) that reacts with rat embryonic fibers in immunofluorescence and with embryonic myosin heavy chains in Western blotting; established by fusion of NSO myeloma cells with spleen cells from a BALB/c mouse immunized with purified myosin from fetal bovine muscles
References:	Schiaffino et al., Devel. Biol. 127: 1-11 (1988)
Depositor:	Prof. S. Schiaffino, University of Padova, Padova, Italy
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension (some cells adhere partly, but trypsin is not necessary to detach these cells)
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% horse serum + 2 mM L-glutamine
Subculture:	split saturated culture at 1:3 to 1:4 every 2 days
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 25-35 hours
Harvest:	maximal density at about 1.0-1.3 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% horse serum, 10% DMSO at about 4-5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgG ₁ monoclonal antibody
Species:	confirmed as mouse with IEF of NP, PEP B
Cytogenetics:	murine flat-modulated hypertetraploid karyotype with 5% polyploidy; 78-93<4n>; 1-2 dmin present in most cells
Viruses:	ELISA: reverse transcriptase negative

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BF-F3
Cell Type:	mouse hybridoma (anti-rat fiber/myosin)
DSMZ No:	ACC 212
Origin:	this hybridoma cell line was described as secreting a mouse monoclonal antibody (IgM) reacting with adult rat type 2B fibers in immunofluorescence and with rat type 2B MHCs (myosin heavy chain) in Western blotting; established in 1983 by fusion of NSO myeloma cells with spleen cells from a BALB/c mouse immunized with purified myosin from fetal bovine muscles
References:	Schiaffino et al., J. Muscle Res. Cell Motil. 10: 197-205 (1989)
Depositor:	Prof. S. Schiaffino, University of Padova, Padova, Italy
DSMZ Cell Culture Data:	
Morphology:	single round cells in suspension and partially adherent, but trypsin/EDTA is not necessary to detach the cells
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + 2 mM L-glutamine
Subculture:	maintain at $0.2-0.5 \times 10^6$ cells/ml, optimal split ratio of 1:2 to 1:4 every 2-4 days; seed out at ca. $5-6 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of 20-30 hours
Harvest:	maximum density at about 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 6×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgM monoclonal antibody
Species:	confirmed as mouse with IEF of AST, NP, PEP B
Cytogenetics:	murine chromosome morphology
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BF-G6
Cell Type:	mouse hybridoma (anti-rat fiber/myosin)
DSMZ No:	ACC 274
Origin:	described as secreting a mouse monoclonal antibody (IgG ₁) reacting with human and rat embryonic fibers in immunofluorescence and with human and rat embryonic myosin heavy chains (MHCs) in Western blotting; this antibody reacts more weakly also with adult rat 2B fibers in immunofluorescence and adult rat 2B MHCs in Western blotting; moreover, it stains muscle cells in the conduction tissue of newborn rat heart; this cell line was produced by fusing NSO myeloma cells with spleen cells from a BALB/c mouse immunized with purified myosin from bovine fetal muscles
References:	Schiaffino et al., Exp. Cell Res. 163: 211-220 (1986); Gorza et al., J. Mol. Cell. Cardiol. 20: 931-941 (1988); Schiaffino et al., Devel. Biol. 127: 1 (1988); Bottinelli et al., J. Physiol. 437: 655 (1991)
Depositor:	Prof. S. Schiaffino, University of Padova, Padova, Italy
DSMZ Cell Culture Data:	
Morphology:	single round cells in suspension (sometimes with loosely adherent cells, sometimes in clumps)
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% horse serum + 2 mM L-glutamine
Subculture:	optimal split ratio of 1:2 to 1:3 every 2-3 days; seed out at about 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30-40 hours
Harvest:	maximal density at about $0.5-1.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% horse serum, 10% DMSO at about $4-6 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgG ₁ monoclonal antibody
Species:	confirmed as mouse with IEF of AST, MDH, NP, PEP B
Cytogenetics:	murine hypertetraploid karyotype; $82(79-88)<4n>$; 2-3 dm present in most cells
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BFTC-905
Cell Type:	human urinary bladder transitional cell carcinoma
DSMZ No:	ACC 361
Origin:	established from a 51-year-old Chinese woman with a grade III papillary transitional cell carcinoma of the urinary bladder in 1990; patient lived in a blackfoot disease endemic area in Taiwan
References:	Tzeng et al., Anticancer Res. 16: 1797-1804 (1996); Cheng et al., Mol. Carcinogenesis 13: 173-181 (1995)
Depositor:	Dr. C. C. Tzeng, National Cheng Kung University Hospital, Tainan, Taiwan
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently in monolayers, forming large islets
Medium:	80-90% Dulbecco's MEM + 10-20% FBS
Subculture:	split culture before reaching confluency 1:3 to 1:5 once or twice a week using trypsin/EDTA (for 5-10 min); seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 60-70 hours
Harvest:	cell harvest of ca. $10-15 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS and 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays; bacterial contamination was eliminated with Baytril (enrofloxacin) and BM-Cyclin (tiamulin & minocycline)
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH
Cytogenetics:	human flat-moded hyperdiploid/hypotriploid karyotype; 58(52-61)<3n>X/XX-X, -4, +5, -9, -13, -15, -16, -18, -21, -22, -22, +mar, add(1)(p36), i(5p), add(5)(q35), del(9)(p22), del(11)(p11), add(11)(q13), add(13)(p12), add(13)(p12), add(15)(p12), add(19)(p13); i(5p) and loss of 9p associated with transitional cell carcinoma; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BFTC-909
Cell Type:	human kidney transitional cell carcinoma
DSMZ No:	ACC 367
Origin:	established from the sarcomatoid component of a grade III transitional cell carcinoma of the renal pelvis of a 64-year-old man in 1990; patient lived in a blackfoot disease endemic area in Taiwan
References:	Tzeng et al., Anticancer Res. 16: 1797-1804 (1996)
Depositor:	Dr. C.C. Tzeng, National University Medical Center, Tainan, Taiwan
DSMZ Cell Culture Data:	
Morphology:	epithelial-like (spindle) cells growing in monolayers; cells contain considerable amounts of small, dark granules
Medium:	80-90% Dulbecco's MEM + 10-20% FBS
Subculture:	split confluent cultures 1:3 to 1:5 once or twice a week using trypsin/EDTA; seed out at ca. 1-2 x 10 ⁶ cells/80 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of ca. 3-5 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1-2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human flat-modulated hypertriploid karyotype with 32% polyploidy; 66-79<3n>XX, -Y, +1, +3, +5, +7, -9, -11, +12, +17, -18, +19, +20, +21, +mar, add(1)(p34)x2, del(2)(p16), +dic(2;9)(q25;p22)x2, add(3)(p24)x2, add(6)(p23)x2, i(7p)x2, i(8q)x2, del(11)(q?22q23), add(12)(q24), add(19)(q13)x2; 2-4 dmin present in most cells; gain of ch 7 and deletions of 9p typical in TCC; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BHK-21
Cell Type:	Syrian hamster kidney
DSMZ No:	ACC 61
Origin:	BHK-21 is a subclone (clone 13) of the parental line established from the kidneys of five unsexed, one-day-old hamsters in 1961; cells can be used for virus replication studies
References:	MacPherson & Stoker, Virology 16: 147-151 (1962); MacPherson, J. Natl. Cancer Inst. 30: 795-811 (1963)
Depositor:	obtained from ATCC (CCL 10), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	fibroblastoid adherent monolayer
Medium:	90% MEM (with Earle's salts) + 10% FBS + 1x non-essential amino acids
Subculture:	split confluent culture 1:2 to 1:3 every 2-4 days using trypsin/EDTA; seed out at ca. 0.2×10^5 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 32-50 hours
Harvest:	about $20-30 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Fingerprinting:	unique DNA profile with (gtg) ₅ multilocus probe
Species:	confirmed as hamster with IEF of AST, LDH, MDH, MPI, NP, PEP B
Cytogenetics:	near-diploid golden (Syrian) hamster karyotype with 21% tetraploidy; 40-45, XY; hyperdiploid sideline present at 19%; resembles published description
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BHT-101
Cell Type:	human thyroid carcinoma
DSMZ No:	ACC 279
Origin:	established from the lymph node metastasis of a 63-year-old woman with anaplastic papillary thyroid carcinoma; cells were described to not produce hormones, but to be partly positive for thyroglobulin- and thyroxine (T4)
References:	Palyi et al., Virchows Archiv B Cell Pathol. 63: 263-269 (1993)
Depositor:	Prof. I. Palyi, National Institute of Oncology, Budapest, Hungary
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing in monolayers
Medium:	80% Dulbecco's MEM + 20% FBS + 0.5% human serum (+ 0.005 IU/ml TSH + 5 :g/ml bovine insulin - cells grow also without these latter supplements)
Subculture:	split confluent culture 1:2 to 1:3 every 2-3 days using trypsin/EDTA; seed out ca. 0.15-0.2 x 10 ⁶ /ml cells
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 2-3 days
Harvest:	cell harvest of 0.7-1.6 x 10 ⁵ /cm ²
Storage:	frozen with 70% medium, 20% FBS, 0.5% human serum, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, desmin-, endothel-, GFAP+, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, G6PD, LDH, NP
Cytogenetics:	apparently human hypertetraploid karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BHY
Cell Type:	human oral squamous cell carcinoma
DSMZ No:	ACC 404
Origin:	established from the tumor of a 52-year-old Japanese man with highly differentiated squamous cell carcinoma of the lower alveolus which was highly invasive to the mandibular bone and the muscle layer of the oral floor; described as anchorage-independent and to be tumorigenic in nude mice
References:	Kawamata et al., Int. J. Cancer 70: 120-127 (1997)
Depositor:	Dr. H. Kawamata, Tokushima University, Tokushima, Japan
DSMZ Cell Culture Data:	
Morphology:	epithelial-like polygonal or round and flat cells (sometimes spindle-form, very heterogenous) growing in monolayers
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS
Subculture:	split confluent culture 1:2 to 1:3 every 3-5 days using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/80 cm ² in 15 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 70 hours
Harvest:	cell harvest of ca. 7×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-1.5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human flat-moded hypotriploid karyotype with 28% polyploidy; 61-68<3n>XX, -Y, -1, -2, +4, -5, -6, +9, +11, -13, +15, -17, -18, +20, +20, -21, +3-4mar, del(X)(q11q21), add(1)(q11), ?inv(2)(p17q11), add(2)(q11), del(3)(p11p14.3)x2, i(4p), add(4)(q3?1), del(5)(q13), add(5)(p15)x1-2, der(8;10)(q10;q10), add(9)(q34), add(9)(p11), del(10)(p13), del(10)(q23), der(11)t(1;11)(q11;q25), add(12)(p11), add(12)(p11), add(12)(q13-21), add(13)(p11), add(16)(q24), del(16)(p13), add(17)(p12), add(18)(p11), add(19)(q13); 3-10 dmin counted in some cells
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BL-41
Cell Type:	human Burkitt's lymphoma
DSMZ No:	ACC 160
Origin:	established from the tumor tissue of an 8-year-old Caucasian boy with Burkitt's lymphoma; cells were described to be EBV-negative
References:	Lenoir et al., IARC Scientific Publ. 59: 309-318 (1985); Cohen et al., J. Natl. Cancer Inst. 78: 235-242 (1987)
Depositor:	Dr. M. H. Falk, GSF, Institute for Clinical Molecular Genetics and Tumor Genetics, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round cells growing non-adherent in suspension
Medium:	90-95% RPMI 1640 + 5-10% FBS (possibly add 1 mM sodium pyruvate + 50 : M mercaptoethanol)
Subculture:	split saturated culture 1:3 to 1:5 every 2 days; optimal density at about 1.0×10^6 cells/ml; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	saturation density at about 2.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a+, CD80-, CD138+, HLA-DR+, sm/cyIgM+, sm/cyIgG-, sm/cykappa+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human hyperdiploid karyotype with 7% polyploidy; 48(42-49)<2n>XY, +7, -13, +2mar, add(8)(q24), t(8;14)(q24;q32), der(15)t(13;15)(q13;p11), add(17)(p12), subclonal rearrangements at 1q23, 7p22, 11q13
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BL-70
Cell Type:	human Burkitt's lymphoma
DSMZ No:	ACC 233
Origin:	established from the tumor tissue at relapse of a 16-year-old Caucasian boy with Burkitt's lymphoma; cells were described to be EBV-negative
References:	Lenoir et al., IARC Scientific Publ. 60: 309-318 (1985); Cohen et al., J. Natl. Cancer Inst. 78: 235-242 (1987)
Depositor:	Dr. M. H. Falk, GSF, Institut for Clinical Molecular Genetics and Tumor Genetics, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round cells growing non-adherent in suspension
Medium:	90% RPMI 1640 + 10% FBS (possibly add 1 mM sodium pyruvate + 50 : M mercaptoethanol)
Subculture:	optimal split ratio of 1:2 to 1:3 every 2-3 days; seed out at ca. 0.3×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 44 hours
Harvest:	saturation density of about 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD37+, HLA-DR+, sm/cyIgM+, sm/cyIgG-, sm/cykappa+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human hyperdiploid karyotype with 4% polyploidy; 47(42-49)<2n>XY, +7, inv(1)(p21q21), del(2)(q33), t(8;14)(q24;q32), t(12;22)(q21.1;q13.2)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BM-1604
Cell Type:	human prostate carcinoma (derivative of DU-145)
DSMZ No:	ACC 298
Origin:	cytogenetic analysis and DNA fingerprinting at the DSMZ documented that this cell line is identical with the human prostate carcinoma-derived cell line DU-145 (DSM ACC 261), but shows some phenotypical differences; DU-145 was established from the tumor tissue removed from the metastatic central nervous system lesion of a 69-year-old man with prostate carcinoma in 1975
References:	Van Helden et al., Brit. J. Cancer 70: 195-198 (1994)
Depositor:	Dr. E. M. Bey, Highveld Biological Ltd., National Repository for Biological Materials, Sandringham, Kelvin, South Africa
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing as monolayers
Medium:	85% RPMI 1640 + 15% FBS
Subculture:	split confluent culture using trypsin/EDTA; seed out at ca. $2-3 \times 10^6$ cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time highly variable, about 60 hours
Harvest:	cell harvest of about 10×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	same DNA profile as DU-145 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypotriploid karyotype with 22% polyploidy; 59(55-63)<3n>X, -X/Y, -Y, -2, -3, -4, -7, -8, -14, -16, -19, -20, -21, -22, +5mar (inc bisat x2-3), del(1)(p31), del(2)(p12), i(5p), del(6)(q22)/add(6)(q?12), der(6)t(4;6)(q22;p22.3), del(9)(p21), add(10)(q24-q25), add(11)(p11), t(11;?22)(q24;q13)/del(11)(q23), der(12)t(5;12)(q13;p12.3), add(13)(q33)x2, add(13)(q33), add(15)(p11), der(18)t(14;18) (q11;q12.3)x1-2; sideline with del(2)(p23), i(8q); closely related to DU-145
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BONNA-12
Cell Type:	human hairy cell leukemia
DSMZ No:	ACC 150
Origin:	established from the spleen of a 46-year-old man with hairy cell leukemia (HCL) at diagnosis in 1988
References:	Wientjens et al., Leukemia & Lymphoma 5: 415-422 (1991)
Depositor:	Dr. J. C. Kluin-Nelemans, Acad. Ziekenhuis, Leiden, The Netherlands
DSMZ Cell Culture Data:	
Morphology:	lymphoblast-like cells growing as single cells and in clumps in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	saturation density at about $2.0-2.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13(+), CD19+, CD20+, CD34-, CD37+, CD138+, HLA-DR+, sm/cyIgM+, sm/cyIgG-, sm/cykappa+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hyperdiploid karyotype with 4% polyploidy; 47(45-47)<2n>X, -Y, +12, t(9;10)(p12/13;q12); supernumerary ring or acf present in most cells
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BPH-1
Cell Type:	human benign prostate hyperplasia
DSMZ No:	ACC 143
Origin:	prostate epithelial cells from a 68-year-old man with benign prostate hyperplasia; cells were immortalized with SV-40 large T-antigen; cells were described to express cytokeratins 8, 18, and 19 (but not 14) and to metabolize prostatic androgens
References:	Hayward et al., In Vitro Cell Develop. Biol. 31A: 14-24 (1995)
Depositor:	Dr. Simon Hayward, University of California at San Francisco, San Francisco, California, USA
DSMZ Cell Culture Data:	
Morphology:	adherent epithelioid cells growing as monolayers
Medium:	80% RPMI 1640 + 20% FBS + 20 ng/ml testosterone + 5 : g/ml transferrin + 5 ng/ml sodium selenite + 5 : g/ml insulin + trace elements mix (a list can be provided as they are no longer commercially available in a single mix)
Subculture:	split confluent culture 1:3 to 1:8 every 3-7 days using trypsin/EDTA; seed out at 0.1×10^5 cells/cm ² ; we noted that the cells grow equally well without the trace element mix which was recommended by the depositor
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of ca. 0.4×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7(+), cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hypertriploid karyotype with 2.5% polyploidy; 75(69-79)<3n>XXY, -4, +5, +6, +7, -10, +11, -12, +15, +21, +3mar; +dmin; i(3q), add(7)(p11), i(8q), add(10)(p13); sideline with add(13)(p12) and smaller 10p+ marker; matches published karyotype; carries large submetacentric 10p+ marker
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BT-474
Cell Type:	human breast ductal carcinoma
DSMZ No:	ACC 64
Origin:	established from a solid, invasive ductal carcinoma of the breast obtained from a 60-year-old woman; cells were reportedly tumorigenic in athymic mice and were found to be susceptible to mouse mammary tumor virus
References:	Lasfargues et al., J. Natl. Cancer Inst. 61: 967-978 (1978)
Depositor:	Dr. H. Kirchner, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent patches of epithelial cells; individual colonies with a thick center and a thin smooth margin
Medium:	80% RPMI 1640 + 20% FBS + 10 :g/ml bovine insulin + 2 mM L-glutamine
Subculture:	split confluent culture 1:2 to 1:3 every 7-10 days using trypsin/EDTA; cells grow slowly; seed out at ca. $2-3 \times 10^6$ cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 100 hours
Harvest:	about $1.0-1.6 \times 10^5$ cells/cm ² or 20×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, GPDH, LDH, NP
Cytogenetics:	human hypertetraploid karyotype; 104(91-111)<4n>
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BT-B
Cell Type:	human cervix carcinoma (derivative of HELA)
DSMZ No:	ACC 227
Origin:	these cells were thought to be "stablished from the malignant urinary bladder carcinoma of a 66-year-old Caucasian man (stage pTa G3) in 1989"; same donor as for cell line BT-A; however, DNA fingerprinting and cytogenetic analysis at the DSMZ established unequivocally that this cell line must be considered de facto a subclone of HELA
References:	Van der Bosch et al., Exp. Cell Res. 187: 185-192 (1990); Wang et al., Immunol. Lett. 27: 191 (1991)
Depositor:	Prof. A. J. Ulmer, Forschungsinstitut Borstel, Borstel, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 3-4 days using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 43 hours
Harvest:	cell harvest of about 0.2-0.3 x 10 ⁶ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament+, vimentin+
Fingerprinting:	same DNA profile as HELA using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, NP, PEP B
Cytogenetics:	human hypertriploid karyotype with 13% polyploidy; 72(68-74)<3n>XX, -Y, +1, -4, +5, +5, +5, -11, +12, +12, -19, -21, -22, +3mar, der(1)t(1;3)(p11;q11), der(1)t(1;5)(p21;q21), der(2)t(2;5)(p11;q11), del(3)(q13), add(4)(q35), del(5)(p12), i(5p)x3, add(6)(q25/27), i(7q), del(8)(q23), del(10)(p11), inv(12)(q21q24)/der(12)t(12;?)(q21;?)x2, add(19)(p13); mar1 resembles i(Xp); HELA markers present
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BTI-EAA
Cell Type:	insect - <i>Estigmene agrea</i>
DSMZ No:	ACC 176
Origin:	established from the larvae of <i>Estigmene agrea</i> (family Arctiidae, order Lepidoptera) (Bärenspinner)
References:	no reference available
Depositor:	Dr. H. S. Conradt, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	very large cells growing in suspension; most cells are round to oval, 10% have long processes and are spindle shaped
Medium:	80% TC 100 medium + 20% FBS
Subculture:	cells are difficult to culture and grow slowly! split saturated culture 1:4 to 1:5 every 4-5 days
Incubation:	no CO ₂ (keep lid of flask closed) at room temperature (20-25 °C)
Harvest:	cell harvest of about 0.5 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as not murine and not primate with IEF of AST, PEP B

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INFORMATION SHEET FOR CELL LINES

Cell Line:	BV-173
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 20
Origin:	established from the peripheral blood of a 45-year-old man with chronic myeloid leukemia (CML) in blast crisis in 1980; contains the t(9;22) b2-a2 fusion gene
References:	Pegoraro et al., J. Natl. Cancer Inst. 70: 447-453 (1983); review article: Drexler, Leukemia Res. 18: 919-927 (1994); review article: Drexler et al., Leukemia Res. 24: 109-115 (2000)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round to elongated, single cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	cells are difficult to culture! cells appear to grow better in 24-well-plates than in flasks; maintain at 0.5-1.0 x 10 ⁶ cells/ml; cells grow slowly, split 1:2 to 1:3 every 3-4 days; seed out at ca. 1 x 10 ⁶ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	maximum density at 2-3 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13+, CD19+, CD20+, CD34-, CD37-, CD79a-, cyCD79a+, CD80-, CD138-, HLA-DR+, sm/cyIgG-, sm/cyIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of G6PD, MDH, NP
Cytogenetics:	human hyperdiploid karyotype; 47(46-48)<2n>X/XY, -9, +22, +mar, add(1)(q42), add(8)(p23), t(9;22)(q34;q11), der(22)t(9;22)(q34;q11), der(?)t(9;?)(?p11;?); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	C-433
Cell Type:	human Ewing's sarcoma (derivative of RD-ES)
DSMZ No:	ACC 268
Origin:	thought to be "established from the benign giant cell tumor excised from a woman who had swelling and pain in the right ankle (distal tibia)"; however, DNA fingerprinting and cytogenetic analysis at the DSMZ established unequivocally that this cell line must be considered de facto a subclone of cell line RD-ES (ACC 260)
References:	Oreffo et al., Clin. Orthopaedics Related Res. 296: 229-241 (1993)
Depositor:	Dr. L. F. Bonewald, University of Texas, San Antonio, Texas, USA
DSMZ Cell Culture Data:	
Morphology:	cells grow loosely adherent in monolayers; some cells grow as round or polygonal cells in clumps in suspension with cellular processes
Medium:	45% RPMI 1640 + 45% McCoy's 5A + 10% FBS
Subculture:	split saturated culture 1:8 to 1:10 every 3-4 days (without using trypsin)
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 25-30 hours
Harvest:	cell harvest of about 1.4 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	same DNA profile as RD-ES using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human flat-moded hypertetraploid karyotype with 4% polyploidy; 103(92-106)<4n>XX, der(X), -X, +1, +1, +1, +1, +2, +6, +6, -14, +17, +18, +6mar, add(X)(p22), del(1)(p22), del(1)(p32p35)x1, der(1)t(1;1)(p32;q11), der(8)t(4;8)(q11;q24), t(9;12)(p12;q13)x2, add(11)(p13), i(13q)x2-3, add(16)(q13), 0-2 dmin/rings present; t(9;12) is associated with benign epithelial tumors; presence of type-2 EWS-FLI1 hybrid transcripts (394 bp) confirmed by RT-PCR
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	C6
Cell Type:	rat glioma
DSMZ No:	ACC 550
Origin:	the glial cell strain C6 was cloned from a rat glioma which was induced by N-nitrosomethylurea
References:	Benda et al., Science 161: 370-371 (1968)
Depositor:	Prof. B. Hamprecht, University of Tübingen, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblastoid cells forming spindles and growing in monolayers
Medium:	82.5% Ham's F12 + 15% horse serum + 2.5% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:4 to 1:5 twice a week using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/175 cm ² flask; maximal density of ca. $2-0-2.5 \times 10^7$ cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	cell harvest of ca. 10×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Species:	confirmed as rat by PCR species analysis
Cytogenetics:	rat hyperdiploid karyotype with 4% polyploidy; 43(40-44)<2n>

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INFORMATION SHEET FOR CELL LINES

Cell Line:	C6-BU-1
Cell Type:	rat brain tumor
DSMZ No:	ACC 108
Origin:	this clone is a bromodeoxyuridine-resistant mutant derived from cell line C6 (established from a nitrosourea-induced brain tumor in a Wistar-Furth strain rat in 1971)
References:	Hamprecht et al., Hoppe-Seyler's Z. Physiol. Chem. 354: 1633-1641 (1973); Hamprecht, Int. Rev. Cytol. 49: 99-170 (1977)
Depositor:	Dr. B. Hamprecht, Physiologisch-Chemisches Institut, Tübingen, Germany
DSMZ Cell Culture Data:	
Morphology:	spindle-shaped, relatively small cells, growing adherently in monolayers
Medium:	90% Dulbecco's MEM + 10% FBS + 100 :M 5-bromodeoxyuridine
Subculture:	split confluent cultures 1:4 to 1:10 every 4 days using trypsin/EDTA; let the cells not grow to complete confluence; seed out at ca. 4×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of about 30 hours
Harvest:	about $2-3 \times 10^7$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Fingerprinting:	unique Hinf I-(gtg) ₅ DNA profile
Species:	confirmed as rat with IEF of MDH, NP
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	C7
Cell Type:	mouse macrophages
DSMZ No:	ACC 416
Origin:	established from the bone marrow of male adult 129/Sv mice (p53 gene knocked out); it is described that the differentiation potential of C7 cells to osteoclasts is maintained in the presence of M-CSF and is abolished irreversibly by the stimuli known to induce differentiation into other hematopoietic lineages
References:	Miyamoto et al., Biochem. Biophys. Res. Commun. 242: 703-709 (1998)
Depositor:	Dr. S. I. Hayashi, Tottori University, Tottori, Japan
DSMZ Cell Culture Data:	
Morphology:	macrophage-like adherent cells with processes
Medium:	90% alpha-MEM + 10% FBS + 1 ng/ml M-CSF (human or murine)
Subculture:	split confluent culture 1:2 to 1:4 every 4-6 days using trypsin/EDTA after washing twice with PBS; seed out at ca. 0.5×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 80-100 hours
Harvest:	cell harvest of ca. $2-3 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, LDH, NP
Cytogenetics:	murine hypertriploid karyotype with 20% polyploidy; 66(64-67)<3n>; no centric fusion markers present
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CA-46
Cell Type:	human Burkitt's lymphoma
DSMZ No:	ACC 73
Origin:	established from the ascites fluid of a patient with American-type Burkitt's lymphoma; cells were described to be EBNA-negative, to carry the (8;14) translocation and to express bcl-2 and c-myc mRNA
References:	Magrath et al., J. Natl. Cancer Inst. 64: 465-476 and 477-483 (1980)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5\text{--}2.0 \times 10^6$ cells/ml, split 1:2 to 1:3; seed out at 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 35 hours
Harvest:	saturation density of over 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a+, CD80(+), CD138-, HLA-DR+, sm/cyIgM+, sm/cyIgG-, sm/cyIgkappa+, sm/cyIglambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human near diploid karyotype with about 3% triploidy; 46(45-48)<2n>X/XY, dup(1)(q21q32), dup(7)(q12q22), t(8;14)(q24;q32); additional rearrangements were present in a sideline, viz. t(6;13)(p21;q32) and del(11)(p11); a supernumerary dmin was present in about 25% metaphases
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CACO-2
Cell Type:	human colon adenocarcinoma
DSMZ No:	ACC 169
Origin:	established from the primary colon tumor (adenocarcinoma) of a 72-year-old Caucasian man in 1974
References:	Fogh et al., J. Natl. Cancer Inst. 58: 209-214 (1977); Rousset, Biochimie 68: 1035-1040 (1986); Pinto et al., Biol. Cell. 47: 323-330 (1983); Bacher et al., Exp. Cell Res. 200: 97-104 (1992)
Depositor:	Prof. A. Bacher, Technical University of Munich, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial adherent cells; after splitting cells start to grow in colonies
Medium:	80% MEM (with Earle's salts) + 20% FBS + 1x non-essential amino acids
Subculture:	cells grow slowly; split confluent culture 1:6 to 1:10 every 5-7 days using trypsin/EDTA or, if necessary, a cell scraper; after thawing cells might start slowly to take up growth; seed out at ca. 1 x 10 ⁶ cells/80 cm ² in 10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 80 hours
Harvest:	about 5-8 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1-2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in microbiological culture, RNA hybridization, PCR assay
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	unique DNA profile using multiplex PCR at D1S80, D2S44, D17S5 and ApoB
Species:	confirmed as human with IEF of MDH, NP
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CADO-ES1
Cell Type:	human Ewing's sarcoma
DSMZ No:	ACC 255
Origin:	established from the malignant pleural effusion from a 19-year-old Japanese woman diagnosed with right-buttock Ewing's sarcoma with multiple lung metastases; heterotransplantable; capable of anchorage-independent growth; neural differentiation in response to cAMP
References:	Kodama et al., Cancer Genet. Cytogenet. 57: 19-30 (1991); Kodama et al., Jpn. J. Cancer Res. 85: 335-338 (1994)
Depositor:	Dr. K. Kodama, Center for Adult Diseases, Osaka, Japan
DSMZ Cell Culture Data:	
Morphology:	round or fusiform cells growing as adherent or semi-adherent monolayered multicellular spheroids
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:4 to 1:10 every 3-7 days using trypsin/EDTA; after thawing use 20% FBS at first; seed out at ca. 5×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	cell harvest of about 4.0×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament+, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human hyperdiploid karyotype with 8% polyploidy; 47(44-48)<2n>XX, +8, dup(1)(q32q44), i(8q), add(18)(p11.32); t(11;22) absent; corresponds to published karyotype; absence of EWS-FLI1 fusion transcripts confirmed by RT-PCR
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAKI-1
Cell Type:	human kidney carcinoma
DSMZ No:	ACC 142
Origin:	established from the kidney carcinoma (clear cell carcinoma) of a 49-year-old man; tumorigenic in mice
References:	Fogh et al., J. Natl. Cancer Inst. 58: 209-214 (1977); idem, 59: 221-226 (1977)
Depositor:	Dr. H. Kirchner, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like, adherent in monolayers
Medium:	90% McCoy's 5A + 10% FBS
Subculture:	split confluent culture at 1:3 to 1:5 using trypsin/EDTA; upon thawing grow immediately in a 80 cm ² flask; seed out at ca. 1-2 x 10 ⁶ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	about 3-5 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of G6PD, MDH, NP
Cytogenetics:	human flat-moded penta-/hexa-ploid karyotype; 112-149<6n>XXX, -Y, -Y, -Y, rearrangements include add(1)(p22), i(3p), add(3)(p14), der(7)t(7;7)(q34;p14), i(8q), i(10q), add(11)(p11), large isodic marker x2; does not correspond with published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAKI-2
Cell Type:	human kidney carcinoma
DSMZ No:	ACC 54
Origin:	established from the primary renal carcinoma of a 69-year-old man
References:	Fogh et al., J. Natl. Cancer Inst. 58: 209-214 (1977); Fogh et al., J. Natl. Cancer Inst. 59: 221-226 (1977)
Depositor:	Dr. H. Kirchner, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells, growing as adherent monolayer
Medium:	90% McCoy's 5A + 10% FBS
Subculture:	change medium twice every week, split confluent cultures 1:2 to 1:3 weekly using trypsin/EDTA, upon thawing grow cells immediately in 80 cm ² ; seed out at ca. 2 x 10 ⁶ cells/80 cm ² in 10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	cell harvest of ca. 1 x 10 ⁶ cells/25 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human flat-moded pentaploid/hexaploid karyotype with 12% polyploidy; 130(118-132)<6n>XXX, -Y, -Y, -Y, inc, add(1)(p22), i(8q), i(?5q)x2, del(6)(q23), t(14;15)(p10;q10), der(19)t(1;19)(q24;q13), dminx1, large iso der(5q)x2, and submeta marker der(?)t(?)t(?)q11)x2; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAL-120
Cell Type:	human breast adenocarcinoma
DSMZ No:	ACC 459
Origin:	established from the pleural effusion of a 43-year-old woman with metastatic breast adenocarcinoma in 1991
References:	unpublished
Depositor:	Dr. J. Gioanni, Centre Antoine Lacassagne, Nice, France
DSMZ Cell Culture Data:	
Morphology:	heterogenous population with epithelial-like and fibroblast-like adherent cells growing in monolayers
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:2 every 2-3 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/175 cm ² ; maintain at about 2×10^6 cells/175 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	cell harvest of ca. 5×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament+, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Cytogenetics:	human hypertetraploid karyotype with 20% polyploidy 98(88-98)<4n>XXX/XXXX, -1, -2, -2, -4, +5, +5, +6, -9, -9, +10, +11, -12, -12, -13, -13, -14, -15, -16, -16, -20, -21, -22, +16-18mar, + 1-2 dmin, add(X)(p2?2), add(1)(q22), del(3)(q22), add(3)(p1?4), del(4)(p15.2), add(6)(q11)x2, del(7)(q32), add(8)(p12), add(8)(q24), der(8;13)(q10;q10), ins(11;?)(q13;?), add(12)(p1?), add(13)(p11)x1-2, add(14)(p11), der(15;?21)(p1?1;?q12)x3; sideline with i(1)(p10)add(1)(p3?2), i(1q), i(3p), der(16)t(5;16)(q13;q24)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAL-12T
Cell Type:	human non-small cell lung carcinoma
DSMZ No:	ACC 443
Origin:	established from a male patient with a non-small cell lung carcinoma in 1981
References:	Fischel et al., Cancer Chemther. Pharmacol. 28: 448-454 (1991); Olivier et al., Eur. J. Cancer 26: 867-870 (1990)
Depositor:	Dr. J. Gioanni, Centre Antoine Lacassagne, Nice, France
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing in monolayers
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:2 to 1:3 every second day using trypsin/EDTA (incubation with trypsin/EDTA for about 15-20 min); seed out at ca. 1×10^6 cells/80 cm ² in 25 ml medium; maximal density at ca. 12×10^6 /175 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of ca. 8×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP+, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human hypotriploid karyotype with 12% polyploidy; 59(57-59)<3n>X, +1, -2, -4, -5, -6, -6, -7, -11, -12, -13, -16, -16, -18, +20, +4-5mar, der(1;18)(p10;q10), der(1)t(X;1)(?;p?21)t(1;?)(q1?1;?), der(1)(q12->q10::p10->pter), add(3)(p11), add(4)(p11), add(5)(p15), add(9)(q1?)x1-2, add(11)(q24), dup(11)(q13q24), del(12)(p12), i(14q), add(19)(q11), i(21q)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAL-148
Cell Type:	human breast adenocarcinoma
DSMZ No:	ACC 460
Origin:	established from the pleural effusion from a 58-year-old French woman with terminally invasive breast cancer (moderately differentiated ductal adenocarcinoma) after repeated cycles of chemotherapy (including adriamycin and taxol)
References:	cited in: Rostagno et al., Anticancer Res. 14: 2025-2032 (1994)
Depositor:	Dr. J. Gioanni, Centre Antoine Lacassagne, Nice, France
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing as single cells or as clumps either in suspension or, if supplemented with EGF, in monolayers
Medium:	80% Dulbecco's MEM + 20% FBS + 2 mM L-glutamine + EGF (1 : g/100 ml)
Subculture:	split confluent culture 1:4 to 1:10 every 3-5 days by agitation; seed out at ca. $0.2-0.3 \times 10^6$ cells/ml (suspension culture) or $0.4-0.8 \times 10^5$ cells/cm ² (adherent culture); maintain at about $0.2-0.7 \times 10^6$ cells/ml or equivalent; cell numbers may be difficult to ascertain due to growth in clumps
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	cell harvest of ca. 3×10^6 cells/ml or 1×10^6 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 6×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Cytogenetics:	human near-diploid karyotype with 16% polyploidy; 45-48<2n>XX, +1, +7, -8, -13, +20, add(1)(p36), der(1;16)(q10;p10), del(2)(q11q2?2), del(13)(q1?1q1?3), der(16)t(1;16)(q12-21; p11); resembles originator's unpublished karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAL-27
Cell Type:	human tongue squamous cell carcinoma
DSMZ No:	ACC 446
Origin:	established from the poorly differentiated squamous cell carcinoma of the tongue removed from a 56-year-old man before treatment in 1982
References:	Gioanni et al., Eur. J. Cancer Clin. Oncol. 24: 1445-1455 (1988)
Depositor:	Dr. J. Gioanni, Centre Antoine Lacassagne, Nice, France
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently in monolayers
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:4 to 1:10 every 3-6 days using trypsin/EDTA; seed out at ca. $0.1-0.2 \times 10^5$ cells/cm ² ; maximum density of ca. 2×10^5 cells/cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 20-40 hours
Harvest:	cell harvest of ca. 1×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Cytogenetics:	human flat-modulated hypodiploid karyotype with 3% polyploidy; 41(38-45)<2n>X, -Y, -4, +7, -10, -11, -18, -20, -21, -22, +2-3mar, add(3)(q21), i(3q), i(7p), del(8)(p21), i(9q), add(13)(p11), add(14)(p11); sideline with additional der(3) and der(7); resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAL-29
Cell Type:	human urinary bladder transitional cell carcinoma
DSMZ No:	ACC 515
Origin:	established from the primary lesion of a fatally invasive, metastatic transitional cell carcinoma of the bladder (grade IV, stage T2) from an 80-year-old woman before treatment
References:	Cattan et al., Brit. J. Cancer 85: 1412-1417 (2001)
Depositor:	Dr. J. L. Fischel, Centre Antoine Lacassagne, Nice, France
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells growing as monolayers
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:2 to 1:4 every 3-6 days using trypsin/EDTA; seed out at ca. $1.5-2.0 \times 10^6$ cells/175 cm ² flask
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	maximal density of ca. 1.0×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1.5-2.0 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hypertriploid karyotype with 8% polyploidy; 71(63-74)<3n>XX/XXX, -4, +9, -11, -13, -17, +20, +3mar, +1-6 dmin, i(3q), add(7)(q22)x1-2, del(7)(q32)x2, del(9)(p21), del(10)(p13), add(12)(p1?3), add(19)(p13)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAL-33
Cell Type:	human tongue squamous cell carcinoma
DSMZ No:	ACC 447
Origin:	established from the surgically removed fragment of a tongue lesion from a 69-year-old man with moderately differentiated squamous cell carcinoma of the tongue in 1983 (prior to therapy)
References:	Gioanni et al., Eur. J. Cancer Clin. Oncol. 24: 1445-1455 (1988)
Depositor:	Dr. J. Gioanni, Centre Antoine Lacassagne, Nice, France
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherent in monolayers
Medium:	90% Dulbecco's MEM + 10% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:3 every 4-5 days using trypsin/EDTA (for about 10 min); seed out at ca. 4×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	cell harvest of ca. 25×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Cytogenetics:	human hyperdiploid karyotype with 4% polyploidy 49(45-49)<2n>XY/XYX, +7, +9, -18, +20, +mar, add(X)(p22), add(3)(p22-24), i(7p), i(8q), add(9)(p24), del(9)(p27?); resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAL-39
Cell Type:	human vulva squamous cell carcinoma
DSMZ No:	ACC 187
Origin:	established from the primary lesion resected from a 76-year-old woman with bulky, moderately differentiated squamous cell carcinoma of the vulva in 1984; cells were reported to express an elevated number of high affinity EGF receptors, to carry gene amplification at 11q13 and to be tumorigenic in nude mice
References:	Gioanni et al., Int. J. Oncol. 3: 293-297 (1993)
Depositor:	Dr. J. Gioanni, Centre Antoine Lacassagne, Nice, France
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial-like cells growing as monolayer
Medium:	80% Dulbecco's MEM + 20% FBS + 2 mM L-glutamine + 1 mM sodium pyruvate + 0.5 nM hydrocortisone + 1 :g/100 ml EGF
Subculture:	cells are difficult to culture! split confluent culture 1:2 to 1:4 every 5-10 days using trypsin/EDTA (cells are sensitive to over-trypsination)
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 50-70 hours
Harvest:	cell harvest of about 0.8 x 10 ⁵ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 0.75-1.5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human flat-moded hypotriploid karyotype with 27% polyploidy; 59(54-60)<3n>XX, -X, +1, +1, +2, +3, +9, +12, +12, +16, +20, +22, +4mar, del(1)(q22), del(3)(p14), add(4)(p16), del(7)(q32), add(9)(p22), del(12)(q15), add(12)(q24), add(18)(q23); additional 11q rearrangements present in markers; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAL-51
Cell Type:	human breast carcinoma
DSMZ No:	ACC 302
Origin:	established from the pleural effusion metastasis of a 45-year-old woman with progressive breast adenocarcinoma (after radio-, chemotherapy and surgery) in 1985; rare example of tumor cell line with normal karyotype
References:	Gioanni et al., Brit. J. Cancer 62: 8-13 (1990)
Depositor:	Dr. J. Gioanni, Centre Antoine Lacassagne, Nice, France
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells growing in monolayers; the culture contains usually a large amount of cell debris
Medium:	80% Dulbecco's MEM (4.5 g/L glucose) + 20% FBS
Subculture:	split culture 1:2 once or twice a week using trypsin/EDTA before cells reach complete confluency; seed out at ca. 2×10^6 cells/25 cm ² ; cells may grow very slowly initially after thawing or after stringent trypsination
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of >60 hours
Harvest:	cell harvest of ca. $15-20 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human near-diploid karyotype with 28% tetraploidy; $46<2n>XX$, no consistent abnormality detected; rare example of tumor cell line with normal karyotype; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAL-54
Cell Type:	human kidney carcinoma
DSMZ No:	ACC 365
Origin:	established from the malignant pleural effusion of a 75-year-old man with kidney carcinoma in 1987
References:	Gioanni et al., Bull. Cancer 83: 553-558 (1996)
Depositor:	Dr. J. Gioanni, Centre Antoine Lacassague, Nice, France
DSMZ Cell Culture Data:	
Morphology:	epitheloid, adherent cells growing in monolayers
Medium:	85% Dulbecco's MEM + 15% FBS + 4 mM L-glutamine + 1 mM sodium pyruvate + 0.4 :g/ml hydrocortisone + 10 ng/ml EGF
Subculture:	split nearly confluent culture 1:2 to 1:3 about once a week using trypsin/EDTA (it may be difficult to detach the cells from the flask - use double quantity of trypsin/EDTA for ca. 15 min); seed out at ca. 2 x 10 ⁶ cells/75 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 72 hours
Harvest:	cell harvest of ca. 8 x 10 ⁶ cells/75 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hyperdiploid karyotype with 12% polyploidy; 52(52-53)<2n>X, -Y, +2, +3, +7, +7, +8, +12, -13, +15, +16, +17, del(3)(p21), del(5)(p14.2p15.2), dup(7)(q11q22), add(14)(p12); del(3p), gain of ch 7 and loss of Y typical of renal cell carcinoma
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAL-62
Cell Type:	human thyroid anaplastic carcinoma
DSMZ No:	ACC 448
Origin:	established from the thyroid gland (right lobe) of a 70-year-old woman with thyroid anaplastic carcinoma in 1988; described as being tumorigenic in heterotransplanted nude mice
References:	Gioanni et al., Bull. Cancer 78: 1053-1062 (1991)
Depositor:	Dr. J. Gioanni, Centre Antoine Lacassagne, Nice, France
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing in monolayers
Medium:	90-95% Dulbecco's MEM + 5-10% FBS
Subculture:	split confluent culture 1:10 to 1:15 every 3-4 days using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/80 cm ² ; maintain at about 10×10^6 cells/175 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	cell harvest of ca. 28×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin+, endothel-, GFAP-, neurofilament+, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Cytogenetics:	human, flat-modet hypertriploid karyotype with 8% polyploidy; 60-76<3n>XX, -X, +3, +3, -4, +6, +7, +9, -11, -14, -18, +19, +20, +20, +2-6mar; del(1)(p34), der(1)t(1;17)(p34;q21), der(3)add(3)(p11)?dup(3)(q21q22)x2-4, del(6)(q15)x2, i(8q), der(?12)t(1;12)(q11-12;q14), add(17)(q21), add(21)(q22); sideline with del(X)(q24), der(7;8)(q10;q10); resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAL-72
Cell Type:	human osteosarcoma
DSMZ No:	ACC 439
Origin:	established from a chemotherapy-resistant tumor sample obtained from the left knee of a 10-year-old boy with anaplastic osteoblastic sarcoma (Rosen grade IIB) in 1989
References:	Rochet et al., Int. J. Cancer 82: 282-285 (1999)
Depositor:	Dr. J. Gioanni, Centre Antoine Lacassagne, Nice, France
DSMZ Cell Culture Data:	
Morphology:	fibroblastoid cells growing adherently in monolayers
Medium:	90% Dulbecco's MEM + 10% FBS + 2 mM L-glutamine + 1x insulin-transferrin-sodium selenite
Subculture:	split confluent culture 1:3 every 4-5 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	cell harvest of ca. 7×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human, bimodal hyperdiploid/hypertetraploid karyotype with 12% polyploidy; 45-56<2n>X, -Y, +1, +1, -2, -2, +8, +11, -13, -15, +19, -20, -20, -22, -22, +15-20mar, +dmin, add(X)(q21), der(1;19)(q10;q10), add(1)(p3?5)add(1)(q3?), add(3)(q2?9)x2, add(6)(q11), del(7)(q11), add(8)(q24), add(8)(q11), add(11)(p1?), del(11)(q13), add(14)(q3?)x2, add(16)(p11), add(16)(q23), add(19)(p11), add(21)(p11) (hyperdiploid clone); additional rearrangements, including der(1;8)(q10;q10) and idic(8;?)(p10;?) present in hypertetraploid clone
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAL-78
Cell Type:	human chondrosarcoma
DSMZ No:	ACC 449
Origin:	established from the recurrence of a dedifferentiated chondrosarcoma (grade III) of the muscle of the thigh from a 76-year-old man in 1989; tumorigenic chondrosarcomatous differentiation of xenografted tumor
References:	Mazeau et al., Oncol. Rep. 4: 697-700 (1997)
Depositor:	Dr. J. Gioanni, Centre Antoine Lacassagne, Nice, France
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblast-like cells growing as monolayer; the culture contains a large amount of cellular debris
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	cells grow extremely slowly; split confluent culture 1:2 every 10-12 days using trypsin without EDTA; seed out at ca. 1×10^6 cells/25 cm ² ; maximal density at ca. 14×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 240 hours
Harvest:	cell harvest of ca. 7×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAL-85-1
Cell Type:	human breast adenocarcinoma
DSMZ No:	ACC 440
Origin:	established from the relapsing invasive galactophoric breast adenocarcinoma resected from a 35-year-old woman during radiotherapy in 1990; cells were described as expressing the multidrug resistance (MDR) gene
References:	Gioanni et al., Bull. Cancer 80: 472 (1993); Fischel et al., Bull. Cancer 80: 409 (1993)
Depositor:	Dr. J. Gioanni, Centre Antoine Lacassagne, Nice, France
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing in monolayers
Medium:	90% Dulbecco's MEM + 10% FBS + 2 mM L-glutamine + 1 mM sodium pyruvate
Subculture:	split confluent culture 1:2 to 1:3 every 3-4 days using trypsin/EDTA (incubation with trypsin/EDTA for about 10 min); seed out at ca. $0.03-0.04 \times 10^6$ cells/cm ² ; maintain at 0.04×10^6 cells/cm ² ; a significant amount of cell debris is always visible in the culture background which might be due to granula or other intracellular material
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 70 hours
Harvest:	cell harvest of ca. 0.2×10^6 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hypotriploid karyotype with 8% polyploidy; 60(57-65)<3n>XX/XXX, +2, +5, -8, -10, -12, -13, -14, -15, -16, -16, -17, -18, +?20, -21, -22, +2-4mar, +fra, add(X)(q2?), del(X)(q11), add(1)(p34), del(2)(q?11q?21)x1-2, add(2)(p2?3), der(3)?inv(3)(p24q29)add(3)(qter), der(3)del(3)(p13)dup(3)(?q13?q26)del(3)(q26qter), del(3)(q21), add(4)(q2?9), der(4)t(4;9)(p15;q22), der(6)t(6;12)(q22-24;q13), add(7)(p11), del(7)(q32)x2, der(9)dup(9)(p11p24)add(9)(p24), add(11)(q24-25)x1-2, del(12)(p13), der(15;21)(q10;q10), add(17)(p13), add(17)(p13), add(19)(p13), der(?20)t(17;20)(q12;q13)dup(17)(q1?2q2?3), add(21)(p11)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAPAN-1
Cell Type:	human pancreas adenocarcinoma
DSMZ No:	ACC 244
Origin:	established from the liver metastasis of a pancreatic ductal adenocarcinoma in a 40-year-old Caucasian man in 1974; described as growing in nude mice forming metastases and as resistant to 5-FU
References:	Fogh et al., J. Natl. Cancer Inst. 58: 209-214 (1977); Kyriazis et al., Cancer Res. 46: 5810-5815 (1986)
Depositor:	obtained from DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblastic semi-confluent monolayer forming loosely adherent foci
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split 1:2 to 1:4 every 6-10 days using trypsin/EDTA; initial growth after recovery is slow; seed out at ca. $1-2 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 50-100 hours
Harvest:	cell harvest of about 1.5×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	apparently human karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAPAN-2
Cell Type:	human pancreas adenocarcinoma
DSMZ No:	ACC 245
Origin:	established from the tumor of a 56-year-old Caucasian man with pancreas adenocarcinoma in 1975
References:	Kyriazis et al., Cancer Res. 46: 5810-5815 (1986)
Depositor:	obtained from DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial-like cells growing in uniform monolayers
Medium:	85% RPMI 1640 + 15% FBS
Subculture:	split semi-confluent (2/3 to 3/4 confluency) culture 1:10 every 7 days using trypsin/EDTA (for 10-20 min); seed out at ca. $1-2 \times 10^6$ cells/80 cm ² , adherence may require several hours or overnight
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 50-70 hours
Harvest:	cell harvest of about 20×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, PEP B
Cytogenetics:	human hypertriploid karyotype with 6% polyploidy; 68-73<3n>XX, -Y, +1, +1, +4, +5, +7, +11, -12, -14, -17, -18, -22, +3mar; del(1)(p11p33)x2, add(11)(p11), der(11)t(?8;11)(q12;p11)x2, i(14q)x2, add(21)(p11)x2, 1-2DM(24%); sideline with +1, i(1q)x2, del(1)(q11)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAT-13.0B10
Cell Type:	mouse hybridoma (CD39)
DSMZ No:	ACC 472
Origin:	described as secreting a mouse monoclonal antibody (IgG ₁) CD39; the cell line was established in 1988 by fusion of PX63AG8.653 cells with spleen cells of a mouse immunized with EBV-transformed B cells from a patient with MHC class-II deficiency re-expressing class-II antigens
References:	Hadam, Leucocyte Typing VI: 157-159 (1997)
Depositor:	Dr. M. R. Hadam, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	round, single cells growing in suspension and some slightly adherent
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement" for the first 1-2 weeks post-thawing)
Subculture:	split saturated culture 1:2 to 1:4 every 2 days; seed out at 0.3 x 10 ⁶ cells/ml; maintain at 0.2-0.4 x 10 ⁶ cells/ml; best to culture cells for the first 5-7 days in a 12-well-plate with HFCS, then cells may be transferred to flasks; after 14 days, cells become slightly adherent (about two-thirds of them), but can be detached by slapping the flask; the HFCS concentration can be decreased after the initial 2 weeks to about 50% (1:100) or 25% (1:200)
Incubation:	at 37°C with 5-10% CO ₂
Doubling Time:	doubling time of about 25 hours
Harvest:	cell harvest of about 15 x 10 ⁶ cells/175 cm ² in 30 ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgG ₁ monoclonal antibody
Viruses:	ELISA: reverse transcriptase negative

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAT-13.1E10
Cell Type:	mouse hybridoma (CD19)
DSMZ No:	ACC 473
Origin:	described as secreting a mouse monoclonal antibody (IgG ₁) CD19; the cell line was established in 1988 by fusion of PX63AG8.653 cells with spleen cells of a mouse immunized with EBV-transformed B cells from a patient with MHC class-II deficiency re-expressing class-II antigens
References:	
Depositor:	Dr. M. R. Hadam, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing as monolayers, some grow in suspension
Medium:	80% Dulbecco's MEM (4.5 g/L glucose) + 20% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement" for the first 1-2 weeks post-thawing)
Subculture:	split saturated cultures 1:3 every 2 days; seed out at 0.2×10^6 cell/ml; maintain at ca. $0.2\text{--}0.4 \times 10^6$ cells/ml; after thawing cells may start slowly to take up growth
Incubation:	at 37°C with 5-10% CO ₂
Doubling Time:	doubling time of about 30-40 hours
Harvest:	maximal density of about 5×10^6 cells/25 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with sparflloxacin, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgG ₁ monoclonal antibody
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAT-13.6E12
Cell Type:	mouse hybridoma (CD20)
DSMZ No:	ACC 474
Origin:	described as secreting a mouse monoclonal antibody (IgG ₂ a) CD20; the cell line was established in 1988 by fusion of PX63AG8.653 cells with spleen cells of a mouse immunized with EBV-transformed B cells from a patient with MHC class-II deficiency re-expressing class-II antigens
References:	not published
Depositor:	Dr. M. R. Hadam, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing adherently in monolayers or in suspension
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement" for the first 1-2 weeks post-thawing)
Subculture:	split adherent cells (removed by trypsin/EDTA) 1:4 to 1:8 and suspension cells (dislodged mechanically) 1:2 to 1:5 every 2-3 days; seed out at 0.2-0.3 x 10 ⁶ cells/ml; maintain at ca. 0.1-0.8 x 10 ⁶ cells/ml
Incubation:	at 37/C with 5-10% CO ₂
Doubling Time:	doubling time of about 20-30 hours
Harvest:	maximal density of about 1.0 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 8 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgG ₂ a monoclonal antibody

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CAT-13.9C1
Cell Type:	mouse hybridoma (CD23)
DSMZ No:	ACC 475
Origin:	described as secreting a mouse monoclonal antibody (IgG ₂ akappa) CD23; the cell line was established in 1988 by fusion of PX63AG8.653 cells with spleen cells of a mouse immunized with an EBV-transformed human B cell line from a patient with MHC class II deficiency re-expressing class II antigens
References:	not published
Depositor:	Dr. M. R. Hadam, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells, growing singly or in small clumps
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement" for the first 1-2 weeks post-thawing)
Subculture:	split saturated culture 1:2 every 2 days; seed out at about 0.2 x 10 ⁶ cells/ml; maintain between 0.2-0.8 x 10 ⁶ cells/ml
Incubation:	at 37/C with 5-10% CO ₂
Doubling Time:	doubling time of about 48 hours
Harvest:	maximal density of about 1.0 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgG ₂ a monoclonal antibody
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CCRF-CEM
Cell Type:	human T cell leukemia
DSMZ No:	ACC 240
Origin:	established from the peripheral blood of a 3-year-old Caucasian girl with acute lymphoblastic leukemia (ALL) at relapse (terminal) in 1964 (first continuous human T-ALL cell line)
References:	Foley et al., Cancer 18: 522-529 (1965)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly in suspension and as adherent epitheloid cells forming a monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture every 2-4 days; cells form monolayers until confluence and will then go into suspension; after transfer of suspension cells to a new flask, a new monolayer will develop; it may be necessary to use trypsin/EDTA or vigorous pipetting to detach the adherent cells; seed out after thawing at $0.5-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24-30 hours
Harvest:	maximal density at ca. 2×10^6 cells/ml (cells in suspension) or $4-5 \times 10^6$ cells/80 cm ² flask
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-6 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3+, CD4+, CD5+, CD6+, CD7+, CD8-, CD13-, CD14-, CD15+, CD19-, CD33-, CD34-, CD68-, CDw90-, HLA-DR-, TCRalpha/beta+, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, NP
Cytogenetics:	human near-tetraploid karyotype with extensive subclonal variation; 90(88-101)<4n>XX, -X, -X, +20, +20, t(8;9)(p11;p24)x2, der(9)del(9)(p21-22)del(9)(q11q13-21)x2; sideline with +5, +21, add(13)(q3?3), del(16)(q12)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CF-10H5
Cell Type:	mouse hybridoma (CD45)
DSMZ No:	ACC 477
Origin:	described as secreting a mouse monoclonal antibody (IgG ₁) CD45; the cell line was established in 1989 by fusion of PX63AG8.653 cells with spleen cells of a mouse immunized with IL-2-driven T cell lines from patients with high numbers of gamma-delta T cells in the peripheral blood
References:	Hadam, Leucocyte Typing VI: 499-502 (1997)
Depositor:	Dr. M. R. Hadam, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	single round cells growing in suspension, some loosely adherent
Medium:	80% Dulbecco's MEM (4.5 g/L glucose) + 20% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement" for the first 1-2 weeks post-thawing)
Subculture:	split saturated culture 1:3 every 1-2 days; seed out at about 0.2 x 10 ⁶ cells/ml; maintain between 0.2-0.4 x 10 ⁶ cells/ml; after thawing cells may start slowly to take up growth and may never exceed a viability of 80%
Incubation:	at 37/C with 5-10% CO ₂
Doubling Time:	doubling time of about 30-45 hours
Harvest:	maximal density of about 0.6 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgG ₁ monoclonal antibody
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CF-1D12
Cell Type:	mouse hybridoma (CD52)
DSMZ No:	ACC 476
Origin:	described as secreting a mouse monoclonal antibody (IgG ₃) CD52; the cell line was established in 1989 by fusion of PX63AG8.653 cells with spleen cells of a mouse immunized with IL-2-driven T cell lines from patients with high numbers of gamma-delta T cells in the peripheral blood
References:	Hadam, Leucocyte Typing VI: 514-517 (1997)
Depositor:	Dr. M. R. Hadam, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	round, single cells growing in suspension and some slightly adherent
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement")
Subculture:	split saturated cultures 1:3 to 1:4 every 3 days; seed out at 0.3×10^6 cell/ml; maintain at $0.2-0.4 \times 10^6$ cells/ml; cells do not grow without HFCS, but it can be used at only 50% (1:100) of the recommended concentration
Incubation:	at 37°C with 5-10% CO ₂
Doubling Time:	doubling time of about 30-35 hours
Harvest:	maximal density of about $0.6-0.8 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgG ₃ monoclonal antibody
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CGTH-W-1
Cell Type:	human thyroid carcinoma
DSMZ No:	ACC 360
Origin:	established from the sternal metastasis of follicular thyroid carcinoma removed from a 70-year-old Chinese woman in 1993; described to carry IGF-1 receptors
References:	Lin et al., In Vitro Cell Dev. Biol. 31A: 819-821 (1995)
Depositor:	Dr. J. D. Lin, Chang Gung Memorial Hospital, Taipei, Taiwan
DSMZ Cell Culture Data:	
Morphology:	adherent large epitheloid cells with long processes growing in monolayers with multilayer foci
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:4 to 1:10 every 2-5 days using trypsin/EDTA; seed out at ca. $1-4 \times 10^6$ cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 20-25 hours
Harvest:	cell harvest of ca. $2-3 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3.5-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin18-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypertriploid karyotype with 10% polyploidy; 77(72-78)<3n>XX, -X, +1, +3, +3, +6, +8, +16, -17, -21, +5mar, del(X)(q25)x2, del(1)(q22), t(3;10)(p11;q11), ?del(4)(q21), del(6)(p11p21), del(7)(p14), i(9q), ?del(10)(q21/q24), der(11)t(11;14)(q24;q12), del(13)(q31)x2, add(15)(q25)x2, del(17)(p11), ?add(18)(p11); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CHO-DHFR⁻
Cell Type:	Chinese hamster ovary cells
DSMZ No:	ACC 126
Origin:	subclone of the parental line CHO which was derived from the ovary of an adult Chinese hamster in 1957; the cells are dehydrofolate reductase negative (dhfr ⁻ ; Dukx cells); patented cell line
References:	Puck et al., J. Exp. Med. 108: 945 (1958); Urlaub & Chasin, Proc. Natl. Acad. Sci. USA 77: 4216-4220 (1980)
Depositor:	Dr. H. Hauser, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent monolayer
Medium:	90% alpha-MEM + 10% FBS + 2 mM L-glutamine
Subculture:	split confluent cultures 1:3 or higher every 2-4 days using trypsin/EDTA; maximum density at 90 x 10 ⁶ cells/175 cm ² ; seed out at ca. 0.1-1.0 x 10 ⁶ cells/80 cm ² in 10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 20-24 hours
Harvest:	about 10 x 10 ⁶ cells/80 cm ² or 40-70 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Fingerprinting:	unique Hinf I-(gtg) ₅ DNA profile
Species:	confirmed as hamster with IEF of AST, NP
Cytogenetics:	near diploid hamster karyotype with 20% polyploidy; 21(20-22)<2n>; distinctive CHO morphology
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CHO-K1
Cell Type:	Chinese hamster ovary cells
DSMZ No:	ACC 110
Origin:	subclone from parental CHO cell line that was initiated from an ovary biopsy of an adult Chinese hamster in 1957
References:	Puck et al., J. Exp. Med. 108: 945 (1958); Proc. Natl. Acad. Sci USA 53: 288 (1965) and 60: 1275 (1968)
Depositor:	Dr. P. E. Bryant, University of St. Andrews, St. Andrews, Fife, UK
DSMZ Cell Culture Data:	
Morphology:	adherent, fibroblastoid cells, may be induced to undergo additional differentiation
Medium:	90% Ham's F12 + 10% FBS
Subculture:	split confluent cultures 1:3 to 1:10 using trypsin/EDTA; seed out about 10^5 - 10^6 cells in an 80 cm ² flask; cells grow up in 3-4 days to confluence
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	about 10×10^6 cells/80 cm ² or 3×10^6 cells/25 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Fingerprinting:	same Hinf I-(gtg) ₅ DNA profile as CHO-DHFR
Species:	confirmed as hamster with IEF of NP, PEP B
Cytogenetics:	near diploid karyotype with 12% polyploidy; 22(20-23); exhibits distinctive CHO karyology
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CHP-126
Cell Type:	human neuroblastoma
DSMZ No:	ACC 304
Origin:	established from a large retroperitoneal mass excised from a 14-month-old girl with neuroblastoma (stage III) in 1973; N-myc and DDX1 have been reported to be amplified
References:	Schlesinger et al., Cancer Res. 36: 3094-3100 (1976)
Depositor:	obtained from RIKEN Cell Bank (RCB0486), Tsubuka Science City, Ibaraki, Japan
DSMZ Cell Culture Data:	
Morphology:	spheroidal cells growing as clumps in suspension; grows as semi-adherent monolayer on surfaces coated with poly-L-lysine
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	maintain at $0.5\text{--}1.5 \times 10^6$ cells/ml; optimal split ratio 1:2 to 1:3 every 3-5 days; resuspend clumps by gentle pipetting on each occasion; growth may be slow after recovery from freezing (doubling time of ca. 150-200 h) and may be encouraged by plating out on flasks coated with poly-L-lysine
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 100 hours
Harvest:	saturation density of ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5\text{--}20 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, MDH, NP
Cytogenetics:	human near-diploid karyotype; 46(44-46)<2n>X, -X, +mar, del(1)(p36), t(2;14)(q37;q21-23), dup(7)(p15p22), der(11)t(1;11)(q22;q24), del(12)(q24.32), add(13)(p13), add(14)(p12); ca. 20-50 dmin; del(1p32-36) and dmin (MYCN amp) are typical of late-stage neuroblastoma
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CL-11
Cell Type:	human colon carcinoma
DSMZ No:	ACC 467
Origin:	established from the primary tumor at the left colon (a moderately differentiated colorectal carcinoma of TNM stage 3); sister cell line of mucoid growing cell line CL-11M; cells were described as expressing tumor-associated carcino-embryonic antigen (CEA)
References:	Farrell et al., Cytotechnology 34: 205-211 (2000)
Depositor:	Dr. O. Pettengill, Dartmouth Hitchcock Medical Center, Lebanon, NH, USA
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells growing in monolayers and forming islets
Medium:	80% mixture of Ham's F12 + Dulbecco's MEM (at 1:1) + 20% FBS
Subculture:	split confluent culture 1:2 to 1:3 once a week using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ² ; maximal density at ca. 1.5×10^5 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 3-4 days
Harvest:	cell harvest of ca. 12×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human near-diploid karyotype with 12% polyploidy; 46(45-47)<2n>XY, -6, +7, -17, -18, +2-3 mar, +1-4 supernumerary r/dm, add(8)(q24); sideline with idem, +7
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CL-14
Cell Type:	human colon carcinoma
DSMZ No:	ACC 504
Origin:	established from a male patient with a well-differentiated primary sigmoid adenocarcinoma of TNM stage T3N0M0
References:	Farrell et al., Cytotechnology 34: 205-211 (2000)
Depositor:	Dr. O. Pettengill, Dartmouth Hitchcock Medical Center, Lebanon, NH, USA
DSMZ Cell Culture Data:	
Morphology:	strongly adherent epitheloid cells growing in monolayers and forming islets
Medium:	80% mixture of Dulbecco's MEM + Ham's F12 at 1:1 + 20% FBS
Subculture:	split confluent culture 1:2 once a week; it is recommended to use trypsin/EDTA at higher concentration or "Accutase" (PAA, Austria) and incubate for 20-30 min at 37 °C, avoid any scraping; seed out at ca. 1×10^6 cells/25 cm ² ; maintain at ca. $1-2 \times 10^6$ cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 1 week
Harvest:	cell harvest of ca. 2×10^6 cells/25 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hypotetraploid karyotype with 3.5% polyploidy; 78-85<4n>XX, -Y, -Y, -3, -3, -4, -4, +5, +5, +6, -8, +9, +9, -10, -10, +11, -13, -14, -16, -16, -17, -18, -18, -18, -21, -21, -21, -22, +6-12mar, del(1)(p32)x2, i(5p)x3, add(14)(p11)x2, add(17)(p11), add(19)(q13)x2; ring and large isodicentric hsr marker chromosomes present
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CL-34
Cell Type:	human colon carcinoma
DSMZ No:	ACC 520
Origin:	established from a woman with a moderately differentiated primary adenocarcinoma of the right colon (TNM stage 2)
References:	Farrell et al., Cytotechnology 34: 205-211 (2000)
Depositor:	Dr. O. Pettengill, Dartmouth Hitchcock Medical Center, Lebanon, NH, USA
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing as monolayer; cells do not grow to complete confluency
Medium:	80% mixture of Dulbecco's MEM + Ham's F12 at 1:1 + 20% FBS
Subculture:	split confluent culture 1:4 to 1:6 once or twice a week using trypsin/EDTA for 10 min; seed out at ca. 4×10^6 cells/80 cm ² ; after thawing use collagen-coated flasks for the first week
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of ca. 15×10^6 cells/180 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hypodiploid karyotype with 16% polyploidy; 45(42-52)<2n>X/XX, -6; sdi with +3, del(3)(q21); near-normal "unstable" karyotype with many inconsistent structural and numerical changes
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CL-40
Cell Type:	human colon carcinoma
DSMZ No:	ACC 535
Origin:	established from the primary colorectal cancer (TNM stage 3) of the right colon of a woman; cells were described to be well differentiated
References:	Farrell et al., Cytotechnology 34: 205-211 (2000)
Depositor:	Dr. O. Pettengill, Dartmouth Hitchcock Medical Center, Lebanon, NH, USA
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing as monolayers and in aggregates
Medium:	80% mixture of Dulbecco's MEM + Ham's F12 at 1:1 + 20% FBS
Subculture:	split confluent culture 1:4 once a week using the double amount of trypsin/EDTA for up to 20 min (cells are difficult to detach), after detachment clumps of cells appear; to avoid multiple detachments cells can be seeded out at $2-3 \times 10^6$ cells/175 cm ² ; reduced viability immediately after thawing (ca. 50%), cells recover subsequently over the next weeks; maximal density of ca. 12×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 3-4 days
Harvest:	cell harvest of ca. 8×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype with 8% polyploidy; 48(45-49)<2n>XX, +12, +13; no consistent structural abnormality was detected
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CMK
Cell Type:	human acute megakaryocytic leukemia
DSMZ No:	ACC 392
Origin:	established from the peripheral blood of a 10-month-old boy with Down's syndrome and acute megakaryocytic leukemia (AML M7) at relapse in 1985
References:	Sato et al., Brit. J. Haematol. 72: 184-190 (1989); Komatsu et al., Blood 74: 42-48 (1989)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single, multiformed, large cells growing in suspension, sometimes loosely adherent (<3%); 2-3% giant, multinucleated cells
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; split saturated culture 1:2 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	maximal density at ca. $1.0-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD13+, CD19-, CD33+, CD34+, CD41+, CD42(+), CD71+, GlyA+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human flat-modulated hypotetraploid karyotype with 8% polyploidy; 85-90% $4n > XY, -X, -Y, -2, -3, +5, -6, -6, -8, +11, -15, -15, +16, -17, -19, +21, +22, +7-11mar, add(1)(q31), add(1)(p36), add(3)(q11), del(3)(p14)x2-3, add(5)(q11), add(5)(q13), dup(8)(q11q21), add(8)(q13-21), del(8)(q11), del(9)(p21)x2, add(9)(q11)x2, del(10)(q22q24), der(11;17)(q10;q10), der(11)dup(11)(p13p15)t(5;11)(q11;p15)x1-2, del(11)(q23), add(12)(p13)x2, add(17)(p1?), add(18)(q23)x2-3, add(19)(p13), der(20)t(1;20)(q2?5;q1?2)x2, add(22)(q13)$; disomic rearrangement of 20q1 close to common deleted segment, together with additional rearrangements of ch 20 present in markers; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CML-T1
Cell Type:	human T cell leukemia
DSMZ No:	ACC 7
Origin:	established from the peripheral blood of a 36-year-old woman with chronic myeloid leukemia (CML) in blast crisis in 1987; cells were described to express T cell surface markers and to have a bcr-abl translocation (producing the p210 bcr-abl protein)
References:	Kuriyama et al., Blood 74: 1381-1387 (1989); Soda et al., Brit. J. Haematol. 59: 671-679 (1985) (case report); review article: Drexler, Leukemia Res. 18: 919-927 (1994)
Depositor:	Dr. K. Kuriyama, Nagasaki University School of Medicine, Nagasaki, Japan
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension
Medium:	90% Iscove's MDM or RPMI 1640 + 10% FBS
Subculture:	maintain at $0.1-1.0 \times 10^6$ cells/ml; split 1:2 to 1:4 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48-60 hours
Harvest:	maximal density at $2.5-3.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3-, CD4+, CD5+, CD6+, CD7+, CD8+, CD13-, CD19-, CD34-, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human near tetraploid karyotype with diploid sideline; 92<4n>XXX, t(6;7)(q24;q35)x2, del(11)(q22.3)x2
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COLO-206F
Cell Type:	human colon carcinoma
DSMZ No:	ACC 21
Origin:	established from the ascites fluid of a 71-year-old man with colon carcinoma in 1975
References:	not published
Depositor:	Dr. G. E. Moore, Colorado Oncology Foundation, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	mostly round cells; about 80% are adherent cells; about 2-8% spindle-shaped cells; some giant cells
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent cultures about 1:6 once a week using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 2-3 days
Harvest:	about $7-10 \times 10^6$ cells/80 cm ² ;
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2.0-2.5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hyperdiploid karyotype with 10% polyploidy; 72(66-77)<2n>XX, -Y, +1, +3, +3, +5, -6, +7, -8, +9, -10, -13, -18, +20, -21, +3mar, i(1q), der(1)t(1;18)(q11;p11), der(3) t(3;?)(p21;?)x2, del(10)(q23.3), der(12)t(3;12)(q11;p13); a small supernumerary centric fragment or ring present in 18% of cells
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COLO-320
Cell Type:	human colon adenocarcinoma
DSMZ No:	ACC 144
Origin:	established from the tumor mass of a 55-year-old Caucasian woman with a moderately undifferentiated adenocarcinoma of the sigmoid colon in 1977; cells were described to produce several hormones, e.g. serotonin, epinephrine; cells express c-myc mRNA
References:	Quinn et al., Cancer Res. 39: 4914-4924 (1979)
Depositor:	Dr. H. Kirchner, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	undifferentiated, round cells in suspension and loosely adherent in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:2 to 1:3 every 24-48 hours (trypsin/EDTA is not necessary as cells are only loosely adherent); seed out at ca. $0.2-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 38 hours
Harvest:	maximal density at about 7×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, G6PD, MDH, NP
Cytogenetics:	human hyperdiploid karyotype; 53(50-54)<2n>XX, -3, -3, +5, +7, +12, -13, -17, +18, +22, +6mar, t(1;5) (p11;q11), der(5)t(5;?)(p15;?), der(7)t(7;?)(q22;?), der(9)t(9;?)(p11;?), del(9)(p22), del(12)(p12), i(13q), der(14)t(14;?)(q24;?), der(18)t(18;?)(p11;?); variable numbers of supernumerary dmns were present in 90% of cells
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COLO-677
Cell Type:	human small cell lung carcinoma
DSMZ No:	ACC 248
Origin:	established from the tumor-containing left axillary lymph node of a 39-year-old white man with small lung cell carcinoma (T3 N3 M3) in 1989
References:	not published
Depositor:	Dr. G. E. Moore, Colorado Oncology Foundation, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	mostly adherent round cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:4 to 1:6 every 3-4 days using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 40 hours
Harvest:	harvest of about 16×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, cytokeratin-8-, cytokeratin-18-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, PEP B
Cytogenetics:	human highly rearranged hypotriploid karyotype with 17% polyploidy; 62(58-62)<3n>XY, -X/Y, -2, -4, -4, -6, -8, -9, -9, -10, -13, -15, -16, +5mar; add(1)(p21), der(3)t(1;3)(q22;q24), add(4)(q32), add(5)(p11), del(7)(q32), add(11)(p14), add(13)(p11)x2, add(14)(q32), i(14q), add(17)(p12)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COLO-678
Cell Type:	human colon carcinoma
DSMZ No:	ACC 194
Origin:	established from a metastatic lymph node of a 69-year-old man with colon carcinoma in 1984
References:	not published
Depositor:	Dr. G. E. Moore, Colorado Oncology Foundation, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	round or spindle-shaped adherent cells growing as monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:4 every 6-7 days using trypsin/EDTA; seed out at ca. $1-3 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60-70 hours
Harvest:	cell harvest of about $20-25 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype with 4% polyploidy; 57(52-58)<2n>XXY, +X, +3, +7, +8, +10, +11, +12, +15, -17, -18, +22, +4mar, add(3)(p11), amp(7)(q11.23)x2, del(12)(q21/22q24.2)x2; trisomy 7 and 12 are associated with colon carcinoma
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COLO-679
Cell Type:	human melanoma
DSMZ No:	ACC 264
Origin:	established in 1985 from the soft tissue metastasis of a malignant melanoma from a 47-year-old woman
References:	not published
Depositor:	Dr. G. E. Moore, Colorado Oncology Foundation, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblastic, but not limited to monolayer growth
Medium:	90-95% RPMI 1640 + 5-10% FBS
Subculture:	split confluent culture 1:5 to 1:10 using trypsin/EDTA; change medium twice weekly; seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 34 hours
Harvest:	cell harvest of about 25×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, HMB-45(+), neurofilament(+), vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human, flat-moded, hypotetraploid karyotype with 8% polyploidy; 83-92<4n>XXXX, -1, +3, -4, +7, +7, -14, -15, -15, -16, -18, -20, -20, +3mar; del(3)(p11)x1-2, del(3)(q13.3), add(7)(q35)x2, del(9)(p22)x2, add(10)(p11)x1-2, add(10)(p13), add(14)(p11), add(19)(p13.3)x2, add(21)(p11), add(22)(q13)x4; extensive subclonal variation of chs 1, 5 and 7; 9p- recurrent in melanoma
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COLO-680N
Cell Type:	human esophagus squamous cell carcinoma
DSMZ No:	ACC 182
Origin:	derived from the esophagus squamous cell carcinoma of a 57-year-old black woman in 1985 via passage through a nude mouse
References:	not published
Depositor:	Dr. G. E. Moore, Colorado Oncology Foundation, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	adherent, epitheloid cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent cultures 1:2 to 1:3 every 2-3 days using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 60-70 hours
Harvest:	cell harvest of about $0.2-0.4 \times 10^6$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human highly rearranged hyperdiploid/hypotriploid karyotype with 13.5% polyploidy; 53-58<2n>X, -X, +1, +1, -3, -5, -5, +6, +7, +12, -13, -14, -14, -16, -16, -18, -20, -21, -22, +16mar; add(1)(p35)x1-2, add(1)(q11)x1-2, add(1)(p11), add(3)(p11), add(4)(q35), add(7)(q35)x1-2, add(7)(p22)del(7)(q11q21), der(9)t(5;9)(q14;p11), del(11)(p11), add(12)(q24), add(19)(p13), add(19)(q13)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COLO-699
Cell Type:	human lung adenocarcinoma
DSMZ No:	ACC 196
Origin:	established from the pleural fluid of a 57-year-old woman with adenocarcinoma of the lung in 1986
References:	not published
Depositor:	Dr. G. E. Moore, Colorado Oncology Foundation, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	adherent, epitheloid cells growing as monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent cultures every 2-3 days using trypsin/EDTA; seed out about $1-2 \times 10^6$ cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 38 hours
Harvest:	cell harvest of about 12×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, cytokeratin-8-, cytokeratin-18-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+; cells are HMB-45-positive (this marker is highly associated with melanomas - hence the cells may be melanoma-derived)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypertriploid karyotype with 2% polyploidy; 70(62-72)<3n>XX, -X, +1, +3, -4, +6, +7, +7, +8, -9, -10, +12, -14, -16, -17, -17, +18, +20, der(2)t(2;10)(q27;q12), der(7)t(7;?)(p21;?)x2, i(10q), der(11)t(8;11)(q12;q14), idic(16q), der(21)t(21;?)(p12;?)x2; a supernumerary small ring chromosome was present in 28% cells; corresponds to original karyotype with reinterpretation
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COLO-704
Cell Type:	human ovarian adenocarcinoma
DSMZ No:	ACC 198
Origin:	established from the ascites fluid stemming from the colon metastasis of a 46-year-old white woman with ovarian adenocarcinoma in 1986
References:	not published
Depositor:	Dr. G. E. Moore, Colorado Oncology Foundation, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	grows as clusters and single cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; optimal split ratio of 1:2 to 1:3 every 2-3 days; seed out at ca. $0.2-0.3 \times 10^6$ cells/ml
Incubation:	5% CO ₂ (or may be grown in closed flasks), 37 °C
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	saturation density at about 2.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hyperdiploid karyotype with 22% polyploidy; 51(48-53)<2n>XX, +1, +7, +8, +8, -17, +19, +20, i(1q), der(8)t(8;?)(q21;?); dm present in 16% cells; i(1q) is associated with ovarian carcinoma; identification confirmed
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COLO-720L
Cell Type:	human B lymphoblastoid cells (from patient with adenocarcinoma)
DSMZ No:	ACC 206
Origin:	B-lymphoblastoid cell line established from the peripheral blood of a 73-year-old woman with metastatic adenocarcinoma in 1988
References:	not published
Depositor:	Dr. G. E. Moore, Colorado Oncology Foundation, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	lymphoblast-like cells growing in suspension as single cells and in clumps
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂ (or may be grown in closed flasks)
Doubling Time:	doubling time of ca. 20-30 hours
Harvest:	saturation density at about $2.0-2.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD37+, HLA-DR+, sm/cyIgM-, smIg/cyG+, sm/cykappa(+)/+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hyperdiploid karyotype with 10% polyploidy; 47(45-50)<2n>XX, +12, der(5)t(3;5)(p21.3;p15); the 3p21.3 breakpoint is associated with benign adenoma
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COLO-783
Cell Type:	human melanoma
DSMZ No:	ACC 257
Origin:	established in 1990 from the right axillary lymph node of a 48-year-old woman with metastatic malignant melanoma in the right chest
References:	not published
Depositor:	Dr. G. E. Moore, Colorado Oncology Foundation, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblasts, not limited to monolayer growth
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:2 to 1:3 every 1-2 weeks using trypsin/EDTA; seed out at about 2-3 x 10 ⁶ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 5-7 days
Harvest:	cell harvest of about 20 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2-3 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypotriploid karyotype with 20% polyploidy; 65(61-69)<3n>XXX, -4, -5, -9, -10, -12, +2mar, add(4)(p11), del(6)(p21), del(7)(p14)x2
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COLO-800
Cell Type:	human melanoma
DSMZ No:	ACC 193
Origin:	established from the subcutaneous nodule of a 14-year-old boy with melanoma in 1990
References:	not published
Depositor:	Dr. G. E. Moore, Colorado Oncology Foundation, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing as monolayer; some cells look like fibroblasts or neurons
Medium:	90-95% RPMI 1640 + 5-10% FBS
Subculture:	split confluent culture 1:4 to 1:5 two to three times per week using trypsin/EDTA; seed out at ca. 1-2 x 10 ⁶ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours (with 5% FBS)
Harvest:	cell harvest of about 5 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, HMB-45+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypotriploid karyotype with minor hypertriploid clone and 8% polyploidy; 64(50-69)<3n>X, -Y, -X/Y, -4, -5, +7, -8, -9, -10, -11, -12, +13, +13, +14, +15, -16, -19, +20, -21, -21, der(3)t(3;?)(p11;?), der(13q21q)x2, del(14)(?q11-12q24.2)x2, der(15)t(15;?)(q26;?)x2; identity confirmed by marker correspondence
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COLO-818
Cell Type:	human melanoma
DSMZ No:	ACC 81
Origin:	established from malignant melanoma
References:	not published
Depositor:	Dr. G. E. Moore, Colorado Oncology Foundation, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	adherent growth in monolayers, mostly spindle-like, some cells with cellular processes, 1-5% very big cells, no strict contact inhibition
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture about 1:6 once a week using trypsin/EDTA; seed out at about 1×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 60 hours
Harvest:	cell harvest of about 8×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	unique DNA profile using multiplex PCR of minisatellite markers, but identical with COLO-800 (DSM ACC 193)
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypotriploid karyotype with 30% polyploidy; 62(57-63)X,-X,-X,-4,-5,+7,+8,-9,-10,-11,-13,+15,-16,-17,-19,-21,-21,+3mar,1 dm present in 15% cells; add(3)(p11), der(13;21)(p10;10); high level of subclonal rearrangement present; closely related to COLO-800 (DSM ACC 193)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COLO-824
Cell Type:	human breast carcinoma
DSMZ No:	ACC 200
Origin:	established from the pleural fluid of a 52-year-old man with breast carcinoma in 1991
References:	not published
Depositor:	Dr. G. E. Moore, Colorado Oncology Foundation, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	cells grow loosely adherent in monolayers, some cells grow as round cells in clumps in suspension
Medium:	85% RPMI 1640 + 15% FBS
Subculture:	split 1:2 to 1:3 every 2-4 days without use of trypsin/EDTA (cells are only loosely adherent); seed out at ca. $3-5 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60-80 hours
Harvest:	cell harvest of about $15-20 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human highly rearranged hyperdiploid karyotype with 11% polyploidy; 51(48-52)<2n>X, -Y, +1, -2, -6, +7, -10, -13, -15, -19, -21, -22, +12mar, del(1)(p12), der(1)t(1;?)(p12;?), der(6)t(1;6)(q25;q12), der(7)t(7;?)(q32/34;?), der(8)t(8;?)(q24;?), der(12)t(12;?)(p13;?), der(15)t(15;?)(p12;?), der(17)t(17;?)(p12;?); carries small bisatellited marker
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COLO-849
Cell Type:	human melanoma
DSMZ No:	ACC 216
Origin:	derived in 1992 from right axillary lymph nodes of a 43-year-old Caucasian man with metastatic malignant melanoma
References:	Morse & Moore, Cancer Genet. Cytogenet. 69: 108-112 (1993)
Depositor:	Dr. G. E. Moore, Colorado Oncology Foundation, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	fibroblastic cells growing mainly as adherent monolayer and partly in suspension; sedimented cells have a brown color (suggesting production of melanin)
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	split confluent culture 1:4 to 1:10 every 4-7 days using trypsin/EDTA; seed out at ca. $2-3 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	cell harvest of about 18×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2.5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, HMB-45+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypotriploid karyotype; $60 < 3n > X, -Y, -1, +3, -4, -5, +7, -10, -11, -12, -13, -14, -15, -16, -17, -18, +20, +22, der(1)t(1;?)(q11;?), i(3q), idic(3)(3cen->p24::1q11->qter), der(7)t(7;?)(q33;?)x2, der(9)t(9;?)(p11;?)/del(9)(p13)$; the large idic(3) marker and i(3q) derive from an earlier translocation, t(1;3)(q11;p24), present in a sideline; idic marker clearly derives from published translocation
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COS-1
Cell Type:	African green monkey kidney
DSMZ No:	ACC 63
Origin:	established from CV-1 Simian cells (<i>cercopithecus aethiops</i>) which were transformed by an origin-defective mutant of SV-40; cells are host cells for the propagation of pure populations of recombinant SV virus; classified as risk category 1 according to the German Central Commission for Biological Safety (ZKBS)
References:	Gluzman, Cell 23: 175-182 (1981)
Depositor:	Dr. R. Riebe, Friedrich-Loeffler-Institute, Insel Riems, Germany
DSMZ Cell Culture Data:	
Morphology:	fibroblast-like, adherent monolayers
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent cultures 1:3 to 1:10 every 3-7 days using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ² in 10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	about 5×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Fingerprinting:	unique Hinf I-(gtg) ₅ DNA profile (identical to COS-7, DSM ACC 60)
Species:	confirmed as of non-human primate origin with IEF of AST, NP, PEP B
Cytogenetics:	hypotetraploid simian karyotype with 4% polyploidy; 102(95-108)<4n>; structural rearrangements present
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HCV-, HBV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	COS-7
Cell Type:	African green monkey kidney
DSMZ No:	ACC 60
Origin:	derived from CV-1, a simian cell line (<i>cercopithecus aethiops</i>), by transformation with an origin-defective mutant of SV-40; cells were described to support the growth of SV-40 viruses; classified as risk category 1 according to the German Central Commission for Biological Safety (ZKBS)
References:	Gluzman, Cell 23: 175-182 (1981)
Depositor:	Dr. R. Riebe, Friedrich-Loeffler-Institute, Insel Riems, Germany
DSMZ Cell Culture Data:	
Morphology:	fibroblast-like cells growing as monolayers
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:3 to 1:5 every 2-3 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ² ; cells are growing slowly
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35-48 hours
Harvest:	about $2-3 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Fingerprinting:	unique Hinf I-(gtg) ₅ DNA profile (identical to COS-1, DSM ACC 63)
Species:	confirmed as of primate origin with IEF of LDH, MPI, NP, PEP B
Cytogenetics:	simian hypodiploid karyotype with 25% polyploidy; 57(53-58)<2n>
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CPC-N
Cell Type:	human small cell lung carcinoma
DSMZ No:	ACC 306
Origin:	established from the pleural effusion metastasis of a patient with small cell lung carcinoma in 1987 (= CPN)
References:	Fabien et al., Cancer J. 7: 198-201 (1994); Paulin & Charnay, C.R. Acad. Sci. Paris 314: 259-262 (1992)
Depositor:	Dr. C. Paulin, University Hospital of Lyon, Lyon, France
DSMZ Cell Culture Data:	
Morphology:	mostly round cells growing in suspension in very large aggregates preventing exact estimation of cell numbers
Medium:	90% McCoy's 5A + 10% FBS
Subculture:	split saturated culture 1:2 every 3-4 days; seed out 2×10^6 cells in 5 ml; viability remains at max. 60%, large amounts of cell debris may prevail in the flask
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 4 days
Harvest:	maximal density at ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin(+), cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hyperdiploid karyotype with 16% polyploidy; 49(47-51)<2n>XY, +6, +15, +20, add(1)(p36), i(3q), del(6)(q1?3q2?5), der(8)t(8;17)(p23;q21), add(13)(p11)del(13)(q12q22), add(15)(p12), ins(16;8)(p13;HSR?q24); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CRO-AP2
Cell Type:	human B cell lymphoma
DSMZ No:	ACC 48
Origin:	established in 1996 from the diagnostic pre-treatment pleural effusion sampled shortly before death of a 49-year-old HIV+ homosexual man with primary effusion lymphoma (PEL) and previous history of Kaposi's sarcoma; autopsy revealed involvement of serous membranes without formation of solid tumor mass; cells were described to be EBV+, HHV-8+ and to carry a BCL-6 point mutation
References:	Carbone et al., Int. J. Cancer 73: 562-569 (1997); Uphoff et al., Leukemia 12: 1806-1809 (1998); review article: Drexler et al., Leukemia 12: 1507-1517 (1998)
Depositor:	Prof. A. Carbone, Centro di Riferimento Oncologico, Aviano, Italy
DSMZ Cell Culture Data:	
Morphology:	lymphoblast-like cells growing in clumps or as single cells in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture about 1:2 to 1:3 every 2-4 days; maintain at ca. $0.2-0.5 \times 10^6$ cells/ml feeding cultures daily; seed out after thawing at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 20-40 hours
Harvest:	maximal density at ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19-, CD20+, CD34-, CD37+, CD79a-, CD80-, CD138-, HLA-DR+, sm/cyIgGMkappa/lambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human hyperdiploid karyotype with 12% polyploidy; 48(47-48)<2n>X, +X, -Y, +12, add(?Y)(p11), add(1)(q43.1), der(15)t(8;15)(q11;q25), der(22)t(1;22)(q21;q13); sideline with add(9)(q13); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8+, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CRO-AP3
Cell Type:	human B cell lymphoma
DSMZ No:	ACC 275
Origin:	established from the malignant primary effusion lymphoma (PEL) of a 42-year-old HIV+ man at diagnosis (prior to therapy) without a previous history of Kaposi's sarcoma; cells were shown to carry the human herpesvirus 8 (HHV-8), but not the Epstein-Barr virus (EBV) and not the human immunodeficiency virus (HIV)
References:	Carbone et al., Brit. J. Haematol. 102: 1081-1089 (1998); Uphoff et al., Leukemia 12: 1806-1809 (1998); review article: Drexler et al., Leukemia 12: 1507-1517 (1998)
Depositor:	Prof. A. Carbone, Centro di Riferimento Oncologico, Aviano, Italy
DSMZ Cell Culture Data:	
Morphology:	single, round cells growing in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 to 1:3 every second day; seed out at ca. 0.3×10^6 cells/ml; maintain at ca. $0.2-0.3 \times 10^6$ cells/ml; maximal density at ca. 0.6×10^6 cells/ml (maximal density should not be exceeded)
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of ca. 0.4×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD4-, CD10-, CD13-, CD19-, CD20-, CD34-, CD37-, CD79a-, CD80-, CD138-, HLA-DR+, sm/cyIgG-, sm/cyIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, MDH
Cytogenetics:	human flat-modulated hypotetraploid karyotype; 80-88<4n>XX, -Y, -Y, -1, -2, +6, -7, -10, -13, -13, -22, -22, +6mar, i(1q), add(2)(p?), add(3)(q2?7), der(4;9)(q10;q10)x1-2, del(5)(p13p15)x1-2, del(5)(q2?1)x1-2, der(6)t(6;7)(q15;q11)add(7)(q3?5), del(8)(p11), del(9)(p21), add(9)(p11), del(11)(q2?2q2?3)x2, der(12)t(2;12)(p12;q24.32), add(13)(p11), t(14;19)(q11;q13)x2, der(15)t(1;?)(?15)(q21;?)(?;q11), ider(?)t(8;?)(q11;?); extensive subclonal rearrangements present; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8+, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CRO-AP5
Cell Type:	human B cell lymphoma
DSMZ No:	ACC 215
Origin:	established in 1998 from the pleural effusion of a 35-year-old HIV-positive homosexual man with primary effusion lymphoma and a history of Kaposi's sarcoma; cells are HHV-8-positive, but HIV-negative
References:	Carbone et al., Brit. J. Haematol. 102: 1081-1089 (1998); Uphoff et al., Leukemia 12: 806-1809 (1998); review article: Drexler et al., Leukemia 12: 1507-1517 (1998)
Depositor:	Prof. A. Carbone, Centro di Riferimento Oncologico, Aviano, Italy
DSMZ Cell Culture Data:	
Morphology:	single round to pleomorph cells growing in suspension
Medium:	80-85% RPMI 1640 + 15-20% FBS
Subculture:	split saturated culture 1:3 every 3-4 days; maintain at ca. $0.4-0.8 \times 10^6$ cells/ml; seed out after thawing at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35-40 hours
Harvest:	maximal density at ca. 1×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19-, CD20-, CD34-, CD37+, CD79a-, CD80-, CD138-, HLA-DR+, smlgGMkappa/lambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH
Cytogenetics:	human hyperdiploid karyotype with 12% polyploidy; 44-49<2n>X, -Y, +8, +2mar, del(4)(q25), add(8)(p11), dup(12)(q11q14), del(14)(q24); sideline with i(8q); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8+, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CTV-1
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 40
Origin:	established from the peripheral blood of a 40-year-old man with acute monoblastic leukemia (AML M5) at relapse in 1982 (according to original publication allegedly derived from a woman)
References:	Chen et al., Gann (Jpn. J. Cancer Res.) 75: 660-664 (1984)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round or multiformed cells, singly or clustered in suspension
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; seed out at ca. $0.8-1.0 \times 10^6$ cells/ml; split 1:2 to 1:4 every 3-4 days; after thawing, culture should first be started with 20% FBS and best in 24-well-plates
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	maximal density at about $1.0-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3-, CD4-, CD5-, CD6+, CD7+, CD8-, CD13-, CD14(+), CD15+, CD19-, CD33-, CD34-, CD68-, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human near tetraploid karyotype with 9% polyploidy; 92(89-94)<4n>XX/XXY/XXYY, -1, +6, +6, -17, t(1;7)(p34.2;q34)x2, del(3)(p21), i(6q)x2, t(12;16)(q24.32;q11)x2
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	CX-1
Cell Type:	human colon adenocarcinoma (derivative of HT-29)
DSMZ No:	ACC 129
Origin:	established from the tumor tissue of a 44-year-old woman with well-differentiated colon adenocarcinoma in 1984; cell line is a subclone of cell line HT-29 (DSM ACC 299)
References:	Goldin et al., Eur. J. Cancer 17: 129-142 (1981)
Depositor:	obtained from DKFZ Tumorbank (TZB No. 610004), Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	small circular cells, adherent, grow to large colonies in which the single cells are no longer discernible
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:3 to 1:6 every 3-5 days using trypsin/EDTA, seed out about 10 ⁶ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	about 30 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1-2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypertriploid karyotype with 14% polyploidy; 72(65-74)<3n>XX, -X, +4, +7, -8, -8, -9, +11, +11, -13, +15, +17, -18, +2mar, der(4)t(1;4)(p11;p15)t(1;?) (p32-36;?), del(4)(q32), del(7)(p21), i(13q), add(15)(p11), del(17)(p12), add(21)(p11); two small markers present; carries 3 markers similar to those present in HT-29; rearrangements of chs 1, 7, 17 typical of colon carcinoma
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	D-11
Cell Type:	fish rainbow trout liver
DSMZ No:	ACC 77
Origin:	this cell line is a subclone of the original cell line R1 established after elimination of mycoplasma with BM-Cyclin (tiamulin & minocycline)
References:	Ahne, Zbl. Bakt. Hyg. B 180: 480-504 (1985)
Depositor:	Dr. M. Kohlpoth, Akademie für Tierschutz, Neubiberg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, fibroblastoid cells growing as monolayers
Medium:	80% Medium 199 (25 mM HEPES) with Earle's salts + 20% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:2 every 7-10 days using trypsin/EDTA in PBS without $\text{Ca}^{2+}/\text{Mg}^{2+}$ at room temperature for 5-10 min
Incubation:	at room temperature (about 20 °C) without CO_2 (keep lid of flask closed), cells can be stored temporarily (up to 1 week) at 4 °C
Doubling Time:	doubling time of about 80 hours or more
Harvest:	about $5\text{-}15 \times 10^6$ cells/80 cm^2
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays

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INFORMATION SHEET FOR CELL LINES

Cell Line:	D-36
Cell Type:	mouse mast cells
DSMZ No:	ACC 184
Origin:	established from the bone marrow of a DBA/2 mouse in 1986; cells can be used for an IL-10 assay
References:	Schlaak et al., J. Immunol. Methods 168: 49-54 (1994)
Depositor:	Dr. J. F. Schlaak, University of Mainz, Mainz, Germany
DSMZ Cell Culture Data:	
Morphology:	growing in suspension in small clumps
Medium:	90% Iscove's MDM + 10% FBS + 2 mM L-glutamine + 1 mM sodium pyruvate + 1 ng/ml mouse IL-4 + 1 ng/ml mouse IL-3
Subculture:	maintain at about $0.2-8.0 \times 10^5$ cells/ml; split ratio of 1:4 every 2-3 days; viability of cells may drop to about 20% immediately after thawing but recovers after approximately 1 week
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 25 hours
Harvest:	maximum density at about 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, NP
Cytogenetics:	murine near-diploid karyotype with 2% tetraploidy; 40(37-41)<2n>; centric fusion only rarely present
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	D10.G4.1
Cell Type:	mouse T-helper cells
DSMZ No:	ACC 45
Origin:	clone established from an AKR/J(H-2 ^k) mouse; indicator cell line for IL-1
References:	Kaye et al., J. Exp. Med. 158: 836 (1983); Kaye et al., J. Immunol. 133: 2291-2293 (1984)
Depositor:	Dr. H. Quentmeier, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension
Medium:	propagation medium: 90% RPMI 1640 + 10% FBS + 2 mM L-glutamine + 30 U/ml rh or rm IL-2 (CTLL units)
Subculture:	maintain at $5-30 \times 10^4$ cells/ml; split ratio 1:3 to 1:4 every 3 days; cells have to be feedered every two weeks with spleen cells of H-2 ^k mice; feeder medium: 90% RPMI 1640 + 10% FBS + 2 mM L-glutamine + 10% (vol) supernatant of Con A-stimulated mouse spleen cells, 0.05 M methyl- α -D-mannopyranoside, 200 μ g/ml conalbumin; seed out at ca. 3×10^5 cells/ml
Incubation:	at 37 °C with 7.5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	maximal density at $0.6-1.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1.5-2.0 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	half maximal proliferation with about 10 pg rh IL-1 β /ml in MTT assay
Species:	confirmed as mouse with IEF of AST, MDH, NP
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	D3
Cell Type:	mouse hybridoma (anti-chicken N-CAM)
DSMZ No:	ACC 428
Origin:	cell line was established by fusion of mouse NS-1 cells with spleen cells from a BALB/c mouse immunized with wheat germ agglutinin affinity purified chicken brain proteins immobilized on lectin beads; described as secreting a mouse monoclonal antibody (IgG) reacting with the cytoplasmic domain of the 140, 180 and 250 kDa glycosylated neuronal cell surface protein N-CAM, preferentially in neurons of the chicken visual system like retina, tectum and other CNS regions and PNS; also reacting with mouse, goldfish and human N-CAM in paraformaldehyde-fixed cryosections and Western blots
References:	Schlosshauer, J. Neurochem. 52: 82-92 (1989)
Depositor:	Dr. B. Schlosshauer, Natural and Medical Science Institute (NMI), Reutlingen, Germany
DSMZ Cell Culture Data:	
Morphology:	round, single cells growing in suspension and some slightly adherent
Medium:	90% RPMI 1640 + 10% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement")
Subculture:	maintain at ca. 0.5×10^6 cells/ml; split saturated culture 1:5 to 1:10 every 2-3 days (some cells attach to the surface of the culture flask, but can be detached by tapping the flask); seed out at ca. 0.5×10^6 cells/ml; maximum density of ca. $1-2 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 22 hours
Harvest:	cell harvest of ca. $0.5-1.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgG ₂ A monoclonal antibody
Species:	confirmed as murine with IEF of LDH, NP
Cytogenetics:	murine flat-modulated hypertetraploid karyotype; 98-104<4n>; two centric fusion markers present
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DA-1
Cell Type:	mouse lymphoma
DSMZ No:	ACC 51
Origin:	established from a BALB/c mouse induced by MOLV; described to be growth-dependent on IL-3, but to recognize also GM-CSF
References:	Pierce et al., Cell 41: 685-693 (1985)
Depositor:	Dr. E. Schmitt, Institute of Immunology, University of Mainz, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension
Medium:	95% Iscove's MDM + 5% FBS + 2 mM L-glutamine + 2.5×10^{-5} M mercaptoethanol + IL-3 (either 30% vol conditioned medium of WEHI-3B cell line (DSM ACC 26) or 20 U/ml mIL-3)
Subculture:	maintain at $0.2-0.4 \times 10^6$ cells/ml; split 1:2 to 1:3 every 1-2 days, cells grow fast. CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximal density at 0.8×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture assays
Species:	confirmed as mouse with IEF of AST, G6PD, NP, PEP B
Viruses:	ELISA: reverse transcriptase negative

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DAN-G
Cell Type:	human pancreas carcinoma
DSMZ No:	ACC 249
Origin:	established from a human individual with pancreas carcinoma in 1985
References:	not published
Depositor:	obtained from DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:2 to 1:4 every 2-4 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 33 hours
Harvest:	cell harvest of about $1-3 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human irregular hypotriploid karyotype with 3.5% polyploidy; 59(52-61)<3n>der(X), -X/Y, -X/Y, -1, -2, -3, -4, -5, +6, +7, -9, -10, -13, -16, -17, -18, -20, -22, -22, +4mar, add(2)(p?16), add(4)(q25), i(5p), add(6)(p22), add(7)(p11/13), der(7)t(3;7)(q11;p11), del(7)(p13), add(8)(p11), add(9)(p21), add(12)(p11), i(13q), der(14)t(14;?22)(q32;q12), add(15)(q26), der(19)t(19;hsr)(p13;?), add(20)(q13)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DAUDI
Cell Type:	human Burkitt's lymphoma
DSMZ No:	ACC 78
Origin:	established from the Burkitt's lymphoma (left orbital biopsy) of a 16-year-old African boy in 1967; cells were described to be EBV-positive and to express mRNA for proto-oncogene bcl-2; this cell line has been assigned by the ZKBS (German Central Commission for Biological Safety) to risk category 2
References:	Klein et al., Cancer Res. 28: 1300-1310 (1968); Nadkarni et al., Cancer 63: 64-79 (1969)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	single round cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; split ratio 1:2 to 1:4 every 2-4 days; cells may grow slowly initially; cell culture may be started in 24-well-plates and later transferred to flasks; seed out at ca. $0.5-1 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30 hours
Harvest:	maximal density at about 1.7×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13(+), CD19+, CD20+, CD34-, CD37+, HLA-DR+, sm/cyIgG-, sm/cyIgM+, sm/cykappa+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP, PEP B
Cytogenetics:	human near diploid karyotype with 20% polyploidy; 46(45-48)<2n>XY/XXY, +7, -9, t(8;14)(q24;q32)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DB
Cell Type:	human B cell lymphoma
DSMZ No:	ACC 539
Origin:	established from ascites of a 45-year-old Caucasian man with diffuse large cell lymphoma
References:	Beckwith et al., J. Natl. Cancer Inst. 82: 501-509 (1990)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round single cells in suspension (partly in clusters or clumps)
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. $0.5\text{-}2.0 \times 10^6$ cells/ml; minimum density at ca. 0.4×10^6 cells/ml; after thawing cultivate in 6-well-plates during the first 2 weeks
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 6×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD4-, CD10+, CD13-, CD19(+), CD20+, CD34-, CD37+, CD38+, CD79a+, cyCD79a+, CD80+, HLA-DR-, sm/cyIgG+, sm/cyIgM-, sm/cykappa-, sm/cylambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypertriploid karyotype; 80(79-81)<3n>XXXXYY, -3, +7, +7, +8, -9, +11, -13, +16, +18, +20, +20, +20, +21, +22, +mar, der(4)t(X;4)(q11;p13), add(5)(q11), t(10;20)(q24;p11), ?inv(13)(q21.2q32), t(14;18)(q32;q21), del(16)(p13.2), der(18)t(14;18)(q32;q21); carries t(14;18) associated with B-cell lymphoma
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DBTRG-05MG
Cell Type:	human glioblastoma
DSMZ No:	ACC 359
Origin:	established from the primary tumor cells of a 59-year-old Caucasian woman with glioblastoma multiforme following irradiation and chemotherapy in 1985; cells were described to express NSE (neuron-specific enolase) and EGF-R (epidermal growth factor receptor)
References:	Kruse et al., In Vitro Cell. Dev. Biol. 28A: 609-614 (1992)
Depositor:	Dr. C. A. Kruse, University of Colorado, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	spindle, fibroblast-like cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:5 twice a week using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of ca. 10×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human flat-modulated hypotetraploid karyotype with 12% polyploidy; 81-88<4n>XXX, -X, -1, -1, -3, -4, -4, +7, +7, -10, -10, -13, -13, -14, -14, -16, +6mar, der(4)t(4;14)(p11;q11), add(5)(p15)x2, add(7)(p22)x3, del(8)(q22), inv(9)(p13q34)x1-2, add(11)(p15)x2, der(11)t(1;11)(q23;q21), del(11)(q13), add(16)(q24)x2; +7 and -10 recurrent in gliomas; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DEL
Cell Type:	human malignant histiocytosis
DSMZ No:	ACC 338
Origin:	established from the pleural effusion of a 12-year-old boy with malignant histiocytosis at diagnosis in 1987; initially established by serial passage in nude mice
References:	Barbey et al., Int. J. Cancer 45: 546-553 (1990); Dirks et al., Leukemia 10: 142-149 (1996); Mason et al., Brit. J. Haematol. 74: 161-169 (1990); Drexler et al., Leukemia 14: 1533-1559 (2000)
Depositor:	Dr. S. Barbey, Hopital Necker, Paris, France
DSMZ Cell Culture Data:	
Morphology:	grows often in clusters in suspension, most cells are polygonal (some cells are round)
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:3 to 1:5 every 2-3 days; seed out at ca. $3-4 \times 10^5$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	maximal density at ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD5-, CD7-, CD10-, CD11b-, CD13+, CD14-, CD15+, CD19-, CD20-, CD25+, CD30+, CD33-, CD34-, CD68-, CD71+, HLA-DR+, smIgG-, smIgM-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human biclonal, hyperdiploid/hypertriploid, karyotype with 6.7% polyploidy; 47(44-49)<2n>XY, +5, +5, +6, -8, +mar, der(5)t(5;6)(q35;p21), t(5;6)(q35;p21), add(10)(q23-24), der(13)t(1;13)(q32;p11)t(1;13)(q21;q34), add(16)(q23), add(19)(p13), 33%; 74(74-77)<3n>XXY, +1, +3, +5, +5, +6, +7, -10, +13, -18, +20, t(5;6)(q35;p21)x2, add(10)(q24)x1-2, der(13)t(1;13)(q32;p11)t(1;13)(q21;q34)x2, add(16)(q23), add(19)(p13), 60%; matches published karyotype, t(2;5) not detectible cytogenetically; expression of NPM-ALK fusion gene was detected by RT-PCR analysis
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DERL-2
Cell Type:	human T cell lymphoma
DSMZ No:	ACC 531
Origin:	established from the peripheral blood of a 30-year-old Caucasian man with hepatosplenic gamma-delta T cell lymphoma (T-cell NHL) in 1995 at disease progression; simultaneous sister cell line of cell line DERL-7
References:	Di Noto et al., Leukemia 15: 1641-1649 (2001); Camera et al., Haematologica 85: 496-501 (2000); Matsuo & Drexler, Leukemia Res. (in press).
Depositor:	Dr. R. Di Noto, University of Naples, Italy
DSMZ Cell Culture Data:	
Morphology:	multiformed cells growing in clusters in suspension
Medium:	80% RPMI 1640 + 20% FBS + 20 ng/ml human IL-2
Subculture:	split saturated culture 1:2 every 2-3 days; seed out at ca. 1.0×10^6 cells/ml, cells grow slowly at first, cultivate in 6- or 12-well-plates for optimal growth; maximal density of ca. 2.5×10^6 cells/ml; CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60 hours
Harvest:	cell harvest of ca. 2.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5-7 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in microbiological culture and PCR assays
Immunology:	CD2+, CD3-, CD4-, CD5-, CD6+, CD7+, CD8+, CD13-, CD19-, CD33-, CD34-, HLA-DR-, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile (identical to that of sister cell line DERL-7)
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype with 8% polyploidy; 48(46-49)<2n>XY, +7, +8, +17, t(1;5)(p13;q32), i(7q), der(10;?17)(q10;q10); sideline with +del(7)(p11); i(7q) is associated with hepatosplenic T-NHL; matches originator's karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DERL-7
Cell Type:	human T cell lymphoma
DSMZ No:	ACC 524
Origin:	established from the bone marrow blood of a 30-year-old Caucasian man with hepatosplenic gamma-delta T cell lymphoma (T-cell NHL) in 1995 at disease progression; simultaneous sister cell line of cell line DERL-2
References:	Di Noto et al., Leukemia 15: 1641-1649 (2001); Camera et al., Haematologica 85: 496-501 (2000); Matsuo & Drexler, Leukemia Res. (in press).
Depositor:	Dr. R. Di Noto, University of Naples, Italy
DSMZ Cell Culture Data:	
Morphology:	small ovoid ("lymphoblastic") cells growing singly or in small clumps in suspension
Medium:	80% RPMI 1640 + 20% FBS + 20 ng/ml human IL-2
Subculture:	split saturated culture 1:2 to 1:3 every 2-4 days; seed out at ca. $0.5\text{-}1.0 \times 10^6$ cells/ml in 12-well-plates for optimal growth, cultures may be continued in flasks after 2-3 weeks following recovery; cultures may contain significant amounts of cell debris; maximal density of ca. 3.0×10^6 cells/ml; CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 3×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5\text{-}10 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in microbiological culture and PCR assays
Immunology:	CD2+, CD3-, CD4-, CD5-, CD6+, CD7+, CD8-, CD13-, CD19-, CD33-, CD34-, HLA-DR-, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile (identical to that of sister cell line DERL-2)
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype; 47(46-47)<2n>XY, +8, i(7q), add(10)(p12.1); sideline with der(1)t(1;16)(q12.2;p13.3); i(7q) is associated with hepatosplenic T-NHL; resembles published karyotype

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DG-75
Cell Type:	human Burkitt's lymphoma
DSMZ No:	ACC 83
Origin:	established from the pleural effusion of a 10-year-old boy with Burkitt's lymphoma (refractory, terminal) in 1975
References:	Ben-Bassat et al., Int. J. Cancer 19: 27-33 (1977)
Depositor:	Dr. H. Ben-Bassat, Hadassah Medical Organization, Jerusalem, Israel
DSMZ Cell Culture Data:	
Morphology:	round to polygonal, single or clustered cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.1-0.5 \times 10^6$ cells/ml; split 1:5 to 1:10 every 24-48 hours; seed out at ca. 0.5×10^6 /ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	saturation density at $0.75-1.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20(+), CD34-, CD37+, CD138(+), HLA-DR+, sm/cyIgM+, sm/cyIgG-, sm/cykappa+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, PEP B
Cytogenetics:	human near diploid karyotype with 6% polyploidy; $46<2n>XY, t(8;14)(q24;q32)$
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DK-MG
Cell Type:	human glioma
DSMZ No:	ACC 277
Origin:	established from the glioblastoma multiforme (right parietal, at diagnosis) of a 67-year-old woman in 1991
References:	Kruse et al., In Vitro Cell. Dev. Biol. 34: 455-462 (1998)
Depositor:	Dr. C. A. Kruse, University of Colorado, Denver, Colorado, USA
DSMZ Cell Culture Data:	
Morphology:	spindle, fibroblast-like cells growing mostly adherent as monolayer
Medium:	90% RPMI 1640 + 10% FBS + 2 mM L-glutamine
Subculture:	split nearly confluent culture 1:5 every 5-6 days using trypsin/EDTA; seed out at ca. 1.0×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of ca. $4-5 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at $2-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH
Cytogenetics:	human hypodiploid karyotype with 12% polyploidy; 44(42-44)<2n>XX, +7, -9, -10, -12, -14, +mar, del(6)(q13q25), add(9)(p2?2), der(15)t(14;15)(q12;p11), del(16)(p13), del(17)(q25), add(19)(q1?3), variable numbers of dmin detectible in most cells; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DLD-1
Cell Type:	human colon adenocarcinoma
DSMZ No:	ACC 278
Origin:	established from a colorectal adenocarcinoma
References:	Dexter et al., Cancer Res. 39: 1020-1025 (1979)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells growing as monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:5 every 2-3 days using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of ca. 20×10^6 /175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	identical with HCT-15 (DSM ACC 357), otherwise unique DNA profile using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human near-diploid karyotype with tetraploid (19%) and polyploid sidelines (1.9%); 46(43-47)<2n>X/XY, add(2)(p13/p23); corresponds to published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DMBM-2
Cell Type:	mouse macrophages
DSMZ No:	ACC 269
Origin:	established from the bone marrow of a C3H/HeJ mouse
References:	Monner et al., J. Leuk. Biol. 61: 469-480 (1997)
Depositor:	Dr. D. Monner, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing in aggregates, mostly loosely adherent, some cells have long processes
Medium:	80% RPMI 1640 + 20% horse serum + 1 mM sodium pyruvate + 1 mM non-essential amino acids + 1.25 µg/ml vitamin B12 + 75 µM alpha-thioglycerin + 2 mM L-glutamine + 2.5% 10x L-929 supernatant (DSM ACC 2)
Subculture:	split 1:10 every 4-6 days detaching the cells by flushing the adherent cells with medium through a 25G needle (do not try to disaggregate cells by passing through the needle); cells grow best in Corning round plates (100 x 20 mm)
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 50 hours
Harvest:	cell harvest of about 3-5 x 10 ⁶ cells/plate
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 0.5-1.0 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, LDH, NP, PEP B
Cytogenetics:	murine near-diploid karyotype with 10% tetraploidy; 42(37-43)<2n>; centric fusion markers absent
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DND-41
Cell Type:	human T cell leukemia
DSMZ No:	ACC 525
Origin:	established from the peripheral blood of 13-year-old boy with T-acute lymphoblastic leukemia (T-ALL; type III cortical) in 1977; described to carry p15INK4B/p16INK4A deletions and p53 mutation
References:	Drexler et al., Leuk. Res. 9: 209-229 (1985); Drexler, The Leukemia-Lymphoma Cell Line FactsBook, Academic Press, 2000
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly or, occasionally, in clumps
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:2 to 1:4 every 2-4 days; seed out at ca. $0.5-1.0 \times 10^6$ cells/ml; maintain at ca. $0.3-1.0 \times 10^6$ cells/ml; maximal density of ca. 2×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	cell harvest of ca. 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2+, CD3+, CD4(+), CD5(+), CD6(+), CD7+, CD8-, CD13-, CD19-, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-modet near-tetraploid karyotype with 12% polyploidy; 92(86-98)<4n>XXYY, -9, -15, +20, +mar; resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DOHH-2
Cell Type:	human B cell lymphoma
DSMZ No:	ACC 47
Origin:	established from the pleural effusion of a 60-year-old man with refractory immunoblastic B cell lymphoma progressed from follicular centroblastic/centrocytic lymphoma in 1990; cells were described to carry the 14;18 chromosomal translocation (bcl-2 rearranged, mbr) and to express bcl-2 mRNA
References:	Kluin-Nelemans et al., Leukemia 5: 221-224 (1991)
Depositor:	Dr. J. C. Kluin-Nelemans, University Medical Center, Leiden, The Netherlands
DSMZ Cell Culture Data:	
Morphology:	cells grow as single cells or in clumps in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.3-0.7 \times 10^6$ cells/ml; split 1:2 every second day; seed out at ca. $0.5-1.0 \times 10^6$ cells/ml in 24-well-plate
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	maximal density of about $0.7-1.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD37+, CD79a+, CD80+, CD138-, HLA-DR+, sm/cyIgG+, sm/cyIgM-, sm/cykappa-, sm/cylambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hyperdiploid karyotype; 47(43-48)<2n>XY, +7, der(8)t(8;18)(q24;q21), der(14)t(8;14)(q24;q32), der(18)t(14;18)(q32;q21); earliest of a recent series of cell lines with cohabiting 8;14 (Burkitt or diffuse large cell lymphoma) and 14;18 (follicular lymphoma) translocations
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DU-145
Cell Type:	human prostate carcinoma
DSMZ No:	ACC 261
Origin:	established from the tumor tissue removed from the metastatic central nervous system lesion of a 69-year-old man with prostate carcinoma in 1975
References:	Mickey et al., in "Models for Prostate Cancer", G.P. Murphy (ed.), Alan R. Liss, 1980, p. 67-84
Depositor:	obtained from DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells growing as monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:5 every 2-3 days using trypsin/EDTA; seed out at ca. $2-3 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30-40 hours
Harvest:	cell harvest of about 35×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypotriploid karyotype with 12% polyploidy; 62(58-65)<3n>X, -X/Y, -X/Y, -2, -3, +5, -8, -10, -13, +15, +15, -16, -18, -19, -20, -21, -22, +3mar, del(1)(p31), del(2)(p11), i(5p), del(6)(q22), del(9)(p12), del(11)(q23), der(12)t(11;12)(q11;p11), add(13)(q33), add(13)(q33), add(15)(p11)x2, add(16)(q24); closely resembles reported karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DU-4475
Cell Type:	human breast carcinoma
DSMZ No:	ACC 427
Origin:	established from the local metastatic cutaneous nodule of an undifferentiated carcinoma of a 62-year-old woman which developed seven years after radical mastectomy of a poorly differentiated ductal carcinoma of the breast; cells were described to grow readily in athymic nude mice and to produce tumors from which cells could be introduced into culture in vitro; cells reportedly express components of milk fat globules and the bcl-1 gene
References:	Langlois et al., Cancer Res. 39: 2604-2613 (1979); Faust et al., Cancer Res. 52: 2460-2463 (1992)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	large round cells growing singly or in clusters in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 to 1:3 once a week; seed out at ca. 0.2×10^6 cells/ml; maintain at ca. 0.2×10^6 cells/ml; maximum cell density at ca. 0.5×10^6 /ml; due to cluster formation, cells are difficult to count; cultures may contain large amounts of debris and viability may be consistently below 70%
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 80-100 hours
Harvest:	cell harvest of ca. $0.3\text{-}0.4 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP (loss of LDH isoenzymes)
Cytogenetics:	human flat-modulated near-tetraploid karyotype with 12% polyploidy; 88-93<4n>XXXX, +1, +1, -5, -6, +9, -10, -10, +15, +15, -16, -16, +22, +4mar, i(1q)x2, ?add(1)(p35-36)x2, ?i(5p)x2, add(6)(p11), add(6)(p1?), del(6)(q25), add(9)(q35), del(11)(q24)x2, add(15)(p11)x2, add(17)(p1?)x2, del(21)(q22.2)x2; sideline with -20, -20, +del(7)(p11); gain of 1q and loss of 6q typical in breast carcinoma; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	DV-90
Cell Type:	human lung adenocarcinoma
DSMZ No:	ACC 307
Origin:	established from the pleural effusion of a 50-year-old man with lung adenocarcinoma (stage IV) in 1990; cells were described to be tumorigenic in nude mice
References:	Monti et al., Tumori 80: 378-384 (1994)
Depositor:	Dr. F. Monti, Rimini, Italy
DSMZ Cell Culture Data:	
Morphology:	adherent cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS + 2 mM L-glutamine + 1x non-essential amino acids
Subculture:	split confluent culture 1:2 to 1:3 every 3-4 days using trypsin/EDTA; seed out ca. 2×10^6 cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 95 hours
Harvest:	cell harvest of ca. 10×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2.0×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17(+), cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH, PEPB
Cytogenetics:	human near-diploid karyotype with 17% polyploidy; 46(41-47)<2n>X, -Y, +8, add(6)(q27); matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EB-1
Cell Type:	human Burkitt's lymphoma
DSMZ No:	ACC 80
Origin:	established from the maxilla tumor lymph node of a 9-year-old black girl with Burkitt's lymphoma in 1964; cells were described to be EBV-positive
References:	Epstein & Barr, Lancet i: 252-253 (1964); Epstein & Barr, J. Natl. Cancer Inst. 34: 231-240 and 241-253 (1965); Epstein & Achong, J. Natl. Cancer Inst. 34: 241-253 (1965)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round, single or aggregated cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	cells grow very fast; maintain at $0.5-1.0 \times 10^6$ cells/ml; split 1:2 to 1:3
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximal density at $1.0-2.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD37+, HLA-DR+, sm/cyIgG-, sm/cyIgM-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hyperdiploid karyotype with 15% polyploidy; 47(44-48)<2n>XX/XXX, +7, t(8;14)(q24;q32), der(11)t(11;?)(q22;?), der(19)t(19;?)(q13;?); sideline carries additional ch 2
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EBL
Cell Type:	embryonic bovine lung cells
DSMZ No:	ACC 192
Origin:	established from lung of a 7-month-old bovine fetus; frozen at approximately the 110th passage; useful for detection of pasteurella multocida toxine
References:	Bechmann & Schöss, Dtsch. Tierärztl. Wschr. 95: 283-285 (1988) and 98: 310-312 (1992); Rutter & Luther, Vet. Rec. 114: 393-396 (1984)
Depositor:	Dr. V. Öppling, Paul-Ehrlich-Institute, Langen, Germany
DSMZ Cell Culture Data:	
Morphology:	rectangular-like adherent cells growing as monolayers
Medium:	85% RPMI 1640 or MEM (with Earle's salts) + 15% FBS
Subculture:	split confluent culture every 2-3 days using trypsin/EDTA; for the test cells should be starved and split once a week; seed out about 1×10^6 cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Harvest:	cell harvest of about 15×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2.5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as bovine with IEF of G6PD, NP
Cytogenetics:	apparently bovine karyotype with bimodal ploidy distribution; 42(40-42)<2n>/80(77-84)<4n>; structurally rearranged with apparent reduction of ch number by centric fusion followed by tetraploidization
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	ECV-304
Cell Type:	human urinary bladder carcinoma (derivative of T-24)
DSMZ No:	ACC 310
Origin:	described as being "spontaneously immortalized from the umbilical cord vein of a new-born Japanese boy in 1984"; cells were described to have a high proliferative potential in the absence of any growth factors; however, DNA fingerprinting at DSMZ showed clearly that this cell line is in reality a derivative of human urinary bladder carcinoma cell line T-24
References:	Takahashi et al., In Vitro Cell Dev. Biol. 26: 265-274 (1990); ibid, 27A: 766-768 (1991); ibid, 28A: 380-382 (1992); MacLeod et al., Int. J. Cancer 83: 555-563 (1999); Dirks et al., In Vitro Cell Dev. Biol. 35: 558-559 (1999)
Depositor:	Dr. K. Takahashi, Tokyo Medical College, Tokyo, Japan
DSMZ Cell Culture Data:	
Morphology:	adherent growth as cobblestone monolayers
Medium:	80-90% Medium 199 (with HEPES buffer + Earle's salts) + 10-20% FBS
Subculture:	optimal split ratio of 1:10 to 1:20 once or twice a week using trypsin/EDTA; seed out ca. 1-2 x 10 ⁴ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 2 days
Harvest:	cell harvest of max. 10 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	same DNA profile as T-24 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of LDH, MDH, NP
Cytogenetics:	human hypertriploid karyotype with 6.7% polyploidy; 78(75-81)<3n>XX, -X, +1, -4, +5, +5, +5, +7, -8, -9, -10, +14, -15, -15, +17, +18, +20, +20, +20, +21, +22, +4-6mar; add(1)(p32)x1-2, i(1p), der(2q;5q), der(3)t(3;11)(p25;q13), i(5p)x2, inv(7)(p14q31), add(7)(q22), del(9)(p22), add(11)(p15), der(14)t(7;14)(q21;q24), add(18)(p11)x2, add(?19) (q11); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EFE-184
Cell Type:	human endometrium carcinoma
DSMZ No:	ACC 230
Origin:	established from the ascitic fluid of a 69-year-old Caucasian woman with endometrial carcinoma in relapse in 1985; cells were received at passage #91, frozen at DSMZ at passage #99
References:	Abstract 2.070.09 in 15th International Cancer Congress, Hamburg, 1990
Depositor:	Prof. Dr. F. Hölzel, Women's University Hospital Eppendorf, Hamburg, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture at 1:4 about once a week using trypsin/EDTA; cells grow very slowly; seed out about $1 \times 10^6/175 \text{ cm}^2$
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 92 hours
Harvest:	cell harvest of about $4\text{-}5 \times 10^6$ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 0.8×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP(+), neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human flat-moded hypertriploid karyotype with 1-2 supernumerary dm in most cells and 23% polyploidy; 80(72-81)<3n>XX, -X, +1, +2, -4, -5, +6, +8, +10, -13, +17, +19, +20, +21, +6mar; der(1)t(1;?)(p11;?)x2, del(2)(q24)x2, der(2)t(2;3)(q11;p11), del(12)(p13), der(16)t(16;?)(q21/23;?), der(17)t(7;17)(q11;p11), add(22)(p11)x2
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EFM-19
Cell Type:	human breast carcinoma
DSMZ No:	ACC 231
Origin:	established from the pleural effusion of a 50-year-old Caucasian woman with breast carcinoma (ductal type, histological grading II) in 1979; cells were received at passage #53, frozen at DSMZ at passage #56
References:	J. Natl. Cancer Inst. 73: 313-321 (1984); In Vitro 20: 157-166 (1984); J. Clin. Endocrinol. Metabol. 60: 1243-1249 (1985)
Depositor:	Prof. Dr. F. Hölzel, Women's University Hospital Eppendorf, Hamburg, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid, growing in clusters at first, then as monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:2 to 1:3 twice a week using trypsin/EDTA; seed out at ca. 2×10^6 cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 64 hours
Harvest:	cell harvest of about $1.5\text{--}2.0 \times 10^6$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human highly rearranged hypotriploid karyotype with 5.8% polyploidy; 62(58-65)<3n>X, der(X), apparent ch 17 nullisomy and preponderance of marker chromosomes, e.g. add(1)(p11), idel(1)(q11q22), i(1p), i(2p), add(4)(p11), add(4)(p11), etc.; resembles original description
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EFM-192A
Cell Type:	human breast carcinoma
DSMZ No:	ACC 258
Origin:	established from the pleural effusion of the left side from a 46-year-old Caucasian woman with metastasizing breast adenocarcinoma in 1985; sister cell lines EFM-192B and C show differences in molecular aspects; erbB-2 gene is amplified and overexpressed; cells were received at passage #42, frozen at DSMZ at passage #47
References:	Rau et al., J. Cancer Res. Clin. Oncol. 120 (suppl.): R108 (1994)
Depositor:	Prof. Dr. F. Hölzel, Women's University Hospital Eppendorf, Hamburg, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid, adherent growing as monolayer
Medium:	80% RPMI 1640 + 20% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:2 to 1:3 once a week using trypsin/EDTA; seed out at ca. 2×10^6 cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 9 days
Harvest:	cell harvest of about $2-4 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	same DNA profile as EFM-192B/EFM-192C using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, NP, MDH
Cytogenetics:	human flat-moded near-triploid karyotype with 20% polyploidy; 68(60-70)<3n>XXXX, +X, -1, -4, +9, -12, -17, -17, -18, -22, +6mar, add(1)(p22), add(1)(p?32), add(2)(q35/37), add(3)(q27/29), i(15q), add(17)(p12), dminx1 present at 38%
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EFM-192B
Cell Type:	human breast carcinoma
DSMZ No:	ACC 308
Origin:	established from the pleural effusion of the left side (14 days after EFM-192A) of a 46-year-old Caucasian woman with breast adenocarcinoma; cells were described to overexpress the <i>erbB-2</i> gene; sister cell lines EFM-192A and -192C show differences in molecular aspects; cells were received at passage #31, frozen at DSMZ at passages #35-40
References:	Rau et al., J. Cancer Res. Clin. Oncol. 120 (Suppl): R-108 (1994) (abstract)
Depositor:	Prof. Dr. F. Hölzel, Women's University Hospital Eppendorf, Hamburg, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid adherent cells growing as monolayers
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	subcultivation before reaching confluence; split 1:2 to 1:3 once a week using trypsin/EDTA; seed out at ca. $0.5-1.0 \times 10^5$ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 4-5 days
Harvest:	cell harvest of ca. $2-4 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2.5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament, vimentin-
Fingerprinting:	same DNA profile as EFM-192A/EFM-192C using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of LDH, MDH, NP
Cytogenetics:	human heteroploid hypotriploid karyotype with 13% polyploidy; 62(58-62)<3n>XXX/XX, -X, +2, +4, +6, -7, -8, -8, -10, -11, -12, +14, +16, -17, -17, -17, -18, -18, -19, -20, -21, -22, +8mar, add(1)(p22-31), add(2)(q?35), del(3)(p22), add(5)(p15), del(6)(q25), add(10)(p11), add(13)(q32), add(14)(p12), add(15)(p12), der(15)t(10;15)(q11;p12), der(20)t(1;20)(q24;q12/13), der(21)t(7;21)(q11;p12)x2; most cells carry a small ring and 1-2 large submetacentric 15p+ markers; closely related to EFM-192A; apparent ch 17 nullisomy
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EFM-192C
Cell Type:	human breast carcinoma
DSMZ No:	ACC 311
Origin:	established from the pleural effusion of the left side (14 days after EFM-192A) of a 46-year-old Caucasian woman with breast adenocarcinoma; sister cell lines EFM-192A and -192B show differences in molecular aspects; erbB-2 gene is amplified and overexpressed; cells were received at passage #50, frozen at DSMZ at passage #56
References:	Rau et al., J. Cancer Res. Clin. Oncol. 120 (Suppl): R-108 (1994) (abstract)
Depositor:	Prof. Dr. F. Hölzel, Women's University Hospital Eppendorf, Hamburg, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid adherent cells growing as monolayers
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	optimal split ratio of 1:2 to 1:3 once a week using trypsin/EDTA; seed out at ca. $5-8 \times 10^4$ cells/cm ² or 2×10^6 cells/25 cm ² flask
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 4 days
Harvest:	cell harvest of ca. $2-4 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	same DNA profile as EFM-192A/EFM-192B using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	highly rearranged human hypohexaploid karyotype with 10% polyploidy; 122-138<6n>XXXXX/XXXXXX; related to EFM-192A and EFM-192 B
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EFO-21
Cell Type:	human ovary cystadenocarcinoma
DSMZ No:	ACC 235
Origin:	established from the ascitic fluid of a 56-year-old Caucasian woman with dedifferentiated serous cystadenocarcinoma of the ovary in 1979; cells were received at passage #152, frozen at DSMZ at passage #164
References:	Simon et al., J. Natl. Cancer Inst. 70: 839-845 (1983)
Depositor:	Prof. Dr. F. Hölzel, Women's University Hospital Eppendorf, Hamburg, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid adherent cells growing in monolayers
Medium:	80% RPMI 1640 + 20% FBS + 2 mM L-glutamine + 1x MEM non-essential amino acids + 1 mM sodium pyruvate
Subculture:	split confluent culture once a week using trypsin/EDTA; seed out about $1-2 \times 10^6$ cells/175 cm ² , after thawing cells might at first grow very slowly
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 43 hours
Harvest:	cell harvest of about 6×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 0.5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	apparently human karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EFO-27
Cell Type:	human ovary adenocarcinoma
DSMZ No:	ACC 191
Origin:	established from the solid omental metastasis of a mucinous papillary adenocarcinoma of the ovary of a 36-year-old woman in 1979; initially diploid, later tetraploid; growth in nude mice; cells were received at passage #48, frozen at DSMZ at passage #53-57
References:	Simon et al., J. Natl. Cancer Inst. 70: 839-845 (1983); Kunzmann et al., Cancer Genet. Cytogenet. 28: 201-212 (1987)
Depositor:	Prof. Dr. F. Hölzel, Women's University Hospital Eppendorf, Hamburg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial-like cells growing as monolayers
Medium:	80% RPMI 1640 + 20% FBS + 2 mM L-glutamine + 1x MEM non-essential amino acids + 1 mM sodium pyruvate
Subculture:	split confluent culture 1:2 to 1:5 every 2-5 days using trypsin/EDTA (incubate for 10-20 minutes with trypsin/EDTA); seed out at ca. 1.5×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 50-80 hours
Harvest:	cell harvest of about $0.3-0.6 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1.5-2.5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human flat-moded hypotetraploid karyotype with 17% polyploidy; 78-91<4n>XXXX, -4, -10, -13, -13, -19, +mar; i(9q), 16qh+x2; extensive subclonal rearrangements, e.g. der(5)t(5;7)(q31.2;q11)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EGI-1
Cell Type:	human bile duct carcinoma
DSMZ No:	ACC 385
Origin:	established from the solid tumor of a 52-year-old Caucasian man with advanced stage malignant bile duct carcinoma (patient had not received any chemotherapy prior to cell explantation) in 1984 and three consecutive passages in female nu/nu mice (NMRI); primary tumor histology: large cell adenocarcinoma of low differentiation, metastasizing (ascites in patient); human character of cell line confirmed by karyotype in passage #34 of in vitro culture (1986); cells were received at passage #74, frozen at DSMZ at passage #95
References:	Scherdin et al., Immunobiology 175: 1-143 (1987), abstract B.21
Depositor:	Prof. Dr. F. Hölzel, Women's University Hospital Eppendorf, Hamburg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent polymorphic (epitheloid or dendritic) cells growing in monolayers with multilayer foci
Medium:	90% Dulbecco's MEM + 10% FBS + 1x MEM amino acids (both essential and non-essential) + 2 mM L-glutamine
Subculture:	split confluent culture 1:3 to 1:6 every 3-5 days using trypsin/EDTA (more than 10 min trypsin/EDTA treatment might be required on confluent cultures); seed out at ca. $1-2 \times 10^6$ cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of >45 hours
Harvest:	cell harvest of ca. $10-12 \times 10^6$ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP, PEP B
Cytogenetics:	human hypertriploid with 8% polyploidy; 70(68-72)<3n>XYY, -8, -10, +11, -12, -15, +21, +5mar, del(4)(q22), i(5q), i(6p)x2, add(18)(q23)x2, del(21)(q22); rearrangements of 18q and 21q implicated in gall bladder carcinoma
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EHEB
Cell Type:	human chronic B cell leukemia
DSMZ No:	ACC 67
Origin:	established from the peripheral blood of a 69-year-old woman with B-CLL (chronic lymphocytic leukemia) prior treatment by EBV-transformation in 1988; cells were described to be EBV+, CD5+ and to express mRNA of bcl-2, bcl-3 and c-myc
References:	Saltman et al., Leukemia Res. 14: 381-387 (1990)
Depositor:	Dr. K. Guy, Western General Hospital, Edinburgh, UK
DSMZ Cell Culture Data:	
Morphology:	lymphoblastoid, round to polygonal, single or clustered cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.1-0.5 \times 10^6$ cells/ml; split at ca. 1:2 every 2-3 days; cells may grow slowly initially; use initially 24-well plate culture dishes and 20% FBS; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-70 hours
Harvest:	maximal density at about 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-6 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD34-, CD37+, CD38+, CD79a-, cyCD79a+, CD80+, CD138+, HLA-DR+, cyIgG+, cyIgM-, cykappa+, cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human hypodiploid karyotype with 1.6% polyploidy; 45(42-46)<2n>X, -X; no structural rearrangements detected
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	ELM-I-1
Cell Type:	mouse erythroleukemia
DSMZ No:	ACC 44
Origin:	established from an X-irradiated C3H/HeNSIc mouse; cells were described to be stroma-independent, responsive to erythropoietin and to produce hemoglobin (augmented by IL-3)
References:	Itoh et al., Leukemia Res. 12: 471-478 (1988); Shiozaki et al., Leukemia Res. 14: 287-291 (1990)
Depositor:	Dr. K. J. Mori, Niigata University, Niigata, Japan
DSMZ Cell Culture Data:	
Morphology:	large, round, single cells in suspension
Medium:	90% alpha-MEM + 10% FBS
Subculture:	maintain at $0.1-0.5 \times 10^6$ cells/ml; split 1:3
Incubation:	at 37 °C with 5% CO ₂
Harvest:	saturation density of about $0.5-1.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of MDH, NP, PEP B
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EM-2
Cell Type:	human chronic myeloid leukemia in blast crisis
DSMZ No:	ACC 135
Origin:	established from the bone marrow of a 5-year-old Caucasian girl with Philadelphia-chromosome positive chronic myeloid leukemia (CML) in second relapse in 1980 (after bone marrow transplantation day +28); sister cell line of EM-3 (DSM ACC 134)
References:	Keating et al., Clin. Haematol. 1: 1021-1029 (1987); Raskind et al., Cancer Genet. Cytogenet. 25: 271-284 (1987); Keating, Bailliere's Clin Haematol 1: 1021-1029 (1987); review article: Drexler et al., Leukemia Res. 24: 109-115 (2000)
Depositor:	Dr. A. Keating, Toronto General Hospital, Ontario, Canada
DSMZ Cell Culture Data:	
Morphology:	single, round to oval cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.2-1.0 \times 10^6$ cells/ml; split ratio of 1:2
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling of ca. 35-40 h
Harvest:	maximum density at about $1.0-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4+, CD13+, CD14-, CD15+, CD19-, CD33+, CD34(+), CD68(+), HLA-DR-
Fingerprinting:	same DNA profile as EM-3 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypertriploid karyotype with hypotetraploid sideline; 74(70-86)<3n>X, -X, -X, +3, +4, +6, +6, +6, +8, -9, +11, -14, -14, +15, +17, -19, +21, +22, +mar; der(5)t(5;?)(q13-15;?), der(9)t(9;22)(q34;q11), i(17q)x2; in some cells up to three copies of der(9) are present; the karyotype overlaps that of EM-3 (DSM ACC 134) and corresponds with those reported for both EM-2 and EM-3
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EM-3
Cell Type:	human chronic myeloid leukemia in blast crisis
DSMZ No:	ACC 134
Origin:	established from the bone marrow of a 5-year-old Caucasian girl with Philadelphia chromosome-positive CML (chronic myeloid leukemia) in second relapse in 1980 (after bone marrow transplantation day +47); sister cell line of EM-2 (DSM ACC 135)
References:	Keating et al., Clin. Haematol. 1: 1021-1029 (1987); Raskind et al., Cancer Genet. Cytogenet. 25: 271-284 (1987); Keating, Bailliere's Clin Haematol 1: 1021-1029 (1987); review article: Drexler et al., Leukemia Res. 24: 109-115 (2000)
Depositor:	Dr. A. Keating, Toronto General Hospital, Ontario, Canada
DSMZ Cell Culture Data:	
Morphology:	single, round to oval cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.5 \times 10^6$ cells/ml; subculture at a 1:2 to 1:3 dilution every 48 hours; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	grows up to about 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4-, CD13+, CD14-, CD15+, CD19-, CD33+, HLA-DR-
Fingerprinting:	same DNA profile as EM-2 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypertriploid karyotype; 73(69-77)<3n>X, -X, -X, t(9;22)(q34;q11), i(17q)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EOL-1
Cell Type:	human acute myeloid (eosinophilic) leukemia
DSMZ No:	ACC 386
Origin:	established in 1984 at diagnosis from the peripheral blood of a 33-year-old man with acute myeloid (eosinophilic) leukemia (AML) following hypereosinophilic syndrome; we found that EOL-1 carries an MLL partial tandem duplication
References:	Saito et al., Blood 66: 1233-1240 (1985); Mayumi, Leukemia Lymphoma 7: 243-250 (1992); Quentmeier et al., Leukemia 17: 980-981 (2003)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	(relatively small) round cells growing in suspension, singly or in small clusters
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5\text{--}2.0 \times 10^6$ cells/ml; split 1:2 to 1:3 every 3-4 days; seed out at ca. 0.5×10^6 cells/ml; start culture with 20% FBS
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60 hours
Harvest:	maximum density of ca. 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4+, CD11b+, CD13+, CD14-, CD15+, CD19-, CD32+, CD33+, CD34-, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, MDH, NP
Cytogenetics:	human hyperdiploid karyotype with 7.5% polyploidy; 50(48-51)<2n>XY, +4, +6, +8, +19, del(9)(q22); matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EPLC-272H
Cell Type:	human epidermoid lung carcinoma
DSMZ No:	ACC 383
Origin:	established from a surgical lung specimen of a 57-year-old Caucasian man who underwent surgery for undifferentiated squamous cell carcinoma of the lung (without having received prior radiation or chemotherapy) in 1986
References:	Bepler & Koehler, Cancer Genet. Cytogenet. 84: 39-45 (1995)
Depositor:	Dr. G. Bepler, Duke University Medical Center, Durham, North Carolina, USA
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells (without sharp contours) growing adherently as monolayers
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split confluent culture 1:2 to 1:3 every 3-5 days using trypsin/EDTA (incubate for >10 min); seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60 hours
Harvest:	cell harvest of ca. $5-7 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human hypotriploid karyotype with 12% polyploidy; 66(59-70)<3n>XX, -Y, -4, +7, +8, -18, -19, -20, -21, +6mar, add(1)(p32), del(2)(q33), der(2;11)(q10;q10), add(3)(p25), i(5p), i(5q), add(6)(q11), add(7)(q33)x2-3, add(7)(p15), del(8)(p12-21), del(10)(p14)x1-2, add(11)(q24)x2, add(12)(p11-12), add(13)(q33), add(14)(p12), i(15q), der(17)t(3;17)(p11;p12), add(19)(q13), add(21)(p12), dmx2; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	ESS-1
Cell Type:	human endometrial stromal sarcoma
DSMZ No:	ACC 461
Origin:	established from an endometrial stromal sarcoma of the uterus of a 76-year-old Caucasian woman; cells were described as demonstrating a mainly mesenchymal phenotype with signs of epithelial characteristics within foci and an extensively multilayered growth pattern
References:	Fuzesi et al., Cancer Genet. Cytogenet. 84: 85-88 (1995); Gunawan et al., Int. J. Cancer 77: 424-428 (1998)
Depositor:	Dr. B. Gunawan, University Hospital Göttingen, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent cells growing in monolayers, in some regions in multilayers
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split confluent culture 1:2 every second week using trypsin/EDTA, but change the medium frequently; seed out at ca. 1×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 230 hours
Harvest:	cell harvest of ca. 8×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin+, endothel-, GFAP+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Cytogenetics:	human karyotype with hyper-diploid/tetraploid and -octaploid clones present in approx. equal numbers; 47(45-47)<2n>XX, +19, der(3)t(3;6)(q27;q11-12), add(6)(q27), add(19)(p11); sideline with i(2q); resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	EVSA-T
Cell Type:	human breast carcinoma
DSMZ No:	ACC 433
Origin:	established from the malignant ascitic effusion from a woman with metastatic breast carcinoma; described as being estrogen receptor-negative and progesterone receptor-positive
References:	Lippmann et al., Cancer Res. 36: 4595-4601 (1976); Davidson et al., Mol. Endo. 1: 216-223 (1987); Borras et al., Cancer Letters 120: 23-30 (1997)
Depositor:	Dr. G. Leclercq, Institut Jules Bordet, Bruxelles, Belgium
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells growing in monolayers
Medium:	90% MEM (with Earle's salts) + 10% FBS + 2 mM L-glutamine + 25 mM HEPES buffer
Subculture:	split semi-confluent culture 1:3 every 3-4 days using trypsin/EDTA, dissociate cells with a pipette (10 ml) during trypsinization; seed out at ca. 2×10^6 cells/25 cm ² in 10 ml medium; maximal density at ca. 0.6×10^6 /cm ² ; after thawing, viability may be low (about 50%) for the first week
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of maximal 0.5×10^6 /cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO + 25 mM HEPES buffer at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-8+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypotriploid karyotype with 4% polyploidy; 62(55-64)<3n>XXX, +1, -2, -5, -5, +6, -8, -9, -10, -13, -13, -14, -15, -16, -17, -18, -20, -22, +6-8mar, add(X)(p11), der(X)t(X;9)(q21;q11), add(1)(p1?1), der(1)del(1)(p36)dup(1)(q22q32), del(1)(q11), der(1;4)(q10;q10)del(4)(q1?13q?23), der(2)t(2;5)(p11;q13)add(2)(q3?5), der(3)t(3;5)(p14;q11), del(4)(q22), del(6)(q25q27), del(6)(p11), add(6)(p23), del(7)(p14), add(8)(q24), ider(8)(q10)ins(8;?)dup(8)(?q21q24)add(8)(q24), add(11)(p15), add(12)(p13), del(12)(p13), dup(13)(q21q33), add(21)(p11), add(21)(p11), large submetacentric der(8) hsr marker present
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	F-36P
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 543
Origin:	established at diagnosis from the pleural effusion of a 68-year-old man with acute myeloid leukemia (AML M6) secondary to myelodysplastic syndrome (MDS, subtype refractory anemia with excess of blasts, RAEB) in 1989; cells were described to be proliferatively responsive to GM-CSF and IL-3
References:	Chiba et al., Blood 78: 2261-2268 (1991); Drexler et al., Leukemia 11: 701-708 (1998)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	polymorph cells growing as adherent cells and singly in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:3 to 1:8 every 2-5 days; seed out at ca. $0.3-0.5 \times 10^6$ cells/ml; maintain at ca. 1.0×10^6 cells/ml; maximum density at ca. 1.8×10^6 cells/ml. Upon cytokine withdrawal cells do not die but undergo a stationary survival status. CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24-36 hours
Harvest:	cell harvest of ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD13+, CD14-, CD15-, CD19-, CD33+, CD34+, CD41-, CD42-, CD71+, GlyA+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-modulated hypodiploid karyotype with 6% polyploidy; 41(40-43)<2n>XY, -3, -13, -16, -19, -21, -22, +2mar, ins(X;13)(p11;q?q?), del(3)(p13p24), add(5)(q11), dup(5)(q11q35), der(6)add(6)(p23)dup(6)(q25q27), der(7)add(7)(p13)add(7)(q12), add(9)(q35), der(9)del(9)(q11q34)inv(9)(p2?4;q34), der(10)t(8;10)(q24;p14)t(10;7)(q25;q12), del(11)(q14q21), der(21)t(10;21)(p14;p12); sideline with dup(1)(p35p36), dup(2)(q1?q2?), del(3)(p13p24); matches published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	F1-652
Cell Type:	mouse hybridoma (anti-fiber/myosin)
DSMZ No:	ACC 309
Origin:	described as secreting a mouse monoclonal antibody (IgG ₁) reacting with embryonic fibers in rodent and human muscle and with embryonic myosin heavy chain on Western blots; the cell line was established in 1985 by fusion of X63AG8.653 cells with spleen cells from a BALB/c mouse that had been immunized with human skeletal muscle myosin
References:	Cho et al., J. Cell Sci. 107: 2361-2371 (1994)
Depositor:	Dr. S. M. Hughes, King's College, London, England
DSMZ Cell Culture Data:	
Morphology:	round large cells, lightly adherent (can be detached without trypsin)
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + 2 mM L-glutamine + 1 mM sodium pyruvate + 1x non-essential amino acids
Subculture:	split saturated culture 1:3 to 1:5 once or twice a week; cultivate optimally between 0.1-1.0 x 10 ⁶ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30-36 hours
Harvest:	maximal density of about 2.0 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1-2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgG ₁ monoclonal antibody
Species:	confirmed as murine with IEF of AST, NP, LDH
Cytogenetics:	murine hypertetraploid karyotype; 87(72-90)<4n>; carries 2-3 centric fusion markers and 1-2 dm
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	F4/4.K6
Cell Type:	mouse T cells
DSMZ No:	ACC 52
Origin:	established from a bovine insulin-specific CD4 ⁺ T cell line (C57Bl/10 x C57Bl/10.BR ^{H2^b} x H-2 ^k); cell line can be used as assay for IL-4
References:	Schmitt et al., Eur. J. Immunol. 20: 1709-1715 (1990)
Depositor:	Dr. E. Schmitt, Institute of Immunology, Mainz, Germany
DSMZ Cell Culture Data:	
Morphology:	single or clustered cells in suspension
Medium:	CG medium + 50 U/ml mouse IL-4 (strictly without FBS)
Subculture:	maintain at 0.2-0.4 x 10 ⁶ cells/ml; split 1:2; upon thawing of frozen cells, cells must be washed thoroughly as already tiny amounts of FBS will lead to adherence
Incubation:	at 37 °C with 5% CO ₂
Harvest:	saturation density at about 0.5 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture assays

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INFORMATION SHEET FOR CELL LINES

Cell Line:	F9
Cell Type:	mouse embryonal carcinoma
DSMZ No:	ACC 112
Origin:	initiated from a testicular teratocarcinoma of a strain 129 mouse
References:	Bernstine et al., Proc. Natl. Acad. Sci. USA 70: 3899-3903 (1973); Strickland et al., Cell 15: 393-403 (1978)
Depositor:	obtained from ATCC (CRL 1720), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	adherent cells with undifferentiated morphology growing in clumps
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS
Subculture:	presoak flasks with 0.1% gelatin at 4 /C for 2 hours, wash 3x with PBS; split confluent culture 1:10 to 1:20 every 2-4 days using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 /C with 5% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	about 70×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, MDH, NP, PEP B
Cytogenetics:	murine near-diploid karyotype with 6% tetraploidy; 38(37-40)<2n>; 1-2 centric fusion markers present
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	FDCP-1
Cell Type:	mouse bone marrow
DSMZ No:	ACC 368
Origin:	established from the long-term bone marrow culture of B6D2F1 mice; cells are IL-3-dependent; their differentiation is blocked; this subclone is responsive to G-CSF (FDCP means "factor dependent cell-Paterson" after the Paterson Institute, Manchester, UK)
References:	Dexter et al., J. Exp. Med. 152: 1036-1047 (1980)
Depositor:	Prof. T. M. Dexter, Paterson Institute of Cancer Research, Manchester, UK
DSMZ Cell Culture Data:	
Morphology:	single or clustered cells growing in suspension
Medium:	70% Iscove's MDM + 20% horse serum + 10% vol conditioned medium of WEHI-3B cell line (DSM ACC 26)
Subculture:	split saturated culture 1:2 to 1:3 every 2-3 days; seed out at ca. 5×10^5 cells/ml, first 3-4 days after thawing should not be splitted. CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	maximal density at ca. 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, LDH, NP
Cytogenetics:	murine hypodiploid karyotype; $32 < 2n >$; carries 8 centric fusion markers
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	FDCP-Mix cl.A4
Cell Type:	mouse bone marrow
DSMZ No:	ACC 401
Origin:	established from the long-term bone marrow cultures of BDF1 mice; cells are mIL-3-dependent (FDCP means "factor dependent cell-Paterson" after the Paterson Institute, Manchester, UK); it is described that cells can be maintained in a primitive state or induced to differentiate
References:	Spooncer et al., Nature 310: 228-230 (1984); Spooncer et al., Differentiation 31: 111-118 (1986); Dexter et al., J. Exp. Med. 152: 1036-1047 (1980); Ihle & Askew, Int. J. Cell Cloning 7: 68-91 (1989)
Depositor:	Prof. T. M. Dexter, Patterson Institute of Cancer Research, Manchester, UK
DSMZ Cell Culture Data:	
Morphology:	round, single cells in suspension, sometimes in small clusters
Medium:	80% Iscove's MDM + 20% FBS + 10 ng/ml murine IL-3 (or 10-20% conditioned medium of cell line WEHI-3B, DSM ACC 26)
Subculture:	split saturated culture 1:2 every 3-4 days; seed out at ca. 1.0×10^6 cells/ml; after thawing, cells are initially very difficult to culture and viability may be initially rather low (20-30%); hence it is advisable to culture the cells first in 24-well-plates and to exchange the medium, adding fresh mIL-3, already about 5-8 hours after thawing and then daily. CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 70-80 hours
Harvest:	maximum density at about 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of LDH, NP
Cytogenetics:	murine near-diploid karyotype with 10% polyploidy; 40(38-40)<2n>; fusion markers absent
Viruses:	ELISA: reverse transcriptase positive

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	FLK-BLV-044
Cell Type:	ovine embryonal kidney cells
DSMZ No:	ACC 153
Origin:	established from the embryonal kidney cells of an ovine in 1976; cells are susceptible to infection with BLV (bovine leukemia virus) leading to the formation of syncytia
References:	
Depositor:	Dr. R. Riebe, Friedrich-Loeffler-Institute, Insel Riems, Germany
DSMZ Cell Culture Data:	
Morphology:	large fibroblast-like cells growing as monolayers
Medium:	90% MEM (with Earle's salts) + 10% FBS ^o tional: + 850 mg/L NaHCO ₃ + 120 mg/L sodium pyruvate + 1x non-essential amino acids]
Subculture:	split confluent culture 1:2 to 1:8 every 3-5 days using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of about 4 x 10 ⁶ /80 cm ²
Storage:	frozen with 82.5% medium, 10% FBS, 7.5% DMSO at about 0.2-0.4 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as ovine with IEF of AST, LDH, NP
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	FU-OV-1
Cell Type:	human ovarian carcinoma
DSMZ No:	ACC 444
Origin:	established post-hysterectomy from the tumor tissue of a 65-year-old Japanese woman with malignant ovarian carcinoma (stage: FIGO IIIc; histology: poorly differentiated serous papillary adenocarcinoma) in 1989
References:	Emoto et al., Pathol. Res. Pract. 195: 237-242 (1999)
Depositor:	Dr. M. Emoto, Fukuoka University Medical School, Fukuoka, Japan
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells growing in monolayers
Medium:	80-90% mixture of Ham's F12 + Dulbecco's MEM (at 1:1) + 10-20% FBS
Subculture:	split confluent culture 1:2 to 1:3 once a week using trypsin/EDTA; seed out at ca. 1×10^6 cells/175 cm ² in 50 ml medium; maximal density at ca. 10×10^6 /175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 120 hours
Harvest:	cell harvest of ca. $5-10 \times 10^6$ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Cytogenetics:	human flat-moded hypotriploid karyotype with 12% polyploidy; 58-62<3n>X, -X, -X, -1, -2, -3, -4, -7, +9, -10, -12, -13, -14, -16, -17, -17, -18, -19, -20, -20, -22, +fra, +dm, del(2)(p23), inv(3)(p25q29), del(4)(q21/22), add(5)(p11), add(6)(q27), add(9)(p11), add(11)(p15)x2, add(11)(q24), add(14)(p11), add(14)(p11), add(16)(p13), add(19)(q13)x2, add(?21)(p11)x2; resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	G/G
Cell Type:	mouse urinary bladder
DSMZ No:	ACC 224
Origin:	established from an adult female C3H/E1 mouse; non-tumorigenic urine bladder cell line
immortalized	in vitro
References:	De Boer et al., Int. J. Cancer 54: 1022-1027 (1993)
Depositor:	Dr. W. I. de Boer, Erasmus University, Rotterdam, The Netherlands
DSMZ Cell Culture Data:	
Morphology:	epithelial, adherent cells, growing in up to 3 layers
Medium:	45% Ham's F10 + 45% Dulbecco's MEM + 10% FBS + 5 :g/ml bovine insulin + 5 :g/ml human transferrin + 50 nM hydrocortisone + 5 ng/ml selenite
Subculture:	split confluent culture 1:3 to 1:10 every 3-7 days using trypsin/EDTA
Incubation:	at 37 /C with 5% CO ₂
Doubling Time:	doubling time of about 35-50 hours
Harvest:	harvest of about 0.5 x 10 ⁵ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1.5-2.5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, NP
Cytogenetics:	murine near-tetraploid karyotype with 3.6% polyploidy; 80/81(76-83)<4n>; up to 3 (sub)metacentric markers present in most cells arising by centric fusion
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	GAMG
Cell Type:	human glioma
DSMZ No:	ACC 242
Origin:	established from the biopsy specimens from a 42-year-old woman with glioma (histologically defined as glioblastoma)
References:	Tysnes et al., Anticancer Res. 13: 1325-1330 (1993); Akslen et al., Anticancer Res. 8: 797-803 (1988)
Depositor:	Dr. O. B. Tysnes, University of Bergen, Haukeland Hospital, Bergen, Norway
DSMZ Cell Culture Data:	
Morphology:	adherent, large, spindle-like cells growing in monolayers
Medium:	90% Dulbecco's MEM + 10% FBS + 2 mM L-glutamine + 4x MEM non-essential amino acids
Subculture:	split confluent culture 1:5 to 1:6 every 4-6 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ² flask
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of about 40-50 hours
Harvest:	cell harvest of about 10×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypotriploid karyotype with 17% polyploidy; 66(61-67)<3n>XXX, -4, +5, +7, -10, -13, -19, -22, +dm x1-2, der(1)t(1;4)(q11;p11), del(3)(p14), der(10)t(3;10)(p14;q13), del(17)(p11), del(18)(p11); additional subclonal rearrangements, add(2)(q11), add(7)(q11), i(9q); simultaneous +7, -10, -22 and dm recurrent in gliomas
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	GDM-1
Cell Type:	human acute myelomonocytic leukemia
DSMZ No:	ACC 87
Origin:	established in 1979 from the peripheral blood of a 66-year-old woman with a Philadelphia chromosome negative myeloproliferative disorder (resembling chronic myeloid leukemia, CML, in blast crisis) after transformation to refractory acute myelomonocytic leukemia (AML)
References:	Ben-Bassat et al., Leukemia Res. 6: 743-752 (1982)
Depositor:	Dr. J. Minowada, Fujiasaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	single round to oval cells in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	cells are very difficult to culture! best to culture initially in 24-well-plates; seed out at ca. 1×10^6 cells/24-well; maintain at $0.5-0.8 \times 10^6$ cells/ml; cells grow very slowly, split carefully
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60-80 hours
Harvest:	maximum density at 1.2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4+, CD13-, CD14-, CD15+, CD19-, CD33+, CD34+, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hyperdiploid karyotype; 48(46-48)<2n>XX, +8, +13, -16, +mar, t(2;11)(q36;q13), del(6)(q21), t(7;?)(q35;?), del(12)(p13)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	GH3
Cell Type:	rat pituitary tumor
DSMZ No:	ACC 469
Origin:	derived in 1965 from a pituitary tumor induced in a female Wistar/Furth rat following repeated transplantation with the MtT/W5 tumor (originally induced in a female F/Fu rat by cranial X-irradiation)
References:	Takemoto et al., Cancer Res. 22: 917-924 (1962); Tashjian et al., Endocrinol. 82: 342-352 (1968)
Depositor:	Dr. M. L. Thieulant, University of Rennes, France
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently in semi-confluent monolayers
Medium:	82.5% Ham's F10 + 15% horse serum + 2.5% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:2 to 1:5 every 2-5 days, cells may be detached using trypsin/EDTA or mechanically by agitation; seed out at ca. 4×10^6 cells/80 cm ² ; maintain at $0.3-0.5 \times 10^5$ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60-90 hours
Harvest:	maximal density of 2×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with ciprofloxacin (Ciprobay), then negative in microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as rat by cytogenetics and PCR analysis with Alu, globin and rRNA primers
Cytogenetics:	hypertriploid rat karyotype with 20% polypoidy; 68-71<3n>

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INFORMATION SHEET FOR CELL LINES

Cell Line:	GH4-C1
Cell Type:	rat pituitary tumor
DSMZ No:	ACC 482
Origin:	established from a pituitary tumor of a 7-month-old female Wistar/Furth rat; cell line GH4 was subcloned in order to obtain a cell line secreting high amounts of growth hormones
References:	Takemoto et al., Cancer Res. 22: 917-924 (1962); Tashjian et al., Endocrinol. 82: 342-352 (1968)
Depositor:	Dr. M. L. Thieulant, University of Rennes, France
DSMZ Cell Culture Data:	
Morphology:	about 70% of the cells grow adherently, 30% grow in suspension (some single, others in clusters)
Medium:	82.5% Ham's F10 + 15% horse serum + 2.5% FBS
Subculture:	split confluent culture 1:2 every 2-3 days, cells using TNE (40 mM TrisCl pH 7.5, 1 mM EDTA pH 7.5, 150 mM NaCl); seed out at ca. 0.5×10^6 cells/ml; maintain at $0.3-0.5 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	maximal density of 0.8×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with enrofloxacin (Baytril), then negative in microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as rat by PCR analysis with Alu, globin and rRNA primers
Cytogenetics:	rat karyotype

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INFORMATION SHEET FOR CELL LINES

Cell Line:	GIRARDI HEART C2
Cell Type:	human cervix carcinoma (derivative of HELA)
DSMZ No:	ACC 116
Origin:	these cells were cloned from the GIRARDI HEART cell line (ATCC CCL 27, passage #623) in 1981 and were thought to "represent human heart cells"; however, isoenzyme analysis (Nelson-Rees et al., Science 191: 96-97, 1976) and DNA fingerprinting at the DSMZ established unequivocally that this cell line must be considered de facto a subclone of HELA; cells were described to be sensitive to various toxins and hormones
References:	Milczewski, in Proc. Vth Int. Symp. Mycotoxins & Phycotoxins, 1982
Depositor:	Dr. K. E. von Milczweski, Bundesanstalt für Milchforschung, Institut für Mikrobiologie, Kiel, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial cells growing as monolayers
Medium:	45% Medium 199 (with Hank's salts) and 45% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:6 every 3-4 days using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Harvest:	about 10 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2-3 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed same DNA profile as HELA
Species:	confirmed as human with IEF of AST, MDH
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	GIRARDI HEART C7
Cell Type:	human cervix carcinoma (derivative of HELA)
DSMZ No:	ACC 121
Origin:	these cells were cloned from the GIRARDI HEART cell line (ATCC CCL 27, passage #623) in 1981 and were thought to "represent human heart cells"; however, isoenzyme analysis (Nelson-Rees et al., Science 191: 96-97, 1976) and DNA fingerprinting at the DSMZ established unequivocally that this cell line must be considered de facto a subclone of HELA; cells were described to be sensitive to various toxins and hormones
References:	Milczewski, in Proc. Vth Int. Symp. Mycotoxins & Phycotoxins, 1982
Depositor:	Dr. K. E. von Milczweski, Bundesanstalt für Milchforschung, Institut für Mikrobiologie, Kiel, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial cells growing as monolayers
Medium:	45% Medium 199 (with Hank's salts) and 45% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:6 every 3-4 days using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Harvest:	about 10 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2-3 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	same DNA profile as HELA with multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	GM-7373
Cell Type:	bovine endothelial aorta
DSMZ No:	ACC 109
Origin:	established from a male bovine fetus at 24-week gestation; vascular endothelium transformed with benzo(a)pyrene in vitro; cell line described as tumorigenic
References:	Grinspan et al., J. Cell. Physiol. 114: 328-338 (1983)
Depositor:	Prof. Presta, University of Brescia, Brescia, Italy
DSMZ Cell Culture Data:	
Morphology:	cells grow as adherent multilayer
Medium:	80% MEM (with Earle's salts) + 20% FBS + non-essential amino acids + MEM vitamins
Subculture:	split confluent culture 1:2 to 1:3 every 3-6 days using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30-60 hours
Harvest:	cell harvest of about 0.8×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2.0×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Fingerprinting:	unique DNA profile with (gtg) ₅ multilocus probe
Species:	the species human, mouse and rat were excluded with IEF of LDH, NP, PEP B
Cytogenetics:	hypodiploid bovine karyotype with 15% polyploidy; 58(54-60)<2n>; centric fusion and large submetacentric marker; corresponds to published karyotype
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	GMS-10
Cell Type:	human glioblastoma
DSMZ No:	ACC 405
Origin:	established from the malignant brain tumor of a 49-year-old man with glioblastoma in 1995
References:	Halfter et al., unpublished
Depositor:	Dr. H. Halfter, University of Münster, Münster, Germany
DSMZ Cell Culture Data:	
Morphology:	fibroblast-like cells growing as monolayers
Medium:	80-85% Dulbecco's MEM + 10-15% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:3 to 1:5 every 3-6 days using trypsin/EDTA; seed out at ca. $0.5-1.0 \times 10^6$ cells/80 cm ² in 15 ml medium
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 50-70 hours
Harvest:	cell harvest of ca. 5×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $0.5-1.0 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, NP
Cytogenetics:	human flat-moded hypertriploid karyotype with 11% polyploidy; 69-76<3n>XXYY, +Y, +2, +3, +9, -10, -12, +15, +15, -18, +20, +20, -22, +2-4mar, +1-4dmin, del(2)(p11), add(2)(q31;hsr), der(4)t(3;4)(p21;p15.3), add(4)(p1?), del(6)(q23), add(7)(q11), add(9)(p23), add(13)(p11), add(15)(p11)x3, add(19)(q13), add(21)(p11)x3; same hsr present in mar and add(2)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	GOS-3
Cell Type:	human astrocytoma/oligodendroglioma
DSMZ No:	ACC 408
Origin:	established from the tumor material of brain from a 55-year-old man with a mixed astro-oligodendroglioma (grade II/III); described to express neuroepithelial markers GFAP, vimentin and S-100 protein as well as pp60 ^c SRC], but not the neuro-specific variant pp60 ^c SRCN] and to express neuropoietic factors CNTF, LIF and their receptors; LIF and oncostatin M induce growth inhibition and differentiation
References:	Halfter et al., Growth Factors 15: 135-147 (1998); Halfter et al., J. Neuro-Oncol. 39: 1-18 (1998)
Depositor:	Dr. H. Halfter, University of Münster, Münster, Germany
DSMZ Cell Culture Data:	
Morphology:	fibroblastoid, adherent cells growing as monolayer
Medium:	90% Dulbecco's MEM + 10% FBS + 4 mM L-glutamine
Subculture:	split confluent culture 1:2 to 1:3 every third day using trypsin/EDTA; maintain at 7-10 x 10 ⁶ cells/175 cm ² ; seed out at ca. 1 x 10 ⁶ cells/25 cm ² in 10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of ca. 10-15 x 10 ⁶ cells/175 cm ² ; maximal density of 17 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human hypodiploid karyotype with 20% polypoidy; 44(40-45)<2n>XXY/XXYY, -6, +7, -10, -14, -22, der(1)del(1)(p21)t(1;9)(q42;q34), der(9)t(9;16)(p13;p11)t(1;9)(q42;q34), der(16)t(9;16)(p13;p11), der(18)t(6;18)(p12;q12.2); gain of 7 and loss of 10 and 22 typical in low-grade astrocytoma-oligodendroglioma; gains of sex chromosomes (particularly Y) are very rare in tumor cells
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	GRANTA-519
Cell Type:	human B cell lymphoma
DSMZ No:	ACC 342
Origin:	established from the peripheral blood taken in 1991 at relapse of a high-grade B-NHL (leukemic transformation of mantle cell lymphoma, stage IV) diagnosed in a 58-year-old Caucasian woman with previous history of cervical carcinoma
References:	Jadayel et al., Leukemia 11: 64-72 (1997); review: Drexler & MacLeod, Leukemia Res. 26: 781-787 (2002)
Depositor:	Dr. D. Jayadel, Institute of Cancer Research, Sutton, Surrey, UK
DSMZ Cell Culture Data:	
Morphology:	spheroid cells growing in suspension, singly or in clumps
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + 2 mM L-glutamine
Subculture:	maintain at $0.5\text{--}1.5 \times 10^6$ cells/ml; split ratio of 1:4 every 3 days; seed out at ca. 1.0×10^6 cells/ml
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of about 49 hours
Harvest:	maximum density of about 2.6×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13(+), CD19+, CD20+, CD30(+), CD34-, CD37+, CD79a+, cyCD79a+, CD80+, CD138+, HLA-DR+, sm/cyIgG-, sm/cyIgM+, sm/cykappa-, sm/cylambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human hypodiploid karyotype with 8% polyploidy; 44(39-44)<2n>XX, -12, -17, -18, +mar, add(1)(p22), del(3)(p14p23), i(8p), i(8q), add(9)(p22)x1-2, t(11;14)(q13;q32), add(13)(p12), add(18)(q21); sideline with two copies of der(14) and der(9); carries t(11;14) and rearrangement at 9p22 associated with cyclin D1 activation and deletion of p15/p16; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	H-1184
Cell Type:	human small cell lung carcinoma
DSMZ No:	ACC 502
Origin:	established in 1985 from the lymph node from a 42-year-old white man with metastatic small cell lung carcinoma prior to treatment; corresponds to NCI-H1184; matched B-lymphoblastoid cell line (B-LCL) is available
References:	Carney et al., Cancer Res. 45: 2913-2923 (1985); Phelps et al., J. Cell. Biochem. Suppl. 24: 32-91 (1996)
Depositor:	Drs. A. K. Virmani & A. F. Gazdar, University of Texas, Southwest Medical Center, Dallas, TX, USA
DSMZ Cell Culture Data:	
Morphology:	small epitheloid cells growing adherently in monolayers or in suspension as clusters
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 every 5-8 days using trypsin/EDTA; seed out at ca. 0.1×10^6 cells/cm ² in a small flask of 25 cm ² ; maintain at ca. $0.1-1.0 \times 10^6$ cells/cm ² ; after recovery change medium sparingly
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-70 hours
Harvest:	maximum density of ca. 1.2×10^6 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype with 8% polyploidy; 57(53-59)<2n>XY/XXY, +1, +3, +5, +8, +9, -10, -10, +11, +12, -14, +16, +20, +22, +6mar, add(1)(p34), add(1)(p11), add(3)(p14), add(3)(q21-22), add(5)(p11), add(5)(q11), add(8)(q21), add(11)(q25), add(12)(q24)x1-2, der(12)t(11;12)(q11;q24), add(13)(p11), del(?13)(q12q21), del(16)(p12p13), der(?22)t(14;22)(q21;q13)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	H-1339
Cell Type:	human small cell lung carcinoma
DSMZ No:	ACC 506
Origin:	established from the pleural effusion of a 49-year-old white woman in 1986 prior to treatment for extensive small cell lung cancer which responded only partially to chemotherapy
References:	Phelps et al., J. Cell. Biochem. Suppl. 24: 32-91 (1996)
Depositor:	Drs. A. K. Virmani & A. F. Gazdar, University of Texas, Southwest Medical Center, Dallas, TX, USA
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells growing in monolayers
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split confluent culture 1:2 to 1:10 every 5-10 days using trypsin/EDTA; seed out at ca. 0.25×10^5 cells/cm ² (about 2 million cells per 80 cm ² flask); maintain at ca. $0.1-0.3 \times 10^5$ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-90 hours
Harvest:	maximum harvest of ca. 1.5×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human highly rearranged flat-modulated karyotype with 12% polyploidy; $80-95 < 4n > XX, -X, -X$; rearrangements include: add(1)(q11), der(1)del(1)(q22)trp(1)(q11q22), der(4;8)(q10;q10), add(7)(q1?1), del(7)(q32), add(8)(p11), der(9)t(1;9)(q11;q13), add(10)(p11), add(11)(p1?), der(13;21)(q10;q10), add(14)(p11), add(16)(q23), add(17)(p12), der(18;21)(q10;q10)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	H-1963
Cell Type:	human small cell lung carcinoma
DSMZ No:	ACC 540
Origin:	derived from the lung of a 56-year-old black man with small cell lung carcinoma prior to treatment in 1988
References:	Phelps et al., J. Cell Biochem. Suppl. 24: 32-91 (1996)
Depositor:	Drs. A. K. Virmani & A. F. Gazdar, University of Texas, Southwest Medical Center, Dallas, TX, USA
DSMZ Cell Culture Data:	
Morphology:	cells grow in clumps in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split culture 1:2 every 3 days; seed out at ca. 0.5×10^6 cells/ml; cell number can only be estimated due to the formation of large aggregates
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 3-4 days
Harvest:	cell harvest of ca. $1-2 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human bimodal hypodiploid/hypotetraploid karyotype with 3% polyploidy; 44(42-48)<2n>X, -Y, -5, -10, -13, +18, -19, +2mar, add(1)(q32), ins(2;hsr)(p21-23;?), add(3)(p11), del(4)(q24-26), del(10)(p13-14), add(18)(p11)x2, del(20)(q12), der(22)(t1;22)(q12;q13), dmin present in some cells
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	H-209
Cell Type:	human small cell lung carcinoma
DSMZ No:	ACC 499
Origin:	established in 1979 from the bone marrow aspirated from a 55-year-old white man with small cell lung carcinoma prior to treatment; corresponds to NCI-H209; described as expressing neuroendocrine biochemical markers, e.g. neuron-specific enolase, brain creatine kinase, L-DOPA decarboxylase and bombesin
References:	Carney et al., Cancer Res. 45: 2913-2923 (1985); Phelps et al., J. Cell. Biochem. Suppl. 24: 32-91 (1996)
Depositor:	Drs. A. K. Virmani & A. F. Gazdar, University of Texas, Southwest Medical Center, Dallas, TX, USA
DSMZ Cell Culture Data:	
Morphology:	spherical-to-ovoid cells growing singly in large clumps
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2-4 days; seed out at ca. $0.7-1.0 \times 10^6$ cells/ml; maintain at ca. $0.5-1.5 \times 10^6$ cells/ml; note that cells grow slowly after thawing out during which time clumps should be resuspended in original medium to encourage growth
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-70 hours
Harvest:	maximum density of ca. 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-6 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype with 8% polyploidy; 47(44-48)<2n>XY, -4, -5, +6, +7, +8, -10, -14, -15, -17, +0-2mar, r/dm, add(1)(p32-34), add(3)(p11), add(4)(q24-25), add(5)(p1?), add(6)(q14)x1-2, add(8)(p1?), add(17)(p1?); extensive subclonal variation
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	H-2171
Cell Type:	human small cell lung carcinoma
DSMZ No:	ACC 544
Origin:	derived from the pleural effusion of a 50-year-old white male with extensive small cell lung carcinoma
References:	after chemotherapy in 1989; corresponds to NCI-H2171 Phelps et al., J. Cell Biochem. Suppl. 24: 32-91 (1996); Virmani et al., Genes Chromosomes Cancer 21: 308-319 (1998)
Depositor:	Drs. A. K. Virmani & A. F. Gazdar, University of Texas, Southwest Medical Center, Dallas, TX, USA
DSMZ Cell Culture Data:	
Morphology:	large round cells growing singly or in clumps in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2-4 days; seed out at ca. 1.0×10^6 cells/ml; maintain at $0.7\text{--}2.0 \times 10^6$ cells/ml; after recovery the growing is slow
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35-60 hours
Harvest:	maximum density at 3.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 10×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypodiploid karyotype with 8% polyploidy; 42-44<2n>XY, -4, -5, -8, -10, -11, -13, -15, -17, -19, +20, +4-5mar, add(1)(p36), der(2)ins(2;?)(p2?:?)(t(2;2)(q37;p24), add(3)(p11), i(3q), del(6)(q25q26), del(9)(p11), der(12)add(12)(p11)del(12)(q24), add(14)(p11), i(15q), add(18)(p11)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	H25B10
Cell Type:	mouse hybridoma (anti hepatitis B)
DSMZ No:	ACC 98
Origin:	established by fusion of P3-NSI/1-AG4-1 mouse myeloma cells with spleen cells from a mouse immunized with hepatitis B surface antigen; a monoclonal antibody (IgG ₁) against hepatitis B surface antigen is produced; patented cell line (= ATCC CRL 8017)
References:	
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single round to oval cells in suspension
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS
Subculture:	cells are difficult to culture! maintain at $0.5-1.0 \times 10^5$ cells/ml; split 1:4 every 2-3 days
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximal density at about $1-2 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, MDH
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HAP-T1
Cell Type:	hamster pancreas carcinoma
DSMZ No:	ACC 222
Origin:	established from the BHP-induced pancreatic carcinoma of a Syrian golden hamster (GN strain) in 1983
References:	Gastroenterol. Jpn. 23: 183-194 (1988)
Depositor:	Dr. S. Saito, Toyama Medical and Pharmaceutical University, Toyama, Japan
DSMZ Cell Culture Data:	
Morphology:	epitheloid monolayer
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture at 1:3 every 3 days or at 1:6 every 5-6 days using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 25-30 hours
Harvest:	cell harvest of about 50-60 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Fingerprinting:	unique DNA profile with (gtg) ₅ -probe, heterozygous with human single locus probe pYNH24
Species:	human origin was ruled out with IEF of AST, LDH, MDH, PEP B
Cytogenetics:	near diploid hamster karyotype with tetraploid sideline (18%) and 18% polyploidy; 44(40-45)<2n>X/XX; paucity of homologous chromosomes indicates a high level of rearrangement; hamster markers present; resembles published karyotype
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HC-1
Cell Type:	human hairy cell leukemia
DSMZ No:	ACC 301
Origin:	spontaneously immortalized peripheral blood leukocytes from a 56-year-old white man with hairy cell leukemia (HCL) at diagnosis; cells were described to grow in soft agar and to be tumorigenic in irradiated nude mice
References:	Schiller et al., Leukemia 5: 399-407 (1991)
Depositor:	Dr. J. Schiller, Veterans Administration Hospital, Madison, Wisconsin, USA
DSMZ Cell Culture Data:	
Morphology:	lymphoblastoid cells growing as single cells or in clumps, in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 to 1:5 every 2-4 days; maintain at $0.3-0.6 \times 10^6$ cells/ml; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 45 hours
Harvest:	saturation density of ca. $1.0-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD25-, CD34-, CD37+, CD79a+, CD80+, CD138+, HLA-DR+, sm/cyIgG-, sm/cyIgM-, sm/cykappa+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human near-tetraploid karyotype with 3% polyploidy; 92(89-92)<4n>XXYY, add(2)(p13->23); differs from published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HCC-1143
Cell Type:	human breast carcinoma
DSMZ No:	ACC 517
Origin:	established from a primary, invasive, ductal breast carcinoma (TNM stage IIa, tumor size 5, grade 3, without lymph node metastasis) from a 52-year-old white woman prior to therapy in 1994; a paired EBV+ B-lymphoblastoid cell line (B-LCL HCC-1143BL) is available; reported to express EGP2, CK19 but neither estrogen nor progesterone receptors are expressed
References:	Gazdar et al., Int. J. Cancer 78: 766-774 (1998)
Depositor:	Drs. A. K. Virmani & A. F. Gazdar, University of Texas, Southwest Medical Center, Dallas, TX, USA
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing as adherent monolayers
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split confluent culture 1:2 to 1:3 every 4-7 days using trypsin/EDTA (for 10-20 min); seed out at ca. $1.0\text{--}1.7 \times 10^6$ cells/80 cm ² flask
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-60 hours
Harvest:	cell harvest of about $0.4\text{--}0.9 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1.5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human highly rearranged flat-modified hypertriploid karyotype with 8% polyploidy; 74-78<3n>XXX, -2, +3, -4, -6, +7, +7, +7, -9, -10, -11, -11, -13, -15, -17, -19, +20, +14-20 mar, +2 dmin, add(X)(p11.3), add(X)(p22)add(X)(q11), add(X)(q25-27), del(1)(p22), del(1)(q13)x1-2, del(3)(q12)x2, del(4)(q11), add(4)(p15), del(5)(q13), der(6)t(1;6)(q22;q24), add(7)(p14), del(7)(p14), del(7)(p11), add(7)(p11), ins(7;?X)(q11;?), add(7)(q3?), del(8)(p1?), del(9)(p22), add(11)(q23), add(12)(q24), add(13)(p11), der(14;21)(q10;q10), add(14)(p11), add(15)(p11), add(15)(p11), add(16)(q24), i(18q), add(22)(p11)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HCC-15
Cell Type:	human non-small cell lung carcinoma
DSMZ No:	ACC 496
Origin:	established from the tumor of a 47-year-old man with non-small lung cell carcinoma (subtype squamous carcinoma); matched EBV+ B-lymphoblastoid cell line (B-LCL) is available (HCC-15BL)
References:	Virmani et al., Genes Chromosomes Cancer 21: 308-319 (1998); Girard et al., Cancer Res. 60: 4894-4906 (2000)
Depositor:	Drs. A. K. Virmani & A. F. Gazdar, University of Texas, Southwest Medical Center, Dallas, TX, USA
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells growing as monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:4 to 1:6 every 4-5 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	cell harvest of ca. 8×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypodiploid karyotype with 16% polyploidy; 37(34-38)<2n>X, -X, -4, -13, -14, -17, -18, -19, -20, -22, +1-2mar, del(3)(q13), der(5;19)(q10;q10)add(19)(q13), add(9)(p2?), del(10)(p12-13), add(12)(q24.3), del(20)(?p12), add(22)(q1?); sideline with add(8)(p12), del(7)(q21)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HCC-1599
Cell Type:	human breast carcinoma
DSMZ No:	ACC 503
Origin:	established in 1995 from the primary tumor cells of a 44-year-old woman with breast carcinoma (ductal type; TNM stage IIIA, grade 3); matched EBV+ B-lymphoblastoid cell line (B-LCL) is available (HCC-1599BL)
References:	Gazdar et al., Int. J. Cancer 78: 766-774 (1998)
Depositor:	Drs. A. K. Virmani & A. F. Gazdar, University of Texas, Southwest Medical Center, Dallas, TX, USA
DSMZ Cell Culture Data:	
Morphology:	cells grow as aggregates in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:2 every 4 days; seed out at ca. $3-5 \times 10^6$ cells in 5 ml; it might be of advantage to start culture initially in a 6-well-plate, otherwise cells can be grown in flasks; cell numbers can only be estimated as the cells aggregate in large clumps which are rather difficult to disperse; replace half the medium every 2nd day, cells should then be dispersed by pipetting
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 4-5 days
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human highly rearranged flat-modet hypertetraploid karyotype; 92-105<4n>XX, -X, -X, -2, -5, -5, -5, +7, -9, +10, -11, -13, -13, -14, -15, -16, -17, -17, +18, +19, -20, -20, -20, -21, -22, -22, -22, +27-30mar, del(X)(q23), add(1)(p36)x2, der(1;3)(q10;q10)del(1)(p35), del(1)(p35), add(2)(q3?), add(3)(q2?)x2, add(4)(q2?)x2, der(4;14)(q10;q10), add(5)(p15), add(6)(p23), add(6)(q25)x2, add(7)(p11), add(7)(p22)x2, add(7)(q31), add(7)(p11), add(8)(p11), add(8)(q24), add(9)(p22), add(9)(p11), add(10)(q24)x2, add(11)(p12), add(11)(q2?), del(11)(q13), i(13q), add(14)(p11), del(16)(p13.2), add(16)(q22), add(16)(q24), add(19)(q13)x2, add(19)(q13.2), add(21)(p11)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HCC-1937
Cell Type:	human breast carcinoma
DSMZ No:	ACC 513
Origin:	established from a primary ductal carcinoma of the breast mammary gland from a 24-year-old Caucasian woman in 1995; cells were described to be homozygous for the BRCA1 mutation; a paired EBV+ B-lymphoblastoid cell line (B-LCL HCC-1937BL) is available
References:	Tomlison et al., Cancer Res. 58: 3237-3242 (1998)
Depositor:	Drs. A. K. Virmani & A. F. Gazdar, University of Texas, Southwest Medical Center, Dallas, TX, USA
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial cells growing as monolayer, about 5-10% are giant vacuolated cells
Medium:	85-90% RPMI 1640 + 10-15% FBS
Subculture:	split confluent culture 1:2 to 1:4 every 2-5 days using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ² flask
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of about 5×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HCC-33
Cell Type:	human small cell lung carcinoma
DSMZ No:	ACC 487
Origin:	established from the pleural effusion of a 52-year-old man with small cell lung carcinoma; matched EBV+ B-lymphoblastoid cell line (B-LCL) is available (HCC-33BL)
References:	Virmani et al., Genes Chromosomes Cancer 21: 308-319 (1998); Girard et al., Cancer Res. 60: 4894-4906 (2000)
Depositor:	Drs. A. K. Virmani & A. F. Gazdar, University of Texas, Southwest Medical Center, Dallas, TX, USA
DSMZ Cell Culture Data:	
Morphology:	round to ovoid cells growing singly or in clumps
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2-5 days; seed out at ca. 1.0×10^6 cells/ml; maintain at $0.5-1.5 \times 10^6$ cells/ml; after thawing, cell clumps should be carefully resuspended to encourage growth
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	cell harvest of ca. 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+ (dot-like expression, no regular filaments), desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-modulated hypotriploid karyotype with 8% polyploidy; 61-68<3n>XX, -Y, +1, +1, +3, -10, -12, -13, +17, -20, -22, +3-5 mar, ider(1)(q10)del(1)(?q12?q43)x2, add(1)(p36), del(1)(p32)x1-2, dup(2)(p23p24), der(3;17)(q10;q10)x1-2, add(8)(p11), del(9)(q21), der(12)t(11;12)(q13.4;q24.3), add(13)p11; sideline with add(5)(p11), add(15)(p11), add(17)(p1?), del(22)(q12); variable dmin present in most cells; dup(2) includes MYCN locus
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HCC-366
Cell Type:	human non-small cell lung carcinoma
DSMZ No:	ACC 492
Origin:	established from the tumor of an 80-year-old woman with non-small cell lung carcinoma (subtype adeno-squamous carcinoma); cells were described to carry fractional allelic loss with loss of heterozygosity at various hot spots; matched EBV+ B-lymphoblastoid cell line (B-LCL) is available (HCC-366BL)
References:	Oie et al., J. Cell. Biochem. Suppl. 24: 24-31 (1996); Virmani et al., Genes Chromosomes Cancer 21: 308-319 (1998); Girard et al., Cancer Res. 60: 4894-4906 (2000)
Depositor:	Drs. A. K. Virmani & A. F. Gazdar, University of Texas, Southwest Medical Center, Dallas, TX, USA
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells growing as monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:2 to 1:3 once a week using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ² ; maximal density at ca. 0.3×10^5 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 72-96 hours
Harvest:	cell harvest of ca. 4×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7(+), cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hyperdiploid karyotype with 12% polyploidy; 48-56<2n>XX, +1, +1, -2, +4, +7, +8, +11, -13, -15, -18, +7-9mar, add(X)(p11), add(1)(p22), add(1)(q11)x1-2, add(4)(q31)x1-2, del(8)(p12), del(8)(q13), add(10)(p13), add(14)(p11), add(17)(p12)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HCC-44
Cell Type:	human non-small cell lung carcinoma
DSMZ No:	ACC 534
Origin:	established from the lung of a 54-year-old woman with non-small cell lung cancer of the adenocarcinoma type
References:	Virmani et al., Genes Chromosomes Cancer 21: 308-319 (1998); Oie et al., J. Cell. Biochem. Suppl. 24: 24-31 (1996)
Depositor:	Drs. A. K. Virmani & A. F. Gazdar, University of Texas, Southwest Medical Center, Dallas, TX, USA
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells growing as monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:5 twice a week using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/80 cm ² flask; maximal density of ca. 8×10^6 cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 6×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with sparfloxacin, then negative in microbiological culture and PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hyperpentaploid karyotype with 8% polyploidy; 120-124<5n>XX, +1, +1, +3, +3, -4, -4, -5, +7, +7, +7, -9, -9, -10, +11, -18, -21, -21, +22, +22, +22, +7-14mar, der(1)t(X;1)(q11;p11)x2, del(1)(q21)x2, i(1q), t(1;3)(p11;q11), add(2)(q?11), del(2)(p11), add(3)(q28), add(3)(p11), i(5p), add(8)(p2?), add(8)(q2?), add(8)(q24), der(9;19)(q10;q10)x2, add(10)(q2?5), add(11)(p13), der(11)add(11)(p1?1)add(11)(q25), add(12)(p13), add(16)(p11), add(17)(p1?2), add(18)(p11), del(20)(q12)x1-2, add(20)(q13)x2; 2-6 dmin present in 80% cells
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HCT-15
Cell Type:	human colon adenocarcinoma
DSMZ No:	ACC 357
Origin:	established from a colorectal adenocarcinoma; HCT-15 is a sister cell line of the cell line DLD-1 (DSMZ ACC 278)
References:	Dexter et al., Cancer Res. 39: 1020-1025 (1979)
Depositor:	Dr. R. A. F. MacLeod, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing as monolayer with contact inhibition
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 3-5 days using trypsin/EDTA; seed out 2×10^6 cells in ca. 7 ml medium in a 80 cm ² flask; maximal density at ca. 50×10^6 cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 20-25 hours
Harvest:	harvest of ca. 25×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin-
Fingerprinting:	same DNA profile as DLD-1 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, LDH, MPI
Cytogenetics:	human pseudodiploid karyotype with 16% polyploidy; $46<2n>XYq+, t(8;17)(p23;q12/21), inv(11)(p14.2;q13)$; stable karyotypic differences cf DLD-1; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HD-MY-Z
Cell Type:	human Hodgkin's lymphoma
DSMZ No:	ACC 346
Origin:	established in 1991 from the pleural effusion (rich in Hodgkin/Reed-Sternberg cells) from a 29-year-old patient with nodular sclerosing Hodgkin's disease (stage IIIB) refractory to therapy
References:	Bargou et al., J. Exp. Med. 177: 1257-1268 (1993); review article: Drexler, Leukemia Lymphoma 9: 1-26 (1993)
Depositor:	Dr. R. Bargou, Max Delbrück Centrum für Molekulare Medizin, Berlin, Germany
DSMZ Cell Culture Data:	
Morphology:	variable morphology; semi-adherent, spheroid to fusiform cells growing in clumps in mono- and multilayers, occasional giant cells present
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.2-0.75 \times 10^6$ cells/ml or $0.8-1.6 \times 10^5$ cells/cm ² ; optimal split ratio of 1:3 to 1:8 every 2-5 days; cells may be dislodged mechanically without trypsin/EDTA; seed out at ca. $0.3-0.5 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	saturation density at ca. 1.5×10^6 cells/ml or 5×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3-, CD10+, CD13+, CD14(+), CD15+, CD19-, CD21-, CD25-, CD30-, CD34+, CD68+, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human highly rearranged hyperdiploid karyotype with 24% polyploidy; 52-55<2n>X/Xdel(X)(p11), +1, +2, +5, +6, +8, +9, -17, -17, -18, +19, +20, +21, +22, +2mar, der(1)t(1;3)(q32;p21), der(2)t(2;?10)(p25;q22), add(3)(q27), del(4)(q22), i(5p), i(5q), der(5)t(5;8)(p14.1;p12), del(6)(q15), add(7)(p15), add(7)(p11), del(8)(q11), der(9)add(9)(p24)add(9)(q21), der(10)add(10)(p11)add(10)(q22), add(11)(q24), i(13q), der(13)t(7;13)(q11;p13), add(18)(p13), add(19)(q13.1); additional rearrangements present in subclones include der(1)t(1;2)(q11;p25), del(3)(q21), add(16)(q24)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HDLM-2
Cell Type:	human Hodgkin's lymphoma
DSMZ No:	ACC 17
Origin:	established from the pleural effusion of a 74-year-old man with Hodgkin's disease (nodular sclerosing, stage IV) in 1982
References:	Drexler et al., Leukemia Res. 10: 487-500 (1986); idem, Leukemia 2: 371-376 (1988); idem, Int. J. Cancer 43: 1083-1090 (1989); review article: Drexler, Leukemia Lymphoma 9: 1-25 (1993); MacLeod et al., Leukemia 14: 1803-1814 (2000)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round to polygonal cells in suspension; heterogeneity in cell size, 15-100 μ m or larger; many bi- and polynucleated cells
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	maintain at high density (about $0.5-1.0 \times 10^6$ cells/ml); exchange culture medium and add 10-25% new medium; seed out at ca. 1×10^6 cells/ml in a 24-well-plate
Incubation:	at 37 $^{\circ}$ C with 5% CO ₂
Doubling Time:	doubling time of about 70-75 hours
Harvest:	maximal density of $2.0-2.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2+, CD3-, CD4(+), CD5-, CD6-, CD7-, CD8-, CD13-, CD14-, CD15+, CD19-, CD21-, CD25+, CD30+, HLA-DR+, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	remarkable human hypodiploid cell line with large number of rearrangements involving all chromosomes except ch 10; 36(35-38)<2n>X, der(Y), breakpoints at: 1p21, 1q11, 2p14, 2q37, 3q21, 3p14, 3q11, 3q25.2, 3q27, 4q25, 5q11, 5q35, 6q15/16, 7p14, 8q21.1, 8q23, 8q24.2, 9p11, 11q13, 12p12, 12q24.3, 13p11, 13q14, 14q21, 14q23/24.2, 15p11, 15q13, 17p11, 17q25, 18p11, 19p13, 19q13, 21q21, 22q33, Ycen, Yq12
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HDQ-P1
Cell Type:	human breast carcinoma
DSMZ No:	ACC 494
Origin:	established in 1995 from a primary ductal infiltrating breast carcinoma of a 50-year-old Caucasian woman; cells are described to be estrogen- and progesterone-receptor negative, and to possess a point mutation in p53 and a five-fold amplification of c-myc
References:	Wang et al., Cancer Genet. Cytogenet. 120: 58-78 (2000)
Depositor:	Dr. B. Tetu, CHUQ, Quebec, Canada
DSMZ Cell Culture Data:	
Morphology:	epithelial cells growing adherently in monolayers
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:2 to 1:3 every 4-5 days using trypsin/EDTA (cells are difficult to detach from the flask - incubate with trypsin/EDTA for more than 20 min); cells detach faster applying "Accutase" (PAA, Austria); seed out at ca. 1×10^6 cells/80 cm ² ; maximal density at ca. 4×10^4 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 72 hours
Harvest:	cell harvest of ca. 4×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human highly-rearranged flat-moded hypertetraploid/hypopentaploid karyotype with 13% polyploidy; 92-107<4n>XX, -X, -X, +1, -3, -3, -3, +5, +5, +5, +6, +7, +7, -8, +10, -13, -13, -14, -15, -15, -16, -16, -17, -17, -18, -18, -18, -19, -19, -20, -20, -21, -21, -22, -22, -22, +28-29mar, add(X)(p2?)x2, add(1)(p36)x2, add(1)(p35), add(1)(q1?2)x2, der(2)t(2;3)(p16;q11), add(5)(p11q3?4)x2, add(5)(q11), der(5)t(5;?3)(q1;?;p11), der(5)t(5;5)(p1?;q1?4), i(5p), add(6)(p11), add(7)(p14), add(7)(p11)x2, add(7)(q11)x2, add(8)(q21), del(9)(p23), del(9)(p11), i(10p), del(11)(q22), der(12)t(7;12)(q11;q24), add(14)(p11)x2, add(15)(p11), add(16)(p13), add(16)(q23), add(17)(p11), del(17)(p12), add(18)(p11), add(19)(p13), add(19)(q13), del(22)(q12.2); HDQ-P1 markers
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HEL
Cell Type:	human erythroleukemia
DSMZ No:	ACC 11
Origin:	established from the peripheral blood of a 30-year-old man with erythroleukemia (AML M6) in relapse (after treatment for Hodgkin's disease) in 1980; cells were described to be capable of spontaneous and induced globin synthesis
References:	Martin et al., Science 216: 1233-1235 (1982)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round, large to occasionally giant polynucleated, single cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.1-0.5 \times 10^6$ cells/ml, split 1:3 to 1:5 every 3 days, cells start to grow after about 4 days; seed out at ca. 5×10^5 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 36 hours
Harvest:	maximal cell density at 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, PCR assays
Immunology:	CD3-, CD13+, CD14-, CD15+, CD19-, CD33+, CD41+, CD42+, CD71+, GlyA+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypotriploid karyotype with 2.3% polyploidy; 63(60-64)<3n>XYY, -2, -9, -10, -10, -11, -14, -16, -16, -17, -19, +20, +21, +2mar, del(2)(q32), t(3;6)(p13;q16), der(5)t(5;17)(q10;q10), der(6)t(1;6)(p13;p21), der(7)add(7)(p14;q32), add(8)(p21), der(9)t(9;?)(?;11)(p24;?)(?;q13), del(11)(q13), add(15)(p11), del(20)(q13), r(20)(p11q11), dup(21)(q11q22.3-qter), psu dic(22;9)t(9;?)(?;22)(p24;?)(?;p11-13); resembles published karyotype; carries also masked 5q- and 20q- consistent with AML M6/MDS
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HELA
Cell Type:	human cervix carcinoma
DSMZ No:	ACC 57
Origin:	established from the epitheloid cervix carcinoma of a 31-year-old black woman in 1951; later diagnosis changed to adenocarcinoma; first aneuploid, continuously cultured human cell line
References:	Scherer et al., J. Exp. Med. 97: 695 (1953); Gey et al., Cancer Res. 12: 264 (1952)
Depositor:	obtained from ATCC (CCL 2), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing in monolayers
Medium:	90% MEM (with Earle's salts) + 10% FBS + 2 mM L-glutamine + non-essential amino acids (cells also grow well in 90-95% RPMI 1640 + 5-10% FBS)
Subculture:	split confluent culture 1:4 to 1:6 every 3-5 days using trypsin/EDTA; cells reach confluence quickly; seed out at ca. $1-2 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of ca. $5-15 \times 10^6$ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of G6PD, MDH, NP
Cytogenetics:	human hypertriploid/hypotetraploid karyotype with 15% polyploidy; 82(77-84)<4n>XXX, -X, -2, -3, -4, +5, +5, +5, -7, -8, -11, -13, -14, -15, -16, -17, -18, -19, -21, -22, +mar; der(1)t(1;3)(p11;q11), add(2)(q37), del(3)(q13), der(3)t(3;5)(q11;q11), i(5p)x4, del(6)(q15), der(9)t(1;9)(p11;p11), add(12)(q23), i(15q), der(19)t(13;19)(q22;p13)x1-2, add(21)(p11), del(?22)(q12)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HELA-S3
Cell Type:	human cervix carcinoma
DSMZ No:	ACC 161
Origin:	cell line is a subclone of its parent HELA (DSM ACC 57) derived in 1955
References:	Puck et al., J. Exp. Med. 103: 273-284 (1956)
Depositor:	Dr. J. Wehland, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial, adherent cells growing as monolayer
Medium:	90% Ham's F12 + 10% FBS
Subculture:	split confluent culture 1:5 to 1:8 every 5-7 days (cells grow slowly) using trypsin/EDTA; seed out at ca. $2-3 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 37 hours
Harvest:	harvest of about $7-10 \times 10^6$ cells/80 cm ² flask
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	confirmed as HELA-derivative, but otherwise unique DNA profile using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human flat-moded hypotriploid karyotype with 32% polyploidy; 68(53-68)<3n>XX, -X, +1, -2, -4, +5, +8, +12, -13, -14, -20, -21, -22, -22, +4mar; del(1)(p35), der(1)t(1;3) (p11;q11), der(3)t(3;5)(q11;q11), i(5p)x2, del(6)(q23), add(6)(q27), del(7)(q32), i(8q), add(8)(q24), del(9)(p22p23), t(11;19)(q13;q13), add(12)(q23)x2, add(13)(p12), i(16q), add(19)(p13), add(21)(p11), i(22q); high level of spontaneous aberrations; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HEP-3B
Cell Type:	human hepatocellular carcinoma
DSMZ No:	ACC 93
Origin:	established from the tumor tissue of an 8-year-old black boy in 1976; description: cells contain integrated hepatitis B virus genome; cells were described to produce a variety of proteins, e.g. alpha-fetoprotein, albumin, transferrin, alpha2-macroglobulin, alpha1-antitrypsin, haptoglobin and others; patented cell line; this cell line has been assigned by the ZKBS (German Central Commission for Biological Safety) to risk category 2
References:	Aden et al., Nature 282: 615-616 (1979); Knowles et al., Science 209: 497-499 (1980)
Depositor:	Dr. H. J. Grub, Medical School of Freiburg, Freiburg, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing as monolayers
Medium:	90% MEM (with Earle's salts) + 10% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:4 to 1:10 every 3-5 days using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Harvest:	about 5 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at 1 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, PEP B
Cytogenetics:	human hypotriploid stemline with near tetraploid side line (6%); 63 (60-64)<3n>XX, -Y, -1, -1, -2, -3, -4, -8, -9, -13, +der(11)t(1;11)(q11;q14)x2, +der(?15)t(1;?15)(q11;q24), i(7p)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HEP-G2
Cell Type:	human hepatocellular carcinoma
DSMZ No:	ACC 180
Origin:	established from the tumor tissue of a 15-year-old Argentine boy with hepatocellular carcinoma in 1975; cell line is patented; cells were described to not harbor a hepatitis B virus genome; cells reportedly produce a variety of proteins: alpha-fetoprotein, albumin, alpha2-macroglobulin, alpha1-antitrypsin, transferrin, alpha1-antichymotrypsin, haptoglobin, ceruloplasmin, plasminogen, complement (C3, C4), C3 activator, fibrinogen, alpha1-acid glycoprotein, alpha2-HS glycoprotein, β -lipoprotein, retinol binding protein
References:	Aden et al., Nature 282: 615-616 (1979); Knowles et al., Science 209: 497-499 (1980)
Depositor:	obtained from ATCC (HB 8065), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial-like cells growing as monolayers and in small aggregates
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:2 to 1:4 every 3-6 days using trypsin/EDTA; seed out at ca. $2-3 \times 10^6$ cells/25 cm ² ; after thawing, initially to be cultured with 20% FBS
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	cell harvest of about $25-30 \times 10^6$ cells/80 cm ² ,
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hyperdiploid karyotype; 52(47-54)<2n>XY, +2, +14, +17, +20, +2mar, t(1;21)(p22.2;p11-12), i(17q)/der(17)t(17;17)(p11;q11)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HEPA 1-6
Cell Type:	mouse hepatoma
DSMZ No:	ACC 175
Origin:	cell line was derived from the BW7756 tumor that arose in a C57L mouse; cells were described to secrete several liver-specific products (e.g. albumin, alpha1-antitrypsin, alpha-fetoprotein, amylase); cells may grow in serum-free medium when supplemented with dexamethasone and selenium (= ATCC CRL 1830)
References:	Darlington et al., J. Natl. Cancer Inst. 64: 809-819 (1980); Darlington, Meth. Enzymol. 151: 19-38 (1987)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial cells growing as monolayer
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS
Subculture:	split confluent culture 1:4 to 1:10 every 3-6 days using trypsin/EDTA; cells grow rather quickly; seed out at ca. $1-2 \times 10^6$ cells/25 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of about 18 hours
Harvest:	about $10-15 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2.5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, MDH, NP, PEP B
Cytogenetics:	murine hypotetraploid karyotype with centric fusion and 6% polyploidy; flat-moded, 53-70<4n>
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HH-16 cl.2/1
Cell Type:	rat mammary fibrosarcoma
DSMZ No:	ACC 485
Origin:	established from the ascitic fluid of a rat with a tumor which was produced by injecting cultured cells of a chemically-induced (dimethylbenzanthracene) mammary tumor; rat: strain Sprague-Dawley, female, 240 days old, adult; the original tumor was induced by single feeding of DMBA on day 45 after birth; sister cell line of HH-16 cl. 4 (DSM ACC 358) received at passage #90, frozen at passage #97; cells were described to produce invasive fibrocarcinomas in S.D. rats upon s.c. injection with remote metastases
References:	Hölzel et al., Cell Biol. Int. Rep. 7: 555-556 (1983); Steffen et al., Cancer Res. 48: 7212-7218 (1988); Scherdin et al., Oncogene 5: 1619-1627 (1990); Steffen et al., Genes Chromosomes Cancer 4: 46-57 (1992)
Depositor:	Prof. Dr. Fritz Hölzel, Women's University Hospital Eppendorf, Hamburg, Germany
DSMZ Cell Culture Data:	
Morphology:	fibroblastoid growing in a criss-cross pattern
Medium:	90% RPMI 1640 + 10% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:4 every 3-4 days using trypsin/EDTA (NB: cells are sensitive to trypsin/EDTA and should not be incubated for more than 5 min at room temperature); seed out at ca. $1-2 \times 10^6$ cells/80 cm ² ; maintain at $4-5 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	cell harvest of ca. $7-8 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as rat by cytogenetics and PCR analysis with Alu, globin and rRNA primers
Cytogenetics:	rat near diploid karyotype with 11% polyploidy; 39-43<2n>

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HH-16.cl.4
Cell Type:	rat mammary tumor
DSMZ No:	ACC 358
Origin:	established from the ascitic fluid of a Sprague-Dawley rat (adult female, 240-days-old) with a rat mammary tumor in 1977; tumor was produced by injection of cultured cells from DMBA-induced mammary tumor (DMBA = 9,10-di-methyl-1,2-benzanthracene); derived from identical tumor as HH-16 cl.2/1 (DSM ACC 485); received at passage #50, frozen at passage #58; cells were described to produce invasive adenocarcinomas in S.D. rats upon s.c. injection with remote metastases in lymph nodes and lung
References:	Hölzel et al., Cell Biol. Int. Rep. 7: 555-556 (1983); Scherдин et al., Oncogene 5: 1619-1627 (1990)
Depositor:	Prof. Dr. Fritz Hölzel, Women's University Hospital Eppendorf, Hamburg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS + 2 mM L-glutamine + 1 mM sodium pyruvate
Subculture:	split confluent culture 1:5 to 1:10 twice a week using trypsin/EDTA; seed out at ca. $2-3 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-80 hours
Harvest:	cell harvest of ca. $1.0-1.5 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as rat with IEF of AST, NP
Cytogenetics:	rat flat-moded near-tetraploid karyotype with 10% polyploidy; 79-84<4n>
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HKT-1097
Cell Type:	hamster kidney tumor
DSMZ No:	ACC 445
Origin:	established from diethylstilbesterol-induced renal tumors of a 14-month-old male Syrian Golden Hamster in 1997; cells were described as expressing estrogen receptors, glial fibrillary acidic protein and other markers of the neuroectodermal lineage and as being responsive to estrogen
References:	Laurent et al., In Vitro Cell. Dev. Biol. 35A: 339-345 (1999); Brohee et al., In Vitro Cell. Dev. Biol. 36: 640-649 (2000)
Depositor:	Dr. G. Laurent, University of Mons-Hainaut, Mons, Belgium
DSMZ Cell Culture Data:	
Morphology:	fibroblastoid cells growing in monolayers
Medium:	95% Dulbecco's MEM (4.5 g/L glucose) + 5% FBS + 2 mM L-glutamine + 25 mM HEPES
Subculture:	split confluent culture 1:20 to 1:50 every 3-4 days using trypsin/EDTA; seed out at ca. 0.05×10^6 cells/cm ² ; maintain at 0.2×10^6 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 19 hours
Harvest:	cell harvest of ca. 0.3×10^6 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Cytogenetics:	hyperdiploid hamster karyotype with 10% polyploidy; s = 45(39-46)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HL-60
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 3
Origin:	established from the peripheral blood of a 35-year-old woman with acute myeloid leukemia (AML FAB M2) in 1976; cells can be used for induction of differentiation studies; described to be responsive to DMSO, phorbol ester TPA and other reagents and to carry amplified c-myc gene; present cells are apparently tetraploid derivatives of hypodiploid original where c-myc was amplified in dmin (instead of hsr)
References:	Collins et al., Nature 270: 347-349 (1977); Gallagher et al., Blood 54: 713-733 (1979); Dalton et al., Blood 71: 242-247 (1988); Collins, Blood 70: 1233-1244 (1987) (review on HL-60)
Depositor:	Dr. E. Porfiri, The Royal Free Hospital, Department of Haematology, London, UK
DSMZ Cell Culture Data:	
Morphology:	round, single cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml, split 1:2 to 1:5 every 1-2 days; seed out at about 1×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 25 hours
Harvest:	maximal density of $1.5-2.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4+, CD13+, CD14-, CD15+, CD19-, CD33+, CD34-, HLA-DR-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human flat-moded hypotetraploid karyotype with hypodiploid sideline and 1.5% polyploidy; 82-88<4n>XX, -X, -X, -8, -8, -16, -17, -17, +18, +22, +2mar, ins(1;8)(p?31;q24hsr)x2, der(5)t(5;17)(q11;q11)x2, add(6)(q27)x2, der(9)del(9)(p13)t(9;14)(q?22;q?22)x2, der(14)t(9;14)(q?22;q?22)x2, der(16)t(16;17)(q22;q22)x1-2, add(18)(q21); sideline with: -2, -5, -15, del(11)(q23.1q23.2); c-myc amplicons present in der(1) and in both markers
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HN
Cell Type:	human oral squamous cell carcinoma
DSMZ No:	ACC 417
Origin:	established from the tumor of a 60-year-old Japanese man with moderately differentiated squamous cell carcinoma of the soft palate which was invasive into muscle layer; cell line was established from a cervical lymph node metastasis which occurred seven years after initial treatment of the primary tumor; the tumor metastasized not only to the lungs, but also to the brain; described as anchorage-independent and tumorigenic in nude mice
References:	Kawamata et al., Int. J. Cancer 70: 120-127 (1997)
Depositor:	Dr. H. Kawata, Tokushima University, Tokushima, Japan
DSMZ Cell Culture Data:	
Morphology:	epithelial-like polygonal or round and flat cells (sometimes spindle-form, very heterogeneous) growing in multilayers
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS
Subculture:	split confluent culture 1:5 every 5 days using trypsin/EDTA; seed out at ca. 1.5×10^6 cells/80 cm ² in 15 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of ca. $8-10 \times 10^6$ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, NP
Cytogenetics:	human flat-modulated hypotriploid karyotype with 4% polyploidy; 60-64<3n>XX, -Y, +1, -2, -4, +5, -6, +7, -10, -12, -13, -13, -17, -18, -18, -21, -22, +3-5mar, add(X)(p22), der(1)add(1)(p36)ins(1;?)(q11;?), t(4;14)(q21;p11), i(4p)x1-2, del(5)(q31), i(5p), add(6)(p23), del(6)(q15), add(7)(q31), der(8)t(8;11)(p11;p11)add(8)(q24), add(11)(q11), add(11)(q24), ins(12;?)(q21;?)dup(12)(q11q24.32), dup(13)(q11q32), der(14)t(8;14)(q11;p11), i(14q), add(15)(p11), add(15)(p11), add(?15)(p11), der(17)t(13;17)(q13;q22), add(19)(p13); sideline with del(1)(p11), del(7)(q11)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HPB-ALL
Cell Type:	human T cell leukemia
DSMZ No:	ACC 483
Origin:	established from the peripheral blood of a 14-year-old Japanese boy with ALL and thymoma at diagnosis in 1973
References:	Morikawa et al., Int. J. Cancer 21: 166-170 (1978)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round to polymorph cells growing singly or in clusters in suspension
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2-3 days; seed out at ca. 1×10^6 cells/ml; after thawing, we recommend to use initially 24-well-plates and 20% FBS, cells may need 3-5 days to recover post-thawing
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	maximum density of about 4×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2+, CD3+, CD4+, CD5+, CD6+, CD7+, CD8+, CD10+, CD13-, CD15+, CD19-, CD21+, CD34-, CD38+, HLA-DR-, TCRalpha/beta+, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	pseudodiploid human karyotype with 8% polyploidy; 46<2n>XY, der(1)t(1;16)(q22;p11-12)add(16)(p13), del(2)(p24), del(3)(p11), der(5)t(1;5)(q22-24;q32-33), r(16)(?p12?q12); sideline with -3 instead of del(3); resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HPD-1NR
Cell Type:	hamster pancreas ductal adenocarcinoma
DSMZ No:	ACC 314
Origin:	established from the pancreatic ductal adenocarcinoma of a female Syrian (Golden) hamster induced by treatment with BHP (chemical carcinogen); sister cell line of HPD-2NR (DSM ACC 293); described to be homotransplantable (i.e. in other syngeneic hamsters) and to carry a mutation in exon 1 of the c-Ki-ras oncogene
References:	Mori et al., Int. J. Pancreatol. 16: 171-177 (1994)
Depositor:	Dr. T. Mori, Nara Medical University, Nara, Japan
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial cells growing in monolayers (about 2-5% very large cells)
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:4 every 5-6 days using trypsin/EDTA; seed out at about 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60 hours
Harvest:	cell harvest of about $6-7 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Fingerprinting:	identical DNA profile with HPD-1NR multilocus probe (gtg) ₅
Species:	apparently hamster according to IEF of AST, LDH, MDH, NP, PEP B
Cytogenetics:	near triploid highly rearranged karyotype with 36% polyploidy; 60-66<3n>XX, -X; hamster markers present; resembles published karyotype
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HPD-2NR
Cell Type:	hamster pancreas ductal adenocarcinoma
DSMZ No:	ACC 293
Origin:	established from the pancreatic ductal adenocarcinoma of a female Syrian (Golden) hamster induced by treatment with BHP (chemical carcinogen); sister cell line of HPD-1NR (DSM ACC 314); described to be homotransplantable (i.e. in other syngeneic hamsters) and to carry a mutation in exon 1 of the c-Ki-ras oncogene
References:	Mori et al., Int. J. Pancreatol. 16: 171-177 (1994)
Depositor:	Dr. T. Mori, Nara Medical University, Nara, Japan
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial cells growing as monolayers
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split confluent culture 1:3 to 1:8 every 2-6 days using trypsin/EDTA; seed out at about 0.4×10^5 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	cell harvest of ca. $1.5\text{--}2.0 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2\text{--}3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Fingerprinting:	unique and heterozygous DNA profile with pYNH24 single locus probe
Species:	apparently hamster according to IEF of AST, LDH, MDH, NP, PEP B
Cytogenetics:	near triploid highly rearranged karyotype with 29% polyploidy; 58-67<3n>XX, -X; hamster markers present; resembles published karyotype
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HSB-2
Cell Type:	human T cell leukemia
DSMZ No:	ACC 435
Origin:	established in 1966 from the peripheral blood of an 11-year-old boy with lymphosarcoma that had progressed to acute lymphoblastic leukemia (ALL); cell line was established and initially passaged in newborn hamsters; cells were described as carrying the t(1;7)(p34;q34) involving the LCK and TCRB genes and a submicroscopic del(1)(p32) leading to the SIL-SCL fusion gene; one of the oldest immature T cell lines, also known in the literature as CCRF-HSB-2
References:	Adams et al., Cancer Res. 28: 1121-1125 (1968); Adams et al., Exp. Cell Res. 62: 5-10 (1970)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round to polygonal cells, growing singly or in dense clumps in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:3 every 2-3 days; seed out at ca. $0.3-0.4 \times 10^6$ cells/ml; maintain at 0.4×10^6 cells/ml; maximal density at ca. 1.3×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 25 hours
Harvest:	cell harvest of ca. 0.8×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3-, CD4-, CD5+, CD6+, CD7+, CD8+, CD13-, CD15+, CD19-, CD34-, CD71+, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, PEP B
Cytogenetics:	human pseudodiploid karyotype with 4% polyploidy; 46(42-46)<2n>XY, t(1;7)(p34;q35), carries t(1;7) effecting LCK-TCRB fusion, resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HT-1080
Cell Type:	human fibrosarcoma
DSMZ No:	ACC 315
Origin:	established from the biopsy from a fibrosarcoma of a 35-year-old Caucasian man in 1972; patient did not receive any chemo- or radiotherapy
References:	Rasheed et al., Cancer 33: 1027-1033 (1974)
Depositor:	Dr. M. Wirth, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells growing as monolayers
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture about 1:10 every 3 days using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/80 cm ² + 7-10 ml medium
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 25×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human near-diploid karyotype with 24% polyploidy; 46<2n>XY, der(5)t(5;5)(p15;q32), der(11)t(10;11)(q24;q11); sideline with -13, del(1)(p32), del(1)(q24); 5p15 is a putative recurrent breakpoint in fibrosarcoma
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HT-1376
Cell Type:	human urinary bladder carcinoma
DSMZ No:	ACC 397
Origin:	established from a transurethral resection of invasive, moderately pleomorphic (grade 3) bladder transitional cell carcinoma of a 58-year-old Caucasian woman who had not received chemo- or radiotherapy; cells were received at passage #43 and were frozen at DSMZ at passage #56
References:	Rasheed et al., J. Natl. Cancer Inst. 58: 881-890 (1977); Grimm et al. Urol Res. 23: 293-300 (1995)
Depositor:	Dr. W. A. Schulz & Dr. B. J. Schmitz-Dräger, Urologische Klinik, Düsseldorf, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid adherent cells growing in monolayers
Medium:	85% Dulbecco's MEM (4.5 g/L glucose) + 15% FBS + 2 mM L-glutamine
Subculture:	split confluent cultures 1:5 once a week using trypsin/EDTA; seed out at ca. 1.5×10^6 cells/80 cm ² ; maximal density at ca. 60×10^6 cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60 hours
Harvest:	cell harvest of ca. 40×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1.5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human flat-moded pentaploid karyotype with 8% polyploidy; 111(102-116)<5n>XX/XXX, +3, +3, +5, +5, -6, +7, +8, +8, -12, -13, -13, -14, -14, -15, -15, -15, -15, -16, -17, -17, -19, -19, +20, -22, -22, -22, +6-8mar; del(2)(q25), add(3)(q27)x2, del(3)(p11p14), i(5q), add(5)(p15), del(5)(p13), add(5)(q1?3;hsr), del(6)(q25), add(7)(p22), del(7)(q11), add(8)(p23)x2, i(9q)x2, del(9)(p21), del(10)(q24), add(11)(p15), add(12)(p13)x1-2, del(12)(p11), der(13)t(13;?)(p11;?)(?;1)(?;q11), add(17)(p1?), add(18)(p11); sideline with i(5p); 3p, 5p/q10, +7, 8p, 9p, 10q, 11p, 17p rearrangements associated with bladder TCC; matches published karyotypes
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HT-29
Cell Type:	human colon adenocarcinoma
DSMZ No:	ACC 299
Origin:	established from the primary tumor of a 44-year-old Caucasian woman with colon adenocarcinoma in 1964; described to be heterotransplantable forming well-differentiated grade I tumors
References:	Fogh & Trempe in Human Tumor Cells in Vitro (ed. Fogh J.), Plenum Press, New York, 1975, pp. 115-159
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, epitheloid cells growing as monolayers and in large colonies
Medium:	90% McCoy's 5A + 10% FBS
Subculture:	split confluent culture 1:3 to 1:6 every 3-6 days using trypsin/EDTA; seed out at about 0.1-0.2 x 10 ⁵ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 40-60 hours
Harvest:	cell harvest of about 25 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH
Cytogenetics:	human hypertriploid karyotype with 17.5% polyploidy; 71(63-74)<3n>XX, -X, +1, +2, +7, -8, -9, +11, -13, -13, -14, +15, -18, +19, -21, -22, +4-6mar, del(3)(p24), del(4)(q22)x1-2, add(6)(q25), add(7)(p15), i(13q), add(19)(q13); sideline with i(17q), t(17;19)(p10;q10); resembles published karyotype and to that of CX-1 which is a subclone; carries very large isochromosomal ABR marker
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HUP-T3
Cell Type:	human pancreas carcinoma
DSMZ No:	ACC 259
Origin:	established from the malignant ascites of a 66-year-old Japanese man with pancreas carcinoma in 1984
References:	Nishimura et al., Int. J. Pancreatol. 13: 31-41 (1993)
Depositor:	Dr. S. Saito, Toyama Medical and Pharmaceutical University, Toyama, Japan
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently in monolayers and clusters
Medium:	90% MEM (with Earle's salts) + 10% FBS + 1% non-essential amino acids + 1% sodium pyruvate
Subculture:	split confluent culture 1:10 once a week using trypsin/EDTA; seed out about 1×10^6 cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Harvest:	about 14×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1.4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, cytokeratin-7(+), desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human flat-moded hypodiploid karyotype with 13% polyploidy; 39(36-40)<2n>XY, -6, -8, -9, -10, -12, -13, -17, -17, -19, -20, +3mar (2 rings), del(4)(p15), add(11)(q22.3;HSR), add(12)(q24;HSR), add(19)(p13), add(21)(p12); gene amplification suggested by large HSR; ch 17 nullisomy
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	HUP-T4
Cell Type:	human pancreas carcinoma
DSMZ No:	ACC 223
Origin:	established from the malignant ascites of a 60-year-old Japanese man with pancreas carcinoma in 1985
References:	Nishimura et al., Int. J. Pancreatol. 13: 31-41 (1993)
Depositor:	Dr. S. Saito, Toyama Medical and Pharmaceutical University, Toyama, Japan
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently in monolayers
Medium:	80% MEM (with Earle's salts) + 20% FBS + 1% non-essential amino acids + 1% sodium pyruvate
Subculture:	optimal split ratio of 1:2 to 1:3 every 2-5 days using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 38 hours
Harvest:	cell harvest of about 1-2 x 10 ⁵ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, cytokeratin-7+, desmin-, endothel-, GFAP-, neurofilament-, vimentin(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human near-triploid karyotype with 11% polyploidy and 1-2 supernumerary dm; 69/70(66-73)<3n>X, ?del(X)/XY, -Y, -2, -4, +6, +7, +8, -9, -9, +11, +16, +mar; inv(3)(p22q24), i(8q), idic(9)(pter), add(17)(p12); 2 medium size ring ch present at 10%
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	IEC-6
Cell Type:	rat normal small intestine
DSMZ No:	ACC 111
Origin:	normal rat epithelial cells from the small intestine; cells were described to synthesize fibronectin and collagen; no continuous cell line: about 10-15 doublings after thawing of the stock cells
References:	Quaroni et al., J. Cell. Biol. 80: 248-265 (1979)
Depositor:	obtained from ATCC (CRL 1592), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	adherent, transparent, epitheloid cells growing as monolayers (polymorphy of cell forms)
Medium:	45% Dulbecco's MEM (4.5 g/L glucose) and 45% RPMI 1640 + 10% FBS + 0.1 U/ml insulin
Subculture:	split confluent cultures 1:2 to 1:6 about twice a week using trypsin/EDTA; seed out at ca. 1-2 x 10 ⁶ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 50 hours
Harvest:	about 16 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1-2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as rat with IEF of AST, MDH, MPI, NP
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	IGR-1
Cell Type:	human melanoma
DSMZ No:	ACC 236
Origin:	established from the metastatic tumor in a groin lymph node of a 42-year-old man with malignant melanoma (primary tumor histology: NM)
References:	Aubert et al., J. Natl. Cancer Inst. 64: 1029-1040 (1980)
Depositor:	Dr. C. Aubert, Institut Paoli Calmettes, Marseille, France
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial-like cells growing as monolayers (dendritic branching cells until confluence, upon contact inhibition smooth and round)
Medium:	85-90% Dulbecco's MEM + 10-15% FBS
Subculture:	split confluent culture 1:3 every 5 days using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of about 75-100 hours
Harvest:	cell harvest of about 30×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2.0-2.5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin- , cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin- , endothel-, GFAP-, HMB-45+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human flat-moded, hypertriploid karyotype with 10% polyploidy; 71(68-75)<3n>der(X), del(X), -Y, -2, +5, -8, +9, -10, -11, +5mar, add(X)(p21.3), del(X)(q22), add(1)(p36), der(6)t(6;9)(q22.3;q21.1)x2, add(8)(p21), add(9)(p13), del(9)(p21)x2, del(9)(q11), der(11)t(7;11)(q11;p11), add(13)(p12), add(17)(p12); loss of 9p associated with melanoma
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	IGR-37
Cell Type:	human melanoma
DSMZ No:	ACC 237
Origin:	established from the lymph node metastasis (groin) of a 26-year-old man with malignant melanoma (primary tumor histology: SSM level IV); same patient as cell line IGR-39 (DSM ACC 239); described as very tumorigenic in nude mice
References:	Aubert et al., J. Natl. Cancer Inst. 64: 1029-1040 (1980)
Depositor:	Dr. C. Aubert, Institut Paoli Calmette, Marseille, France
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing as monolayers
Medium:	85% Dulbecco's MEM + 15% FBS
Subculture:	split confluent culture 1:2 every 7-14 days using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 108 hours
Harvest:	cell harvest of about 2-3 x 10 ⁵ /cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, HMB-45+, neurofilament-, vimentin+
Fingerprinting:	same DNA profile as IGR-39 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human flat-moded hypertriploid karyotype with 11% polyploidy; 78(67-78)<3n>XXY, +1, +3, +5, +7, +7, +8, +9, -10, -12, -13, -14, -15, -17, +18, +20, +20, +2mar, der(1)t(1;?21)(p13;q11)x2, del(9)(p13p21), der(10)t(9;10)(q11/21;q23/25), i(13q)x1-2, der(16)t(1;16)(p11/13;q21/23)x2, der(19)t(17;19)(q12;p13)x1-2; rearrangements at 1p12/22, 19p13 and +7 recur in malignant melanoma
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	IGR-39
Cell Type:	human melanoma
DSMZ No:	ACC 239
Origin:	established from the primary (achromic) cutaneous tumor (left thigh) of a 26-year-old man with malignant melanoma (primary tumor histology: SSM level IV); same patient as cell line IGR-37 (DSM ACC 237); described as tumorigenic in nude mice
References:	Aubert et al., J. Natl. Cancer Inst. 64: 1029-1040 (1980)
Depositor:	Dr. C. Aubert, Institut Paoli Calmettes, Marseille, France
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing in monolayers
Medium:	85% Dulbecco's MEM + 15% FBS
Subculture:	split subconfluent or confluent culture 1:4 every 3-4 days using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ² ; optimal density of about 4×10^6 /175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30 hours
Harvest:	maximal density of about $6-8 \times 10^6$ /175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	same DNA profile as IGR-37 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human flat-moded tetra-/pentaploid karyotype; 104-112<4n>XXY; rearrangements include der(1)t(1;?21)(p13;q11), der(16)t(1;16)(p13;q23) also present in the autologous sister line IGR-37
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	IM-9
Cell Type:	human B lymphoblastoid cells (from patient with multiple myeloma)
DSMZ No:	ACC 117
Origin:	established from the bone marrow of a woman with multiple myeloma in 1967; cells were described to synthesize IgG and to have receptors for insulin and calcitonin; cells express bcl-2 mRNA; cells appear to be rather "EBV+ B-lymphoblastoid cells" than myeloma-derived cells (see also Pellat-Deceunynk et al., Blood 86: 4001-4002, 1995), hence NOT a malignant myeloma cell line
References:	Buell et al., Ann. N.Y. Acad. Sci. 190: 221-234 (1972); van Boxel et al., Nature 251: 443-444 (1974); review article: Drexler & Matsuo, Leukemia Res 24: 681-703 (2000)
Depositor:	obtained from ATCC (CCL 159), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	lymphoblastoid, single or clumped cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $1.0-1.75 \times 10^6$ cells/ml, split 1:2 to 1:3 every 2-3 days; seed out at ca. 1×10^6 cells/ml, do not culture below 0.5×10^6 cells/ml; as the medium becomes quickly acidic (yellow) without a corresponding cell growth, it is advisable to change the supernatant medium daily
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 45 hours
Harvest:	maximal cell density of $2-3 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD34-, CD37-, CD79a-, cyCD79a+, CD80+, CD138(+), HLA-DR+, sm/cyIgG+, sm/cyIgM-, sm/cykappa+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP, PEP B
Cytogenetics:	near diploid; 46(44-46)<2n>XX; apparently normal karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	IMR-32
Cell Type:	human neuroblastoma
DSMZ No:	ACC 165
Origin:	established from an abdominal mass (diagnosed as neuroblastoma) of a 1-year-old Caucasian boy in 1967
References:	Tumilowicz et al., Cancer Res. 30: 2110 (1970)
Depositor:	Dr. J. Wehland, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	fibroblast-like adherent cells growing as monolayer
Medium:	80% RPMI 1640 + 20% FBS + 1x non-essential amino acids
Subculture:	split confluent culture 1:2 every 3-5 days (trypsin should not be used, cells can be detached by tapping the flask or pipetting); seed out at ca. $0.5\text{--}1.0 \times 10^6$ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	about $1\text{--}2 \times 10^6$ cells/ml or 0.5×10^6 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, neurofilament+, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, PEP B
Cytogenetics:	human hyperdiploid karyotype; 49(44-51)<2n>XY, +1, +6, +12, der(1)t(1qter->1p34::hsr::17q22->qter)x2, add(16)(q24); sideline with del(3)(p14), add(7)(q35), del(9)(q34), add(19)(q13); NB associated changes include: N-myc amplified in hsr regions and unbalanced t(1;17); matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	IPC-298
Cell Type:	human melanoma
DSMZ No:	ACC 251
Origin:	established from the primary tumor (right cervical) of a 64-year-old woman with cutaneous melanoma
References:	Aubert et al., Int. J. Cancer 54: 784-792 (1993)
Depositor:	Dr. C. Aubert, Institut Paoli Calmette, Marseille, France
DSMZ Cell Culture Data:	
Morphology:	adherent, spindle-like cells (a few epithelioid cells) growing as confluent monolayer
Medium:	90-95% RPMI 1640 + 5-10% FBS
Subculture:	split confluent culture 1:8 to 1:10 every 4-5 days using trypsin/EDTA; seed out at ca. $2-5 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30 hours
Harvest:	cell harvest of about $20-30 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, HMB-45+, neurofilament+, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human flat-moded hypertriploid karyotype with 13% polyploidy; 74(74-88)<3n>XX, -X, +1, -2, +3, -4, +6, +7, +9, -11, -11, +12, -13, -14, -15, +16, +21, +21, +4mar, dmin in 34% cells, del(3)(p14)x2, del(3)(q26.3)x2, del(6)(p23), del(7)(p14), add(7)(p22;HSR), der(9)t(9;11)(p21;q11), add(12)(p13)x3, del(12)(p13), del(17)(p12)x1-2; sidelines with additional numerical and structural ch 7 rearrangements
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	IPL-LD-65Y
Cell Type:	insect - gypsy moth (<i>Lymantria dispar</i>)
DSMZ No:	ACC 181
Origin:	established from the larvae of a gypsy moth (species <i>Lymantria dispar</i> , family <i>Lymantriidae</i> , order <i>Lepidoptera</i>) (Schwammspinner)
References:	no reference available
Depositor:	Dr. H. S. Conradt, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	large cells; up to 30% grow adherent with processes; suspension cells are round to oval
Medium:	80% TC 100 medium + 20% FBS
Subculture:	cells are difficult to culture and grow slowly! split confluent culture 1:4 every 4-5 days using a (plastic) cell scraper
Incubation:	without exposure to CO ₂ (keep lid of flask closed) at room temperature (20-25 /C)
Harvest:	cell harvest of up to 10 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as not murine and not primate with IEF of AST, MDH, PEP B

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INFORMATION SHEET FOR CELL LINES

Cell Line:	J-774A.1
Cell Type:	mouse monocytes-macrophages
DSMZ No:	ACC 170
Origin:	established from a tumor in a female BALB/c mouse in 1968; cells were described to synthesize lysozyme and interleukin-1 and to have receptors for immunoglobulin and complement
References:	Ralph et al., J. Exp. Med. 143: 1528-1533 (1976)
Depositor:	Dr. J. Wehland, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	relatively small cells growing adherently; adherent cells will round up after a few days in culture
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:3 to 1:4 about twice a week (prior to reaching confluence); cells detach spontaneously if allowed to exceed confluence; cells may be detached with a rubber policeman or using TEN (40 mM TrisCl pH 7.5, 1 mM EDTA pH 7.5, 150 mM NaCl) (not trypsin/EDTA) for quantitative harvesting or for seeding out of cells collected from the supernatant for passaging; use always new culture flasks, otherwise cells are difficult to detach; seed out at ca. $1-2 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	about 4.5×10^7 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, MDH, NP, PEP B
Cytogenetics:	murine hypotetraploid karyotype; 78(70-81)<4n>; in most cells a centric fragment was present; mouse chromosome morphology confirmed
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JAR
Cell Type:	human choriocarcinoma
DSMZ No:	ACC 462
Origin:	established from the trophoblastic tumor of the placenta of a 24-year-old Caucasian woman
References:	Pattillo et al., In Vitro 6: 398-399 (1971)
Depositor:	Dr. B. Ugele, University Hospital, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing adherent in monolayers
Medium:	85% RPMI 1640 + 15% FBS
Subculture:	split confluent culture 1:10 every 3-4 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 20 hours
Harvest:	cell harvest of ca. $10-15 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Cytogenetics:	human flat-modulated near-triploid karyotype with 27% polyploidy; 60-70<3n>XXY, +1, -2, -5, -7, -8, -9, -13, -13, -14, -15, -17, -19, -21,+12mar, +acf, add(X)(p22), del(1)(p3?p3?), add(1)(q11), der(1)add(1)(p11)add(1)(q4?), inv(3)(p25.2q24), add(3)(q13), add(4)(q?31), del(4)(q2?2), add(9)(p21), add(12)(q24), i(13)(q10)dup(13)(q11q34), add(16)(q24), add(17)(p12), add(18)(p11.2); sideline with add(2)(p2?), del(8)(q22), add(21)(p11); resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JEG-3
Cell Type:	human choriocarcinoma
DSMZ No:	ACC 463
Origin:	established from a human gestational choriocarcinoma which was taken at autopsy from a cerebral metastasis; cells were initially propagated in hamster cheek pouches by serial passage prior to cultivation in culture; serial sister cell line of cell line BEWO (DSM ACC 458)
References:	Pattillo et al., Cancer Res. 28: 1231-1236 (1968); Kohler & Bridson, J. Clin. Endocrinol. 32: 683-687 (1971)
Depositor:	Dr. B. Ugele, University Hospital, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing as monolayer and carrying many intracellular dark granula
Medium:	90% Ham's F12 + 10% FBS
Subculture:	split confluent culture 1:10 every 3-4 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ² ; maintain at ca. 10×10^6 cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 24 hours
Harvest:	cell harvest of ca. 25×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	isame DNA profile as BEWO using multiplex PCR of minisatellite markers
Cytogenetics:	human hypertriploid karyotype with 12% polyploidy; 71(69-72)<3n>XXY, +2, +5, +9, -11, -13, -18, +4-6mar, ?del(X)(p11.4), der(4)t(4;11)(p16;q13), add(6)(q27)x1-2, add(7)(q32), add(7)(p22), i(13q), add(15)(p11), add(19)(p13); sideline with add(8)(p1?), add(10)(q21), ?del(18)(q22); resembles published karyotype and (marginally) that of BEWO
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JJN-3
Cell Type:	human plasma cell leukemia
DSMZ No:	ACC 541
Origin:	established from the bone marrow of a 57-year-old woman with plasma cell leukemia (IgA1kappa) at diagnosis in 1987; JJN-3 is a subclone of the parental cell line JJN-1; plasma cell leukemia is related to multiple myeloma
References:	Jackson et al., Clin. Exp. Immunol. 75: 93-99 (1989); Drexler & Matsuo, Leukemia Res. 24: 681-703 (2000) (review article)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single round to oval cells showing a morphological range from mononuclear to occasionally multinucleated cells
Medium:	40% Dulbecco's MEM + 40% Iscove's MDM + 20% FBS
Subculture:	split saturated culture 1:3 to 1:6 every 2-3 days; seed out at ca. $0.2-0.4 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 20-35 hours
Harvest:	cell harvest of ca. 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19-, CD20-, CD34-, CD37-, CD79a-, cyCD79a+, CD80-, CD138+, sm/cyIgG-, sm/cyIgM-, sm/cykappa+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hypotriploid karyotype with 9% polyploidy; 58-67<3n>XX, +1, -2, +5, +8, +8, -9, -11, -12, -13, -15, -17, +20, add(1)(p22), der(1)t(1;?3)(q41;p26)x2, add(3)(p26), add(5)(p15)x1-2, i(5p), del(6)(q13), del(7)(q32), der(7)t(7;11)(q36;q13), add(8)(p1?)x2, t(12;19)(p13;q13), der(14)add(14)(p11)t(14;16)(q32;q23), der(14)t(14;16)(q32;q23), der(16)t(14;16)(q32;q23)x2, del(22)(q12); carries two copies of t(14;16) associated with c-MAF activation; resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JK-1
Cell Type:	human chronic myeloid leukemia in blast crisis
DSMZ No:	ACC 347
Origin:	established from the biopsy material of the shoulder tumor of a 62-year-old man with chronic myeloid leukemia (CML) in erythroid blast crisis in 1987; described to produce hemoglobin
References:	Okuno et al., Cancer 66: 1544-1551 (1990); review article: Drexler, Leukemia Res. 18: 919-927 (1994); review article: Drexler et al., Leukemia Res. 24: 109-115 (2000)
Depositor:	Dr. Y. Okuno, Kyoto University Hospital, Kyoto, Japan
DSMZ Cell Culture Data:	
Morphology:	round single cells growing in suspension (cell pellet shows reddish color hemoglobin)
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	maintain at $0.3-1.0 \times 10^6$ cells/ml; split 1:2 to 1:3 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	maximal density of ca. 1.2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD13+, CD19-, CD41+, CD42+, GlyA+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human hyperdiploid karyotype without significant polyploidy; 48(45-48)<2n>XY, +8, +22, t(1;7)(q10;q10), der(9)t(9;22)(q34;q11), der(22)t(9;22)(q34;q11)x2; carries two Ph; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JMSU-1
Cell Type:	human urinary bladder carcinoma
DSMZ No:	ACC 505
Origin:	derived from malignant ascitic fluid of a 75-year-old man with urinary bladder carcinoma in 1988; described as being of epithelial origin with morphologically distinct cells (polygonal to spindle-shaped) and as expressing cytokeratin-19 fragment
References:	Morita et al., Urol. Res. 23: 143-149 (1995); Morita et al., Eur. Urol. 32: 237-244 (1997)
Depositor:	Dr. T. Morita, Jichi Medical School, Tochigi, Japan
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:2 to 1:4 every 2-4 days using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ² ; maximal density at ca. 0.3×10^6 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 10×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hypertriploid karyotype with 12% polyploidy; 69-74<3n>X/XX, -Y, -Y, +1, +1, +2, -5, +9, -12, +13, -15, -15, -16, -17, +19, -22, +9mar, add(1)(q32), add(2)(q3?), del(4)(q31q33), del(6)(q25), add(7)(p11), del(8)(p22), del(9)(p22), del(10)(p12), del(10)(q22), add(11)(p12), add(13)(p11), add(16)(p13), i(17q), add(19)(p13); resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JOSK-I
Cell Type:	human histiocytic lymphoma (derivative of U-937)
DSMZ No:	ACC 155
Origin:	supposedly "established from the peripheral blood of a 72-year-old woman with acute myelomonocytic leukemia (AML FAB M4) in 1983"; however DSMZ DNA fingerprinting and cytogenetic analysis established unequivocally that JOSK-I, JOSK-M and U-937 are identical; thus, JOSK-I is clearly a derivative of U-937
References:	Ohta et al., Cancer Res. 46: 3067-3074 (1986)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	single, round to polygonal cells growing in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.3\text{--}1.0 \times 10^6$ cells/ml; split ratio of 1:2 to 1:5; seed out at ca. $0.2\text{--}1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 28-35 hours
Harvest:	saturation density at about $1.5\text{--}2.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4+, CD13+, CD14-, CD15+, CD19-, CD33+, CD68+, HLA-DR+
Fingerprinting:	same DNA profile as U-937 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hyperdiploid karyotype with 3.5% polyploidy; 56(52-56)<2n>Xder(X), +2, +4, +6, +7, +8, +12, -15, +17, +19, +20, +mar, der(X)t(X;3)(q11;p13)t(1;3)(q11;p25), del(1)(p11p22), t(1;5)(p21.1;q23), t(2;18)(p17;q23), del(4)(q22), add(6)(p23), del(7)(q21)/der(7)t(7;18)(q11.23;p11), der(10)t(10;11)(p14;q22.3)/der(10)t(10;11)(p14;q22.3)dup(10)(q25.1), der(11)t(10;11)(p14;q22.3), add(13)(p13), der(17)t(17;18)(p11;q12); (mar resembles Ph); carries uncommon 10;11 translocation associated with AML M5
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JOSK-M
Cell Type:	human histiocytic lymphoma (derivative of U-937)
DSMZ No:	ACC 30
Origin:	supposedly "established from the peripheral blood of a 37-year-old man with chronic myeloid leukemia (CML) in myelomonocytic blast crisis in 1984"; however, DSMZ DNA fingerprinting and cytogenetic analysis established unequivocally that JOSK-I, JOSK-M and U-937 are identical; thus, JOSK-M is clearly a derivative of U-937
References:	Ohta et al., Cancer Res. 46: 3067-3074 (1986)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	single, round to oval cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.2-0.3 \times 10^6$ cells/ml, subculture at a 1:4 to 1:5 dilution
Incubation:	at 37 °C with 5% CO ₂
Harvest:	cells grow up to about 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD3-, CD13+, CD14-, CD15+, CD19-, CD33+, HLA-DR+
Fingerprinting:	same DNA profile as U-937 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of G6PD, MDH, NP
Cytogenetics:	human hyperdiploid with 4% polyploidy; 55(51-57)<2n>XXY, +X, +3, +7, +8, +19, +20, +3mar, t(1;5;7)(p21-22;q35;p11), der(3)t(1;3)(q21;q26-27), der(6)t(6;?)(p25;?), der(10)t(10;?)(p13;?), del(11)(q22-23), der(12)t(12;?)(p13;?), der(15)t(15;?)(q24;?)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JTC-15
Cell Type:	rat hepatoma
DSMZ No:	ACC 329
Origin:	established in 1963 from the ascites of a rat with hepatoma induced with 4-dimethylaminoazobenzene
References:	Takaoka et al., Japan. J. Exp. Med. 39: 267-275 (1969)
Depositor:	obtained from RIKEN Cell Bank (RCB0063), Tsubuka Science City, Ibaraki, Japan
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial cells growing in monolayers
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:5 to 1:8 every 4-5 days using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 30×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as rat with IEF of MDH, NP
Cytogenetics:	rat hyperdiploid/hypotriploid karyotype with 33% polyploidy; 51(51-58)<2n>; secondary constriction marker present; resembles published karyotype
Viruses:	ELISA: reverse transcriptase positive

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JTC-27
Cell Type:	rat hepatoma
DSMZ No:	ACC 330
Origin:	established from the transplantable rat ascites hepatoma AH-601 induced by feeding with aminoazotoluene; the hepatoma was passaged by intraperitoneal transplantation through JAR-2 rats
References:	Huh et al., Jap. J. Exp. Med. 42: 249-262 (1972)
Depositor:	obtained from RIKEN Cell Bank (RCB0065) Tsubuka Science City, Ibaraki, Japan
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial-like cells growing in monolayers
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	split confluent culture 1:6 to 1:8 twice a week using trypsin/EDTA; seed out at ca. 2×10^4 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 37 hours
Harvest:	cell harvest of maximally $2.5-3 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as rat with IEF of AST, MDH, NP
Cytogenetics:	rat hyperdiploid karyotype with 15% polyploidy; 47(45-49)<2n>
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JURKAT
Cell Type:	human T cell leukemia
DSMZ No:	ACC 282
Origin:	established from the peripheral blood of a 14-year-old boy with acute lymphoblastic leukemia (ALL) at first relapse in 1976; often this cell line is called "JM" (JURKAT and JM are derived from the same patient and are sister clones), occasionally JM may be a subclone with somewhat divergent features
References:	Schneider et al., Int. J. Cancer 19: 621-626 (1977)
Depositor:	Dr. Jun Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly or in clumps in suspension
Medium:	90% RPMI 1640 + 10% FBS + 2 mM L-glutamine
Subculture:	optimal split ratio of about 1:2 to 1:3 every 2-3 days; seed out after thawing at about 1×10^6 cells/ml; maintain at $0.5-1.5 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 25-35 hours
Harvest:	maximal density at about 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2+, CD3+, CD4(+), CD5+, CD6+, CD7+, CD8-, CD13-, CD19-, CD34+, TCRalpha/beta+, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human flat-moded hypotetraploid karyotype with 7.8% polyploidy; 87(78-91)<4n>XX, -Y, -Y, -5, -16, -17, -22, add(2)(p21)/del(2)(p23)x2; sideline with additional der(5)t(5;10)(q11;p15), del(9)(p11)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JURL-MK1
Cell Type:	human chronic myeloid leukemia in blast crisis
DSMZ No:	ACC 532
Origin:	established from the peripheral blood of a 73-year-old man with chronic myeloid leukemia (CML) in blast crisis in 1993; cells carry the Philadelphia chromosome; JURL-MK1 and JURL-MK2 are sister cell lines derived from the same patient
References:	Di Noto et al., Leukemia 11: 1554-1564 (1997)
Depositor:	Dr. R. Di Noto, University of Naples, Italy
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly or in small clusters in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2-4 days; seed out at ca. $0.5-1.0 \times 10^6$ cells/ml; maintain at ca. $0.2-2.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of ca. 2.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5-8 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD4-, CD13+, CD14+, CD15+, CD19-, CD30+, CD33+, CD34+, CD41+, CD42+, CD116+, CD117+, CD123+, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile, sister cell line of JURL-MK2
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypodiploid karyotype with 4% polyploidy; 45(41-45)<2n>XY, -9, add(1)(p35), der(9)del(9)(p21.2)t(9;22)(q34;q11), der(11)t(11;9;22)(p15;q34;q11), add(12)(p13)x2, add(17)(p13), der(22)t(9;22)(q34;q11); carries t(9;22) effecting BCR-ABL fusion (two copies); does not appear to match published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JURL-MK2
Cell Type:	human chronic myeloid leukemia in blast crisis
DSMZ No:	ACC 533
Origin:	established from the peripheral blood of a 73-year-old man with chronic myeloid leukemia (CML) in blast crisis in 1993; cells carry the Philadelphia chromosome; JURL-MK1 and JURL-MK2 are sister cell lines derived from the same patient
References:	Di Noto et al., Leukemia 11: 1554-1564 (1997)
Depositor:	Dr. R. Di Noto, University of Naples, Italy
DSMZ Cell Culture Data:	
Morphology:	round single cells growing in suspension, heterogenous in size, occasional giant cells
Medium:	85% RPMI 1640 + 15% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 3-4 days; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. $0.2-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 72 hours
Harvest:	cell harvest of ca. 2.3×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD4-, CD13+, CD14+, CD15-, CD19-, CD30+, CD33+, CD34+, CD41+, CD42+, CD116+, CD117+, CD123+, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile, sister cell line of JURL-MK1
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypotetraploid karyotype with 3.5% polyploidy; 81(77-82)<4n>XXYY, -2, -7, -12, +13, -14, -14, -16, -18, -19, -22, +1-3mar, add(9)(q1)x2, der(9)del(9)(p21.2)t(9;22)(q34;q11)x2, der(11)t(11;9;22)(p15;q34;q11)x2, add(17)(p13)x1-2, der(22)t(9;22)(q34;q11)x2; carries two copies of Ph effecting BCR-ABL fusion together with additional FISH signals on 11p; does not match published karyotype but iteratively resembles JURL-MK1 (sister cell line)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JVM-13
Cell Type:	human chronic B cell leukemia
DSMZ No:	ACC 19
Origin:	established from the peripheral blood of a male patient with B-prolymphocytic leukemia (PLL) at diagnosis; cell line was established by EBV-transformation during treatment with phorbol ester TPA
References:	Melo et al., Clin. Exp. Immunol. 73: 23-28 (1988)
Depositor:	Dr. E. Matutes, Royal Marsden Hospital, London, UK
DSMZ Cell Culture Data:	
Morphology:	lymphoblastoid cells growing in clusters in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; split saturated culture 1:3 every 3 days; seed out at ca. $0.4-0.5 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 70 hours
Harvest:	maximum density at about $1-2 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a+, CD80+, CD138-, HLA-DR+, sm/cyIgM+, sm/cyIgG-, sm/cykappa-, sm/cylambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hyperdiploid karyotype with 4% tetraploidy; 47(44-48)<2n>X, +8, -13, t(2;?)(q35;?), t(8;13)(p12;q13), t(8;?)(p12;?)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JVM-2
Cell Type:	human chronic B cell leukemia
DSMZ No:	ACC 12
Origin:	established from the peripheral blood of a 63-year-old woman with B-prolymphocytic leukemia (B-PLL) at diagnosis; cell line was established by EBV-transformation during treatment with phorbol ester TPA; cells were described to have immunoglobulin light chain restriction (lambda)
References:	Melo et al., Int. J. Cancer 38: 531-538 (1986); Melo et al., Clin. Exp. Immunol. 73: 23-28 (1988)
Depositor:	Dr. E. Matutes, Royal Marsden Hospital, London, UK
DSMZ Cell Culture Data:	
Morphology:	lymphoblastoid, round to polygonal cells growing in clusters in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.1-0.5 \times 10^6$ cells/ml; cells might grow slowly; split 1:2 to 1:4 every 2-4 days; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-70 hours
Harvest:	maximal density at about 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a-, cyCD79a+, CD80+, CD138+, HLA-DR+, sm/cyIgM+, sm/cyIgG-, sm/cykappa-, smlambda-, cylambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human pseudodiploid karyotype with 1.8% polyploidy; 46(42-46)<2n>XX, t(1;15)(q32.3;q22.1), der(8)t(3;8)(q13;p21), t(11;14)(q13;q32), dup(16)(p11p13.2); resembles published karyotype; carries t(11;14) associated with cyclin D1 activation
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	JVM-3
Cell Type:	human chronic B cell leukemia
DSMZ No:	ACC 18
Origin:	established from the peripheral blood of a 73-year-old man with B-prolymphocytic leukemia (PLL) at diagnosis; cell line was established by EBV-transformation during treatment with phorbol ester TPA; cells express mRNA of the proto-oncogenes bcl-2 and bcl-3
References:	Melo et al., Int. J. Cancer 38: 531-538 (1986); Melo et al., Clin. Exp. Immunol. 73: 23-28 (1988); Robinson et al., J. Clin. Pathol. 38: 897-903 (1985)
Depositor:	Dr. E. Matutes, Royal Marsden Hospital, London, UK
DSMZ Cell Culture Data:	
Morphology:	lymphoblastoid cells growing in clumps in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.1-0.5 \times 10^6$ cells/ml; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	maximum density at $1.0-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a+, CD80+, CD138-, HLA-DR+, sm/cyIgM+, sm/cyIgG-, sm/cykappa+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human near tetraploid karyotype without clear mode; 87-93<4n>XXYY, +3, -7, -8, -10, +12, +12, -14, -19, -19, -20, -20; appears to be derivative of original hyperdiploid line which exhibited trisomy 12
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	K-562
Cell Type:	human chronic myeloid leukemia in blast crisis
DSMZ No:	ACC 10
Origin:	established from the pleural effusion of a 53-year-old woman with chronic myeloid leukemia (CML) in blast crisis in 1970; cells can be used as highly sensitive targets in in-vitro natural killer assays; cells produce hemoglobin; cells carry the Philadelphia chromosome with a b3-a2 fusion gene
References:	Lozzio et al., J. Natl. Cancer Inst. 50: 535-538 (1973); Lozzio et al., Blood 45: 321-334 (1975); review article: Drexler, Leukemia Res. 18: 919-927 (1994)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round large, single cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.1-0.5 \times 10^6$ cells/ml; split 1:3 to 1:5 every 3 days; seed out at ca. $0.3-0.5 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	maximal density at $1.0-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, PCR assays
Immunology:	CD3-, CD13+, CD19-, CD34-, CD41(+), CD42+, CD45+, CD71+, GlyA(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypotriploid karyotype without sharp mode; 61-68<3n>XX, -X, -3, +7, -13, -18, +3mar, del(9)(p11/13), der(14)t(14;?)(p11;?), der(17)t(17;?)(p11/13;?), der(?18)t(15;?18)(q21;q12), del(X)(p22); two markers appear from FISH to have arisen from Ph
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KARPAS-1106P
Cell Type:	human B cell lymphoma
DSMZ No:	ACC 545
Origin:	established from the pleural effusion of a 23-year-old woman with mediastinal lymphoblastic B cell non-Hodgkin's lymphoma (B-NHL) at relapse in 1984
References:	Nacheva et al., Blood 84: 3422-3428 (1994); Tamura et al., Int. J. Cancer 78: 100-105 (1998)
Depositor:	Dr. A. Karpas, University of Cambridge, Cambridge, UK
DSMZ Cell Culture Data:	
Morphology:	spherical cells growing in clumps in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. 0.5×10^6 cells/ml; after thawing, it is recommended to culture the cells in 24-well-plates for the first week
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60-100 hours
Harvest:	cell harvest of ca. $1-2 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a+, cyCD79a+, CD80+, CD138-, HLA-DR(+), sm/cyIgG+, sm/cyIgM-, sm/cyKappa-, sm/cyLambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype with 3% polyploidy; 49(45-51)<2n>XXX, +X, +9, +20, del(X)(q25)/i(Xp), der(X)t(X;?)(p23;?)(?;13)(?;q11), del(2)(p11p13.1), der(3)t(2;3)(p25;p14), i(9p), dup(12)(q11q14.1), del(14)(q11q13), del(15)(q11q15), add(18)(q21), del(20)(q13)x2, add(21)(p13); BCL2 and MALT1 rearrangements not detected; closely matches published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KARPAS-299
Cell Type:	human T cell lymphoma
DSMZ No:	ACC 31
Origin:	established from the peripheral blood of a 25-year-old man with T cell non-Hodgkin lymphoma in 1986; now classified as "CD30+ anaplastic large cell lymphoma" (ALCL); cells carry the NPM-ALK fusion gene
References:	Fisher et al., Blood 72: 234-240 (1988); Mason et al., Brit. J. Haematol. 74: 161-168 (1990)
Depositor:	Dr. A. Karpas, University of Cambridge, Cambridge, UK
DSMZ Cell Culture Data:	
Morphology:	round to polygonal cells, growing singly or in small clusters in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $1.0\text{--}2.0 \times 10^6$ cells/ml; split 1:3 every 2-3 days; seed out at ca. 1×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	maximal density of about $2\text{--}3 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3-, CD4+, CD5+, CD6-, CD7-, CD8-, CD10-, CD13+, CD14-, CD15-, CD19-, CD20+, CD25+, CD30+, CD33-, CD34-, CD71+, HLA-DR+, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypodiploid karyotype with 14% polyploidy; 44(42-45)<2n>XY, -10, -22, t(1;17)(q22;p11), t(2;5)(p23;q35), del(6)(q23), der(13)t(13;?)(p12;?), der(14)t(14;22)(p12;q11), der(19)t(19;?)(q13;?), del(22)(q12); expression of NPM-ALK fusion gene was detected by RT-PCR analysis
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KARPAS-422
Cell Type:	human B cell lymphoma
DSMZ No:	ACC 32
Origin:	established from the pleural effusion of a 73-year-old woman with B cell non-Hodgkin lymphoma (intraabdominal, diffuse large cell lymphoma, refractory, terminal) in 1987; carries t(14;18) IGH-BCL2 fusion gene (breakpoint in major breakpoint region, MBR)
References:	Dyer et al., Blood 75: 709-714 (1990)
Depositor:	Dr. A. Karpas, University of Cambridge, Cambridge, UK
DSMZ Cell Culture Data:	
Morphology:	round to polygonal cells, growing singly or in small clusters in suspension
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	cells are difficult to culture! maintain at $0.5-1.0 \times 10^6$ cells/ml; split 1:2 every 3-4 days; seed out with 20% FBS at about 1×10^6 cells/ml, possibly no cell growth during first week (we suggest to use first 24-well culture plates); upon thawing viability drops to about 50%
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60-90 hours
Harvest:	maximal density at about 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a+, CD80-, CD138+, HLA-DR+, cyIgM-, cyIgG+, cyIgkappa+, cyIglambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP, PEP B
Cytogenetics:	human hyperdiploid karyotype with 10% polyploidy; 47(44-48)<2n>XX, +14, t(2;10)(p23;q22), t(4;11)(q21;q24), t(4;16)(q21;p13), der(14)t(14;18)(q32;q21)x2
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KARPAS-45
Cell Type:	human T cell leukemia
DSMZ No:	ACC 105
Origin:	established from the bone marrow of a 2-year-old boy at diagnosis of acute lymphoblastic leukemia (ALL) prior to therapy in 1972; cells were reported as carrying the MLL-AFX and AFX-MLL fusion genes; retrospectively, the original diagnosis was T-ALL
References:	Karpas et al., Leukemia Res. 1: 35-49 (1977); Smith et al., Lancet i: 74-77 (1973) (case report); Borkhard et al., Oncogene 14: 195-202 (1997)
Depositor:	Dr. A. Karpas, MRC Centre, Cambridge, UK
DSMZ Cell Culture Data:	
Morphology:	spherical cells growing as clumps in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated cultures 1:2 to 1:3 every 2-3 days; seed out at ca. $0.4-0.5 \times 10^6$ cells/ml; optimal cell density at ca. $0.5-2.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-100 hours
Harvest:	maximal cell density at ca. 2.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2+, CD3-, CD4+, CD5+, CD6(+), CD7+, CD8+, CD13-, CD19-, CD34-, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hypotetraploid karyotype with 8% polyploidy; 87(84-88)<4n>XX/XXY, -Y, -3, +6, -14, -18, +mar, t(X;11)(q13;q23), der(X)t(X;11)(q13;q23), t(1;5)(q21;q12.2)x2, del(4)(q22), der(X)t(X;11)(q13;q23)t(11;14)(p13;q11); sideline with del(16)(q22); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KARPAS-620
Cell Type:	human plasma cell leukemia
DSMZ No:	ACC 514
Origin:	established from the peripheral blood of a 77-year-old woman at diagnosis of plasma cell leukemia (IgGkappa) in 1987; plasma cell leukemia is related to multiple myeloma
References:	Nacheva et al., Brit. J. Haematol. 74: 70-76 (1990); review article: Drexler & Matsuo, Leukemia Res. 24: 681-703 (2000)
Depositor:	Dr. A. Karpas, University of Cambridge, Cambridge, UK
DSMZ Cell Culture Data:	
Morphology:	single round to polymorph cells growing in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 every 2-4 days; seed out at ca. $0.5-1.0 \times 10^6$ cells/ml in 24-well-plates; maintain at ca. 0.5×10^6 cells/ml; maximal density of ca. 1.5×10^6 cells/ml; it is recommended to culture the cells in 24-well-plates, viability may drop to 60-70% after thawing, recovered cells will not reach more than 85-90% viability
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60 hours
Harvest:	maximal density of ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in microbiological culture, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19-, CD20+, CD34-, CD37+, CD79a-, cyCD79a+, CD80-, CD138+, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypotriploid karyotype; 64(61-66)<3n>XX, -X, -X, -1, -2, -3, -4, +7, -9, -10, -13, +15, +16, +18, +19, -21, +mar, t(1;11)(q32;q13), add(7)(p15), der(8)t(8;11)(q24;q13)x2, der(?8)(1qter->q11::14q32->q11::8p23->q24::11q13->qter), del(9)(p24), add(13)(q3?2), der(14)t(1;14)(q11;q32), der(14)t(8;14)(q24;q32)x2; IGH signals detected in both ders(8)t(8;11) at 8q24 (MYC); matches published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KASUMI-1
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 220
Origin:	established from the peripheral blood of a 7-year-old Japanese boy with acute myeloid leukemia (AML FAB M2) (in 2nd relapse after bone marrow transplantation) in 1989; cells carry the t(8;21) ETO-AML1 fusion gene
References:	Asou et al., Blood 77: 2031-2036 (1991); review article: Drexler et al., Leukemia 9: 480-500 (1995)
Depositor:	Dr. N. Kamada, Hiroshima University, Hiroshima, Japan
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly or in small clumps in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture about 1:2 to 1:3 every 3 days; optimal density at about 0.5×10^6 cells/ml; viability may drop below 50% within the first 24 hours after thawing; we recommend to start culture with a cell concentration of $>1 \times 10^6$ cells/ml and to cultivate initially in 24-well plates in the presence of 20% FBS; a significant amount of cell debris is always visible in the background which might be due to granula or other intracellular material
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48-72 hours
Harvest:	saturation density at about 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4+, CD13+, CD14-, CD15+, CD19-, CD33+, CD34+, CD38+, CD68-, CD71+, HLA-DR(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, PEP B
Cytogenetics:	human hypodiploid karyotype; $45<2n>X, -Y, -9, -13, -16, +3mar, t(8;21)(q22;q22), der(9)t(9;?)(p22;?), der(15)t(9;15)((?q11;p11)$; carries both partners of 8;21 translocation associated with AML (mainly FAB M2)
Viruses:	ELISA: reverse transcriptase positive; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KASUMI-2
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 526
Origin:	established from the bone marrow of a 15-year-old boy with acute lymphoblastic leukemia (ALL FAB L2) in 1990 at diagnosis
References:	unpublished (Asou et al.)
Depositor:	Dr. M. Asou, Hiroshima University, Hiroshima, Japan
DSMZ Cell Culture Data:	
Morphology:	small round cells growing singly or in small clumps in suspension, some loosely adherent
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:2 to 1:6 every 2-3 days; seed out at ca. 1.5×10^6 cells/ml, start culture in 24-well-plates in the presence of 20% FBS (for the first 1-2 weeks); maintain at ca. $0.5\text{--}2.0 \times 10^6$ cells/ml; maximal density of ca. 7×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 7×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 8×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD10+, CD13+, CD19+, CD20+, CD34-, CD37-, CD38+, CD79a-, cyCD79a+, CD80-, CD138-, HLA-DR+, sm/cyIgG-, smIgM-, cyIgM+, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypodiploid karyotype with 3% polyploidy; 45(44-46)<2n>X, -Y, del(2)(p23.2/p21p23.1), t(3;14)(q27-28;q24), del(6)(q21), i(7q), del(13)(q1?3q1?4), der(19)t(1;19)(q23;p13); sideline with idem 46,XY; carries der(19) partner of t(1;19) with E2A rearrangement confirmed by FISH
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KB
Cell Type:	human cervix carcinoma (derivative of HELA)
DSMZ No:	ACC 136
Origin:	originally described as "epidermoid carcinoma established from the mouth of a Caucasian man in 1954"; however, as shown by DSMZ DNA fingerprinting (see below) this cell line must be considered an HELA subclone (human cervix carcinoma, see HELA DSM ACC 57)
References:	Eagle et al., Proc. Soc. Exp. Biol. Med. 89: 362 (1955); Lavappa et al., Nature 259: 211-213 (1976); Lavappa et al., In Vitro 14: 469-475 (1978)
Depositor:	Dr. J. Wehland, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent cells, epithelial-like, growing as monolayers
Medium:	90% MEM (with Earle's salts) + 10% FBS + MEM non-essential amino acids + 0.075% Na-bicarbonate
Subculture:	split confluent culture 1:3 to 1:5 every 3-6 days using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Harvest:	about 5-10 x 10 ⁶ cells/80 cm ² flask
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	same DNA as HELA using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypertriploid karyotype with 8% polyploidy; 76(73-78)<3n>
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KB-3-1
Cell Type:	human cervix carcinoma (derivative of HELA)
DSMZ No:	ACC 158
Origin:	the parental continuous cell line HELA was established from the primary cervix carcinoma of a 31-year-old black woman 1951; KB-3-1 is a derivative of HELA (DSM ACC 57) and is used in multiple drug resistance studies giving rise to drug-resistant mutants (e.g. KB-V1 resistant to vinblastine; KB-CH ^R -8-5 resistant to colchicine); cells express c-myc mRNA
References:	Fojo et al., Cancer Res. 45: 3002-3007 (1985); Akiyama et al., Somatic Cell. Mol. Genet. 11: 117-126 (1985)
Depositor:	Dr. M. Gottesman, NIH, Bethesda, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	epithelial-like, adherent cell line growing in monolayers
Medium:	90% Dulbecco's MEM + 10% FBS + 1 mM sodium pyruvate
Subculture:	split confluent culture up to 1:10 using trypsin/EDTA; seed out at ca. 1×10^6 cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 40 hours
Harvest:	about $6-10 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	same DNA profile as HELA using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypertriploid karyotype with 1.8% polyploidy; 76(73-78)<3n>XX, -X, -1, -3, -4, +6, +7, +8, -9, -18, -20, +11mar, der(1)t(1;?)(p11;?), der(3)t(3;?)(p11;?), der(3) t(3;?)(p14;?), der(3p;5q), del(6)(q21), del(7)(q31), der(8)t(8;?11) (q24;q11), der(9) t(9;?)(q34;?), der(12)t(12;?)(p12;?), der(19)t(19;?)(q13;?); supernumerary single dm present in 11% cells; 3 or more HELA markers present
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KB-V1
Cell Type:	human cervix carcinoma (derivative of HELA)
DSMZ No:	ACC 149
Origin:	KB-V1 is a multidrug-resistant (mdr) subclone derived in 1985 from KB-3-1 (DSM ACC 158) which itself is a derivative of the 1951 established HELA (DSM ACC 57) (31-year-old black woman); cells express strongly mdr1 mRNA and protein
References:	Roninson et al., Proc. Natl. Acad. Sci. USA 83: 4538-4564 (1986); Shen et al., J. Biol. Chem. 261: 7762-7770 (1986); Akiyama et al., Somat. Cell Genet. 11: 117-126 (1985)
Depositor:	Dr. M. Gottesman, National Institutes of Health, Bethesda, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	epitheloid-like, sometimes round cells growing in loosely adherent monolayers when vinblastine is present
Medium:	85% Dulbecco's MEM + 15% FBS + 200-1000 ng/ml vinblastine
Subculture:	cells are difficult to culture in the presence of vinblastine! split ratio of confluent cultures 1:3 to 1:10; split every 2-4 days, dependent on the vinblastine concentration, using trypsin/EDTA; upon thawing, first grow the cells without vinblastine, then in increasing amounts; seed out at ca. 3×10^6 cells/80 cm ² in 10 ml medium
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of about $6-10 \times 10^6$ /80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	same DNA profile as HELA using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human flat-moded hypotriploid karyotype with 13% polyploidy; 71-77<3n>XX, -X, +1, +3, +5, +5, +5, +7, +7, +8, +8, -9, -11, +12, +15, -16, +17, -18, -19, +21, -22, +mar, der(1;3)(q10;p10), del(3)(q13), der(3;5)(p10;q10), i(5p)x3, del(7)(q10), der(7)add(7)(p1?)t(7;11)(q3?5;q11), add(8)(q12), del(9)(p24), add(9)(q34), add(12)(p13), add(12)(q1?4), add(15)(p11), add(22)(p11)x2; drug selection maintains variable numbers of dmin visible in 26% cells as well as smaller particulate chromatin in most remaining cells; karyotype closely related to HELA
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KCI-MOH1
Cell Type:	human pancreas adenocarcinoma
DSMZ No:	ACC 498
Origin:	established from the moderately differentiated adenocarcinoma of the head of the pancreas from a 74-year-old African-American man in 1996; patient was treated with pancreato-duodenectomy; cells were described as being tumorigenic in SCID mice
References:	Mohammad et al., Pancreas 16: 19-25 (1998)
Depositor:	Dr. R. M. Mohammad, Wayne State University, Detroit, MI, USA
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:2 to 1:3 every 3-4 days using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/80 cm ² ; maintain at ca. $3-4 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 70 hours
Harvest:	maximal density at ca. 7×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with enrofloxacin (Baytril), then negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypotriploid karyotype with 8% polyploidy; 61(57-63)<3n>X/XX, -Y, +1, -2, -6, +8, -10, +11, -12, -13, -14, -16, -18, -21, -21, +mar, add(8)(p2 ?)x2, i(11p), del(11)(p12), add(13)(p11), add(17)(p12); differs from published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KCL-22
Cell Type:	human chronic myeloid leukemia in blast crisis
DSMZ No:	ACC 519
Origin:	established from the pleural effusion of a 32-year-old woman with Philadelphia chromosome-positive CML in blast crisis in 1981; described to contain the t(9;22) b2-a2 fusion gene and a p53 mutation
References:	Kubonishi et al., Gann 74: 319-322 (1983); Kubonishi & Miyoshi, Int. J. Cell Cloning 1: 105-117 (1983); Drexler et al., Leukemia Res. 23: 207-225 (1999)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round, single cells in suspension, sometimes in small clusters
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:5 to 1:10 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. $0.2-0.5 \times 10^6$ cells/ml; maximal density of ca. 2.0×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	cell harvest of about $1.0-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4-, CD13+, CD14+, CD15+, CD19-, CD25-, CD33+, CD34+, CD54+, HLA-DR-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype with 3.3% polyploidy; 51(47-51)<2n>X, -X, +1, +6, +8, +8, +14, +22, del(1)(p22), t(9;22)(q34;q11), add(17)(p12-13), i(21q), der(22)t(9;22)(q34;q11); carries Phx2; matches published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KE-37
Cell Type:	human T cell leukemia
DSMZ No:	ACC 46
Origin:	established from a 27-year-old man with acute lymphoblastic leukemia (ALL) in 1979
References:	Drexler et al., Leukemia Res. 9: 209-229 (1985)
Depositor:	Prof. R. Berger, Institute of Molecular Genetics, Paris, France
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension
Medium:	90-92.5% RPMI 1640 + 7.5-10% FBS
Subculture:	split saturated culture 1:2 to 1:3 every second day; maintain at about $1-1.5 \times 10^6$ cells/ml; seed out at ca. $1-2 \times 10^6$ cells/ml; initially after thawing, cells should be cultured with 20% FBS for the first 1-2 weeks; maximum density at ca. 3×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	cell harvest of ca. 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5-8 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD1a+, CD2+, CD3-, cyCD3+, CD4+, CD5+, CD6+, CD7+, CD8-, CD13-, CD19-, CD34-, CD90+, HLA-DR-, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human flat-modulated hypotetraploid karyotype; 91(86-92)<4n>XXYY, +8, -14, -14, t(7;12)(q32-33;p12-13)x2, der(8)t(8;14)(q24;q11)x4, der(14)t(8;14)(q24;q11)x2; resembles published karyotype; carries 4 copies of der(8)t(8;14) associated with T-ALL; normal ch 14 present
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KELLY
Cell Type:	human neuroblastoma
DSMZ No:	ACC 355
Origin:	established from a patient with neuroblastoma; ca. 100-fold genomic amplification of N-myc in der(17)hsr
References:	Schwab et al., Nature 305: 245-248 (1983); Preis et al., Cancer Res. 48: 6530-6534 (1988)
Depositor:	Dr. R. A. F. MacLeod, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, round to fusiform with polar neuritic processes, growing in mono- and multilayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:10 every 3-6 days using trypsin/EDTA; seed out at ca. 0.2-0.8 x 10 ⁵ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	cell harvest of ca. 2.5 x 10 ⁵ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4-5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, neurofilament+, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human near-diploid karyotype with 20% polyploidy; 46-47<2n>X, -7, -8, +16, +22, +2mar, del(1)(p34), t(2;7)(p12;q11), add(4)(p15.3), add(5)(p15), add(9)(q34), add(10)(q24), add(11)(q24), add(14)(q32), ins(17;2)(q21;p24 hsr)x2; del(1p34) and 2p24 hsr are typical of late-stage neuroblastoma; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KG-1
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 14
Origin:	established from the bone marrow of a 59-year-old man with erythroleukemia that developed into acute myeloid leukemia (AML) at relapse in 1977; cells were described to form colonies in soft-agar in response to colony-stimulating factor
References:	Koeffler & Golde, Science 200: 1153-1154 (1978); Koeffler & Golde, Blood 56: 344-350 (1980); Koeffler, Blood 62: 709-721 (1983)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round, single cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.2-1.0 \times 10^6$ cells/ml; split 1:2 to 1:4 every 2-3 days; seed out at ca. 1×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 38 hours
Harvest:	maximum density at $1.0-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, PCR assays
Immunology:	CD3-, CD4-, CD13+, CD14-, CD15+, CD19-, CD33+, CD34+, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypodiploid karyotype with 4.5% polyploidy; 45(42-47)<2n>X/XY, -4, +8, +8, -12, -17, -20, +2mar, der(5;17)(q10;q10)del(5)(q?11q?13), dup(7)(q12q33), del(7)(q22q35), i(8q)x2, der(8)t(6;8)(p11;q22), der(8)t(8;12)(p11;q13), der(11)t(1;11)(q13-21;p11-p13)), der(16)t(?12;16)(?p13;q13/21); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KG-1a
Cell Type:	human acute myeloid leukemia (derivative of KG-1)
DSMZ No:	ACC 421
Origin:	established from the bone marrow of a 59-year-old man with erythroleukemia that developed into acute myeloid leukemia (AML); KG-1a is a subclone of AML cell line KG-1 (DSM ACC 14) described as "undifferentiated variant"; corresponds to ATCC CCL-246.1
References:	Koeffler et al., Blood 56: 265-273 (1980)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	mostly single round or 1-2% giant cells growing in suspension, some cells are loosely adherent
Medium:	80% alpha-MEM (or RPMI 1640) + 20% FBS
Subculture:	maintain at $0.5\text{--}1.5 \times 10^6$ cells/ml; split 1:2 every 2-3 days; seed out at ca. 1.0×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	maximum density at ca. 2.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5\text{--}7 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4-, CD13+, CD14-, CD15+, CD19-, CD33+, CD34+, HLA-DR+
Fingerprinting:	same DNA profile as KG-1 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of LDH, NP
Cytogenetics:	human near-diploid karyotype with 5% polyploidy; 46(44-47)<2n>X/XY, +8, +11, -12, -17, -20, der(5;17)(q10;q10)del(5)(q?11q?13), del(7)(q22q35), i(8q), der(8)t(8;12)(p11;q13)add(8)(q24), i(11q), add(16)(q13/21), der(19)t(14;19)(q11;q13), add(20)(p13); sideline with -18; resembles published karyotype; closely related to, but distinct from KG-1
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KM-H2
Cell Type:	human Hodgkin's lymphoma
DSMZ No:	ACC 8
Origin:	established from the pleural effusion of a 37-year-old man with Hodgkin's disease (mixed cellularity progressing to lymphocyte depletion; stage IV at relapse) in 1974
References:	Kamesaki et al., Blood 68: 285-292 (1986); review article: Drexler, Leukemia Lymphoma 9: 1-25 (1993)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round to polygonal cells in suspension; heterogeneity in cell size, bi- and polynucleated cells
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.3\text{--}1.0 \times 10^6$ cells/ml; split 1:2 to 1:4 every second day; seed out at ca. 0.8×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 48 hours
Harvest:	saturation density at about 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2- CD3-, CD13(+), CD14(+), CD15+, CD19-, CD21+, CD25-, CD30+, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypotriploid karyotype with 3.5% polyploidy; 59(56-60)<3n>der(X)der(Y), -X, -1, -1, -2, -2, -3, -4, -5, -5, -6, -8, -9, -9, -10, -10, -10, -11, -12, -13, -14, -15, -15, -16, -16, -18, -19, -20, -21, -22, +25mar; add(1)(q32), add(3)(q26), add(3)(q26), del(4)(q26q31), i(4q), add(5)(p11), add(6)(q14), add(7)(p14), add(7)(q34), del(7)(q31), add(8)(q24), del(8)(p12), del(11)(q23), i(13q), der(13)t(11;13)(p13;q13), add(14)(p12)x2, del(17)(p12), i(18q); apparently typical HD karyotype with extraordinary level of rearrangements including several HSR-bearing markers with ectopic rDNA; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KMOE-2
Cell Type:	human acute erythremia
DSMZ No:	ACC 37
Origin:	established from the peripheral blood of a 2-year-old girl with acute erythremia (Di Gugliemo's disease, erythroblastosis) at diagnosis in 1978; cells were described to produce hemoglobin
References:	Okano et al., J. Cancer Res. Clin. Oncol. 102: 49-55 (1981)
Depositor:	Dr. A. Kulozik, University of Ulm Medical School, Ulm, Germany
DSMZ Cell Culture Data:	
Morphology:	single, large, round to oval cells in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	maintain at $0.2-0.3 \times 10^6$ cells/ml; seed out at ca. $0.3-0.4 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling of ca. 40-50 hours
Harvest:	maximal density of 0.6×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD13+, CD19-, CD34-, CD41(+), CD42+, CD71+, GlyA(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypotetraploid karyotype (mode weak) with 6% polyploidy; 82(80-88)<4n>XX, -X, -X, -10, -13, -14, -16, -18, -20, -22, +3mar, der(1)t(1;?)(q21;?)x2, del(7)(p15)x2, del(9)(p12), i(17q); karyotype appears to represent derivative of that originally reported
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KMS-12-BM
Cell Type:	human multiple myeloma
DSMZ No:	ACC 551
Origin:	established from the bone marrow of a 64-year-old woman with multiple myeloma in 1988
References:	Ohtsuki et al., Brit. J. Haematol. 73: 199-204 (1989); review article: Drexler & Matsuo, Leukemia Res. 24: 681-703 (2000)
Depositor:	Prof. M. Namba, Okayama University, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	mostly single cells in suspension, 1-2% giant cells
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. $0.5-1.0 \times 10^6$ cells/ml; first week after thawing cultivate in 24-well-plate, initially cells grow slowly
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60 hours
Harvest:	cell harvest of ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19-, CD20+, CD34-, CD37+, CD79a-, cyCD79a+, CD80-, CD138+, HLA-DR-, cyIgG-, cyIgM-, cykappa-, cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hypertriploid karyotype with 3% polyploidy ; 69-79<3n>XX, -X, +1, +1, -4, +5, +6, +8, +9, -10, +13, +14, -15, -16, +17, -18, +19, +20, +21, -22, +4-5mar, der(1)t(1;4)(q11;q32), del(1)(p21)x2, der(1;8)(q10;q10)x2, add(4)(q32)x1-2, der(5;17)(p10;q10)x2, add(6)(q2?5), der(9)add(9)(p24)t(9;11)(q34;q13), der(9)t(9;11)(q34;q13), der(11)t(11;14)(q13;q32)x2-3, idic(?;13)(?;q32-33),x2, der(14)t(11;14)(q13;q32), der(17)t(11;17)(q21.2;q22)t(11;14)(q13;q32)x2, add(19)(q13), add(19)(q13); related to published karyotype; carries semi-cryptic t(11;14) with IGH/CCND1 rearrangement
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KOPN-8
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 552
Origin:	established from the peripheral blood of a 3-month-old girl with B cell precursor acute lymphoblastic leukemia (BCP-ALL) of type B-III in 1977; cells were described to carry the t(11;19)(q23;p13) leading to the MLL-ENL fusion gene
References:	Nakazawa et al., Clin. Haematol. 20: 189 (1978); Matsuo & Drexler, Leukemia Res. 22: 567-579 (1998) (review)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single round rather small cells growing in suspension
Medium:	90-95% RPMI 1640 + 5-10% FBS
Subculture:	split saturated culture 1:3 to 1:5 every 3 days; seed out at ca. $0.3-0.5 \times 10^6$ cells/ml; maintain at ca. $0.5-1.5 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of ca. 2.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in microbiological culture, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20(+), CD34-, CD37+, CD38+, CD79a-, cyCD79a+, CD80-, CD138-, HLA-DR+, sm/cyIgG-, smlgM-, cyIgM+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypodiploid karyotype with 4% polyploidy; 45(42-45)<2n>XX, -14, t(8;13)(q24;q21.2), t(11;19)(q23;p13), der(13)t(13;14)(p11;q11); carries t(11;19) effecting MLL-ENL fusion; matches published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KPL-1
Cell Type:	human breast carcinoma
DSMZ No:	ACC 317
Origin:	established from the pleural fluid of a 50-year-old Japanese woman with ductal carcinoma of the breast with a predominant intraductal component in 1992; cells were described to express estrogen, but not progesteron receptors and to secrete tissue polypeptide antigen
References:	Kurebayashi et al., Brit. J. Cancer 71: 845-853 (1995)
Depositor:	Dr. J. Kurebayashi, Kawasaki Medical School, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	adherent growing in monolayers (very dense colonies)
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:8 to 1:10 once a week using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of ca. 5×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human highly rearranged hypertriploid karyotype with 4% polyploidy; 77(70-82)<3n>XX, -X, -1, +2, -11, -13, +14, +15, -17, -18, -19, -20, -20, -21, -22, +15mar; add(X)(q28), add(2)(q35), add(3)(p25)x2, add(5)(p15), add(7)(p14), add(7)(p14), add(8)(p12), add(9)(p22), add(10)(p13)x2, add(19)(q13); matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KU-19-19
Cell Type:	human urinary bladder transitional cell carcinoma
DSMZ No:	ACC 395
Origin:	established from the invasive transitional cell carcinoma of the bladder (grade 3, pT3b) of a 76-year-old man in 1993; described as secreting several hematopoietic cytokines and chorionic gonadotropin
References:	Tachibana et al., Cancer Res. 55: 3438-3443 (1995); Steube et al., Biochem. Biophys. Res. Commun. 242: 497-501 (1998)
Depositor:	Dr. M. Tachibana, University of Keio, Tokyo, Japan
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing adherently in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:5 to 1:10 twice a week using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of ca. 15×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human hypertriploid karyotype; 74(70-75)<3n>XXX, +1, +5, +6, -7, +9, -11, +12, +15, +16, -17, -18, +19, +21, -22, +2mar; add(1)(p32-34)/del(1)(p34), add(4)(q35)x1-2, del(5)(q14.2), add(6)(q11), del(11)(p14), add(12)(q24)/der(12)t(12;17)(q24;q12-22), add(19)(q13); hypertriploid derivative of original hypotetraploid clone; 4q+ and 19q+ markers confirm identity
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KU-812
Cell Type:	human chronic myeloid leukemia in myeloid blast crisis
DSMZ No:	ACC 378
Origin:	established from the peripheral blood of a 38-year-old man with chronic myeloid leukemia (CML) in myeloid blast crisis; described to express basophilic features
References:	Kishi, Leukemia Res. 9: 381-390 (1985); review article: Drexler, Leukemia Res. 18: 919-927 (1994); review article: Drexler et al., Leukemia Res. 24: 109-115 (2000)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single round cells growing in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	maintain at $0.4-0.8 \times 10^6$ cells/ml; split 1:2 every 3-4 days; seed out at ca. 0.5×10^6 cells/ml; cells are difficult to grow; after thawing, cells may have a lag phase of 1-2 weeks before reassuming vigorous growth
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 80-100 hours
Harvest:	maximum density of ca. 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD11b(+), CD13+, CD14-, CD15-, CD32+, CD33+, CD41(+), CD42-, CD61-, CD68+, CD71+, HLA-DR-, GlyA+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human hypotriploid karyotype with 8% polyploidy; 61(58-62)<3n>XYY, -2, -3, -5, +6, -7, +8, -10, -12, -16, -17, -18, +19, -20, t(9;22)(q34;q11)x2, i(11q), i(17q); Ph (two copies) together with i(17q) and additional chs 8 and 19; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KYSE-140
Cell Type:	human esophageal squamous cell carcinoma
DSMZ No:	ACC 348
Origin:	derived from the moderately differentiated invasive esophageal squamous cell carcinoma resected from middle intra-thoracic esophagus of a 54-year-old Japanese man prior to treatment (depth of invasion was a ₁); described to carry p53 mutation
References:	Shimada et al., Cancer 69: 277-284 (1992); Shimada et al., Brit. J. Surg. 80: 605-607 (1993); Kanda et al., Int. J. Cancer 58: 291-297 (1994); Tanaka et al., Int. J. Cancer 65: 372-376 (1996)
Depositor:	Dr. Y. Shimada, Kyoto University School of Medicine, Kyoto, Japan
DSMZ Cell Culture Data:	
Morphology:	adherent cells growing as monolayer with the tendency to pile up
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 2-4 days using trypsin/EDTA (cells may need five or more minutes to detach from the flask; prior to using trypsin, cells should be washed twice with PBS); seed out at ca. 2-3 x 10 ⁶ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	cell harvest of ca. 20-30 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1-3 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hyperdiploid/hypotriploid karyotype with 19% polyploidy; 59(56-64)<2n>Xder(X), -Y, add(X)(p1?), del(1)(q32), add(1)(q43), i(3q), add(2)(p2?), del(3)(q22), add(4)(p11), del(4)(q11), add(5)(q1?)x1-2, ins(6;?)(q1?), add(7)(q35)x2, add(8)(q24), del(9)(p11), add(9)(q34), add(11)(p15), add(12)(q24), add(13)(p11), i(13q), add(18)(p12), add(19)(p13), inc.; many additional structural and numerical rearrangements, including multiple insertions, producing an overwhelming preponderance of marker chromosomes
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KYSE-150
Cell Type:	human esophageal squamous cell carcinoma
DSMZ No:	ACC 375
Origin:	established from the poorly differentiated esophageal squamous cell carcinoma resected from upper (cervical) esophagus of a 49-year-old Japanese woman after receiving radiotherapy (tumor was invading contiguous structures); described as carrying amplified oncogenes, c-erb-B (8x) and cyclin D1 (4x) and producing tumors in nude mice
References:	Shimada et al., Cancer 69: 277-284 (1992); Shimada et al., Brit. J. Surg. 80: 605-607 (1993); Kanda et al., Int. J. Cancer 58: 291-297 (1994); Tanaka et al., Int. J. Cancer 65: 372-376 (1996)
Depositor:	Dr. Y. Shimada, Kyoto University School of Medicine, Kyoto, Japan
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing as adherent monolayer
Medium:	49% RPMI 1640 + 49% Ham's F12 + 2% FBS
Subculture:	split confluent culture 1:20 every 3-4 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 20-25 hours (but only 12.5 hours when serum is increased to 10%)
Harvest:	cell harvest of ca. 1×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human hypertriploid karyotype with 12% polyploidy; 70(68-75)<3n>XderX, -X, +1, -3, -4, -8, -9, +11, -14, -15, -16, -18, -21, +12mar (4xbisat), add(X)(q27), inv(1)(p34q32), del(1)(p32), del(1)(q11), del(7)(q32), del(8)(p21), i(9q), add(9)(q34), add(11)(p11), der(11)add(11)(p1?)add(11)(q24), del(1)(p11), del(12)(q21), del(15)(q23q25), add(19)(q13)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KYSE-180
Cell Type:	human esophageal squamous cell carcinoma
DSMZ No:	ACC 379
Origin:	established from the well differentiated esophageal squamous cell carcinoma resected from middle intra-thoracic esophagus of a 53-year-old Japanese man prior to treatment (tumor was invading contiguous structures)
References:	Shimada et al., Cancer 69: 277-284 (1992); Shimada et al., Brit. J. Surg. 80: 605-607 (1993); Kanda et al., Int. J. Cancer 58: 291-297 (1994); Tanaka et al., Int. J. Cancer 65: 372-376 (1996)
Depositor:	Dr. Y. Shimada, Kyoto University School of Medicine, Kyoto, Japan
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing as adherent monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 3-5 days using trypsin/EDTA; seed out at ca. $2-3 \times 10^6$ cells/80 cm ² in 10 ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	cell harvest of ca. 30×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2.5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human hyperdiploid karyotype with 11% polyploidy; 55(50-57)<2n>XX, +1, +1, +3, +7, +7, +8, -14, +16, -17, -17, -18, -18, +5-8mar, del(X)(q23), add(1)(q32), der(1;6)(q10;p10), der(1;13)(q10;q10), add(3)(p26), add(5)(q25), add(6)(p11), del(6)(q23), add(7)(p11), del(7)(p11), add(7)(q35), i(7q), del(8)(p22), add(10)(p11), add(11)(q11), del(11)(q11q14), add(13)(p11), add(13)(q33), add(14)(p11), add(15)(p11), add(21)(p11)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KYSE-270
Cell Type:	human esophageal squamous cell carcinoma
DSMZ No:	ACC 380
Origin:	derived from well differentiated invasive esophageal squamous cell carcinoma resected from middle intra-thoracic esophagus of a 79-year-old Japanese man prior to treatment (depth of invasion was not beyond the muscularis propria); described to carry p53 mutation
References:	Shimada et al., Cancer 69: 277-284 (1992); Shimada et al., Brit. J. Surg. 80: 605-607 (1993); Kanda et al., Int. J. Cancer 58: 291-297 (1994); Tanaka et al., Int. J. Cancer 65: 372-376 (1996)
Depositor:	Dr. Y. Shimada, Kyoto University School of Medicine, Kyoto, Japan
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing in monolayers
Medium:	49% RPMI 1640 + 49% Ham's F12 + 2% FBS
Subculture:	split confluent culture 1:5 every second days using trypsin/EDTA (incubation for 10-15 min); seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	cell harvest of ca. 10×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, MDH, NP
Cytogenetics:	human hypotriploid karyotype with 16% polyploidy; 66(64-67)<3n>XX/XXX, +1, -2, +3, -5, -6, +7, +8, -9, -10, -15, -15, -17, -18, -19, -20, -22, +5mar (1bisat), ?del(X)(q25), add(1)(p1?), add(1)(p2?), i(1p), der(1)t(1;2)(q13;?)(?;q11), add(2)(q37), del(2)(q23), add(3)(p2?6)x2, i(5p), add(5)(p15), del(7)(p11), add(8)(q24), add(16)(q24), add(18)(q23), add(19)(p13), add(20)(q12), add(21)(p11)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KYSE-30
Cell Type:	human esophageal squamous cell carcinoma
DSMZ No:	ACC 351
Origin:	derived from well differentiated invasive esophageal squamous cell carcinoma resected from middle intra-thoracic esophagus of a 64-year-old Japanese man prior to treatment; cell line established from tumor cells heterotransplanted into athymic mice; described to be heterotransplantable in athymic mice and to carry p53 mutation and amplification of c-erbB, c-myc and cyclin D1
References:	Shimada et al., Cancer 69: 277-284 (1992); Shimada et al., Brit. J. Surg. 80: 605-607 (1993); Kanda et al., Int. J. Cancer 58: 291-297 (1994); Tanaka et al., Int. J. Cancer 65: 372-376 (1996)
Depositor:	Dr. Y. Shimada, Kyoto University School of Medicine, Kyoto, Japan
DSMZ Cell Culture Data:	
Morphology:	adherent, epitheloid with long processes growing in monolayers
Medium:	45% RPMI 1640 + 45% Ham's F12 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:10 every 2-6 days using trypsin/EDTA; seed out at ca. $1-4 \times 10^6$ cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 0.7×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	highly rearranged hypertriploid karyotype with 8% polyploidy; 66-74<3n>Xder(X), -Y, +1, +1, -5, +7, -9, -10, -15, +16, -18, -18, -19, -21, -21, -21, -22, -22, +12-15mar, t(X;2)(p11;p11), ins(1;1)(p34/36;q21q32), add(1)(q11), der(1)add(1)(p21)(q22), del(2)(q24), der(2)t(2;4)(p11;q11)t(2;4)(q37;q11), del(3)(p14), add(3)(p11), add(4)(q11)x2, der(4)t(1;4)(p11;q11), add(5)(q11), add(5)(q11), add(6)(p21), i(7q), del(7)(p14), add(7)(p14), der(8)t(7;8)(q11;p11), add(9)(p24), i(10q), del(11)(q11)x1-2, add(11)(p13), del(12)(p12.2), i(13q), add(14)(q32), add(15)(p11), add(16)(p1?), add(17)(p1?), add(19)(q13.1)x2, inc.; extreme level of rearrangement
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KYSE-410
Cell Type:	human esophageal squamous cell carcinoma
DSMZ No:	ACC 381
Origin:	established from the poorly differentiated invasive esophageal squamous cell carcinoma resected from the cervical esophagus of a 51-year-old Japanese man prior to treatment (tumor invasion into the adventitia was obvious); described to overexpress hst-1 (= heparin-binding growth factor) and cyclin D1
References:	Shimada et al., Cancer 69: 277-284 (1992); Shimada et al., Brit. J. Surg. 80: 605-607 (1993); Kanda et al., Int. J. Cancer 58: 291-297 (1994); Tanaka et al., Int. J. Cancer 65: 372-376 (1996)
Depositor:	Dr. Y. Shimada, Kyoto University School of Medicine, Kyoto, Japan
DSMZ Cell Culture Data:	
Morphology:	epitheloid adherent cells growing in monolayers
Medium:	90-95% RPMI 1640 + 5-10% FBS
Subculture:	split confluent culture 1:5 to 1:10 twice a week using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 45 hours
Harvest:	cell harvest of ca. 15×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human flat-moded hypotetraploid karyotype with 4% polyploidy; 79-91<4n>Xder(X), -10, -18, -19, -21, -22, -22, +6mar, add(X)(qter), add(1)(p11)/del(1)(p13), add(3)(q21)x1-2, add(5)(p13), der(6)t(4;6)(q12;q14)x1-2, dup(7)(p12p14), der(7)del(7)(p12p21)t(7;9)(q33;q10), add(8)(p12)x1-2, add(9)(q34), add(10)(p14), add(11)(p11), add(12)(q24)x1-2, add(12)(p11-12)x2, del(12)(q14), i(13q)x2, i(14q)x1-2, add(14)(p11), add(19)(p13), add(?20)(q13), add(21)(p11)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KYSE-450
Cell Type:	human esophageal squamous cell carcinoma
DSMZ No:	ACC 387
Origin:	established from the well differentiated invasive esophageal squamous cell carcinoma resected from middle intra-thoracic esophagus of a 59-year-old Japanese man prior to treatment (depth of invasion was not beyond the submucosa); described as carrying p53 mutation and amplification of the oncogene c-myc
References:	Shimada et al., Cancer 69: 277-284 (1992); Shimada et al., Brit. J. Surg. 80: 605-607 (1993); Kanda et al., Int. J. Cancer 58: 291-297 (1994); Tanaka et al., Int. J. Cancer 65: 372-376 (1996)
Depositor:	Dr. Y. Shimada, Kyoto University School of Medicine, Kyoto, Japan
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently as monolayers
Medium:	45% RPMI 1640 + 45% Ham's F12 + 10% FBS
Subculture:	split confluent culture 1:8 to 1:10 every 3-4 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 5×10^6 cells/25 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hyperdiploid karyotype with 6.7% polyploidy; 57(49-57)<2n>, -Y, +1, +3, +3, +7, +12, +17, -18, +7mar, der(1)t(1;6)(p22;q11), add(1)(p32/34), del(2)(p23), del(3)(p22), del(4)(q21), add(5)(q35), del(6)(q25), add(7)(p22), add(8)(p23), add(9)(q34), del(11)(q14q21), add(12)(q14), del(12)(p11p12), der(14)t(8;14)(q21;p11), del(17)(p11), der(19)t(1;19)(q11;q12); sideline with add(11)(q?23); extensive subclonal rearrangements of ch 7
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KYSE-510
Cell Type:	human esophageal squamous cell carcinoma
DSMZ No:	ACC 374
Origin:	established from the well-differentiated esophageal squamous cell carcinoma with moderate invasion of the adventitia resected from cervical esophagus of a 67-year-old Japanese woman after cisplatin and radiotherapy; described as carrying p53 mutation and amplification of c-myc, hst-1, cyclin D1, and to induce well-differentiated tumors in nude mice
References:	Shimada et al., Cancer 69: 277-284 (1992); Shimada et al., Brit. J. Surg. 80: 605-607 (1993); Kanda et al., Int. J. Cancer 58: 291-297 (1994); Tanaka et al., Int. J. Cancer 65: 372-376 (1996)
Depositor:	Dr. Y. Shimada, Kyoto University School of Medicine, Kyoto, Japan
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing as monolayer with multilayer foci
Medium:	90% RPMI 1640 (or mixture of RPMI 1640 + Ham's F12 at 1:1) + 10% FBS
Subculture:	split confluent culture 1:5 to 1:20 every 4-8 days using trypsin/EDTA; seed out at ca. 0.5-3.0 x 10 ⁶ cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 1.5 x 10 ⁵ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-8+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH
Cytogenetics:	human hypertriploid karyotype with 8% polyploidy; 73(66-73)<3n>XXX, -X, +1, +1, -2, -2, +3, -4, +5, +6, +7, -14, -14, +19, -21, -21, +22, +9mar, add(X)(p22.2), add(1)(q21), ider(1)(q10)t(1;8)(q21;q11), add(2)(q11), del(3)(q21), del(3)(p1?), add(5)(p14), i(8p), der(8)add(8)(p11)add(8)(q24), i(9p), add(9)(p11), add(11)(p15), add(11)(q13)x2, add(14)(p11), add(15)(p11)x2, der(18)add(18)(p11)dup(18)(q11q23)add(18)(q23), der(18)add(18)(p11)add(18)(q2?3), der(18)add(18)(p11)add(8)(q2?1), del(19)(p13)x2, der(19)t(10;19)(q11;q13), add(21)(p11), del(22)(q13)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KYSE-520
Cell Type:	human esophageal squamous cell carcinoma
DSMZ No:	ACC 371
Origin:	established from the moderately differentiated invasive esophageal squamous cell carcinoma resected from lower intra-thoracic esophagus of a 58-year-old Japanese woman prior to treatment (tumor had invaded the adventitia); described as carrying p53 mutation and amplification of the oncogene c-myc
References:	Shimada et al., Cancer 69: 277-284 (1992); Shimada et al., Brit. J. Surg. 80: 605-607 (1993); Kanda et al., Int. J. Cancer 58: 291-297 (1994); Tanaka et al., Int. J. Cancer 65: 372-376 (1996)
Depositor:	Dr. Y. Shimada, Kyoto University School of Medicine, Kyoto, Japan
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently as monolayers, occasional giant cells present (ca. 1%)
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:6 to 1:8 every 3-4 days using trypsin/EDTA; seed out at ca. $0.5-1.0 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. $4-5 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human flat-modulated hypotriploid karyotype with 8% polyploidy; 58-60<3n>XX, -X, +1, +1, -2, -2, -4, -5, -6, +7, -8, -10, -10, -11, -12, -12, -13, -14, -15, -16, -17, -18, +20, +20, -21, -21, -22, -2, +7-8mar, add(X)(p11), add(X)(q11), add(1)(q11), der(1)del(1)(q12)t(1;7)(p36;q21), der(1)del(1)(q12)t(1;5)(p36;q11), del(1)(q21), del(1)(p34), add(2)(q37), del(3)(q13), add(4)(p11), add(4)(p15), add(5)(q11)x1-2, add(6)(q1?), t(6;15)(p10;q10)x1-2, del(7)(q11)x1-2, add(7)(p22), add(9)(p23), der(11)add(11)(p15)t(11;12)(q24;q11), add(11)(p15), add(16)(q2?3), add(17)(q24), add(18)(p11), add(19)(p13)del(22)(q13)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	KYSE-70
Cell Type:	human esophageal squamous cell carcinoma
DSMZ No:	ACC 363
Origin:	established from the poorly differentiated invasive esophageal squamous cell carcinoma resected from middle intra-thoracic esophagus of a 77-year-old Japanese man prior to treatment (depth of invasion was not beyond the muscularis propria)
References:	Shimada et al., Cancer 69: 277-284 (1992); Shimada et al., Brit. J. Surg. 80: 605-607 (1993); Kanda et al., Int. J. Cancer 58: 291-297 (1994); Tanaka et al., Int. J. Cancer 65: 372-376 (1996)
Depositor:	Dr. Y. Shimada, Kyoto University School of Medicine, Kyoto, Japan
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 4-6 days using trypsin/EDTA (cells may need five or more minutes to detach from the flask; prior to using trypsin, cells should be washed twice with PBS); seed out at ca. $0.5-1.0 \times 10^6$ cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	cell harvest of ca. 0.3×10^6 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hypotriploid karyotype with 8% polyploidy; 60(54-63)<3n>XXX, -Y, -2, -3, -3, -4, -6, +7, -8, -9, -10, -13, -14, -15, -16, -16, -16, -17, -18, -18, -21, -22, +10mar, del(X)(q25), add(1)(p11), add(2)(q33), del(2)(q33), del(3)(q13), add(4)(q35), add(5)(p15), del(5)(q14), add(7)(q36), del(7)(p11), del(8)(p22.2), qdp(9)(p11p24), add(10)(q25), add(11)(p?13), add(19)(p13), i(?22p)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	L-1210
Cell Type:	mouse lymphocytic leukemia
DSMZ No:	ACC 123
Origin:	established from the methylcholanthrene-induced tumor in a DBA strain mouse (subline 212, 8-month-old female) in 1949; the cells were propagated by heterotransplantation into DBA/2 mice; an in-vitro suspension culture was first reported in 1966; cells are used for cytotoxicity studies
References:	Law et al., J. Natl. Cancer Inst. 10: 179-192 (1949); Moore et al., J. Natl. Cancer Inst. 36: 405-421 (1966)
Depositor:	obtained from ATCC (CCL 219), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	single round, lymphoblast-like cells in suspension
Medium:	90% Dulbecco's MEM + 10% horse serum
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; split 1:2 to 1:4 every 1-2 days; seed out at ca. $0.5-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	maximum density at about 2.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% horse serum, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, MDH, NP, PEP B
Cytogenetics:	murine hypodiploid karyotype; 39(36-42), with about 75% polyploidy; about 77% of cells carry one or more rearrangements/aberrations
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	L-1236
Cell Type:	human Hodgkin's lymphoma
DSMZ No:	ACC 530
Origin:	established from the peripheral blood of a 34-year-old man with Hodgkin's disease (mixed cellularity, stage IV, refractory, terminal, third relapse) in 1994
References:	Wolf et al., Blood 87: 3418-3428 (1996); Kanzler et al., Blood 87: 3429-3436 (1996); Joos et al., Int. J. Cancer 103: 489-495 (2003)
Depositor:	Dr. J. Wolf, University Hospital Köln, Germany
DSMZ Cell Culture Data:	
Morphology:	variable morphology with semi-adherent spheroid to fusiform cells growing in clumps in mono- and multilayers, occasional giant cells
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2-3 days, adherent cells can be detached by tapping the flask; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. $0.3-0.5 \times 10^6$ cells/ml; maximal density of ca. 0.6×10^6 cells/ml; a daily exchange of spent medium might be of advantage
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of ca. 0.6×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD2-, CD3-, CD13-, CD14-, CD15+, CD19-, CD21-, CD25-, CD30+, CD34-, CD54+, CD80+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-modulated hypotriploid karyotype with 17% polyploidy; 57-66<3n>XX, -Y, +3, -4, -10, +11, -14, -18, +19, -20, -21, -22, +4-6mar, add(1)(p22), t(1;14)(p34;q22), der(1;9)(q10;p10), der(1)add(1)(p34)dup(1)(q21q43), der(1)add(1)(p2?)del(1)(q21), add(2)(q ?22), inv dup(2)(p16p23)x2, der(3)t(3;9)(q27;q21), add(6)(p23), add(6)(q22-24), add(7)(p22), der(7)t(7;7)(p22;q22), inv(7)(p11q35)add(7)(p11), add(8)(q24), der(8)t(8;1)(q24;p32)t(1;14)(p34;q22), del(9)(q21), del(10)(q12), add(11)(p1 ?), del(11)(q13), der(12)t(12;17)(p13;q13)t(17;Y)(q22;q11), del(12)(q21-22), add(13)(p11), inv(14)(p13q32.32)del(14)(q22;q32.32)x2, t(15;20)(q22;p11-12), add(16)(q23-24), del(17)(q23)x2, add(19)(q13); resembles originator's karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	L-363
Cell Type:	human plasma cell leukemia
DSMZ No:	ACC 49
Origin:	established from the peripheral blood of a 36-year-old woman with plasma cell leukemia (IgG) in 1977; cells were described to be EBNA-negative and to express mRNA for proto-oncogene bcl-2; plasma cell leukemia is related to multiple myeloma
References:	Diehl et al., Blut 36: 331-338 (1978)
Depositor:	Prof. V. Diehl, University Hospital of Köln, Köln, Germany
DSMZ Cell Culture Data:	
Morphology:	single, large, round or oval cells in suspension
Medium:	85% RPMI 1640 + 15% FBS
Subculture:	maintain at $0.2-0.5 \times 10^6$ cells/ml; split 1:2 to 1:3 every 3-4 days; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 25 hours
Harvest:	maximum density at 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19-, CD20-, CD34-, CD37-, CD79a-, cyCD79a+, CD80-, CD138+, HLA-DR+, sm/cyIgG-, sm/cyIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, PEP B
Cytogenetics:	human hyperdiploid karyotype with 11% polyploidy; 49(45-50)<2n>X, -X, -X, -6, +7, +8, +11, +11, +14, +19, -22, t(5;8)(q12/13;q24.32), der(8)t(5;8)(q12/13;q24.32), der(8)t(7;8)(q11.2;p22), del(11)(q13)x2, add(14)(q32), del(17)(p12), add(19)(q13); sideline with add(13)(q34); 11q13 and 14q32 breakpoints recurrent in PCL; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	L-428
Cell Type:	human Hodgkin's lymphoma
DSMZ No:	ACC 197
Origin:	established from the pleural effusion of a 37-year-old woman with Hodgkin's lymphoma (stage IVB, nodular sclerosis, refractory, terminal) in 1978
References:	Schaadt et al., Blut 38: 185-190 (1979); Schaadt et al., Int. J. Cancer 26: 723-731 (1980); review article: Drexler, Leukemia Lymphoma 9: 1-26 (1993)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	single cells in suspension or as clusters; polymorph cells, some cells have "spikes", 1-5% giant multinucleated cells
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.2-0.5 \times 10^6$ cells/ml; split culture 1:2 every 2-3 days; after an initial slow start cells grow quickly; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	saturation density at about $0.5-1.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3-, CD13-, CD14-, CD15+, CD19-, CD21(+), CD25-, CD30+, CD34-, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human flat-modulated hypertetraploid karyotype with 10% polyploidy; 94(75-99)<4n>XXXX, +2, -5, +6, -9, -9, +12, -13, +17, +4mar, +2r/dmin, del(1)(q13), der(2)t(2;4)(p14;q21)x2-3, add(5)(p11), add(6)(q24/25)x2, add(7)(q35)x2, del(7)(p14), add(9)(p22), del(11)(q22.3), del(12)(q21)x1-2, add(13)(p11)x2, add(14)(q32)x2, der(16)t(16;?)(q24;?)t(1;?)(q11;?), del(17)(p12), del(21)(q21.2)x2; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	L-5178-Y
Cell Type:	mouse T cell lymphoma
DSMZ No:	ACC 320
Origin:	established from an 8-month-old female DBA/2 mouse with T cell lymphoma in 1985
References:	Evans et al., Proc. Natl. Acad. Sci. USA 83: 4379-4383 (1986) and 84: 7562-7566 (1987)
Depositor:	obtained from DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	grows as single or aggregated round cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:5 to 1:10 every 3 days; seed out at $0.1-0.5 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 20 hours
Harvest:	max. density at about 2.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4.5-5.0 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, LDH, MDH, NP
Cytogenetics:	murine near diploid karyotype with 13% polyploidy; 39(37-42)<2n>; 1 centric fusion marker
Viruses:	ELISA: reverse transcriptase positive

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German Collection of Microorganisms and Cell Cultures
Dept. Human and Animal Cell Cultures

Mascheroder Weg 1b, D-38124 Braunschweig, Germany

Tel. +49-531-2616.161, FAX +49-531-2616.150

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<http://www.dsmz.de>

INFORMATION SHEET FOR CELL LINES

Cell Line:	L-540
Cell Type:	human Hodgkin's lymphoma
DSMZ No:	ACC 72
Origin:	established from the bone marrow of a 20-year-old woman with Hodgkin's disease (nodular sclerosis; stage IVB, pre-terminal stage)
References:	Diehl et al., J. Cancer Res. Clin. Oncol. 101: 111-124 (1981); Diehl et al., Cancer Treat. Rep. 66: 615-632 (1982); review article: Drexler, Leukemia Lymphoma 9: 1-25 (1993)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 every 3 days, exchange spent medium every 1-2 days; seed out at ca. 0.5 x 10 ⁶ cells/ml; after thawing culture first in 24-well-plates
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 70 hours
Harvest:	saturation density at about 1.5 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5-7 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2+, CD3-, CD13-, CD14-, CD15+, CD19-, CD21-, CD25+, CD30+, CD34-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH
Cytogenetics:	human flat-moded hypotriploid karyotype with 16% polyploidy; 59-64<3n>XX, -X, -1, +2, -6, +7, -8, -9, -10, -11, -13, -14, -16, -17, -18, +22, +5-6mar, der(1;5)(q10;p10)x1-2, der(2;12)(q10;q10), der(2)t(2;?8)(q3?7;q2?1)x1-2, ins(5;?)(p11;?), add(7)(p1?), add(8)(q24)x1-2, add(9)(p1?), add(11)(q24), der(15)t(1;15)(p21;p13)x2, add(19)(p11), add(21)(p13)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	L-929
Cell Type:	mouse connective tissue fibroblast
DSMZ No:	ACC 2
Origin:	established from the normal subcutaneous areolar and adipose tissue of a male C3H/An mouse; used as target in TNF detection assays
References:	Earle, J. Natl. Cancer Inst. 4: 165-212 (1943); Sanford et al., J. Natl. Cancer Inst. 9: 229-246 (1948)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	fibroblasts growing as monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent cultures 1:5 to 1:10 using trypsin (do not use trypsin/EDTA); after 2-3 days monolayer will be confluent; split 2-3 times a week; seed out initially at about $1.0\text{--}2.0 \times 10^6$ cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 21-24 hours
Harvest:	ca. $4\text{--}8 \times 10^6$ cells/25 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1\text{--}2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, PCR assays
Species:	confirmed as mouse with IEF of AST, MDH, PEP B
Cytogenetics:	murine hypertriploid karyotype; 61-67, 12-16 centric fusion markers present
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	L138.8A
Cell Type:	mouse mast cell
DSMZ No:	ACC 114
Origin:	established from bone marrow mast cells of a female BALB/c mouse in 1984; cells are IL-3 responsive and secrete IL-6
References:	Hültner et al., Immunology 67: 408-413 (1989); Hültner & Möller, Exp. Hematol. 18: 873-877 (1990)
Depositor:	Dr. L. Hültner, GSF, Institute for Experimental Hematology, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing in suspension; during exponential growth cells aggregate in small clusters
Medium:	80-90% RPMI 1640 + 10-20% FBS + 20-50 U/ml IL-3 or 5-10% vol conditioned medium of WEHI-3B cell line (DSM ACC 26)
Subculture:	cells are initially (after thawing) difficult to grow! after thawing seed at different concentrations in 24-well plates, later maintain at $0.1-1.0 \times 10^6$ cells/ml, do not exceed 1×10^6 cells/ml; split 1:4 to 1:10 about 1-2 times per week. CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximal density at about 2.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 7×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, LDH, MDH, NP
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LAMA-84
Cell Type:	human chronic myeloid leukemia in blast crisis
DSMZ No:	ACC 168
Origin:	established from the peripheral blood of a 29-year-old woman with chronic myeloid leukemia (CML) (treated 5 years with busulfan from 1979-1984) one month after onset of myeloid-megakaryocytic blast crisis; cells were described to express megakaryocytic and erythroid markers and to respond to induction of differentiation with various reagents; parental cell line of subclone LAMA-87 (DSM ACC 270)
References:	Seigneurin et al., Exp. Hematol. 15: 822-832 (1987); Champelovier et al., Leukemia Res. 18: 903-918 (1994); review article: Drexler et al., Leukemia Res. 24: 109-115 (2000); review article: Drexler, Leukemia Res. 18: 919-927 (1994)
Depositor:	Prof. D. Seigneurin, University Hospital of Grenoble, Grenoble, France
DSMZ Cell Culture Data:	
Morphology:	single, round to polymorph cells growing in suspension, some cells are loosely adherent
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; optimal split ratio of 1:2 to 1:3 every 2-4 days; dislodge any adhering cells by shaking culture flask; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	saturation density at about 2.0×10^6 cells/ml ¹
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4-, CD11b+, CD13+, CD14-, CD15-, CD19-, CD33+, CD34-, CD41-, CD42-, HLA-DR(+)
Fingerprinting:	unique DNA profile using multiplex PCR at D1S80, D2S44, D17S5 and ApoB
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypertriploid karyotype with 3.6% polyploidy; 73/74(69-77)<3n>XX, -X, +1, -2, +5, +6, +8, +13, -14, +17, +17, -18, +22, +mar, del(7)(p15), der(9)t(9;22)(q34;q11)x2, i(11q), add(13)(q33), del(17)(p12), der(22)t(9;22)(q34;q11)x4; carries Ph (4 copies); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LAMA-87
Cell Type:	human chronic myeloid leukemia in blast crisis (derivative of LAMA-84)
DSMZ No:	ACC 270
Origin:	the parental cell line LAMA-84 (DSM ACC 168) has been established from the peripheral blood of a 29-year-old woman with chronic myeloid leukemia (CML) in blast crisis; subclone LAMA-87 was obtained after subcutaneous transplantation of LAMA-84 cells into estrone-treated nude mice
References:	Champelovier et al., Leukemia Res. 18: 903-918 (1994); Champelovier et al., Exp. Hematol. 25: 958-965 (1997); review article: Drexler, Leukemia Res. 18: 919-927 (1994); review article: Drexler et al., Leukemia Res. 24: 109-115 (2000)
Depositor:	Dr. P. Champelovier, Centre Hospitalier Universitaire, Grenoble, France
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly or in small clusters in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 every 2-3 days; spent medium should be exchanged every 24-48 hours even without splitting of the cells; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	maximal density at ca. 1×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD13+, CD15-, CD19-, CD33+, CD34-, CD41-, CD42-, GlyA-, HLA-DR-
Fingerprinting:	unique DNA profile using multiplex PCR of minisatellite markers, but identical with LAMA-84 (DSM ACC 168)
Species:	confirmed as human with IEF of LDH, NP
Cytogenetics:	human flat-moded hypotriploid karyotype; 60-64<3n>XX, -X, -2, -3, -7, +8, -10, -13, -14, -15, -16, -18, +19, -20, +22, der(9)t(9;22)(q34;q11)x2, del(12)(p12), del(12)(q14), add(19)(p13), add(?19)(p11), der(22)t(9;22)(q34;q11)x4; four copies of Ph present in most cells; resembles published karyotype (distinguishable from LAMA-84 by 12p- marker)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LAT
Cell Type:	ovine aortic tissue cells
DSMZ No:	ACC 349
Origin:	established from the aortic tissue cells of a lamb; cells were described to be susceptible to Visna Maedi Virus
References:	not published
Depositor:	Dr. M. Ferrari, Istituto Zooprofilattico Sperimentale, Brescia, Italy
DSMZ Cell Culture Data:	
Morphology:	fibroblastoid, growing in monolayers, forming adherent clumps upon reaching confluency
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split near-confluent culture 1:5 every 4-5 days using trypsin/EDTA; seed out at ca. 1.5×10^6 cells/80 cm ² ; optimal density at $1.0\text{--}4.5 \times 10^4$ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of 30-50 hours
Harvest:	cell harvest of ca. 5×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS and 10% DMSO at about 1.5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	apparently ovine (different from primate and murine IEF patterns) with IEF of AST, LDH, NP
Cytogenetics:	ovine diploid karyotype with 27% polyploidy; $54 < 2n > XY$, no consistent abnormality detected
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LCL-HO
Cell Type:	human B lymphoblastoid cells (from patient with melanoma)
DSMZ No:	ACC 185
Origin:	this lymphoblastoid cell line was established by EBV transformation of peripheral blood lymphocytes; cell line was submitted as autologous to cell line MEL-HO (DSM ACC 62) that was derived from a woman with primary melanoma in 1976; however, cytogenetic and DNA fingerprinting analysis showed clearly that LCL-HO and MEL-HO are not related
References:	not published
Depositor:	Dr. H. W. L. Ziegler-Heitbrock, University of Munich, Institute of Immunology, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	lymphoblastoid-like cells growing in suspension singly or in clumps
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5\text{--}1.0 \times 10^6$ cells/ml; optimal split ratio of 1:2 to 1:3 every 2-4 days; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	saturation density at about $2.0\text{--}2.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13+, CD19+, CD20+, CD34-, CD37+, CD38+, CD79a-, cyCD79a+, CD80+, CD138+, HLA-DR+, sm/cyIgG+, smIgM-, smkappa-, smlambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human diploid karyotype with 2% polyploidy and near triploid sideline (4%); 46(45-47)<2n>XX, der(12)t(12;?6)(q24.32;p21.2); additional pseudodiploid subclones are present with other structural and numerical rearrangements
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LCL-WEI
Cell Type:	human B lymphoblastoid cells (from patient with melanoma)
DSMZ No:	ACC 218
Origin:	Epstein-Barr virus-positive cell line established from peripheral blood in 1977 from male patient with melanoma
References:	not published
Depositor:	Dr. H. W. L. Ziegler-Heitbrock, Institute of Immunology, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	lymphoblastoid cells growing singly and in large clumps in suspension
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	maintain at about $0.3-1.0 \times 10^6$ cells/ml; optimal split ratio of 1:3 to 1:8 every 2-4 days
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30-40 hours
Harvest:	saturation density at about 2.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 6×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD37+, HLA-DR+, sm/cyIgG+, sm/cyIgM-, sm/cykappa-/+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human near-tetraploid karyotype with diploid (36%) sideline and 4% polyploidy; 90(88-94)<4n>XX/XXY/XXYY; no consistent structural aberrations
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LCLC-103H
Cell Type:	human large cell lung carcinoma
DSMZ No:	ACC 384
Origin:	established from the pleural effusion of a 61-year-old Caucasian man with large cell lung carcinoma with giant cells who had received chemo- and radiotherapy; described to be PAS negative, to exhibit remarkable stroma formation and to overexpress the proto-oncogene c-myc; received at DSMZ at passage #18, frozen at passage #34
References:	Bepler et al., Differentiation 37: 158-171 (1988); Bepler et al., Cancer Genet. Cytogenet. 84: 39-45 (1995)
Depositor:	Dr. G. Bepler, Duke University Medical Center, Durham, North Carolina, USA
DSMZ Cell Culture Data:	
Morphology:	pleomorphic, gigantic, adherent cells growing as monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent cultures 1:2 to 1:3 twice a week using trypsin/EDTA; seed out at ca. 1.5×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	cell harvest of ca. $9-13 \times 10^6$ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human, highly rearranged flat-moded hexaploid/heptaploid karyotype with 4% polyploidy; 130-160<7n>XXY, -X, -Y, -Y, +1, +1, -2, -2, -3, -6, -6, -8, -8, -9, -9, -9, -10, -13, -13, -13, -14, -14, +15, +16, +16, -17, -18, -19, -19, -20, -20, -20, -20, +18-25mar, del(1)(p22.2), add(1)(q11-21), i(1p)x2, add(3)(p23-24)x2, del(5)(q11q15), i(5q)x2, add(5)(p?), add(6)(q11)x2-3, add(7)(q34)x2, del(7)(q22)x2, i(8q), del(9)(p21)x2-3, del(10)(q23-24)x2, der(10;13)(q10;q10)x2, add(11)(q24-25)x2, del(11)(q23)x2, add(13)(p11), add(16)(q24); variable number of dmin present; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LCLC-97TM1
Cell Type:	human large cell lung carcinoma
DSMZ No:	ACC 388
Origin:	cell line represents the xenotransplant of a primary tumor which was obtained from a 44-year-old Caucasian man with large cell lung carcinoma who did not have any prior therapy; described to express c-myc
References:	Bepler et al., Differentiation 37: 158-171 (1988)
Depositor:	Dr. G. Bepler, Duke University Medical Center, Durham, North Carolina, USA
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells, densely packed in islets, growing as monolayers
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split confluent culture 1:3 to 1:5 once or twice a week using trypsin/EDTA before reaching complete confluency; medium (culture supernatant) should be exchanged at least every second day (even without splitting the cells); seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of >50 hours
Harvest:	cell harvest of ca. 10×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human hyperdiploid/hypotriploid karyotype; 58(50-60)<3n>X, -X/Y, -X/Y, +1, +1, +5, +5, -7, -10, -11, +12, -13, -15, -15, -15, -17, -18, -18, -19, -20, -21, -21, -22, der(1;7)(q10;q10)x2, der(1;15)(p10;q10)x2, der(3;6)(q10;q10), der(5;18)(p10;p10), der(5;18)(q10;q10), der(6;10)(p10;q10), i(7p)x2, i(8p), i(8q)x2, der(11)t(1;11)(q11;q24), der(14)t(7;14)(q12;q32), der(14)t(7;14)(q12;q32)add(14)(p11), der(14)t(7;14)(q12;q32)t(14;21)(p10;q10); carries an unprecedented cluster of whole-arm translocations; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LF-CL2A
Cell Type:	rat hepatocarcinoma
DSMZ No:	ACC 195
Origin:	a primary liver carcinoma was induced by dimethyl-aminoazobenzene (DAB) in a Wistar-Commentry rat; the hepatoma was transferred to a syngeneic rat and a cell line was established and subcloned in 1966
References:	Hatzfeld et al., Cancer Res. 38: 16-22 (1978); Cassigena et al., Int. J. Cancer 50: 238-245 (1992); LaFarge-Frayssinet et al., in: Liver Cells and Drugs (ed. A. Guillouzo, John Libbey Eurotext, 1988, Vol. 164, p. 465-471)
Depositor:	Dr. C. Frayssinet, CNRS, Villejuif, France
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing as monolayers
Medium:	90-95% Dulbecco's MEM + 5-10% FBS
Subculture:	split confluent culture about every 2 days using trypsin/EDTA; cells grow quickly; seed about 1 x 10 ⁶ cells into a 175 cm ² flask
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 15 hours
Harvest:	cell harvest of about 30 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2.5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as rat with IEF of AST, G6PD, MDH, PEP B
Cytogenetics:	rat flat-moded near-triploid karyotype; 58-67<3n>; large submetacentric marker present in most cells
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LN-405
Cell Type:	human astrocytoma
DSMZ No:	ACC 189
Origin:	glioblastoma cells established from an astrocytoma tumor (grade IV) of a 62-year-old woman in 1986; cells were described as expressing mRNA for both TGF- β 1 and G-TSF/TGF- β 2 (but only the latter polypeptide was secreted) and expressing GFAP protein
References:	Bodmer et al., J. Immunol. 143: 3222-3229 (1989)
Depositor:	Dr. De Tribolet, University Hospital Lausanne, Lausanne, Switzerland
DSMZ Cell Culture Data:	
Morphology:	fibroblastic, grows as monolayers in culture
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:2 to 1:3 every 4-6 days using trypsin/EDTA; cells grow rather slowly
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 60 hours
Harvest:	cell harvest of about 3-4 x 10 ⁶ cells/175 cm ² or 1.5 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1.0-1.5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, G6PD, MDH
Cytogenetics:	human hypertetraploid karyotype with 37% polyploidy and with 1-3 supernumerary dm present in most cells; 102-108<4n>XXXX, +1, +2, +3, +4, +5, +7, -10, +11, +13, -16, -16, -16, +19, +20, +20, +21, +22, +8mar, del(1)(q41)x2, der(1)t(1;3)(p32;p11), add(4)(p11), add(7)(p14), add(10)(p13), i(13q)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LNCAP
Cell Type:	human prostate carcinoma
DSMZ No:	ACC 256
Origin:	established from the left supraclavicular lymph node metastasis from a 50-year-old man with prostate carcinoma in 1977; cells were described to be androgen-sensitive
References:	Horoszewicz et al., in "Models for Prostate Cancer" (ed. G.P. Murphy), Alan R. Liss, New York, 1981, p. 115-132; Horoszewicz et al., Cancer Res. 43: 1809-1818 (1993); Gibas et al., Cancer Genet. Cytogenet. 11: 399-404 (1984)
Depositor:	Prof. M. Motta, Universita di Milano, Milan, Italy
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblastoid cells growing in aggregates and as single cells
Medium:	85-90% RPMI 1640 + 10-15% FBS
Subculture:	split semi-confluent culture 1:4 once a week using trypsin/EDTA (occasionally cells can be detached by tapping); seed out at ca. $1-2 \times 10^6$ cells/25 cm ² ; after thawing and trypsinization cells may need 1-2 days to become adherent again; due to the strong cell aggregation, it might be difficult to determine an exact cell titer
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 60 hours
Harvest:	cell harvest of about $10-15 \times 10^6$ cells/80 cm ² flask
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypotetraploid karyotype with 6% polyploidy; 88(84-93)<4n>XXYY, -2, -4, -6, -6, -13, -13, +2mar, t(1;15)(p22.3;q22.3)x2, del(2)(p16p23), del(10)(q24)x2, der(16)t(6;16)(p22.1;q22)x2; del(10)(q24) is recurrent in prostate carcinoma; corresponds to published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LOU-NH91
Cell Type:	human lung squamous cell carcinoma
DSMZ No:	ACC 393
Origin:	established from the lower right lung lobe removed two months after diagnosis of highly differentiated squamous cell lung carcinoma with metastasis to one lymph node (no visceral involvement) in a 64-year-old woman (without any prior therapy)
References:	Drouin et al., Genes Chromosomes Cancer 7: 209-212 (1993)
Depositor:	Dr. F. Viguié, Hopital Hotel Dieu, Paris, France
DSMZ Cell Culture Data:	
Morphology:	adherent ovoid-to-epitheloid cells growing in monolayers
Medium:	80% RPMI 1640 + 20% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:2 to 1:4 every 5-10 days using trypsin/EDTA (cells require longer than usual to detach after trypsinization, ca. 5 min); seed out at ca. $0.4-0.8 \times 10^6$ cells/175 cm ² ; doubling time may initially be longer after recovery from N ₂
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 80 hours
Harvest:	cell harvest of ca. 0.1×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hyperdiploid karyotype with declining hyperhaploid sideline (2%) and 9% polyploidy; 50-53<2n>XX/XXXX, +5, +5, +7, +7, -13, +17, -18, +22, +22, +1-2mar, del(4)(q27), ?add(22)(q13)x2 (major clone); 54<2n>XXXX, +5, +5, +7, +7, +22, +22 (minor clone); 27<n>XX, +5, +7, +22; matches published karyotype; unstable karyotype with recent gain of structural rearrangements
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LOUCY
Cell Type:	human T cell leukemia
DSMZ No:	ACC 394
Origin:	established from the peripheral blood of a 38-year-old woman with T cell acute lymphoblastic leukemia, T-ALL (chemotherapy-resistant, remission was never achieved)
References:	Ben-Bassat et al., Cancer Genet. Cytogenet. 49: 241-248 (1990)
Depositor:	Prof. H. Ben-Bassat, Hadassah University Hospital, Jerusalem, Israel
DSMZ Cell Culture Data:	
Morphology:	round, rather small cells growing singly in suspension
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	maintain at $0.5\text{--}1.0 \times 10^6$ cells/ml; split saturated culture 1:2 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60 hours
Harvest:	maximal density at ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3\text{--}4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3+, CD4-, CD5+, CD6+, CD7+, CD8-, CD13-, CD19-, CD34-, TCRalpha/beta-, TCRgamma/delta+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human hypodiploid karyotype with 16% polyploidy; $45\text{--}2n\text{--}X, -X, \text{del}(1)(p3?2p3?4), \text{del}(5)(q14\text{--}15q34\text{--}35), \text{t}(16;20)(p1?1;q1?3)$; sideline with $\text{del}(6)(q23)$; 1p32, 1p34 and 6q- recurrent in T-ALL; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LOVO
Cell Type:	human colon adenocarcinoma
DSMZ No:	ACC 350
Origin:	established from the metastatic nodule resected from a 56-year-old Caucasian man with colon adenocarcinoma in 1972; cells were described to produce CEA
References:	Drewinko et al., Exp. Cell. Res. 101: 414-416 (1976); Drewinko et al., J. Nat. Cancer Inst. 61: 75-83 (1978); Drewinko et al., Cancer Res. 39: 2630-2636 (1979)
Depositor:	Dr. R. Supino, Istituto Nazionale Tumori, Milano, Italia
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial-like cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:5 twice a week using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of ca. 35×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, desmin-, endothel-, GFAP-, neurofilament-, vimentin(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hyperdiploid karyotype with 16% polyploidy; 48/49<2n>XY, +7, +1-2mar, t(2;12)(q21;p11); +7 recurrent in colon carcinoma; 1-2 small satellited markers present; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LP-1
Cell Type:	human multiple myeloma
DSMZ No:	ACC 41
Origin:	established from the peripheral blood of a 56-year-old woman with multiple myeloma (IgG, EBV-negative) in leukemic transformation (refractory, terminal) in 1986
References:	Pegoraro et al., Blood 73: 1020-1027 (1989); review article: Drexler & Matsuo, Leukemia Res 24: 681-703 (2000)
Depositor:	Dr. L. Pegoraro, University of Turin, Turin, Italy
DSMZ Cell Culture Data:	
Morphology:	single, elongated, snake-like cells in suspension
Medium:	80-90% Iscove's MDM + 10-20% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml, cells grow slowly, split 1:2 every 2-3 days; viability may be low initially after thawing; it is recommended to cultivate the cells in 24-well-plates during the first week and to exchange medium without dilution; seed out at ca. 0.7×10^6 cells/well; cells may be partly (lightly) adherent but can be flushed off by pipetting
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of 35-80 hours (depending on cell density)
Harvest:	maximal density at about $1.0-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19-, CD20-, CD34-, CD37-, CD79a-, cyCD79a+, CD80-, CD138-, HLA-DR+, sm/cyIgG+, sm/cyIgM-, sm/cykappa-, sm/cylambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human flat-moded aneuploid hypertriploid karyotype; 79(72-79)<3n>XX, -X,-1, +3, +4, +5, +6, +8, +11, +12, -13, +14, +15, +16, +18, -19, +20, +21, +22, +dminx1-4, add(1)(p11-21), del(3)(q22), der(4)add(4)(p1?)t(4;11)(11;?)(q33;q13)(q24;?)x2, der(5)t(1;5)(p11;q35), del(6)(q15)x1-2, del(8)(p21)x1-2, add(9)(q34-35)x1-2, t(12;13)(p12;q11)x2; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	LXF-289
Cell Type:	human lung adenocarcinoma
DSMZ No:	ACC 265
Origin:	established from the lung adenocarcinoma of a 63-year-old man
References:	Hilbert et al., J. Cancer Res. Clin. Oncol. 115: S50 (1989)
Depositor:	obtained from DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing as monolayer
Medium:	90% Ham's F10 + 10% FBS
Subculture:	split confluent culture 1:4 every 3 days using trypsin/EDTA; seed out at ca. 1.5×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 50 hours
Harvest:	cell harvest of about 15×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human flat-moded near triploid karyotype with 4% polyploidy; 69(62-76)<3n>XY, -X/Y, -1, +6, +9, -13, -14, -18, -19, +22, +3mar, add(1)(p21), i(1q), der(2)t(1;2)(p21;p23)x2, add(4)(p11/15), i(6p), del(12)(q14), i(13q), add(14)(p11), add(15)(p11), del(17)(p11)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	M-07e
Cell Type:	human acute megakaryoblastic leukemia
DSMZ No:	ACC 104
Origin:	established from the peripheral blood of a 6-month-old girl with acute megakaryoblastic leukemia (AML M7) at diagnosis in 1987; subline of the growth factor-independent cell line M-07; M-07e cells respond proliferatively to GM-CSF, IFN- α , IFN- β , IFN- γ , IL-2, IL-3, IL-4, IL-6, IL-15, NGF, SCF, TNF- α , TPO; we have noted that the cell line can become very quickly cytokine-independent (within 3-4 weeks), presumably due to outgrowth of independent cells
References:	Avanzi et al., Brit. J. Haematol. 69: 359-366 (1988); Avanzi et al., J. Cell. Physiol. 145: 458-464 (1990); Drexler et al., Leukemia 11: 701-708 (1997)
Depositor:	Dr. L. Pegoraro, University of Turin, Turin, Italy
DSMZ Cell Culture Data:	
Morphology:	single round cells in suspension, a few cells are lightly adherent
Medium:	80-90% Iscove's MDM or RPMI 1640 + 10-20% FBS + IL-3 (10 ng/ml) or GM-CSF (10 ng/ml) or 10-20% vol conditioned medium of cell line 5637 (DSM ACC 35)
Subculture:	cells are difficult to culture! maintain at $0.5-1.0 \times 10^6$ cells/ml, split 1:2 to 1:3 every two days; cells grow slowly; seed out at about 4×10^5 cells/ml. CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 32-46 hours
Harvest:	cell harvest of a. 0.5×10^6 cells/ml; maximal density at ca. 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD13+, CD14-, CD15(+), CD19-, CD33+, CD34-, CD41+, CD42+, GlyA-, HLA-DR-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human near-diploid karyotype; 46(45-46)<2n>XX, t(11;21)(p11;p13), add(13)(p13), add(22)(p13)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	M1
Cell Type:	mouse myeloid leukemia
DSMZ No:	ACC 331
Origin:	bone marrow from SL mouse with spontaneous myeloid leukemia was injected into isogenic mice and the cell line was established from the ensuing hyperplastic lymph nodes 14 days later by recloning on soft agar; homotransplantable and forms colonies on soft agar (granulocytic/macrophage differentiation inducible by IL-3, LPS or dexamethasone)
References:	Ichikawa, J. Cell. Physiol. 74: 223-234 (1969)
Depositor:	obtained from RIKEN Cell Bank (RCB0169), Tsubuka Science City, Ibaraki, Japan
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly in suspension
Medium:	80-90% mixture of MEM with Earle's salts + Dulbecco's MEM (at 1:1) + 10-20% FBS + 2x non-essential amino acids + 2x vitamins
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; optimal split ratio of 1:2 to 1:4 every 3-5 days
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 25-50 hours
Harvest:	saturation density at ca. 2.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as murine with IEF of MDH, NP
Cytogenetics:	murine hypodiploid karyotype with 20% polyploidy; 30(29-30)<2n>XY, 6-8 centric fusion markers present
Viruses:	ELISA: reverse transcriptase positive

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	M3E3/C3
Cell Type:	Syrian hamster fetal lung
DSMZ No:	ACC 340
Origin:	established from the lung of a Syrian golden hamster fetus on day 15 after gestation in 1981; primary culture cells were treated with hydroxy-proline; cells were described as premature epithelial cells with the ability to differentiate into mucus-producing pulmonary cells
References:	Emura et al., J. Cancer Res. Clin. Oncol. 104: 133-144 (1982)
Depositor:	Prof. Dr. M. Emura, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS + 1 mM sodium pyruvate
Subculture:	split confluent culture 1:3 to 1:6 every 3-4 days using trypsin/EDTA; seed out at $1-2 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	cell harvest of ca. 8×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as hamster with IEF of AST, MPI, NP
Cytogenetics:	near-diploid Syrian hamster karyotype with 10% polyploidy; 43-47<2n>
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MB-020
Cell Type:	insect - cabbage moth (<i>Mamestra brassicae</i>)
DSMZ No:	ACC 79
Origin:	established from the larvae of the butterfly <i>Mamestra brassicae</i> (family Noctuidae, subfamily Noctuinae) (Kohleule)
References:	Lehmann & Weilepp, Arch. Phytopathol. Pflanzenschutz 25: 387-401 (1989)
Depositor:	Dr. H. U. Leistner, Institut für Pathogendiagnostik, Aschersleben, Germany
DSMZ Cell Culture Data:	
Morphology:	polymorphic round cells, partly adherent
Medium:	SF-900 Insect-X Press (protein-free medium) (no FBS)
Subculture:	cells are difficult to culture! split confluent cultures 1:2 every day (cells grow partly adherent, but trypsin is not necessary); maintain at room temperature (about 20 °C) without CO ₂ exposure (keep lid of flask closed); confluent cultures can be stored at 4 °C for up to one week without loss of viability
Incubation:	at 20 °C without CO ₂ (at room temperature; keep lid of flask closed)
Harvest:	about 5-10 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS and 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as not murine and not human/primate with IEF of AST, PEP B

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MB-021
Cell Type:	insect - cabbage moth (<i>Mamestra brassicae</i>)
DSMZ No:	ACC 85
Origin:	established from the larvae of the butterfly <i>Mamestra brassicae</i> (family Noctuidae, subfamily Noctuinae) (Kohleule)
References:	Lehmann & Weilepp, Arch. Phytopathol. Pflanzenschutz 25: 387-401 (1989)
Depositor:	Dr. H. U. Leistner, Institut für Pathogendiagnostik, Aschersleben, Germany
DSMZ Cell Culture Data:	
Morphology:	polymorphic round cells, partly adherent
Medium:	85% Grace's Insect Medium (Modified) + 15% FBS
Subculture:	split confluent cultures 1:2 about twice a week, but without using trypsin (cells grow partly adherent, but can be detached without trypsin); maintain at room temperature (about 20 °C) without CO ₂ exposure (keep lid of flask closed); confluent cultures can be stored at 4 °C for up to one week without loss of viability
Incubation:	at 20 °C without CO ₂ (at room temperature, keep lid of flask closed)
Harvest:	about 5-7 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MB-03
Cell Type:	insect - cabbage moth (<i>Mamestra brassicae</i>)
DSMZ No:	ACC 97
Origin:	established from the larvae of the butterfly <i>Mamestra brassicae</i> (family Noctuidae, subfamily Noctuinae) (Kohleule)
References:	Lehmann & Weilepp, Arch. Phytopathol. Pflanzenschutz 25: 387-401 (1989)
Depositor:	Dr. H. U. Leistner, Institut für Pathogendiagnostik, Aschersleben, Germany
DSMZ Cell Culture Data:	
Morphology:	polymorphic round cells, partly adherent
Medium:	90% Grace's Insect (Modified) Medium + 10% FBS
Subculture:	split cultures 1:2 to 1:3 every 3-5 days (cells grow partly adherent, but trypsin is not necessary); maintain at room temperature (about 20 °C) without CO ₂ exposure (keep lid of flask closed); confluent cultures can be stored at 4 °C for up to one week without loss of viability
Incubation:	at 20 °C without CO ₂ (at room temperature, keep lid of flask closed)
Harvest:	about 5-10 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MB-04
Cell Type:	insect - cabbage moth (<i>Mamestra brassicae</i>)
DSMZ No:	ACC 122
Origin:	established from the larvae of the butterfly <i>Mamestra brassicae</i> (family Noctuidae, subfamily Noctuinae) (Kohleule)
References:	Lehmann & Weilepp, Arch. Phytopathol. Pflanzenschutz 25: 387-401 (1989)
Depositor:	Dr. H. U. Leistner, Institut für Pathogendiagnostik, Aschersleben, Germany
DSMZ Cell Culture Data:	
Morphology:	polymorphic round cells, partly adherent
Medium:	90% Grace's Insect (Modified) Medium + 10% FBS
Subculture:	cells are difficult to culture! split cultures 1:2 every 2-3 days (cells grow partly adherent, but trypsin is not necessary); maintain at room temperature (about 20 °C) without CO ₂ exposure (keep lid of flask closed); confluent cultures can be stored at 4 °C for up to one week without loss of viability; after thawing cells might have a lag phase of up to 4 weeks
Incubation:	at 20 °C (room temperature) without CO ₂ (keep lid of flask closed)
Harvest:	about 5-7 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed to be not of murine or primate origin with IEF of AST, PEP B

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MB-L11
Cell Type:	insect - cabbage moth (Mamestra brassicae)
DSMZ No:	ACC 113
Origin:	established from the larvae of the butterfly Mamestra brassicae (family Noctuidae, subfamily Noctuinae) (Kohleule)
References:	Lehmann & Weilepp, Arch. Phytopathol. Pflanzenschutz 25: 387-401 (1989)
Depositor:	Dr. H. U. Leistner, Institut für Pathogendiagnostik, Aschersleben, Germany
DSMZ Cell Culture Data:	
Morphology:	polymorphic round cells, partly adherent
Medium:	90% Grace's Insect (Modified) Medium + 10% FBS
Subculture:	split confluent cultures 1:2 to 1:3 every 3-5 days (cells are partly adherent, but trypsin is not necessary); maintain at room temperature (about 20 /C) without CO ₂ exposure (keep lid of flask closed); confluent cultures can be stored at 4 /C for up to one week without loss of viability
Incubation:	at 20 /C without CO ₂ (at room temperature, keep lid of flask closed)
Harvest:	about 5-10 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	cells are not of murine or primate origin as detected with IEF of AST, PEP B

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MB-L2
Cell Type:	insect - cabbage moth (<i>Mamestra brassicae</i>)
DSMZ No:	ACC 76
Origin:	established from the larvae of the butterfly <i>Mamestra brassicae</i> (family Noctuidae, subfamily Noctuinae) (Kohleule)
References:	Lehmann & Weilepp, Arch. Phytopathol. Pflanzenschutz 25: 387-401 (1989)
Depositor:	Dr. H. U. Leistner, Institut für Pathogendiagnostik, Aschersleben, Germany
DSMZ Cell Culture Data:	
Morphology:	polymorphic round cells, partly adherent
Medium:	90% Grace's Insect (Modified) Medium + 10% FBS
Subculture:	split confluent cultures 1:2 to 1:3 every 3-5 days (cells are partly adherent, but trypsin is not necessary); maintain at room temperature (about 20 °C) without CO ₂ exposure (keep lid of flask closed); confluent cultures can be stored at 4 °C for up to one week without loss of viability
Incubation:	at 20 °C without CO ₂ (at room temperature, keep lid of flask closed)
Harvest:	about 5-10 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MC-116
Cell Type:	human B cell lymphoma
DSMZ No:	ACC 82
Origin:	established from the pleural effusion of a patient with B cell lymphoma (undifferentiated lymphoma); cells were described to be EBV-negative, to carry the 8;14 translocation and to express mRNA of the proto-oncogene bcl-2
References:	Magrath et al., J. Natl. Cancer Inst. 64: 465-475 and 477-483 (1980)
Depositor:	obtained from ATCC (CRL 1649), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	round to elongated, lymphoblastoid, single or clustered cells in suspension
Medium:	80-85% RPMI 1640 + 15-20% FBS
Subculture:	maintain at $1-2 \times 10^6$ cells/ml; split ratio of 1:2 to 1:3; seed out at ca. 1×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximal density at 2.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD3-, CD10(+), CD13(+), CD19+, CD20+, CD34-, CD37+, CD138-, HLA-DR+, sm/cyIgM+, sm/cyIgG-, sm/cykappa-, sm/cylambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human near diploid karyotype with 1.5% polyploidy; 46(37-46)<2n>XY, dup(1)(q21q32), t(8;14)(q24;q32), del(10)(q23)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MC3T3-E1
Cell Type:	mouse embryo/fetus calvaria fibroblasts
DSMZ No:	ACC 210
Origin:	established from the calvaria of an embryo/fetus C57BL/6 mouse; described to differentiate to osteoblasts and to produce collagen
References:	Kodama et al., Jpn. J. Oral Biol. 23: 899-901 (1981); J. Cell Biochem. 69: 191-198 (1983)
Depositor:	obtained from the RIKEN Cell Bank (RCB1126), Tsubuka Science City, Ibaraki, Japan
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblast-like cells growing as monolayer
Medium:	80-90% alpha-MEM + 10-20% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:3 to 1:4 once or twice a week using trypsin/EDTA; seed out at about 1-2 x 10 ⁶ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 2-3 days
Harvest:	cell harvest of ca. 6-7 x 10 ⁶ cells/80 cm ² or 9 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at ca. 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, MDH, NP
Cytogenetics:	murine hypotetraploid karyotype with 11.5% polyploidy; 72(68-80)<4n>; centric fusion absent
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MCF-7
Cell Type:	human breast adenocarcinoma
DSMZ No:	ACC 115
Origin:	established from the pleural effusion of a 69-year-old Caucasian woman with metastatic mammary carcinoma (after radio- and hormone therapy) in 1970; cells were described of being positive for cytoplasmic estrogen receptors and having the capability to form domes
References:	Soule et al., J. Natl. Cancer Inst. 51: 1409-1413 (1973)
Depositor:	obtained from ATCC (HTB 22), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing as monolayers
Medium:	90% RPMI 1640 + 10% FBS + MEM non-essential amino acids + 1 mM sodium pyruvate + 10 : g/ml bovine insulin
Subculture:	split confluent culture 1:2 to 1:4 every 2-6 days using trypsin/EDTA; seed out at ca. $0.1-0.2 \times 10^5$ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours (range 30-72 hours)
Harvest:	about $20-25 \times 10^6$ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypotetraploid karyotype with 8% polyploidy; 86(80-87)<4n>XX, -X, -X, -1, -1, +6, +7, -9, -10, -10, -12, -13, -13, +14, -17, -17, -18, -18, -20, -20, -20, -22, +6-8mar, add(X)(q27), add(2)(q35-37)x1-2, add(5)(p15), del(6)(q25)x1-2, add(7)(p11), add(7)(p14;hsr), add(10)(q23), del(11)(q21q23), add(15)(p11), i(15q), add(16)(q24), add(19)(q13), add(19)(p13)x1-2; corresponds to published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MDA-MB-453
Cell Type:	human breast carcinoma
DSMZ No:	ACC 65
Origin:	established from an effusion of a 48-year-old woman with breast carcinoma in 1976
References:	Cailleau et al., In Vitro 14: 911-915 (1978); Cailleau et al., Cancer Res. 39: 919-922 (1979)
Depositor:	Dr. H. Kirchner, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like, round, loosely attached monolayers
Medium:	90% Leibovitz's L-15 medium + 10% FBS
Subculture:	cells are difficult to culture! split confluent culture at 1:2 to 1:6 using trypsin/EDTA; cells can partially be detached by tapping the flask; cells may need several hours to readhere again; seed out at ca. 4×10^6 cells/80 cm ²
Incubation:	at 37 °C, no exposure to CO ₂ (keep lid of flask closed)
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	about $25-35 \times 10^6$ cells/80 cm ²
Storage:	frozen with 85% medium, 10% FBS, 5% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human hypotetraploid karyotype with 6% polyploidy and near diploid sideline (2%); 89(85-91)<4n>XXX, -X; highly rearranged leading to extensive marker formation, including der(5)t(5;?14)(p11;q11)x2; dm present in 12% cells
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MDBK
Cell Type:	bovine kidney cells
DSMZ No:	ACC 174
Origin:	cells were derived from a kidney of a normal adult steer in 1957; cells are BVD-free (bovine diarrhea virus)
References:	Madin & Darby, Proc. Soc. Exp. Biol. Med. 98: 574 (1958)
Depositor:	Dr. R. Riebe, Friedrich-Loeffler-Institut, BFA, Insel Riems, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial-like cells
Medium:	95% Dulbecco's MEM + 5% FBS
Subculture:	optimal split ratio at 1:4 to 1:6 using trypsin/EDTA; cells grow within 2-3 days to confluence; seed out $1-2 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	cell harvest of about $5-10 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as bovine with IEF of AST, G6PD, NP
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	ME-1
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 537
Origin:	established from the peripheral blood of a 40-year-old man with acute myeloid leukemia (AML) FAB M4eo at second relapse in 1988; cells were described to carry the inv(16)(p13q22) leading to the fusion gene CBFB-MYH11
References:	Yanagisawa et al., Blood 78: 451-457 (1991); Van Dongen et al., Leukemia 13: 1901-1928 (1999)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 every 3-5 days; seed out at ca. $0.5-1.0 \times 10^6$ cells/ml; maintain at ca. $0.5-1.0 \times 10^6$ cells/ml; minimum density at ca. 0.5×10^6 cells/ml; cells should be cultured in 24-well-plates
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 3-5 days
Harvest:	maximum cell density of ca. 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD4+, CD13+, CD14-, CD15+, CD16-, CD19-, CD33+, CD34+, CD68-, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Cytogenetics:	presence of CBFB-MYH11 fusion gene confirmed by RT-PCR
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MEC-1
Cell Type:	human chronic B cell leukemia
DSMZ No:	ACC 497
Origin:	established in 1993 from the peripheral blood of a 61-year-old Caucasian man with chronic B cell leukemia (B-CLL in polymphocytoid transformation to B-PLL); serial sister cell line of MEC-2 (DSM ACC 500)
References:	Stacchini et al., Leukemia Res. 23: 127-136 (1999)
Depositor:	Prof. F. Caligaris-Cappio, Hospital Mauriziano Umberto, Turin, Italy
DSMZ Cell Culture Data:	
Morphology:	round to polymorphic cells growing in suspension, singly or partly in small aggregates, a few cells are slightly adherent
Medium:	90% Iscove's MDM + 10% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml; after thawing, during the first week cells should be kept with 20% FBS initially; culture in 24-well-plate is of advantage; maintain at about $0.5\text{--}2.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	cell harvest of about 2.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a-, cyCD79a+, CD80-, CD138-, HLA-DR+, cy/sIgG-, cy/sIgM+, cy/smkappa+, cy/smlambda-
Fingerprinting:	same DNA profile as MEC-2 using multiplex PCR of minisatellite markers
Species:	confirmed as human by cytogenetics
Cytogenetics:	human near-diploid karyotype with 10% polyploidy; 46(44-47)<2n>XY, -2, +7, -12, +1-2mar, t(1;6)(q22-23;p21), add(7)(q11), der(10)(10pter->q22::?:2q11->qter), del(17)(p11); small acf/mar present in most cells; resembles published karyotype
Viruses:	PCR: EBV+, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MEC-2
Cell Type:	human chronic B cell leukemia
DSMZ No:	ACC 500
Origin:	established in 1994 from the peripheral blood of a 62-year-old Caucasian man with chronic B cell leukemia (B-CLL in polymphocytoid transformation to B-PLL); serial sister cell line of MEC-1 (DSM ACC 497)
References:	Stacchini et al., Leukemia Res. 23: 127-136 (1999)
Depositor:	Prof. F. Caligaris-Cappio, Hospital Mauriziano Umberto, Turin, Italy
DSMZ Cell Culture Data:	
Morphology:	polymorphic cells growing in suspension in large clumps; clumps can easily be segregated
Medium:	90% Iscove's MDM + 10% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2-3 days; seed out at ca. $0.3-0.5 \times 10^6$ cells/ml; after thawing, during the first week cells should be kept with 20% FBS initially; culture in 24-well-plate is of advantage; maintain at about 0.5×10^6 cells/ml; viability may never exceed 80%
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	maximal density at about 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a+, cyCD79a+, CD80+, CD138(+), HLA-DR+, cy/smlgG-, cy/smlgM+, cy/smkappa+, cy/smlambda-
Fingerprinting:	same DNA profile as MEC-1 using multiplex PCR of minisatellite markers
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hypertriploid karyotype with 7% polyploidy; 75-80<3n>XX/XXY, -6, -9, -10, +11, +12, -13, +15, -17, -21, +10mar; add(X)(p22)x2, inv(12)(p12.2?q24.1)x2, add(18)(q2?1); karyotypically distinct from MEC-1 (as reported)
Viruses:	PCR: EBV+, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MEG-01
Cell Type:	human chronic myeloid leukemia in megakaryocytic blast crisis
DSMZ No:	ACC 364
Origin:	established from the bone marrow of a 55-year-old man with chronic myeloid leukemia (CML) in megakaryocytic blast crisis in 1983
References:	Ogura et al., Blood 66: 1384-1392 (1985); Ogura et al., Blood 72: 49-60 (1988); review article: Drexler, Leukemia Res. 18: 919-927 (1994); review article: Drexler et al., Leukemia Res. 24: 109-115 (2000)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	growing as single cells in suspension, some cells lightly adherent
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:3 to 1:4 every 3 days; seed out at ca. 0.3×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	maximal density at ca. 1.3×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD13+, CD14+, CD15+, CD19-, CD33+, CD34(+), CD41-, CD42-, CD61-, CD68(+), CD71-, GlyA-, HLA-DR-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hyperdiploid karyotype with 12% polyploidy; 54(53-56)<2n>XY, +6, +19, +19, +21, +3-4mar, t(1;15)(p13;p13), ?inv(3)(p25q26), i(4q), add(5)(p15), der(9)t(9;22)(q34;q11)x2, add(10)(p14), dup(13)(q13q33-34), add(14)(p11), der(22)t(9;22)(q34;q11); r/dmin x1 present at 57%; sideline with i(5)(q10)del(5)(q11q13) instead of add(5); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MEL-745A cl. DS19
Cell Type:	mouse erythroleukemia
DSMZ No:	ACC 501
Origin:	cells were transformed by Friend virus
References:	Singer et al., Proc. Natl. Acad. Sci. USA 71: 2668-2670 (1974)
Depositor:	Dr. D. B. Krimer, Centro de Investigaciones Biologicas, Madrid, Spain
DSMZ Cell Culture Data:	
Morphology:	round, single or in small clusters, growing in suspension, about 5% giant cells
Medium:	80% Dulbecco's MEM + 20% FBS + 2 mM L-glutamine
Subculture:	split saturated culture 1:8 to 1:10 every 3 days; seed out at ca. $5 \cdot 10^4$ cells/ml; maintain at about $0.05\text{--}0.5 \times 10^6$ cells/ml; during continuous culture, the cell density should not be higher than 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 20 hours
Harvest:	maximal density at about 0.8×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Species:	confirmed as murine by beta-globin and alu PCR
Cytogenetics:	murine hypertetraploid karyotype

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MEL-HO
Cell Type:	human melanoma
DSMZ No:	ACC 62
Origin:	established from the primary tumor of a woman with melanoma in 1976; cells express mRNA of the oncogene c-myc
References:	Holzmann et al., Int. J. Cancer 41: 542-547 (1988)
Depositor:	Dr. H. W. L. Ziegler-Heitbrock, Institute of Immunology, University of Munich, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells with long, narrow, often branching cell bodies
Medium:	90-95% RPMI 1640 + 5-10% FBS
Subculture:	split confluent culture 1:5 to 1:10 using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	about $10\text{--}12 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1\text{--}2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, desmin-, endothel-, GFAP-, HMB-45+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, PEP B
Cytogenetics:	human hypertriploid karyotype with 1.5% polyploidy; 74(72-76)<3n>XX, -X, +3, +7, +7, -8, -10, -11, +15, -16, +17, +18, +20, +21, +mar; del(1)(q42), del(3)(p21), i(7q)x2, del(9)(p13), der(15)t(15;?)(q21;?), i(17q), del(X)(q11)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MEL-JUSO
Cell Type:	human melanoma
DSMZ No:	ACC 74
Origin:	established from the primary tumor of a 58-year-old woman with melanoma in 1977
References:	Ziegler-Heitbrock et al., Cancer Res. 45: 1344-1350 (1985); Johnson et al., Eur. J. Immunol. 11: 825-831 (1981)
Depositor:	Dr. H. W. L. Ziegler-Heitbrock, Institute of Immunology, University of Munich, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:5 every 3-4 days using trypsin/EDTA; seed out at ca. 3×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 28 hours
Harvest:	about 10×10^6 cells/80 cm ² flask
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, HMB-45+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MFE-280
Cell Type:	human endometrium adenocarcinoma
DSMZ No:	ACC 410
Origin:	established from tumor tissue from a 77-year-old woman with recurrence of endometrial carcinoma (adenomatous, partly papillary, grade G3) in 1990; described as forming heterotransplantable tumors in nude mice and as carrying receptors for progesterone; received at passage #56, frozen at DSMZ at passage #60
References:	Hackenberg et al., J. Cancer Res. Clin. Oncol. 123: 669-673 (1997); Hackenberg et al., Anticancer Res. 18: 1153-1158 (1998)
Depositor:	Dr. R. Hackenberg, University Women's Hospital, Marburg, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently in monolayers
Medium:	40% RPMI 1640 + 40% MEM (with Earle's salts) + 20% FBS + 2 mM L-glutamine + 1x insulin-transferrin-sodium selenite
Subculture:	split confluent culture 1:2 to 1:6 every 3-5 days using trypsin/EDTA; seed out at $1-3 \times 10^6$ cells/25 cm ² ; about 4 hours after thawing, floating cells and medium should be decanted into a second flask
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60-90 hours
Harvest:	cell harvest of ca. $2-3 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hyperdiploid karyotype with 24% polyploidy; 52(48-52)<2n>X, -X, +1, +1, -2, +9, -10, -12, -13, -16, -17, -17, -18, -19, -20, -22, +18mar, add(1)(p34), add(1)(p3?), add(1)(q11), add(3)(q13), add(3)(q27-29), add(4)(p14), der(9)t(9;13)(p11;q11), add(15)(p11); near diploidy and gain of 1q typical in endometrial carcinoma
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MFE-296
Cell Type:	human endometrium adenocarcinoma
DSMZ No:	ACC 419
Origin:	established from the primary tumor of a 68-year-old Caucasian woman with moderately differentiated endometrial adenocarcinoma (grade G2) in 1991; cells were described to express androgen receptors (androgen was reported to inhibit cell proliferation), but no estrogen receptors; cells were described to be tumorigenic in nude mice
References:	Hackenberg et al., Int. J. Cancer 57: 117-122 (1994); Hackenberg et al., J. Cancer Res. Clin. Oncol. 123: 669-673 (1997)
Depositor:	Dr. R. Hackenberg, University Women's Hospital, Marburg, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like polymorphic cells growing adherently in monolayers
Medium:	40% RPMI 1640 + 40% MEM (with Earle's salts) + 20% FBS + 2 mM L-glutamine + 1x insulin-transferrin-sodium selenite
Subculture:	split confluent culture 1:3 to 1:5 once or twice a week using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C without CO ₂ (keep lid of flask closed)
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	cell harvest of ca. 10×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, NP
Cytogenetics:	human diploid karyotype with 20% polyploidy; 46<2n>XX, no consistent abnormality detected; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MFE-319
Cell Type:	human endometrium adenocarcinoma
DSMZ No:	ACC 423
Origin:	established from the primary adenosquamous endometrium carcinoma (grading G1-G2) of an 81-year-old woman in 1992; described to form poorly differentiated tumors in nude mice
References:	Hackenberg et al., J. Cancer Res. Clin. Oncol. 123: 669-673 (1997)
Depositor:	Dr. R. Hackenberg, University Women's Hospital, Marburg, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells growing as monolayer
Medium:	40% RPMI 1640 + 40% MEM (with Earle's salts) + 20% FBS
Subculture:	split confluent culture 1:2 to 1:3 every 5 days using trypsin/EDTA (cells are difficult to detach from the flask; incubation with trypsin/EDTA of 15 min or longer at 37 °C is recommended); seed out at ca. 1-2 x 10 ⁶ cells/25 cm ² ; maximum density of ca. 20 x 10 ⁶ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 100 hours
Harvest:	cell harvest of ca. 15 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypodiploid karyotype with 4% polyploidy; 45(43-46)<2n>X, -X, t(13;18)(q14.3-21;q11-12); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MFM-223
Cell Type:	human breast carcinoma
DSMZ No:	ACC 422
Origin:	established from the pleural effusion of a post-menopausal breast cancer patient without any prior treatment (tumor: widespread ductal mammary carcinoma T4 N2 M1 with histological grade III); cells were described to express androgen, estrogen and progesterone receptors
References:	Hackenberg et al., Cancer Res. 51: 5722-5727 (1991)
Depositor:	Dr. R. Hackenberg, University Women's Hospital, Marburg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent cells growing as monolayer with reticulate growth pattern
Medium:	85% MEM (with Earle's salts) + 15% FBS + 2 mM L-glutamine + 1x insulin-transferrin-sodium selenite
Subculture:	cells grow very slowly; split confluent culture 1:2 every 3-4 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ² in 10 ml medium
Incubation:	at 37 °C without CO ₂ (keep lid of flask closed)
Doubling Time:	doubling time of ca. 96 hours
Harvest:	cell harvest of ca. 15×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-8+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human highly rearranged near-diploid karyotype with 12% polyploidy; 46(45-46)<2n>X, -X, -2, -5, -6, +7, -9, -10, +11, +12, -14, -15, -16, -17, -18, -19, -22, +10mar, add(5)(p15), der(7)?inv(7)(q1?q2?)dup(7)(q31q35), add(7)(q11;hsr), der(7)add(7)(p22)add(7)(q31-32), add(8)(p22-23), add(8)(q?24), del(10)(q2?), add(11)(q22), dup(12)(q13q14), add(13)(p11), add(15)(p11), add(17)(p1?), add(19)(q13), add(21)(p11), add(?22)(q13); carries multiple hsr including ch 7 material; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MH-7777A
Cell Type:	rat hepatoma
DSMZ No:	ACC 345
Origin:	established from the liver tumor of a Buffalo rat; tumor cells were propagated in the hind leg
References:	Vedel et al., Nucl. Acid Res. 11: 4335-4354 (1983); Fritsch et al., FEBS Letters 376: 159-163 (1995)
Depositor:	Drs. M. Fritsch & C. Heidrich, Free University of Berlin, Berlin, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, polymorph round cells growing in monolayers
Medium:	90% Dulbecco's MEM + 10% horse serum + 4 mM L-glutamine + 10 nM insulin + 10 ⁻⁸ M dexamethasone
Subculture:	split nearly confluent culture 1:4 every 3-4 days using trypsin/EDTA, cells should not be grown to full confluency as they tend to detach; seed out at ca. 3 x 10 ⁶ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of ca. 1.5 x 10 ⁷ /80 cm ²
Storage:	frozen with 70% medium, 20% horse serum, 10% DMSO at about 2-3 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as rat with IEF of MDH, NP, PEP B
Cytogenetics:	near-tetraploid rat karyotype with 17% polyploidy; 73-82<4n>
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MH1C1
Cell Type:	rat hepatoma
DSMZ No:	ACC 106
Origin:	a clonal strain of epithelial cells was derived from a transplantable Morris hepatoma (#7795, Buffalo strain) in 1967; cells were described to synthesize and to secrete serum albumin and serum complement component C9
References:	Richardson et al., J. Cell. Biol. 40: 236-247 (1967); Ferro et al., Cell Biochem. Function 6: 245-250 (1988); Schiaffonati et al., Exp. Cell Res. 196: 330-336 (1991)
Depositor:	Dr. M. Ferro, Institute of General Pathology, University of Genova, Genova, Italy
DSMZ Cell Culture Data:	
Morphology:	adherent cells growing in big clusters, different cellular forms
Medium:	85-90% Dulbecco's MEM + 10-15% FBS
Subculture:	split confluent culture 1:2 to 1:4 about once or twice per week using trypsin/EDTA; seed out at ca. 2×10^6 cells/75 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 33-60 hours
Harvest:	cell harvest of about 20×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as rat with IEF of AST, NP
Cytogenetics:	near-diploid rat karyotype with 10% polyploidy; 44(40-46)<2n>
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MHEC5-T
Cell Type:	mouse heart endothelial cells
DSMZ No:	ACC 336
Origin:	spontaneous malignant transformation after homotransplanting MHEC5 NMRI mouse myocardial cells (2-weeks-old), thus forming hemangioendothelioma; cells produce tumors upon injection into histocompatible mice
References:	Plendl et al., Virchows Archiv 426: 619-628 (1995)
Depositor:	Dr. Johanna Plendl, University of Munich, Veterinary Anatomy, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial-like cells growing as monolayers
Medium:	90% Dulbecco's MEM + 10% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:3 to 1:6 every 3-6 days; use of trypsin/EDTA is not necessary (cells can be detached by tapping the flask or repeated pipetting); seed out ca. 0.1×10^5 cells/cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 1×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, MDH, NP
Cytogenetics:	murine flat-modulated hexaploid karyotype; 120(96-122)<6n>; 2-3r/dmin and 3 centric fusion markers
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MHH-CALL-2
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 341
Origin:	established from the peripheral blood of a 15-year-old Caucasian girl with acute lymphoblastic leukemia (cALL) in 1993 (at diagnosis)
References:	Tomeczkowski et al., Brit. J. Haematol. 89: 771-779 (1995)
Depositor:	Dr. J. Tomeczkowski, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	single, relatively small round cells growing in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 every 3-4 days; cells grow rather slowly and require high cell density; seed out at ca. 1×10^6 cells/ml, do not maintain below 1×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 100 hours
Harvest:	maximal density at ca. 3×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13+, CD19+, CD20+, CD37-, HLA-DR+, sm/cyIgG-, sm/cyIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hyperdiploid karyotype with 13% polyploidy; 52(50-52)<2n>XX, +8, +10, +18, +18, +21, +21; hyperdiploidy with tetrasomy 21 associated with pre B-ALL in children; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MHH-CALL-3
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 339
Origin:	established from the bone marrow of an 11-year-old girl with pre B cell acute lymphoblastic leukemia (ALL) in 1993 (at diagnosis)
References:	Tomeczkowski et al., Brit. J. Haematol. 89: 771-779 (1995)
Depositor:	Dr. J. Tomeczkowski, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	single, relatively small round cells growing in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 every 2-3 days; cells grow best in 24-well-plates, seed out at ca. 1×10^6 cells/ml; optimal density at ca. 1×10^6 cells/ml; minimum density at 0.8×10^6 cells/ml, maximum density at ca. 2×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 50 hours
Harvest:	cell harvest of ca. $3-4 \times 10^6$ cells/well of a 24-well-plate
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-6 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD34-, CD37-, CD38+, CD79a+, cyCD79a+, CD80-, CD138+, HLA-DR+, sm/cyIgG-, sm/cyIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP, PEP B
Cytogenetics:	human pseudodiploid karyotype; 46(45-46)<2n>XX, del(6)(q15), der(9)t(9;9)(p21;q11), der(19)t(1;19)(q23;p13); carries t(1;19) primary and 6q- secondary rearrangements associated with pre B-ALL; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MHH-CALL-4
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 337
Origin:	established from the peripheral blood of a 10-year-old Caucasian boy with acute lymphoblastic leukemia (pre B-ALL) at diagnosis in 1993
References:	Tomeczkowski et al., Brit. J. Haematol. 89: 771-779 (1995)
Depositor:	Dr. J. Tomeczkowski, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round cells growing in suspension, partly in clusters
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	optimal split ratio at about 1:2 once a week; seeding out at ca. 1×10^6 cells/ml (initially in a 24-well-plate); culture should not be maintained $<0.6 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 80-100 hours
Harvest:	cell harvest of ca. 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13+, CD19+, CD20+, CD34-, CD37+, CD38+, CD79a-, cyCD79a+, CD80-, CD138-, HLA-DR+, sm/cyIgG-, sm/cyIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypodiploid karyotype with 4% polyploidy; 45(42-46)<2n>X, -Y, r(12)(qter;p13-pter)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MHH-ES-1
Cell Type:	human Ewing's sarcoma
DSMZ No:	ACC 167
Origin:	established from the ascites of a 12-year-old Turkish boy with Ewing's sarcoma of the left pelvis (with peritoneal metastasis)
References:	Pietsch et al., Cancer Genet. 38: 167 (1989) (abstract)
Depositor:	Dr. T. Pietsch, Institute of Neuropathology, University of Bonn, Bonn, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent cells of variable size, single or in small clusters, growing as monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture up to 1:10 every 2-3 days using trypsin/EDTA; cells grow rather fast
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 28 hours
Harvest:	about 25 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	unique DNA profile using multiplex PCR at D1S80, D2S44, D17S5 and ApoB
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hyperdiploid karyotype with 12% polyploidy; 49(46-51)<2n>XXY, +X, +1, +8, +12, -17, dup(1)(q31.1q32.2), del(3)(p11), der(4)t(4;?6)(p14;p11), t(6;7)(q11;p11), der(8)t(8;11)(p12;q13), t(11;22)(q24;q12); presence of type-2 EWS-FLI1 hybrid transcripts (394 bp) confirmed by RT-PCR
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MHH-NB-11
Cell Type:	human neuroblastoma
DSMZ No:	ACC 157
Origin:	established from a neuroblastoma metastasis at an adrenal site of a 4-year-old white boy in 1986; cells were described to express neuron-specific enolase and synaptophysin (but neither GFAP nor S-100 protein) and to carry a 20-fold amplification of N-myc
References:	Pietsch et al., Anticancer Res. 8: 1329-1334 (1988)
Depositor:	Dr. T. Pietsch, University of Bonn, Institute of Neuropathology, Bonn, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, fibroblast-like cells growing as monolayers
Medium:	90% RPMI 1640 + 10% FBS + 2 mM L-glutamine + MEM non-essential amino acids
Subculture:	split confluent culture 1:2 every 4 days using trypsin/EDTA; NB: medium must be changed frequently to prevent acidity which induces cell clumping; optimal dissociation after trypsination is affected by gentle aspiration with a #21 gauge needle and syringe
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 100 hours
Harvest:	cell harvest of about 10 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1-2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, neurofilament+, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human bimodal hyper-, diploid/tetraploid karyotype; 54(48-54)<2n>XY, +1, +1, +2, +6, +7, +8, +13, +20, der(1)t(1;?)(p32;?)x2, der(13)t(13;?2HSR)(q33;?2p23); both the long HSR on ch 13 due to N-myc amplification and the ch 1 short-arm rearrangements are typical of neuroblastoma; identity confirmed
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MHH-PREB-1
Cell Type:	human B cell lymphoma
DSMZ No:	ACC 354
Origin:	established from the lymph node of a 5-year-old Caucasian boy with B cell lymphoblastic non-Hodgkin lymphoma at diagnosis in 1994
References:	not published
Depositor:	Dr. J. Tomeczkowski, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly or in clusters in suspension
Medium:	90-92.5% RPMI 1640 + 7.5-10% FBS
Subculture:	maintain at $0.5-2.0 \times 10^6$ cells/ml; split 1:10 to 1:20 twice a week; seed out at ca. $0.5-1.0 \times 10^5$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 20-40 hours
Harvest:	cell harvest of ca. 2.5×10^6 /ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD37+, CD38+, CD79a+, cyCD79a+, CD80-, CD138-, HLA-DR+, sm/cyIgG-, sm/cyIgM+, sm/cykappa-, sm/cylambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MPI, PEP B
Cytogenetics:	human hyperdiploid karyotype with 12% polyploidy; 48<2n>XYY, +21, ?ins(2;2)(q21;p15p22), t(8;14)(q24;q32)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MKN-45
Cell Type:	human gastric adenocarcinoma
DSMZ No:	ACC 409
Origin:	established from the poorly differentiated adenocarcinoma of the stomach (medullary type) of a 62-year-old woman
References:	Naito et al., Virchows Arch. (Cell Pathol.) 46: 145-154 (1984); Motoyama et al., Acta Pathol. Jpn. 36: 65-83 (1986); Rege-Cambrin et al., Cancer Genet. Cytogenet. 64: 170-173 (1992)
Depositor:	Dr. R. A. F. MacLeod, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	both spindle-shaped or oval cells growing in monolayers and single round cells or clumps in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	cells can be difficult to culture! split semi-confluent culture 1:3 to 1:5 every 5-10 days using trypsin/EDTA (for 5 min or longer); seed out at ca. $1-2 \times 10^6$ cells/80 cm ² ; culture medium may turn acidic (yellow) very quickly, exchange of spent medium is advised, the cells in suspension may be used to seed a new flask
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60 hours
Harvest:	cell harvest of ca. $10-15 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human hypertriploid karyotype with 8% polyploidy; 70-73<3n>XX/XXX, -X, -2, -2, -4, -5, -5, +6, -11, -13, -13, -14, -18, -19, -20, -21, -22, +20mar, add(X)(q1?), der(1)(1pter->p21::1p11->q32::1q44->q11::?), i(2p), der(3)t(3;?9)(p14;q13/21), del(4)(q21), del(4)(q26), del(6)(q1?)x2, del(7)(p11), del(8)(p12), add(8)(q24), del(9)(p11), add(13)(p11), inv(16)(p13q22), add(17)(p1?), add(19)(q13), i(21q); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	ML-1
Cell Type:	human follicular thyroid carcinoma
DSMZ No:	ACC 464
Origin:	established from a recurrent tumor of a dedifferentiated follicular thyroid carcinoma (stage pT4) of a 50-year-old woman; cells were described as taking up iodine and/or glucose and as secreting thyroglobulin, chondroitin sulfate, fibronectin, and triiodothyronine; cells are tumorigenic in nude mice
References:	Schönberger et al., J. Mol. Med. 78: 102-110 (2000)
Depositor:	Dr. J. Schönberger, University Hospital Regensburg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells growing in monolayers
Medium:	80-90% Dulbecco's MEM (4.5 g/L glucose) + 10-20% FBS + 2 mM L-glutamine + 1 mM sodium pyruvate
Subculture:	split confluent culture 1:2 once a week using trypsin/EDTA; seed out at ca. 1×10^6 cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 4-7 days
Harvest:	cell harvest of ca. $25-30 \times 10^6$ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, thyroglobulin+, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Cytogenetics:	human extreme-hypodiploid karyotype with 12% polyploidy; 37(33-38)<2n>XX, -6, -13, -14, -17, -18, -19, -21, add(4)(p16), add(9)(q22)x1-2, der(19)t(9;19)(q1?2;p1?2)/der(9;19)(q10;q10)x1-2; sideline with idem, der(3;19)(?q10;q10), -11
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	ML-2
Cell Type:	human acute myelomonocytic leukemia
DSMZ No:	ACC 15
Origin:	established from the peripheral blood of a 26-year-old man with acute myeloid leukemia (AML M4) at diagnosis of AML (following T-non-Hodgkin lymphoma and T-ALL) in 1978; sister cell lines are ML-1 and ML-3
References:	Herrmann et al., Cancer 46: 1383-1388 (1980); Ohyashiki et al., Cancer Res. 46: 3642-3647 (1986); Ohyashiki et al., Cancer Genet. Cytogenet. 37: 103-200 (1989)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round, single cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; split 1:2 to 1:3 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60 hours
Harvest:	maximum density at $1.0-1.25 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4-, CD13+, CD14-, CD15+, CD19-, CD33+, CD34-, CD68+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human near tetraploid karyotype; 92(84-94)<4n>XX, -Y, -Y, -7, -9, -10, -10, +11, +12, +12, +13, +13, -15, -16, -17, -17, +18, +18, -20, -20, +4mar, der(1)t(1;?)(p21;?)x2, del(6)(q23)x2, der(6)t(6;11)(q27;?q23)x2, ?der(11)t(6;11)(q27;?q23)/del(11)(q23)x2, der(11)t(11;?)(?->11p11->11q23)x2, der(11)t(11;?)(q11-13;?), dup(13)(q32->qter)x2, der(18)t(5;?;18)(q21;?;q11)x2; extensive ch 11 rearrangements (inc. 11q23, typical of M4 and M5 AML) obscure the cytogenetic picture; ML-2 may carry rare t(6;11)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MMQ
Cell Type:	rat pituitary tumor
DSMZ No:	ACC 484
Origin:	cell line is derived from the transplantable 7315a rat pituitary tumor; until establishment of the cell line, the 7315a pituitary tumor was serially transplanted in adult female Buffalo rats since the early 1960s; described as expressing prolactin and dopamine receptors; cells were reported to be responsive to somatostatin, estrogen, progesterone, and acetylcholin
References:	Judd et al., Endocrinology 123: 2341-2350 (1988); Cronin et al., Endocrinology 109: 2160 (1981)
Depositor:	Dr. M. L. Thieulant, CNRS, Rennes, France
DSMZ Cell Culture Data:	
Morphology:	spheroidal cells growing in clumps in suspension
Medium:	90% RPMI 1640 + 7.5% horse serum + 2.5% FBS
Subculture:	optimal split ratio of 1:2 to 1:4 every 2-4 days; seed out at ca. 0.5×10^6 cells/ml; maintain at $0.5-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	maximum density at $1.5-2.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 15% horse serum, 5% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as rat by cytogenetics and PCR analysis with Alu, globin and rRNA primers
Cytogenetics:	rat hyperdiploid karyotype with 6% polyploidy; $49-52<2n>$; high level of spontaneous breakage

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MN-60
Cell Type:	human B cell leukemia
DSMZ No:	ACC 138
Origin:	established from the peripheral blood of a 20-year-old man with acute B cell leukemia (ALL FAB L3) in partial remission in 1981
References:	Roos et al., Leukemia Res. 6: 685-693 (1982)
Depositor:	Dr. G. Roos, University of Umea, Umea, Sweden
DSMZ Cell Culture Data:	
Morphology:	single, round to polygonal cells in suspension
Medium:	90% Ham's F10 + 10% FBS
Subculture:	maintain at $0.2-0.5 \times 10^6$ cells/ml; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48-60 hours
Harvest:	maximal density at 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD34-, CD37+, CD38+, CD79a+, cyCD79a+, CD80+, CD138-, HLA-DR+, sm/cyIgG-, sm/cyIgM+, sm/cykappa-, sm/cylambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human near diploid karyotype; 46(45-47)<2n>XY, dup(1)(q21q41), del(6)(q21), t(8;14)(q24;q32), i(13q)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MOLT-13
Cell Type:	human T cell leukemia
DSMZ No:	ACC 436
Origin:	established from the bone marrow of a 2-year-old girl with T-ALL (T cell acute lymphoblastic leukemia, FAB L2) at relapse in 1983; MOLT-14 (DSM ACC 437) is a simultaneous sister cell line derived from the same patient sample with similar or identical features
References:	Drexler et al., Hematol. Oncol. 7: 115-125 (1989)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round cells with processes, growing in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 every 5-7 days changing medium twice weekly; seed out at ca. 1×10^6 cells/ml in 12-well plates; maintain at a minimum of ca. 1×10^6 cells/ml; FBS may be reduced to 10% gradually; cells rarely exceed 80% viability
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 80-120 hours
Harvest:	maximal density at ca. 3×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3+, CD4-, CD5+, CD6-, CD7+, CD8-, CD13-, CD19-, CD34+, TCRalpha/beta-, TCRgamma/delta+
Fingerprinting:	same DNA profile as MOLT-14 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human pseudodiploid karyotype with 4% polyploidy; 46(44-48)<2n>XX, der(17)t(17;17)(p13;q22.1); resembles published karyotype; karyotypically identical to MOLT-14
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MOLT-14
Cell Type:	human T cell leukemia
DSMZ No:	ACC 437
Origin:	established from the bone marrow of a 2-year-old girl with T-ALL (T cell acute lymphoblastic leukemia, FAB L2) at relapse in 1983; MOLT-13 (DSM ACC 436) is a simultaneous sister cell line derived from the same patient sample with similar or identical features
References:	Drexler et al., Hematol. Oncol. 7: 115-125 (1989)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round, single cells growing in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:4 to 1:5 twice a week; seed out at ca. $0.5-0.7 \times 10^6$ cells/ml; maintain at $0.5-0.6 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	maximal density at ca. $1.0-1.2 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5-10 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2+, CD3+, CD4-, CD5+, CD6-, CD7+, CD8-, CD13-, CD19-, CD34(+), TCRalpha/beta-, TCRgamma/delta+
Fingerprinting:	same DNA profile as MOLT-13 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human pseudodiploid karyotype with 8% polyploidy; 46(44-48)<2n>XX, der(17)t(17;17)(p13;q22.1); resembles published karyotype; karyotypically identical to MOLT-13
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MOLT-16
Cell Type:	human T cell leukemia
DSMZ No:	ACC 29
Origin:	established from the peripheral blood of a 5-year-old girl with T cell acute lymphoblastic leukemia (T-ALL) at relapse in 1984 (sister cell line MOLT-17, DSM ACC 36)
References:	Matsuo et al., Human Cell 1: 263-274 (1988); Drexler et al., Hematol. Oncol. 7: 115-125 (1989)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round to elongated single cells growing in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; split 1:2 to 1:5 twice a week; seed out at ca. $1-2 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	saturation density at $2-2.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5-10 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2+, CD3+, CD4 (clone S3.5)+, CD4 (clone Leu-3a)-, CD5+, CD6+, CD7+, CD8-, CD13-, CD19-, CD34-, TCRalpha/beta+, TCRgamma/delta-
Fingerprinting:	same DNA profile as MOLT-17 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP, PEP B
Cytogenetics:	human near diploid karyotype; 46(43-47)<2n>XX, t(3;11)(p21;p13), t(8;14)(q24;q11); t(8;14)(q24;q11) and 11p13 breakpoint are associated with T-ALL (identical to MOLT-17)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MOLT-17
Cell Type:	human T cell leukemia
DSMZ No:	ACC 36
Origin:	established from the peripheral blood of a 5-year-old girl with Tcell acute lymphoblastic leukemia (T-ALL) at relapse in 1984 (sister cell line MOLT-16, DSM ACC 29)
References:	Drexler et al., Hematol. Oncol. 7: 115-125 (1989)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round to elongated single cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5\text{--}2.0 \times 10^6$ cells/ml; split 1:2 to 1:5
Incubation:	at 37 °C with 5% CO ₂
Harvest:	grows up to 2.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD2+, CD3+, CD4+, CD5+, CD6+, CD7+, CD8-, CD13+, CD19-, TCRalpha/beta-
Fingerprinting:	same DNA profile as MOLT-16 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human near diploid karyotype; 46(43-47)<2n>XX, t(3;11)(p21;p13), t(8;14)(q24;q11); t(8;14)(q24;q11) and 11p13 breakpoint are associated with T-ALL (identical to MOLT-16)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MOLT-3
Cell Type:	human T cell leukemia
DSMZ No:	ACC 84
Origin:	established from the peripheral blood of a 19-year-old man with acute lymphoblastic leukemia (ALL) in relapse in 1971; sister cell line of MOLT-4 (DSM ACC 362)
References:	Minowada et al., J. Natl. Cancer Inst. 49: 891-895 (1972)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round, single cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; split saturated culture 1:2 every second day; seed out at ca. $0.5-0.7 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	maximal density at about $0.7-1.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2+, CD3(+), CD4+, CD5+, CD6+, CD7+, CD8+, CD13-, CD19-, CD34(+), TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	same DNA profile as MOLT-4 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypertetraploid karyotype; 98(94-101)<4n>XXYY, +6, +7, +8, +8, +17, +20, del(6)(q16)x2, der(7)t(7;7)(p15;q11)x2; identity confirmed
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MOLT-4
Cell Type:	human T cell leukemia
DSMZ No:	ACC 362
Origin:	established from the peripheral blood of a 19-year-old man with acute lymphoblastic leukemia (ALL) in relapse in 1971; sister cell line of MOLT-3 (DSM ACC 84)
References:	Minowada et al., J. Natl. Cancer Inst. 49: 891-895 (1972)
Depositor:	Dr. Jun Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round cells growing in suspension, singly or in clusters
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	maintain at ca. $0.5\text{--}1.3 \times 10^6$ cells/ml; seed out at ca. 0.5×10^6 cells/ml; split ratio of ca. 1:2 to 1:3 every 2-3 days
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	maximal density at ca. 1.3×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4\text{--}5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2+, CD3+, CD4+, CD5+, CD6+, CD7+, CD8+, CD13-, CD19-, CD34+, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	same DNA profile as MOLT-3 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human flat-modulated hypertetraploid karyotype; 89-99<4n>XXYY, +4, +7, +8, +20, +20, del(6)(q16)x2, der(7)t(7;7)(p15;q11)x2; closely related to MOLT-3
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MONO-MAC-1
Cell Type:	human acute monocytic leukemia
DSMZ No:	ACC 252
Origin:	established from the peripheral blood of a 64-year-old man with acute monocytic leukemia (AML FAB M5) at relapse in 1985 following myeloid metaplasia; simultaneous sister cell line of MONO-MAC-6 (DSM ACC 124); see also website < www.monocytes.de >
References:	Ziegler-Heitbrock et al., Int. J. Cancer 41: 456-461 (1988); Steube et al., Leukemia Res. 21: 327-335 (1997)
Depositor:	Dr. H. W. L. Ziegler-Heitbrock, Institute of Immunology, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round/multiformed cells in suspension
Medium:	90% RPMI 1640 + 10% FBS + 2 mM L-glutamine + 1x non-essential amino acids + 1 mM sodium pyruvate
Subculture:	maintain at 0.3-1.0 x 10 ⁶ cells/ml; split ratio 1:2 to 1:3 every 2-3 days; seed out at ca. 1 x 10 ⁶ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30-50 hours
Harvest:	maximal density at 1.0-1.5 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5-8 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4-, CD13+, CD14-, CD15+, CD19-, CD33-, CD34+, CD68+, HLA-DR+
Fingerprinting:	same DNA profile as MONO-MAC-6 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human near-diploid karyotype with 10% polyploidy; 49(43-52)<2n>XY, +8, +12, +13, +18, -20, inv dup(3)(q21q27), add(8)(q24), t(9;11)(p22;q23), t(10;12;17)(q24;q13;q11), r(12)(??), del(13)(q1?2q21), t(16;21)(q13;q22.2); diploid version of its sister MONO-MAC-6 with fewer in vitro rearrangements; carries 9;11 translocation (AML M5)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MONO-MAC-6
Cell Type:	human acute monocytic leukemia
DSMZ No:	ACC 124
Origin:	established from the peripheral blood of a 64-year-old man with relapsed acute monocytic leukemia (AML FAB M5) in 1985 following myeloid metaplasia; more information on this cell line on the following website: < www.monocytes.de >; sister cell line of MONO-MAC-1 (DSM ACC 252)
References:	Ziegler-Heitbrock et al., Int. J. Cancer 41: 456-461 (1988)
Depositor:	Dr. H. W. L. Ziegler-Heitbrock, Institute of Immunology, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round/multiformed cells or small clusters of cells in suspension, sometimes loosely adherent; 1-5% giant cells
Medium:	90% RPMI 1640 + 10% FBS + 2 mM L-glutamine + non-essential amino acids + 1 mM sodium pyruvate + 9 µg/ml bovine insulin
Subculture:	maintain at $0.3-1.0 \times 10^6$ cells/ml; split ratio 1:3 to 1:5; avoid lipopolysaccharide contamination which will result in cell clumping, maturation and retarded growth; cells are best maintained in 24-well plates; seed out at ca. 0.3×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60 hours
Harvest:	maximal density at ca. 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD13+, CD14+, CD15+, CD19-, CD33+, CD34-, CD68+, HLA-DR(+)
Fingerprinting:	same DNA profile as MONO-MAC-1 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human flat-modulated hypotetraploid karyotype with near-diploid (8%) and polyploid (17%) sidelines; 84-90<4n>XX/XXX, -Y, +6, +7, -12, -13, -13, -16, -16, +2mar, t(9;11)(p22;q23)x2, add(10)(p11)x2, add(12)(q21), del(13)(q13q14)der(13)t(13;14)(p11;q12)x2, der(17)t(13;17)(q21;p11)x2; first cell line shown to carry t(9;11) characteristic of AML M5
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MS-5
Cell Type:	mouse stromal cells
DSMZ No:	ACC 441
Origin:	established by irradiation of the adherent cells in long-term bone marrow cultures derived from C3H/HeNSlc strain mice
References:	Itoh et al., Exp. Hematol. 17: 145-153 (1989)
Depositor:	Prof. Dr. W. Ostertag, Heinrich-Pette-Institut, Hamburg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblastic cells growing in monolayers
Medium:	90% alpha-MEM + 10% horse serum (or FBS) + 2 mM L-glutamine + 2 mM sodium pyruvate
Subculture:	split confluent culture 1:2 to 1:3 every third day using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/80 cm ² in 25 ml medium; maximal density at ca. 8×10^6 cells/175 cm ² ; caveat: cells may detach spontaneously in aggregates which are difficult to disperse
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of ca. 5×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed by PCR species analysis as murine
Cytogenetics:	murine hypertriploid karyotype

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MSTO-211H
Cell Type:	human mesothelioma
DSMZ No:	ACC 390
Origin:	established from the pleural effusion of a 62-year-old Caucasian man with mesothelioma (malignant biphasic) who did not have any prior therapy
References:	Bepler et al., Differentiation 37: 158-171 (1988)
Depositor:	Dr. G. Bepler, Duke University Medical Center, Durham, North Carolina, USA
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblast-like cells growing as monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:5 every 3 days using trypsin/EDTA; seed out at ca. $2-3 \times 10^6$ cells/80 cm ² in 10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	cell harvest of ca. 15×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human hypertriploid karyotype; 79(77-80)<3n>XYY/XXYY, +2, +4, +7, +7, +7, -10, +12, +13, +14, -15, -15, +16, +16, +16, -17, +18, -22, +1-2mar, 1-20dmin; add(1)(p35-36), add(1)(p34), der(4)t(4;17)(p14;q21)x2, del(6)(q25)x2, del(7)(q11), del(7)(q21.2)x2, inv(9)(p24q34), del(10)(q24), add(14)(p11), add(15)(p11), i?dic(16)(q10)x2; sideline with der(6;9)(p10;q10); high 7p copy number associated with poor prognosis mesothelioma; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MT-3
Cell Type:	human breast carcinoma
DSMZ No:	ACC 403
Origin:	originated from surgical material of a human breast carcinoma; the tumor was passaged in nude mice prior to adaptation to continuous growth in culture vessels
References:	Naundorf et al., Breast Cancer Res. Treatment 23: 87-95 (1992)
Depositor:	Dr. M. C. Bibby, University of Bradford, Bradford, UK
DSMZ Cell Culture Data:	
Morphology:	epithelial cells, after splitting cells start to grow in colonies (not necessarily as monolayer)
Medium:	90% RPMI 1640 + 10% FBS + 2 mM L-glutamine + 1 mM sodium pyruvate
Subculture:	split confluent culture 1:10 every 4-5 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 0.4×10^6 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, NP
Cytogenetics:	human hyperdiploid karyotype with 8% polyploidy; 47(45-48)<2n>X, -X, +7, +22, i(13q), del(22)(q11); trisomy 7 recurrent in breast cancer; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MUTZ-1
Cell Type:	human myelodysplastic syndrome
DSMZ No:	ACC 13
Origin:	established from the peripheral blood of a 5-year-old (Turkish) girl diagnosed with myelodysplastic syndrome (refractory anemia with excess of blasts) and Fanconi anemia in 1994
References:	Steube et al., Leukemia & Lymphoma 25: 345-363 (1997); review article: Drexler, Leukemia Res. 24: 109-115 (2000)
Depositor:	Dr. K. Steube, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	mononuclear cells growing as single cells or in clumps in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split every 2-3 days at 1:3 to 1:8; maintain at $0.2-0.6 \times 10^6$ cells/ml; maximal density at $1.5-2.0 \times 10^6$ cells/ml; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 48-55 hours
Harvest:	cell harvest of about $1.0-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD1-, CD2-, CD3-, CD4-, CD5-, CD7+, CD8-, CD10+, CD11b-, CD13-, CD14-, CD15-, CD19+, CD20-, CD21+, CD25-, CD30-, CD33-, CD34-, CD37+, CD38+, CD68-, CD71+, CD138+, GlyA+, HLA-DR+, TCRalpha/beta-, sm/cyIgG-, cyIgM+, smIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human near-triploid karyotype with 12% polyploidy; 68(68-73)<3n>X, der(X), -X, add(1)(p11), der(2)t(2;?11)(q22;q23), der(3)t(3;5)(q27;q14)t(5;22)(q35;q12)x2, del(3)(p11), del(3)(q11), del(5)(q13/q33)x1-2, der(6)t(3;6)(p23;p21.2)x2, der(7)t(7;11)(q35;q13), der(8)t(1;8)(q11;q24), add(9)(q34), del(9)(p13), der(14)t(14;?)(q23;?), der(14)t(14;?)(p11;?)t(1;?)(q11;?), add(14)(p11), der(15)t(5;15)(p13;p11)x2, der(16)t(7;16)(q21.1;q24), add(18)(p11), add(21)(p11); 1q11 rearrangements typical of transforming Fanconi anemia
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MUTZ-2
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 271
Origin:	established from the peripheral blood of a 62-year-old man with acute myeloid leukemia (AML FAB M2) at diagnosis in 1993; cells are growth factor-dependent and responsive to IL-3, SCF, G-CSF, M-CSF and IFN-gamma
References:	Hu et al., Leukemia 10: 1025-1040 (1996); Drexler et al., Leukemia 11: 701-708 (1997)
Depositor:	Dr. Zhen-Bo Hu, Ontario Cancer Institute, Toronto, Canada
DSMZ Cell Culture Data:	
Morphology:	single cells in suspension, growing partly in clumps
Medium:	70% alpha-MEM + 20% FBS + 20% vol conditioned medium of cell line 5637 (DSM ACC 35) (cells grow well with 50 ng/ml SCF)
Subculture:	optimal split ratio of 1:2 to 1:3 twice a week; seed out at about 0.5×10^6 cells/ml; cells should be grown in 24-well-plates. CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 47 hours
Harvest:	maximum density at about $1.5\text{--}2.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4+, CD5-, CD7-, CD8-, CD10-, CD11b-, CD13+, CD14-, CD15+, CD16-, CD19-, CD20-, CD25+, CD30+, CD33+, CD34+, CD38+, CD41a-, CD42b-, CD56-, CDw65+, CD68+, CD71+, HLA-DR+, GlyA-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human hyperdiploid karyotype with 1.8% polyploidy; 48(46-50)<2n>XY, +8, +10; no structural aberrations detected; +8 is a typical primary change in AML M1
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MUTZ-3
Cell Type:	human acute myelomonocytic leukemia
DSMZ No:	ACC 295
Origin:	established from the peripheral blood of a 29-year-old man with acute myelomonocytic leukemia (AML FAB M4) at diagnosis in 1993
References:	Hu et al., Leukemia 10: 1025-1040 (1996); MacLeod et al., Genes Chromosomes Cancer 16: 144-148 (1996); Drexler et al., Leukemia 11: 701-708 (1997); Masterson et al., Blood 100: 701-703 (2002)
Depositor:	Dr. Zhen-Bo Hu, Ontario Cancer Institute, Toronto, Canada
DSMZ Cell Culture Data:	
Morphology:	round blast-like cells growing as single cells and sometimes in small clumps in suspension
Medium:	70% alpha-MEM + 20% FBS + 10% vol conditioned medium of cell line 5637 (DSM ACC 35) (or 10 ng/ml SCF, GM-CSF, M-CSF or IL-3)
Subculture:	cultivate in 24-well-plates for optimal growth; maintain at $0.5-0.7 \times 10^6$ cells/well; optimal split ratio of 1:2 every 4-6 days; seed out at ca. 1.0×10^6 cells/ml; maintain at $0.4-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 110 hours
Harvest:	saturation density at $1.0-1.5 \times 10^6$ cells/well
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4-, CD5-, CD7-, CD8-, CD10-, CD11b-, CD13+, CD14(+), CD15+, CD16-, CD19-, CD20-, CD25-, CD30(+), CD33-, CD34+, CD38+, CD41a-, CD42b-, CD56-, CDw65+, CD68+, CD71-, HLA-DR+, GlyA-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH, NP
Cytogenetics:	human near-diploid karyotype with 6% tetraploidy; 46(44-48)<2n>XY, t(1;3)(q43;q13)inv(3)(q21q26), t(2;7)(q36;q36)inv(7)(p15q36), t(12;22)(p13;q12); carries t(12;22) recurrent in AML M4; also carries masked inv(3) associated with abnormal megakaryocytopoiesis
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MUTZ-5
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 490
Origin:	established from the peripheral blood of a 26-year-old man with B cell precursor acute lymphoblastic leukemia (BCP-ALL) at relapse in 1998; simultaneous sister cell lines MUTZ-6 and MUTZ-7
References:	Meyer et al., Leukemia 15: 1471-1474 (2001)
Depositor:	Mrs. C. Meyer, DSMZ, Braunschweig
DSMZ Cell Culture Data:	
Morphology:	cells grow singly or in small clusters in suspension, irregular cell forms
Medium:	80% McCoy's 5A + 20% FBS + 2 mM L-glutamine (RPMI 1640 also possible)
Subculture:	split saturated culture 1:2 to 1:3 every 2-4 days; seed out at ca. 1×10^6 cells/ml; after thawing, initial culture in 24-well-plate is of advantage; minimum cell density of 0.25×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 70-100 hours
Harvest:	maximum density at 3.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, 3-, 4-, 5-, 6-, 7-, 8-, 10+, 13-, 14-, 15-, 19+, 20+, 21-, 22+, 25-, 30+, 33(+), 34(+), 37+, 38+, 40+, 41-, 42-, 56-, 71+, cy79a+, cy79b+, 80-, 95+, 115-, 116-, 117-, 119+, 123-, 127+, 135+, 138+, 235a-, HLA-DR+, sm/cyIgGMkl-, TCRabgd-, TdT-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype; $45<2n>X, -Y, t(12;13)(p12;q13-14)$
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	MV4-11
Cell Type:	human acute monocytic leukemia
DSMZ No:	ACC 102
Origin:	established from a 10-year-old boy with acute monocytic leukemia (AML FAB M5) at diagnosis; cells were described to carry a 4;11 translocation; we found that MV4-11 carries an FLT3 internal tandem duplication; patented cell line
References:	Lange et al., Blood 70: 192-199 (1987); Santoli et al., J. Immunol. 139: 3348-3354 (1987); Quentmeier et al., Leukemia 17: 120-124 (2003)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round cells growing in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.2-1.0 \times 10^6$ cells/ml; seed out at ca. 0.6×10^6 cells/ml; split carefully; cells are difficult to culture and grow slowly at first; addition of recombinant GM-CSF or IL-3 (or as 10-20% vol conditioned medium of cell line 5637, DSM ACC 35) or increasing the FBS content to 20% can be of advantage; cultivation in 24-well-plates appears to be better than in large culture flasks
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	saturation density at about 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4(+), CD5-, CD8-, CD10-, CD13+, CD14-, CD15+, CD19-, CD21-, CD25-, CD33+, CD34+, CD37-, CD68+, CD138-, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP, PEP B
Cytogenetics:	human hyperdiploid karyotype; 48(46-48)<2n>XY, +8, +18, +19, -21, t(4;11)(q21;q23)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	N18TG2
Cell Type:	mouse neuroblastoma
DSMZ No:	ACC 103
Origin:	6-thioguanine resistant mutant of cell line N18 that was established from a mouse of strain A/Jax with neuroblastoma C1300; cells were described to form cAMP in response to prostaglandin E1 and to be aneuploid (modal chromosome number of 81)
References:	Minna et al., Nature New Biol. 235: 225-231 (1972); Hamprecht et al., Int. Rev. Cytol. 49: 99-170 (1977)
Depositor:	Dr. B. Hamprecht, Physiologisch-Chemisches Institut, Tübingen, Germany
DSMZ Cell Culture Data:	
Morphology:	round, single, slightly adherent cells, occasionally with short processes
Medium:	90% Dulbecco's MEM + 10% FBS + 100 :M 6-thioguanine
Subculture:	split culture 1:5 to 1:10 every 3-5 days (cells grow adherent, but can be detached without trypsin); seed out at 1×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	about 1.5×10^7 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, MDH, NP, PEP B
Viruses:	ELISA: reverse transcriptase positive

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	N2-261
Cell Type:	mouse hybridoma (anti-fiber/myosin)
DSMZ No:	ACC 322
Origin:	described as secreting a mouse monoclonal antibody (IgG ₁) reacting with neonatal and adult slow and IIA fibers in rat and human and with slow and IIA fast myosin heavy chains in Western blots; the cell line was established in 1985 by fusion of X63AG8.653 cells with spleen cells from a BALB/c mouse that had been immunized with human skeletal muscle myosin
References:	Hughes et al., Dev. Biol. 158: 183-199 (1993)
Depositor:	Dr. S. M. Hughes, King's College, London, England
DSMZ Cell Culture Data:	
Morphology:	single round cells, lightly adherent
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + 2 mM L-glutamine + 1 mM sodium pyruvate + 1x non-essential amino acids
Subculture:	split saturated culture 1:5 to 1:10 every 2-3 days; seed out at about 0.2 x 10 ⁶ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 20 hours
Harvest:	maximal density of about 1.5 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgG ₁ monoclonal antibody
Species:	confirmed as mouse with IEF of MDH, NP
Cytogenetics:	murine hypertetraploid karyotype with 8.3% polyploidy; 88(85-92)<4n>; carries 1-3 centric fusion markers, 1-3 dm
Viruses:	ELISA: reverse transcriptase positive

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	N3-36
Cell Type:	mouse hybridoma (anti-fiber/myosin)
DSMZ No:	ACC 318
Origin:	described as secreting a mouse monoclonal antibody (IgM) reacting with neonatal and adult fast fibers and myosins in rodent and human on Western blots; the cell line was established in 1985 by fusion of X63AG8.653 cells with spleen cells from a BALB/c mouse that had been immunized with human skeletal muscle myosin
References:	Dan-Goor et al., J. Muscle Res. Cell Motility 11: 216-226 (1990)
Depositor:	Dr. S. M. Hughes, King's College, London, England
DSMZ Cell Culture Data:	
Morphology:	single round cells, lightly adherent
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + 2 mM L-glutamine + 1 mM sodium pyruvate + 1x non-essential amino acids
Subculture:	split saturated culture 1:3 to 1:5 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 24 hours
Harvest:	maximal density of ca. 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgM monoclonal antibody
Species:	confirmed as mouse with IEF of AST, LDH, MDH, NP
Cytogenetics:	murine hypertetraploid karyotype, 20% polyploidy; 78-90<4n>; carries large telocentric and two centric fusion markers
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	N4TG3
Cell Type:	mouse neuroblastoma
DSMZ No:	ACC 101
Origin:	6-thioguanine resistant mutant of cell line N4 that was established from a mouse of strain A/Jax with neuroblastoma C1300; cells were described to form increased levels of cAMP in response to prostaglandin E1 which is inhibited by opiate receptors; cells are aneuploid (modal chromosome number of 86)
References:	Amano et al., Exp. Cell Res. 85: 399-408 (1974); Traber et al., FEBS Letters 49: 260-263 (1974)
Depositor:	Dr. B. Hamprecht, Physiologisch-Chemisches Institut, Tübingen, Germany
DSMZ Cell Culture Data:	
Morphology:	flat cell body, slightly adherent, occasionally with short processes
Medium:	90% Dulbecco's MEM + 10% FBS + 100 :M 6-thioguanine
Subculture:	split saturated culture 1:2 to 1:4 every 2-4 days (cells grow slightly adherent, but can be detached without trypsin)
Incubation:	at 37 °C with 5-10% CO ₂
Harvest:	about 10-20 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Fingerprinting:	unique Hinf I-(gtg) ₅ DNA profile
Species:	confirmed as mouse with IEF of MDH, NP, PEP B
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NALM-1
Cell Type:	human chronic myeloid leukemia in blast crisis
DSMZ No:	ACC 131
Origin:	established from the peripheral blood of a 3-year-old girl with chronic myeloid leukemia (CML) in blast crisis in 1975; cells were described to be TdT-positive and to carry the Philadelphia-chromosome t(9;22)
References:	Minowada et al., J. Natl. Cancer Inst. 59: 83-87 (1977); Minowada et al., Leukemia Res. 3: 261-266 (1979); review article: Drexler, Leukemia Res. 18: 919-927 (1994); review article: Drexler et al., Leukemia Res. 24: 109-115 (2000)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	small, single, round cells in suspension
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	cells are difficult to culture! maintain at $1.0\text{-}2.0 \times 10^6$ cells/ml; seed out at ca. 2×10^6 cells/ml; cells grow initially slowly; it might be of benefit to start culture in 24-well plates
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	maximal density at about $3\text{-}5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5\text{-}7 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a-, cyCD79a+, CD80-, CD138+, HLA-DR+, sm/cyIgG-, smIgM-, cyIgM+, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypodiploid karyotype with 5% polyploidy; 45(42-47)<2n>X, -X, der(9)t(9;22)(q34;q12)x1-2, dup(13)(q21), der(22)t(9;22)(q34;q12); identity confirmed
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NALM-19
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 522
Origin:	established from the peripheral blood of a 26-year-old man with "acute undifferentiated leukemia" at diagnosis in 1988
References:	Matsuo et al., Human Cell 4: 257-260 (1991); Matsuo & Drexler, Leukemia Res. 22: 567-579 (1998)
Depositor:	Dr. Jun Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	small single round cells growing in suspension, sometimes in small clumps
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 every 2-3 days; seed out at ca. 1.0×10^6 cells/ml, start culture in 24-well-plates (first 1-3 weeks; cells may start very slowly); maintain at ca. 1.0×10^6 cells/ml; maximal density of ca. $2-3 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-70 hours
Harvest:	cell harvest of ca. $1-2 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 8×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD10-, CD13(+), CD19+, CD20-, CD37(+), CD38+, CD79a-, cyCD79a+, CD80-, CD138(+), HLA-DR+, sm/cyIgG-, sm/cyIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype with 4% polyploidy; 47(45-48)<2n>XY,+5, del(9)(p21.2); revision of published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NALM-6
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 128
Origin:	established from the peripheral blood of a 19-year-old man with acute lymphoblastic leukemia (ALL) in relapse in 1976
References:	Hurwitz et al., Int. J. Cancer 23: 174-180 (1979)
Depositor:	Dr. A. Ganser, Johann-Wolfgang-Goethe University, Frankfurt, Germany
DSMZ Cell Culture Data:	
Morphology:	single, small, round cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $1.0\text{--}2.0 \times 10^6$ cells/ml; split ratio of 1:2 to 1:3 every 3 days; seed out at ca. 1×10^6 cells/ml; minimum density of 0.5×10^6 cells/ml;
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 36 hours
Harvest:	maximal density of $4\text{--}6 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5\text{--}8 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13(+), CD19+, CD34-, CD37-, CD79a-, cyCD79a+, CD80-, CD138+, HLA-DR+, sm/cyIgG-, cyIgM+, smIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, G6PD, MDH, NP
Cytogenetics:	human near diploid karyotype; 46(43-47)<2n>XY, t(5;12)(q33.2;p13.2)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NAMALWA
Cell Type:	human Burkitt's lymphoma
DSMZ No:	ACC 24
Origin:	established from the tumor mass of an African child with Burkitt's lymphoma in 1967; cells express c-myc mRNA
References:	Nadkarni et al., Cancer 23: 64-79 (1969)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	lymphoblastoid, single cells or small clusters in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5\text{--}1.5 \times 10^6$ cells/ml; split 1:2 to 1:3 every 2-3 days; seed out at ca. 1×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 20-30 hours
Harvest:	maximum density at about 2.0×10^6 cells/ml or higher
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD34-, CD37-, CD79a+, CD80-, CD138+, HLA-DR+, sm/cyIgM+, sm/cyIgG-, sm/cykappa-, sm/cylambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human hypodiploid karyotype with 4% polyploidy; 44(43-46)<2n>X, -Y, -5, -6, +7, -10, -13, -21, +3mar, ins(1;?)(p11;?), ins(3)(q11q29), der(5)t(1;5)(q25;q35), del(8)(p22), der(14)t(14;?)(p11;?), der(15)t(15;21)(p11;q11); a supernumerary ring or fragment present in 8% of cells; NAMALWA marker chromosomes present
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NAMALWA.CSN/70
Cell Type:	human Burkitt's lymphoma
DSMZ No:	ACC 70
Origin:	established from the tumor mass of an African child with Burkitt's lymphoma; this variant subline from the original NAMALWA (DSM ACC 24) was described to be characterized by differential expression of immunoglobulin and c-myc genes and of certain cell surface antigens (see reference); cells were reported to harbor a retrovirus of simian origin, SMRV-H
References:	Guy et al., Int. J. Cancer 45: 109-118 (1990)
Depositor:	Dr. K. Guy, Western General Hospital, Edinburgh, UK
DSMZ Cell Culture Data:	
Morphology:	lymphoblastoid, single cells or small clusters, in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximal density at about 2.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD3-, CD10+, CD13+, CD19+, CD37+, HLA-DR+, sm/cyIgM-, sm/cyIgG-
Fingerprinting:	same DNA profile as NAMALWA using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human pseudotriploid karyotype with 2% polyploidy; 69(65-72)<3n>; NAMALWA marker chromosomes present
Viruses:	ELISA: reverse transcriptase positive; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NAMALWA.IPN/45
Cell Type:	human Burkitt's lymphoma
DSMZ No:	ACC 71
Origin:	established from the tumor mass of an African child with Burkitt's lymphoma; this variant subline from the original NAMALWA (ACC 24) was described to be characterized by differential expression of immunoglobulin genes, c-myc genes and cell surface antigens (see reference); cells were
	reported to harbor a retrovirus of simian origin, SMRV-H
References:	Guy et al., Int. J. Cancer 45: 109-118 (1990)
Depositor:	Dr. K. Guy, Western General Hospital, Edinburgh, UK
DSMZ Cell Culture Data:	
Morphology:	lymphoblastoid, single cells with irregular forms or small clusters, in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.5 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximal density of about 2.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD3-, CD13-, CD19+, CD37+, HLA-DR+, sIgM+, sIgG-
Fingerprinting:	same DNA profile as NAMALWA using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypodiploid karyotype; 45(42-49)<2n>X, -Y, -3, +7, ins(1;?)(q11;?), ins(3)(q11;q27-29), der(5)t(5;?)(p15;?), der(12)t(12;?)(q24;?), der(15)t(15;?)(p11;?); NAMALWA marker chromosomes present
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NAMALWA.KN2
Cell Type:	human Burkitt's lymphoma
DSMZ No:	ACC 68
Origin:	established from the tumor mass of an African child with Burkitt's lymphoma; this variant subline from the original NAMALWA (ACC 24) was described to be characterized by differential expression of immunoglobulin and c-myc genes and of some cell surface antigens (see reference);
cells	were reported to harbor a retrovirus of simian origin, SMRV-H
References:	Guy et al., Int. J. Cancer 45: 109-118 (1990)
Depositor:	Dr. K. Guy, Western General Hospital, Edinburgh, UK
DSMZ Cell Culture Data:	
Morphology:	lymphoblastoid, single cells with irregular forms or small clusters, in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.5 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximal density at about $2.0-3.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD3-, CD13-, CD19+, CD37+, HLA-DR+, cyIgM+, smIgM-, sm/cyIgG-
Fingerprinting:	same DNA profile as NAMALWA using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypodiploid karyotype with 7% tetraploidy; 45(41-45)<2n>X, -Y; NAMALWA marker chromosomes present
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NAMALWA.PNT
Cell Type:	human Burkitt's lymphoma
DSMZ No:	ACC 69
Origin:	established from the tumor mass of an African child with Burkitt's lymphoma; this variant subline from the original NAMALWA (ACC 24) was described to be characterized by differential expression of immunoglobulin and c-myc genes and of some cell surface antigens (see reference);
cells	were reported to harbor a retrovirus of simian origin, SMRV-H
References:	Guy et al., Int. J. Cancer 45: 109-118 (1990)
Depositor:	Dr. K. Guy, Western General Hospital, Edinburgh, UK
DSMZ Cell Culture Data:	
Morphology:	lymphoblastoid, single cells with irregular forms or small clusters, in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.5 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximal density at about $2.0-3.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD37(+), HLA-DR+, sm/cyIgM+, sm/cyIgG-
Fingerprinting:	same DNA profile as NAMALWA using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	presence of NAMALWA marker chromosomes confirmed
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NB-4
Cell Type:	human acute promyelocytic leukemia
DSMZ No:	ACC 207
Origin:	established from the bone marrow of a 23-year-old woman with acute promyelocytic leukemia (APL = AML FAB M3) in second relapse in 1989; patented cell line; cells carry the t(15;17) PML-RARA fusion gene
References:	Lanotte et al., Blood 77: 1080-1086 (1991); Duprez et al., Leukemia 6: 1281-1287 (1992); review article: Drexler et al., Leukemia Res. 19: 681-691 (1995)
Depositor:	Dr. M. Lanotte, Centre Hayem, Hopital St. Louis, Paris, France
DSMZ Cell Culture Data:	
Morphology:	single cells in suspension, round and polymorphic cells
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; optimal split ratio at 1:2 to 1:3 every 2-3 days; seed out at ca. $0.5-1 \times 10^6$ cells/ml; viability may be low after thawing, but cells will recover soon when cultured initially with 20% FBS; cells might also grow better in 24-well-plates than in culture flasks
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-48 hours
Harvest:	cell harvest of $1.5-2.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3-, CD4+, CD5-, CD6-, CD7-, CD8-, CD11b-, CD13+, CD14-, CD15+, CD19-, CD33+, CD34-, CD38+, CD68-, HLA-DR-, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypertriploid karyotype with 3% polyploidy; 78(71-81)<3n>XX, -X, +2, +6, +7, +7, +11, +12, +13, +14, +17, -19, +20, +4mar, der(8)t(8;?)(q24;?), der(11)t(11;?)(?->::11p15->11q22.1::11q13->22.1.), der(12)t(12;?)(p11;?), 14p+, t(15;17)(q22;q11-12.1), der(19)t(10;19)(q21.1;p13.3)x2; identity confirmed
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NC-NC
Cell Type:	human B lymphoblastoid cells
DSMZ No:	ACC 120
Origin:	established from peripheral blood lymphocytes of a 27-year-old white Caucasian woman by Epstein-Barr virus (EBV) transformation in 1990; suitable normal control cell line for toxicity or radiosensitivity assays
References:	MacLeod & Bryant, Mutagenesis 7: 285-290 (1992)
Depositor:	Dr. R. A. F. MacLeod, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single, polymorph cells or small clumps in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	maximal density at about 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD3-, CD10-, CD13-, CD19(+), CD20+, CD37+, HLA-DR+, sm/cyIgG+, sm/cyIgM-, sm/cykappa+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human near diploid karyotype with hypodiploid sideline; 45/46(42-46)<2n>XX/X, -X; no structural rearrangements detected
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NCI-H929
Cell Type:	human multiple myeloma
DSMZ No:	ACC 163
Origin:	established from the pleural effusion of a 62-year-old white woman with myeloma (IgAkappa) at relapse; cells were described as follows: cells synthesize high amounts of immunoglobulin, c-myc proto-oncogene is rearranged and c-myc RNA is expressed, cells are EBV-negative; patented cell line
References:	Gazdar et al., Blood 67: 1542-1549 (1986); review article: Drexler & Matsuo, Leukemia Res 24: 681-703 (2000)
Depositor:	obtained from ATCC (CRL 9068), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	single, round to oval cells or clustered in small clumps in suspension
Medium:	80-90% RPMI 1640 + 10-20% FBS + 2 mM L-glutamine + 1 mM sodium pyruvate + 50 :M mercaptoethanol
Subculture:	cells are very difficult to culture; maintain at a minimum of 1.0×10^6 cells/ml; after thawing (7-10 days no growth) cells grow very slowly and viability may be initially low; culture in 24-well or 6-well plates is recommended; seed out at ca. 1×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 70 hours
Harvest:	maximum density at 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19-, CD20-, CD34-, CD37+, CD38+, CD79a-, cyCD79a+, CD80-, CD138+, HLA-DR-, sm/cyIgG-, sm/cyIgM-, smkappa-, cykappa+, sm/cylambda-
Fingerprinting:	unique DNA profile using multiplex PCR at D1S80, D2S44, D17S5 and ApoB
Species:	confirmed as human with IEF of AST, MDH, NP, PEP B
Cytogenetics:	human hypodiploid karyotype with 16% polyploidy; 45(43-46)<2n>X, -X, +8, -10, -13, +mar, dup(1)(q11q25), del(1)(p11p25), del(6)(q25), der(8)t(8;?)(q23;?)x2, del(9)(p13), del(12)(p12)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NEURO-2A
Cell Type:	mouse neuroblastoma
DSMZ No:	ACC 148
Origin:	established from the spontaneous tumor of a strain A albino mouse
References:	Klebe & Ruddle, J. Cell. Biol. 43: 69a (1969); Olmsted et al., Proc. Natl. Acad. Sci. USA 65: 129-136 (1970)
Depositor:	obtained from ATCC (CCL 131), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	about 30% of the cells grow like neuronal cells; 70% are round, loosely attached cells
Medium:	90% Dulbecco's MEM + 10% FBS + 1x non-essential amino acids
Subculture:	split culture 1:4 about once a week using trypsin/EDTA, cells do not grow to confluence and grow rather slowly; seed out at ca. 1×10^6 cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 70 hours
Harvest:	about 8×10^6 cells/75 cm ² ; maximal density at ca. 20×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, MDH, NP, PEP B
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NIH-3T3
Cell Type:	Swiss mouse embryo
DSMZ No:	ACC 59
Origin:	contact-inhibited NIH Swiss mouse embryo; used in DNA transfection assays
References:	Jainchill, J. Virol. 4: 549-553 (1969); Aaronsen & Todaro, J. Cell. Physiol. 72: 141-148 (1968); Copeland et al., Cell 16: 347-356 (1979)
Depositor:	Dr. H. Hauser, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	fibroblast, adherent monolayer
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS (or NCS)
Subculture:	do not allow culture to become completely confluent; split culture at 1:10 every 4 to 6 days using trypsin/EDTA; seed out at ca. 1×10^4 cells/cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 20 hours
Harvest:	cell harvest of about 8×10^4 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, MDH, PEP B
Cytogenetics:	murine hypertriploid karyotype with 3% polyploidy; 68(58-73)<3n>
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NK-92
Cell Type:	human natural killer lymphoma
DSMZ No:	ACC 488
Origin:	established from the peripheral blood of a 50-year-old man with non-Hodgkin's lymphoma (large granular lymphocytic) in 1992; cells were described as having azurophilic granula and strong cytotoxic NK activity
References:	Gong et al., Leukemia 8: 652-658 (1994); review article: Drexler & Matsuo, Leukemia 14: 777-782 (2000); MacLeod et al., Leukemia Res. 26: 1027-1033 (2002)
Depositor:	Dr. H. G. Klingemann, Terry Fox Laboratory, Vancouver, BC, Canada
DSMZ Cell Culture Data:	
Morphology:	cells growing in clumps in suspension
Medium:	75% alpha-MEM + 12.5% FBS + 12.5 % horse serum + 2 mM L-glutamine + 10 ng/ml IL-2
Subculture:	split saturated culture 1:2 every 2 days; seed out at ca. 0.5×10^6 cells/ml; maintain at $0.2-0.6 \times 10^6$ cells/ml; after thawing, it is recommended to first culture the cells in 24-well-plates
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	maximum density of about 0.8×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2+, CD3-, CD4-, CD5-, CD6-, CD7+, CD8-, CD11a+, CD13-, CD34-, CD45+, CD54+, CD56+, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypotetraploid karyotype with 4% polyploidy; 80(75-82)<4n>XXYY, +3, -4, -8, -9, -10, +12, +12, -14, -17, -17, -22, der(3;5)(q10;p10), i(7q), der(7)t(7;7)(p21;q11), der(8)t(8;8)(p23;q13), der(8)t(8;15)(p23.2;q14)x1-2, del(9)(p12p21), der(9)t(9;13)(p12;q11), der(10)t(10;18)(p12;q11)t(10;11)(q24;p13), der(12)t(11;12)(q13.4;q11)x1-2, der(13;19)(q10;q10), del(15)(q14)x1-2, del(16)(q23), der(16)t(11;16)(q22.2;q22), del(20)(?q12q13.1), der(Y)t(Y;17)(p11.3;q11), der(Y)t(Y;17)(p11.3;q11)t(11;17)(p1?2;q25); sideline with der(9;21)(p10;q10)
Viruses:	PCR: EBV+, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NOMO-1
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 542
Origin:	established from the bone marrow of a 31-year-old woman with acute myeloid leukemia (AML FAB M5a) at 2nd relapse; described to show strong lysozyme activity in the cytoplasm, phagocytic activities, and cells are responsive to induction of differentiation with TPA
References:	Kato et al., Acta Haematol. Jpn. 49: 277 (1986); Hamaguchi et al., Blood 77: 94-100 (1991); Uphoff et al., Leukemia 8: 1510-1526 (1994)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single round cells growing in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:2 to 1:4 twice a week; seed out at ca. $0.5-1.0 \times 10^6$ cells/ml; maintain at ca. 0.5×10^6 cells/ml; maximum density at ca. 2×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	cell harvest of ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD4+, CD13+, CD14-, CD15+, CD19-, CD33+, CD34-, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype with 8% polyploidy; 46-47<2n>XX, +8, -13, +mar, add(7)(q32), der(9)del(9)(p11p13-21)t(9;11)(p22;q23), t(9;13)(q13;q11), der(11)t(9;11)(p22;q23); carries reciprocal t(9;11) with rearrangement of MLL; matches published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NRK-49F
Cell Type:	rat kidney fibroblasts
DSMZ No:	ACC 172
Origin:	cell line was cloned from a mixed culture of normal rat kidney cells; cells exhibit contact inhibition and are very sensitive; indicator cells for viral or chemical transformation; this cell line is frequently used for TGF- β bioassays; (= ATCC CRL 1570)
References:	Duc-Nguyen et al., J. Bact. 92: 1133-1140 (1966); DeLarco & Todaro, J. Cell. Physiol. 94: 335-342 (1978); Assoian et al., J. Biol. Chem. 258: 7155-7160 (1983)
Depositor:	Dr. J. Wehland, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	fibroblast-like cells with a fusiform shape
Medium:	95% Dulbecco's MEM + 5% FBS
Subculture:	split culture 1:4 to 1:5 about 2-3 times a week using trypsin/EDTA when the cells have reached 3/4 of confluence; seed out at ca. $0.5-1.0 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	cell harvest of about 3.5×10^6 /80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $0.6-1.0 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as rat with IEF of AST, NP, PEP B
Cytogenetics:	near-diploid rat karyotype with 4% polyploidy; 40(37-42)<2n>
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NRK-52E
Cell Type:	rat kidney epithelial-like cells
DSMZ No:	ACC 199
Origin:	cell line was cloned from a mixed culture of normal rat kidney cells as was NRK-49F (DSM ACC 172); the two cell lines were described to be distinct in growth properties and in transforming abilities; NRK-52E is supposed to be much more stable (= ATCC CRL 1571)
References:	DeLarco & Todaro, J. Cell. Physiol. 94: 335-342 (1978)
Depositor:	Dr. J. Wehland, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing as monolayers
Medium:	90-95% Dulbecco's MEM + 5-10% FBS
Subculture:	split confluent culture 1:3 to 1:5 every 3-5 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 45 hours
Harvest:	cell harvest of about $3-5 \times 10^6$ /80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as rat with IEF of AST, MDH, NP, PEP B
Cytogenetics:	near-diploid rat karyotype with 4% tetraploidy; 41/42(36-42)<2n>XX; hypodiploid sideline with 1q+ marker (18%); near triploid sideline with additional structural and numerical rearrangements (2%); presence of sidelines-subclones may warrant cloning prior to use in transformation assays
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NS20Y
Cell Type:	mouse neuroblastoma
DSMZ No:	ACC 94
Origin:	established from a A/Jax mouse strain with neuroblastoma C1300; cells were described to be aneuploid and to contain high specific activity of choline acetyltransferase
References:	Amano et al., Proc. Natl. Acad. Sci. USA 69: 258-263 (1972)
Depositor:	Dr. B. Hamprecht, University of Tübingen, Tübingen, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent cells growing in monolayers or small clumps, round cell body with cellular processes
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split semi-confluent culture 1:5 to 1:10 using trypsin/EDTA (cells can also be detached by tapping the flask); seed out at ca. 2×10^6 cells/80 cm ² ; maximal density at ca. $5\text{--}10 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of about 1×10^7 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of MDH, NP, PEP B
Cytogenetics:	murine karyotype
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NTERA-2
Cell Type:	human embryonal carcinoma (teratocarcinoma)
DSMZ No:	ACC 527
Origin:	NTERA-2 was cloned from cell line TERA-2 which was derived from a metastatic teratocarcinoma of a 22-year-old Caucasian male
References:	Fogh & Trempe, in "Human Tumor Cells in Vitro", 1975, Plenum Press, pp. 115-159; Andrews et al., Cancer Surveys 2: 41-73 (1983); Andrews et al., Lab. Invest. 50: 147-162 (1984)
Depositor:	Dr. R. Brandt, University of Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells carrying strong granula growing adherent in monolayer
Medium:	85% Dulbecco's MEM + 10% FBS + 5% horse serum
Subculture:	split confluent culture 1:3 to 1:6 1-2 times a week using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ² ; after thawing use collagen-coated flasks for the first week
Incubation:	at 37 °C with 10% CO ₂
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	cell harvest of about 14×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hyperdiploid karyotype; 48-58<2n>XXXY, +1, +16, +20, +22, +3-4mar, add(X)(p11), add(X)(q2?5), der(X;1)(p10;q10), add(1)(p35-36), add(11)(p15), add(13)(q3?1); sideline with i(3q)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NUC-1
Cell Type:	mouse urinary bladder
DSMZ No:	ACC 225
Origin:	established from an adult female C3H/Law mouse; tumorigenic, invasive in vitro immortalized urine bladder cell line, aneuploid (chromosome number 70-75)
References:	De Boer et al., Int. J. Cancer 54: 1022-1027 (1993)
Depositor:	Dr. W. I. de Boer, Erasmus University, Rotterdam, The Netherlands
DSMZ Cell Culture Data:	
Morphology:	spindle cell-like, disorderly arranged cells, growing in multilayers
Medium:	45% Ham's F10 + 45% Dulbecco's MEM + 10% FBS + 5 :g/ml bovine insulin + 5 :g/ml human transferrin + 50 nM hydrocortisone + 5 ng/ml selenite + 10 :M HEPES
Subculture:	split confluent culture 1:10 once a week using trypsin/EDTA
Incubation:	at 37 /C with 5% CO ₂
Doubling Time:	doubling time of about 24 hours
Harvest:	harvest of about 20 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, MDH, NP
Cytogenetics:	hypotetraploid flat-moded murine karyotype with limited centric fusion, dm present in 25% cells; 70(66-75)<4n>
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	NUC-5
Cell Type:	mouse urinary bladder
DSMZ No:	ACC 226
Origin:	established from an adult female C3H/Law mouse; non-tumorigenic in vitro immortalized urine bladder cell line with aberrant growth factor response, aneuploid (chromosome number 70-75)
References:	De Boer et al., Int. J. Cancer 54: 1022-1027 (1993)
Depositor:	Dr. W. I. de Boer, Erasmus University, Rotterdam, The Netherlands
DSMZ Cell Culture Data:	
Morphology:	large and flat regular arranged cells, growing in up to 3 cell layers
Medium:	45% Ham's F10 + 45% Dulbecco's MEM + 10% FBS + 5 :g/ml bovine insulin + 5 :g/ml human transferrin + 50 nM hydrocortisone + 5 ng/ml selenite + 10 :M HEPES
Subculture:	split confluent culture 1:4 every 6-7 days using trypsin/EDTA
Incubation:	at 37 /C with 5% CO ₂
Doubling Time:	doubling time of about 60-75 hours
Harvest:	harvest of 2-3 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1-2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, NP
Cytogenetics:	murine hypotetraploid karyotype with 14% polyploidy and 1-3 dmin in 34% cells; 70(64-74)<4n>; 1-3 centric fusion markers present; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	OCI-AML2
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 99
Origin:	established from the peripheral blood of a 65-year-old man with acute myeloid leukemia (AML FAB M4) in 1986 at diagnosis; cells were described to express the retinoic acid receptor gene
References:	Wang et al., Leukemia 3: 264-269 (1989)
Depositor:	Dr. E. A. McCulloch, Ontario Cancer Institute, Toronto, Canada
DSMZ Cell Culture Data:	
Morphology:	single, round to oval cells in suspension
Medium:	80-90% alpha-MEM + 10-20% FBS
Subculture:	cells are difficult to culture! cells grow slowly; maintain at $0.5-1.5 \times 10^6$ cells/ml; seed out at ca. 0.5×10^6 /ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30-50 hours
Harvest:	maximal density at about 4×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD3-, CD4+, CD13+, CD14(+), CD15+, CD19-, CD33+, CD34-, CD68+, HLA-DR-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human hyperdiploid karyotype with 3.8% polyploidy; 48(43-49)<2n>XY, +6, +8, der(1)inv(1)(p36q31)t(1;6)(q13;p12), der(2)t(2;17)(p23;q24.1)del(2)(q14.2q36), der(3)t(1;3)(p36;p25), ins(3;2)(q21;q14.2q36), t(5;8)(q11.2;q24), der(6)t(1;6)(q31;p12)t(3;6)(q26;q24), inv(12)(p13.3q13.2), t(13;14)(q32/33;q24.2), der(17)t(2;17)(p23;q24.1); sideline with +der(5); carries apparent variant translocations involving several ANLL breakpoints: 1p36, 3q21, 3q26 (megakaryocytic abnormalities), 12p13, 17q24 (FAB-M4)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	OCI-AML5
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 247
Origin:	established from the peripheral blood of a 77-year-old man with acute myeloid leukemia (AML M4) in relapse in 1990; the cells are growth factor-dependent and are described to respond to G-CSF, GM-CSF, IL-3, and FLT3 ligand; we have noted that the cell line can become very quickly cytokine-independent (within 3-4 weeks), presumably due to rapid outgrowth of independent cells
References:	Wang et al., Leukemia 5: 493-499 (1991); Drexler et al., Leukemia 11: 701-708 (1998)
Depositor:	Dr. E. A. McCulloch, Ontario Cancer Institute, Toronto, Canada
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly or in clumps in suspension
Medium:	70% alpha-MEM + 20% FBS + 10% vol conditioned medium of cell line 5637 (DSM ACC 35) (or 10 ng/ml GM-CSF)
Subculture:	maintain at $0.8-1.4 \times 10^6$ cells/ml; optimal split ratio of 1:2 to 1:4 every 2-4 days; seed out at about 0.5×10^6 cells/ml. CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30-50 hours
Harvest:	saturation density at about 2.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-6 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4-, CD13+, CD14-, CD15+, CD19-, CD33+, CD34-, CD68+, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hyperdiploid karyotype; 48(44-48)<2n>XY, +1, +8, der(1)t(1;19)(p13;p13)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	OCI-LY-19
Cell Type:	human B cell lymphoma
DSMZ No:	ACC 528
Origin:	established from the bone marrow of a 27-year-old woman with B-cell Non-Hodgkin's lymphoma (B-NHL) (diffuse large cell lymphoma, DLCL, stage 4B, at relapse) in 1987
References:	Chang et al., Leukemia Lymphoma 19: 165-171 (1995)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single round cells growing in suspension
Medium:	80-90% alpha-MEM + 10-20% FBS
Subculture:	split saturated culture 1:4 to 1:8 every 3-5 days; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. $0.5\text{--}3.0 \times 10^6$ cells/ml; maximal density of ca. 3×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	cell harvest of ca. 3×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20(+), CD37-, CD38+, CD79a+, cyCD79a+, CD80(+), CD138+, HLA-DR+, sm/cyIgG-, sm/cyIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype with 4% polyploidy; 48(46-52)<2n>X, -X, +6, +6, +8, t(4;8)(q3?2;q?24), del(6)(q15)x2, r(8)(??), t(14;18)(q32;q21), add(18)(q23); carries t(14;18) effecting IGH-BCL2 juxtaposition
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	OCI-M1
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 529
Origin:	established from erythroleukemia blasts (AML M6) of a 62-year-old man following a 7-year chlorambucil treatment for chronic lymphocytic leukemia (CLL); described to express many markers of the erythroid lineage as well as several non-erythroid markers
References:	Papayannopoulou et al., Blood 72: 1029-1038 (1988); Tweeddale et al., Blood 74: 572-578 (1989); Hirata et al., Leukemia 4: 365-372 (1990)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly in suspension
Medium:	80-90% Iscove's MDM + 10-20% FBS
Subculture:	split saturated culture 1:2 to 1:4 every second day; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. 0.5×10^6 cells/ml; maximal density of ca. 1.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 1×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD4-, CD13+, CD14-, CD15+, CD19-, CD33+, CD34+, CD41-, CD42-, CD68-, CD71+, GlyA+, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	OMEGA-E
Cell Type:	NIH-3T3 (Swiss mouse embryo) derivative-retroviral packaging cell line
DSMZ No:	ACC 253
Origin:	derived from NIH-3T3 cells; high titer ecotropic helper-free packaging cell line; generated with separate gagpol and ecotropic env expression constructs with minimal sequence overlap and decreased sequence homology achieved by "codon wobbling"; prior to use in nude mice, cells need to be subcloned; classified as risk category S1 according to the German Central Commission for Biological Safety (ZKBS)
References:	Morgenstern et al., Nucleic Acids Res. 18: 3587-3596 (1990)
Depositor:	Dr. W. Dirks, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	cells grow as adherent fibroblastic monolayers
Medium:	90% Dulbecco's MEM + 10% NCS
Subculture:	split confluent culture 1:5 every 3-4 days using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 20-30 hours
Harvest:	cell harvest of about 5-8 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2-3 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of MDH, NP
Cytogenetics:	murine hypertriploid karyotype; 71(63-73)<3n>; 1-2 dm (24% cells)
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	ONCO-DG-1
Cell Type:	human thyroid carcinoma
DSMZ No:	ACC 507
Origin:	established from the malignant tumor of a 49-year-old woman with well-differentiated oxyphilic papillary thyroid carcinoma (PT2, G1, No, Mo); cells were described to be tumorigenic in nude mice
References:	Grimm et al., Virchows Archiv B Cell Path. 62: 97-104 (1992)
Depositor:	Dr. J. Schönberger, University Hospital Regensburg, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells with large vacuoles growing as monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:5 every 3-4 days using trypsin/EDTA (for 10 min); seed out at ca. 1×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of about 10×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1.5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with ciprofloxacin (Ciprobay), then negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, thyroglobulin-, vimentin(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypotriploid karyotype with 8% polyploidy; 59(56-61)<3n>XX, -X, +1, -2, -4, -6, -8, -9, -13, -13, -14, -15, -15, -17, -17, -18, -19, -21, -22, +7-9mar, +0-3 dm, add(1)(p3?5), add(1)(p2?1), add(1)(q11), der(3;15)(p10;q10), add(3)(p14), del(5)(q23), add(6)(q23), der(7;14)(q10;q10), del(9)(p23), add(10)(p11), add(11)(p11)x2, del(?11)(q13q14.2), der(12;18)(q10;q10), add(13)(p11), add(14)(p11), add(16)(q23-24), add(17)(p11), add(21)(p11)x1-2; carries large submetacentric hsr marker
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	OPM-2
Cell Type:	human multiple myeloma
DSMZ No:	ACC 50
Origin:	established from the peripheral blood of a 56-year-old woman with multiple myeloma (IgG lambda) in leukemic phase (relapse, terminal) in 1982; cells express c-myc mRNA
References:	Katagiri et al., Int. J. Cancer 36: 241-246 (1985); review article: Drexler & Matsuo, Leukemia Res 24: 681-703 (2000)
Depositor:	Dr. E. B. Thompson, University of Texas, Galveston, Texas, USA
DSMZ Cell Culture Data:	
Morphology:	single, round to polygonal cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.3-0.7 \times 10^6$ cells/ml; split 1:2 to 1:8 every 2 days; keep at 20% FBS for the first week, slow growth during first 1-2 weeks; seed out at ca. 1×10^6 cells/ml in a 6-well-plate or 24-well-plate
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	maximal density at 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19-, CD20-, CD34-, CD37-, CD79a-, cyCD79a+, CD80-, CD138-, HLA-DR-, sm/cyIgG-, sm/cyIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypertriploid/hypotetraploid karyotype; 78(77-82)<3n>XX-X, +1, +1, +1, +2, +4, +5, -7, +9, +11, -14, +15, +16, +18, +19, -21, +2-3mar, add(1)(p11), add(1)(q11)x2, del(1)(p11)x2, del(1)(q32), t(4;14)(p16;q32.3), der(4)t(4;14)(p16;q32.3), add(5)(q11), add(5)(q15-21), add(5)(q35), der(8)t(1;8)(?q32;q24)x1-2, add(14)(q32.1), der(22)t(1;22)(q11;p13)x2; sideline with del(7)(p15), del(10)(q22); resembles published karyotype; carries cryptic t(4;14) associated with fusion of IGH with FGFR3
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	OTH-74D4
Cell Type:	mouse hybridoma (CD45RA)
DSMZ No:	ACC 478
Origin:	described as secreting a mouse monoclonal antibody (IgG ₁) CD45RA; the cell line was established in 1985 by fusion of PX63AG8.653 cells with spleen cells of a mouse immunized with an EBV-transformed B cell line from a patient with MHC class II deficiency
References:	Hadam, Leucocyte Typing VI: 499-502 (1997)
Depositor:	Dr. M. R. Hadam, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	round, single cells growing in suspension and some slightly adherent
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement" for the first 1-2 weeks post-thawing)
Subculture:	split saturated cultures 1:2 to 1:3 every 2 days; seed out at 0.3×10^6 cell/ml; maintain at $0.2-0.4 \times 10^6$ cells/ml; cells do not grow without HFCS, but it can be used at only 50% (1:100) of the recommended concentration
Incubation:	at 37°C with 5-10% CO ₂
Doubling Time:	doubling time of about 40-50 hours
Harvest:	maximal density of about $0.6-0.8 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgG ₁ monoclonal antibody
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	P-19
Cell Type:	mouse embryonal carcinoma
DSMZ No:	ACC 316
Origin:	established from the teratocarcinoma induced in a C3H/He strain mouse
References:	McBurney et al., Develop. Biol. 89: 503-508 (1982); Rudnicki et al., Develop. Biol. 138: 348-358 (1990)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing as adherent monolayer
Medium:	80-90% alpha-MEM + 10-20% FBS
Subculture:	split confluent culture 1:10 to 1:20 once or twice a week using trypsin/EDTA; seed out at ca. $0.3-0.4 \times 10^5$ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 2-3 days
Harvest:	cell harvest of max. 0.5×10^6 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as murine with IEF of AST, LDH, MDH, NP
Cytogenetics:	murine near-diploid karyotype with 16% polyploidy; 40(31-44)<2n>; one centric fusion marker; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	P-388D1(IL-1)
Cell Type:	mouse monocytes-macrophages
DSMZ No:	ACC 288
Origin:	established from the ascites of a DBA/2 mouse with a lymphoid neoplasm induced by methylcholanthrene and subsequently enriched for adherent (monocyte-macrophage) cells by selection (yielding this IL1-producing subclone); reported to produce IL-1 after stimulation with LPS or PMA, to phagocytize latex articles and to synthesize lysozyme
References:	Dawe et al., Am. J. Pathol. 33: 603 (1957); Koren et al., J. Immunol 114: 894-897 (1975); Coleman et al., Leukemia 13: 1592-1600 (1999)
Depositor:	Dr. U. Groß, University of Würzburg, Institute for Hygiene and Microbiology, Würzburg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent elliptical and round loosely adherent cells growing in monolayers
Medium:	80% RPMI 1640 + 20% horse serum + 3.5 g/L glucose
Subculture:	split confluent culture 1:3 to 1:6 every 3-6 days using trypsin/EDTA; seed out at ca. $0.1-0.3 \times 10^5$ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-60 hours (according to density)
Harvest:	cell harvest of ca. 1.5×10^5 cells/cm ²
Storage:	frozen with 70% medium, 15% horse serum, 5% FBS, 10% DMSO at about $2-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as murine with IEF of LDH, MDH, NP
Cytogenetics:	murine hypertriploid karyotype with 8% polyploidy; 61(57-65)<3n>; 5-8 centric fusion markers
Viruses:	ELISA: reverse transcriptase negative

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	P-815
Cell Type:	mouse mastocytoma
DSMZ No:	ACC 1
Origin:	established from the mastocytoma tumor of a DBA/2 mouse treated with methylcolanthrene; used as target cells for cytotoxic T cell assays; as reported cells exhibit no effector activity in an antibody-dependent cell mediated cytotoxic system
References:	Ralph et al., Biochem. Biophys. Res. Commun. 61: 1268-1275 (1974); Ralph et al., Nature 249: 49-51 (1974); Ralph et al., J. Exp. Med. 143: 1528-1553 (1976); Ralph et al., J. Immunol. 119: 950-954 (1977)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension; some cells may be adherent
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-5 \times 10^5$ cells/ml, cells grow very fast; split 1:5 to 1:10 every 3-4 days; seed out at ca. $1-2 \times 10^6$ cells/10 ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 18-22 hours
Harvest:	maximum cell density at $> 1 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of LDH, NP, PEP B
Cytogenetics:	murine near-diploid karyotype; 41(36-43)<2n>; 1-2 centric fusion markers present
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	P12-ICHIKAWA
Cell Type:	human T cell leukemia
DSMZ No:	ACC 34
Origin:	established from the peripheral blood of a 7-year-old boy with acute lymphoblastic leukemia (ALL)
References:	Kitahara et al., Acta Hematol. Jpn. 41: 140 (1978); Watanabe et al., Cancer Res. 38: 3494-3498 (1978)
Depositor:	Dr. Jun Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	single, polymorph cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.3\text{--}0.9 \times 10^6$ cells/ml; optimal split ratio 1:3 to 1:5
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximum density at about 1.2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD2+, CD3+, CD4+, CD5+, CD6+, CD7+, CD8-, CD13+, TCRalpha/beta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human flat-moded hypotetraploid karyotype with 1.6% polyploidy; 82-84<4n>XX, -Y, -Y, -2, +6, -9, -9, -10, -14, +20, -21, -21, -22, -22, +mar, der(2;9)(p10;q10), del(4)(q25), der(10;22)(q10;q10), add(19)(q13); sideline with N4, add(8)(p1?); matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	P3/NSI/1-AG4-1
Cell Type:	mouse myeloma
DSMZ No:	ACC 145
Origin:	P3/NSI/1-AG4-1 is a non-secreting clone of P3X63AG8 which is derived from P3K, the induced plasmacytoma MOPC-21; cells do not grow in HAT medium, extensively used in cell fusion studies; cells were described to synthesize kappa chain, but do not secrete free light chain
References:	Köhler et al., Eur. J. Immunol. 6: 292-295 (1976)
Depositor:	Dr. G. Köhler, Max-Planck-Institute, Freiburg, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells, single and in small clusters in suspension
Medium:	92.5-95% Iscove's MDM + 5-7.5% FBS + 2 mM L-glutamine
Subculture:	maintain at $0.1-0.8 \times 10^6$ cells/ml; split 1:3 to 1:10 every 1-3 days
Incubation:	at 37 °C with 7.5% CO ₂
Doubling Time:	doubling time of 24 hours
Harvest:	maximal density at $1.2-1.4 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of LDH, NP, PEP B
Cytogenetics:	hypertriploid murine karyotype with 2% polyploidy; 66(62-72)<3n>; 3-4 centric fusion markers present
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PA-TU-8902
Cell Type:	human pancreas adenocarcinoma
DSMZ No:	ACC 179
Origin:	established in 1985 from the primary ductal pancreatic adenocarcinoma (grade II) from a 44-year-old woman; reported as inducing metastasis in nude mice and as secreting proteinases (e.g. urokinase, cathepsin B and D)
References:	Elsässer et al., Virchows Archiv B Cell Pathol. 64: 201-207 (1993)
Depositor:	Dr. H. P. Elsässer, University of Marburg, Marburg, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial adherent cells growing in monolayers
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 5% FBS + 5% horse serum + 2 mM L-glutamine
Subculture:	split confluent culture 1:4 to 1:10 every 3-6 days using trypsin/EDTA; seed out at ca. $2-3 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 25-40 hours
Harvest:	cell harvest of $1.0-1.5 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human flat-moded pseudotetraploid karyotype; 91(83-94)<4n>XXX-X, +5, -6, -6, +7, -10, -10, +12, -13, -13, -14, -19, +20, -22, -22, +5mar, del(4)(q26)x2, der(5)t(5;14)(p11;q11), der(5)t(5;?)(q11;?), der(7)t(7;?)(p14;?), der(9)t(9;11)(p12;q11), der(14)t(?5;14)(?p11;p11)x2, del(17)(p12), der(17)t(17;?)(p11/12;?), der(18)t(18;?)(p12;?), der(19)t(19;?)(q13.1;?), ?i(21q)x3, der(22)t(22;?)(q13;?); carries 19p13.1 recurrent tumor breakpoint; corresponds to vague published description
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PA-TU-8988S
Cell Type:	human pancreas adenocarcinoma
DSMZ No:	ACC 204
Origin:	established in 1985 from the liver metastasis of a primary pancreatic adenocarcinoma from a 64-year-old woman (sister cell line is PA-TU-8988T, DSM ACC 162)
References:	Elsässer et al., Virchows Archiv B Cell Pathol. 61: 295-306 (1992)
Depositor:	Dr. H. P. Elsässer, University of Marburg, Marburg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells growing in monolayers
Medium:	90% Dulbecco's MEM + 5% FBS + 5% horse serum + 2 mM L-glutamine
Subculture:	split confluent culture 1:10 once a week using trypsin/EDTA; seed out at ca. 4×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of about 40-60 hours
Harvest:	cell harvest of about 30×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	same DNA profile as PA-TU-8988T using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human near triploid karyotype with 6% polyploidy; 71(67-77)<3n>XXX, +3, -4, +7, -8, +9, +11, +11, -15, -17, -18, -18, -19, -21, -22, +6mar, del(1)(p21), del(3)(p11p22), del(6)(q26), del(9)(p12), dup(11)(p14->pter), del(12)(p13), add (15)(p12)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PA-TU-8988T
Cell Type:	human pancreas adenocarcinoma
DSMZ No:	ACC 162
Origin:	established in 1985 from the liver metastasis of a primary pancreatic adenocarcinoma from a 64-year-old woman; sister cell line of PA-TU-8988S (DSM ACC 204)
References:	Elsässer et al., Virchows Archiv B Cell Pathol. 61: 295-306 (1992)
Depositor:	Dr. H. P. Elsässer, University of Marburg, Marburg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent cells growing in monolayers
Medium:	90% Dulbecco's MEM + 5% FBS + 5% horse serum + 2 mM L-glutamine
Subculture:	split confluent culture 1:3 to 1:5 every 3-5 days using trypsin/EDTA; seed out at ca. 18×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 22-30 hours
Harvest:	cell harvest of about $20-30 \times 10^6$ cells/75 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	same DNA profile as PA-TU-8988S using multiplex PCR at D1S80, D2S44, D17S5 and ApoB
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human near-triploid karyotype with 2% polyploidy; 68(63-70)<3n>XXX, -4, +6, +6, +8, +11, -15, -16, -17, -19, +20, -21, -22, +mar, der(1)t(1;?)(p11;?), i(6p), del(6)(q23)x2, der(8)t(8;?)(q24.3;?), der(8)t(8;?)(p11;?)x2, der(11)t(11;?)(p13;?)x2, del(12)(p12.3), dup(14)(q11q12), del(17)(p12), der(17)t(17;?)(p13;?); sideline with additional der(3)t(3;?)(q11;?); 1p11 and 8q24 breakpoints previously reported in pancreatic cancer
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PAB-100
Cell Type:	mouse hybridoma (anti-SV40 large T viral antigen)
DSMZ No:	ACC 188
Origin:	established by fusion of NS-1 myeloma cells and spleen cells from a female BALB/c mouse immunized with SVT-2, an SV40-transformed BALB/c mouse cell line; cells reportedly secrete an IgG ₁ antibody recognizing denaturation-sensitive antigenic determinants unique to large T antigen
References:	J. Virol. 34: 752-763 (1980); J. Virol. 37: 478-482 (1981)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single cells growing in suspension
Medium:	80% Dulbecco's MEM + 20% FBS + non-essential amino acids
Subculture:	optimal split ratio 1:2 every 2-3 days; maximum density of about 1.0×10^6 cells/ml should not be exceeded
Incubation:	at 37 °C with 5% CO ₂
Harvest:	cell harvest of about $0.5-1.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, NP, PEP B
Cytogenetics:	predominantly mouse chromosomal morphology with 3 iso-chromosome markers present in most cells; 140(134-155)
Viruses:	ELISA: reverse transcriptase positive

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PAB-122
Cell Type:	mouse hybridoma (anti-p53)
DSMZ No:	ACC 166
Origin:	established by fusion of NS-1 myeloma cells and spleen cells from a male BALB/c mouse immunized with SV40-transformed, 3T3 clone 4 mouse cells; cells were reported to secrete an IgG ₂ b antibody recognizing p53 protein present in nuclei of SV40-transformed cells
References:	Gurney et al., J. Virol. 34: 752-763 (1980)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single cells in suspension
Medium:	80% Dulbecco's MEM + 20% FBS
Subculture:	maintain at $0.5-5 \times 10^5$ cells/ml; split 1:5 every 2-3 days
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximal density at $1-2 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of MDH, NP
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PAB-1620
Cell Type:	mouse hybridoma (anti-wild type p53)
DSMZ No:	ACC 280
Origin:	established in 1982 by immunizing BALB/c mice with VLM tumor cells and fusing splenocytes with SP2/0-Ag14 myeloma cells; described as secreting a mouse monoclonal antibody (IgG ₂ akappa) detecting epitopes of wild type p53 (not mutant or denatured p53), that occur on human, primate, bovine, mouse and rat cells, by ELISA and immunoblotting
References:	Ball et al., EMBO J. 3: 1485-1491 (1984); Hebel et al., Z. Naturforsch. 41C: 94-99 (1986)
Depositor:	Prof. G. Brandner, Institute for Med. Microbiology and Hygiene, Freiburg, Germany
DSMZ Cell Culture Data:	
Morphology:	single round cells growing in suspension, partly slightly adherent
Medium:	90% Dulbecco's MEM + 10% FBS + 4 mM L-glutamine
Subculture:	split saturated culture optimally 1:2 to 1:3 every second day; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. $0.3\text{-}0.4 \times 10^6$ cells/ml; maximal density at ca. 0.7×10^6 cells/ml; some cells attach to the flask surface, but may be detached by tapping the flask; if viability decreases, addition of HFCS ("Hybridoma Fusion and Cloning Supplement") may be useful
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	cell harvest of ca. 0.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of IgG ₂ a monoclonal antibody
Species:	confirmed as mouse with IEF of LDH, NP
Cytogenetics:	murine flat-modet hypertetraploid karyotype; 82-97<4n>; several centric fusion markers present
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PB-1
Cell Type:	mouse pre B cells
DSMZ No:	ACC 241
Origin:	established from the bone marrow of a CBA/C57BL mouse in 1995
References:	Mire-Sluis et al., J. Immunol. Methods 236: 71-76 (2000)
Depositor:	Dr. A. R. Mire-Sluis, National Institute for Biological Standards and Control, Potters Bar, England
DSMZ Cell Culture Data:	
Morphology:	cells are very small and grow mainly in small clusters, but also singly in suspension
Medium:	85% McCoy's 5A + 15% non-heat inactivated FBS + 50 ng/ml human IL-7 + 0.4x MEM vitamins + 1x MEM non-essential amino acids + 0.5x essential amino acids + 1 mM sodium pyruvate + 0.12% sodium bicarbonate + 2 mM L-glutamine + 50 :M mercaptoethanol
Subculture:	cells are difficult to culture! split saturated culture 1:2 every 2-3 days; seed out at ca. 1.0×10^6 cells/ml; after thawing, cells may initially grow rather slowly for about one week; maintain at $1-3 \times 10^6$ cells/ml. CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 50 hours
Harvest:	maximum density at ca. 3×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	IL-7 dependency confirmed by ³ H-thymidine assay
Species:	confirmed as mouse with IEF of AST, NP
Cytogenetics:	murine near-diploid karyotype
Viruses:	ELISA: reverse transcriptase negative

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PC-12
Cell Type:	rat adrenal pheochromocytoma
DSMZ No:	ACC 159
Origin:	established from a transplantable rat adrenal pheochromocytoma in 1976; cells were described to synthesize catecholamines (dopamine, norepinephrine); in response to nerve growth factor (NGF) a neuronal phenotype could be induced reversibly
References:	Greene et al., Proc. Natl. Acad. Sci. USA 73: 2424-2428 (1976); Levi et al., Science 229: 393-395 (1985)
Depositor:	obtained from ATCC (CRL 1721), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	small cells growing in clumps in suspension, adhering poorly to plastic
Medium:	85% RPMI 1640 + 10% horse serum + 5% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 3-4 days; cells grow in suspension without collagen; using collagen (Collagen S- Roche at 30 : g/ml) cells grow adherent, faster and to a higher density; the adherent cells may be detached mechanically without trypsin, for single cells the suspension should be passed several times through a syringe with a 22-gauge needle; seed out at ca. $5-15 \times 10^6$ cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	about $1-2 \times 10^6$ cells/ml or 30×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as rat with IEF of AST, MDH, NP, PEP B
Cytogenetics:	rat hypodiploid karyotype with 20% polyploidy; 39(38-40)<2n>
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PC-3
Cell Type:	human prostate carcinoma
DSMZ No:	ACC 465
Origin:	established from the bone marrow metastasis isolated post-mortem from a 62-year-old Caucasian man with grade IV prostate cancer (poorly differentiated adenocarcinoma) after androgen suppression therapy; described to form tumors in nude mice, to grow in soft agar, and to be unresponsive to androgen treatment
References:	Kaighn et al., Invest. Urol. 17: 16-23 (1979); Pan et al., Cytogenet. Cell Genet. 87: 225-232 (1999)
Depositor:	Dr. J. R. W. Masters, University College London, London, UK (who had obtained the cells from the original investigator)
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing adherently in monolayers or in multilayer foci
Medium:	45% Ham's F12 + 45% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:10 every 3-5 days using trypsin/EDTA; seed out at ca. 1×10^4 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of ca. 0.5×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human highly-rearranged flat-moded hypotetraploid karyotype with 3% polyploidy; 80-90<4n>XX/XXX, -Y, -1, -1, -2, -2, +3, +3, -4, -5, +7, -8, -8, -10, -10, -10, -10, -12, -13, +14, -15, -15, -16, -16, -17, -18, -18, -22, +18-21mar, +1-2dm, del(1)(p34p35), add(1p?32), add(3)(p1?), add(3)(q13), add(3)(q21)x2, del(5)(p1?)x1-2, add(9)(p1?1), del(9)(p21)x1-2, add(12)(q24), add(13)(p11), del(17)(p11), add(19)(p13); sideline with der(3)add(3)(p2?5)add(3q2?), del(4)(q2?4q2?8), add(7)(q11), add(8)(p1?), add(9)(q34); extensive genomic amplification of c-MYC in hsr present in markers; distantly related to published karyotype
Viruses:	PCR: EBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PEER
Cell Type:	human T cell leukemia
DSMZ No:	ACC 6
Origin:	established from the peripheral blood of a 4-year-old girl with T cell acute lymphoblastic leukemia (T-ALL) in second relapse in 1977; cell lines BE-13 (DSM ACC 396) and PEER show identical DNA fingerprints and hence share common origin, presumably due to cross-contamination; however, their diploid and tetraploid karyotypes indicative of earlier and later passage numbers, respectively, suggest that BE-13 is derived from PEER, rather than vice-versa.
References:	Ravid et al., Int. J. Cancer 25: 705-710 (1980); Heyman et al., Leukemia 8: 425-434 (1994)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	small, round cells growing mostly singly in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	cells are very difficult to culture! cells grow best in 24-well-plates; maintain at ca. $0.5-1.0 \times 10^6$ cells/ml; split saturated culture 1:2 every 2-3 days; seed out at ca. 1.0×10^6 cells/ml (culture should not be split during the first 3-4 days after thawing)
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	saturation density at 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3+, CD4+, CD5+, CD6+, CD7+, CD8+, CD13-, CD19-, CD34-, TCRalpha/beta-, TCRgamma/delta+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human pseudodiploid karyotype with 7% polyploidy; 46(42-47)<2n>XX, der(4)?dup ins(4;4)(?p11;q21q25), del(5)(q22q31), del(6)(q13q22), del(9)(p11p22), del(9)(q22); resembles published karyotype; 5q-, 6q-, 9p- and 9q- associated with hematological malignancy
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PF-382
Cell Type:	human T cell leukemia
DSMZ No:	ACC 38
Origin:	established from the pleural effusion of a 6-year-old girl with acute lymphoblastic leukemia (ALL) in second relapse
References:	Pegoraro et al., J. Natl. Cancer Inst. 75: 285-290 (1985)
Depositor:	Dr. L. Pegoraro, University of Turin, Turin, Italy
DSMZ Cell Culture Data:	
Morphology:	single, round to oval cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5\text{--}2.0 \times 10^6$ cells/ml; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 70 hours
Harvest:	maximal density of about $3\text{--}4 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2+, CD3-, CD4+, CD5+, CD6+, CD7+, CD8+, CD13-, CD19-, CD34+, TCRalpha/beta-, TCR gamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human near-diploid karyotype with 9% near-tetraploid and 15% polyploid sidelines; 45/46(43-47)<2n>X/XX, -10, +14; sideline with +add(?15)(p11), add(1)(p32); identity confirmed
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PL-21
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 536
Origin:	established from the peripheral blood of a 24-year-old man with refractory acute promyelocytic leukemia after mediastinal granulocytic sarcoma; however, PL-21 cells do not carry the t(15;17); based on morphology, surface marker expression of CD14/CD68 and expression of the monocyte-specific esterase, PL-21 cells appear to be rather of monocytic origin; we found that PL-21 carries an FLT3 internal tandem duplication
References:	Kubonishi et al., Gann 74: 319-322 (1983); Kubonishi et al., Blood 63: 254-259 (1984); Drexler et al., Leukemia Res. 19: 681-691 (1995) (review); Quentmeier et al., Leukemia 17: 120-124 (2003)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	single cells in suspension; cells have a rather monoblastic morphology without the typical APL granules
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2-3 days; seed out at ca. $0.5-1.0 \times 10^6$ cells/ml; maintain at ca. $0.5-1.0 \times 10^6$ cells/ml; minimum density at ca. 0.3×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 6×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD4+, CD13+, CD14(+), CD15+, CD19-, CD33+, CD34+, CD68+, HLA-DR-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypertetraploid karyotype; 93(83-96)<4n>XXYY, +3, +5, +5, +6, +6, +7, -9, -10, -11, -12, +13, -14, -17, -21, +22, +22, +mar, del(1)(p11), i(5p)x2-3, der(6)t(6;15)(p25;q24), der(13)ins(13;13)q32;q?q?)x3-4; no cytogenetic evidence of PML-RARA fusion; resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PLB-985
Cell Type:	human acute myeloid leukemia (derivative of HL-60)
DSMZ No:	ACC 139
Origin:	allegedly "established from the peripheral blood of a 38-year-old woman with acute myeloid leukemia (AML FAB M4) in relapse in 1985"; DNA fingerprinting showed unequivocally that PLB-985 is a subclone of cell line HL-60
References:	Tucker et al., Blood 70: 372-378 (1987)
Depositor:	Dr. T. A. Rado, University of Alabama at Birmingham, Alabama, USA
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at ca. 0.5×10^6 cells/ml; minimum density at 0.2×10^6 cells/ml; seed out at ca. 1×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	maximal density at about 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD3-, CD13+, CD14-, CD15+, CD19-, CD33+, CD34-, CD68-
Fingerprinting:	same DNA profile as HL-60 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human near diploid karyotype with 4% polyploidy; 46(44-50)<2n>X, -X, +6, dm, dup(1)(q21q41), del(3)(p21), del(6)(q21)x2
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PR-1
Cell Type:	rat pituitary tumor
DSMZ No:	ACC 486
Origin:	established from the pituitary tumor of an ovariectomized F344 rat treated with estrogen for 3 months; described as showing markedly reduced levels of TGF- β 1 and TGF- β type II receptor mRNA; reported as being resistant to antiproliferative effects of TGF- β 1
References:	Pastorcic et al., Cancer Res. 55: 4892-4898 (1995)
Depositor:	Dr. M. L. Thieulant, CNRS, Rennes, France
DSMZ Cell Culture Data:	
Morphology:	round cells, slightly adherent, growing as strings of round cells in branched colonies
Medium:	Dulbecco's MEM + Ham's F12 (1:1) + 2.5% FBS + 10% horse serum
Subculture:	optimal split ratio of 1:2 once a week using trypsin/EDTA; seed out at ca. 3×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 90 hours
Harvest:	cell harvest of ca. 50×10^6 cells/80 cm ²
Storage:	frozen with 65% medium (1:1), 5% FBS, 20% horse serum, 10% DMSO at about 3×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as rat by cytogenetics and PCR analysis with Alu, globin and rRNA primers
Cytogenetics:	rat near-triploid karyotype with 10% polyploidy; 60-68<3n>

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PSI-2
Cell Type:	Swiss mouse embryo - NIH-3T3 derivative (retroviral packaging line)
DSMZ No:	ACC 238
Origin:	subclone of the Swiss mouse embryo cell line NIH-3T3; PSI-2 cells are infected with a replication deficient MLV virus; cells can be used as "packaging cell line"; classified as risk category S1 according to the German Central Commission for Biological Safety (ZKBS)
References:	Mann et al., Cell 33: 153-159 (1983)
Depositor:	Dr. W. Dirks, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblastoid cells growing as monolayers
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:10 to 1:20 every 2-3 days using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 34 hours
Harvest:	cell harvest of about 2-4 x 10 ⁵ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as mouse with IEF of AST, NP
Cytogenetics:	murine hypotetraploid karyotype; centric fragment present at 74%; ring present at 48%; 68/69(64-74)<4n>; a submetacentric marker arising by centric fusion present at 52%
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	PYS
Cell Type:	mouse teratocarcinoma
DSMZ No:	ACC 284
Origin:	the transplantable tumor OTT-6050 arose from a 6-day-old embryo grafted into the testis of a strain 129 mouse; this solid subcutaneous tumor was minced in PBS and mice were injected intraperitoneally; ascites containing embryoid bodies developed and cell cultures were initiated from the embryoid bodies
References:	Lehman et al., J. Cell. Physiol. 84: 13-28 (1974)
Depositor:	Dr. H. Weich, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing as monolayers
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 2-3 days using trypsin/EDTA; seed out at ca. 2-3 x 10 ⁴ cells/80 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	cell harvest of ca. 15 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2-6 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as murine with IEF of LDH, MDH, NP
Cytogenetics:	murine hypotetraploid karyotype; 70(66-72)<4n>; 7-11 centric fusion markers present
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	R1
Cell Type:	fish rainbow trout liver
DSMZ No:	ACC 56
Origin:	established from the liver of a rainbow trout
References:	Ahne, Zbl. Bakt. Hyg. B 180: 480-504 (1985)
Depositor:	Dr. M. Kohlpoth, Akademie für Tierschutz, Neubiberg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, fibroblastoid cells growing as monolayers
Medium:	90% Medium 199 (25 mM HEPES) with Earle's salts with L-glutamine + 10% FBS
Subculture:	cell growth is extremely slow; split confluent culture 1:2 every 7 days using TNE (50 mM Tris, 0.1 M NaCl, 5 mM EDTA, pH 7.5) at +4 /C for 10 min; culture at room temperature (about 20 /C) without CO ₂ (keep lid of flask closed); cells can be stored temporarily (up to 1 week) at 4 /C; seed out at ca. 0.1 x 10 ⁶ cells/ml
Incubation:	at room temperature (about 20 /C) without CO ₂
Doubling Time:	doubling time of about 160 hours
Harvest:	about 5 x 10 ⁶ cells/175 cm ² - saturation density of ca. 10 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated by the depositor with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization assays

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RAJI
Cell Type:	human Burkitt's lymphoma
DSMZ No:	ACC 319
Origin:	established from the left maxilla of a 12-year-old African boy with Burkitt's lymphoma in 1963; first continuous human hematopoietic cell line; classified as risk category 1 according to the German Central Commission for Biological Safety (ZKBS)
References:	Pulvertaft, Lancet i: 238-240 (1964)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round single cells in suspension (partly in clusters or clumps)
Medium:	90-95% RPMI 1640 + 5-10% FBS
Subculture:	optimal split ratio of saturated culture at 1:4 to 1:8 twice a week; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 1-2 days
Harvest:	maximum density at ca. 1×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a+, CD80+, CD138-, HLA-DR+, smIgG(+), cyIgG-, smIgM(+), cyIgM+, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, NP, PEP B
Cytogenetics:	human flat-moded hypotetraploid karyotype with 12% polyploidy; 89(80-91)<4n>XXYY, -4, -9, -12, -16, -18, -21, +3mar, add(4)(q35), add(6)(q27)x2, t(8;14)(q24;q32), der(8)t(8;14)(q24;q32)add(8)(?p12), del(9)(p23), der(14)t(8;14)(q24;q32); carries t(8;14) associated with Burkitt's lymphoma; closely resembles depositor's karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RBL-1
Cell Type:	rat basophilic leukemia
DSMZ No:	ACC 147
Origin:	this rat basophilic leukemia (RBL) cell line was established from chlorethylamine-treated rats; the cells were described to show marked variations in differentiation dependent on the growth rate and the cell cycle phases
References:	Ecclestone et al., Nature New Biology 244: 73-75 (1973); Kulczycki et al., J. Exp. Med. 139: 600-616 (1974)
Depositor:	obtained from ATCC (CRL 1378), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	1/4 adherent (epithelial-like cells), 3/4 in suspension (small round cells)
Medium:	90% RPMI 1640 + 10% FBS (cells may also be grown, but less efficiently, with MEM)
Subculture:	maximal split ratio of 1:4; split 1:3 every 2-3 days; cells grow partly adherent, but trypsin is not needed to detach the cells; seed out at ca. 5×10^6 cells/80 cm ² or $0.3-0.5 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 25 hours
Harvest:	about 46×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Fingerprinting:	unique Hinf I-(gtg) ₅ DNA profile
Species:	confirmed as rat with IEF of AST, NP
Cytogenetics:	74(69-76); rodent marker chromosomes present
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RBL-2H3
Cell Type:	rat basophilic leukemia
DSMZ No:	ACC 312
Origin:	the rat basophilic leukemia (RBL) line was originally obtained by Eccleston et al. and was maintained by serial passage in neonatal Wistar rats; four cell lines (named RBL I-IV) were started from this tumor between 1973 and 1975; the subline RBL-IV HR ⁺ was derived from RBL-IV; further subcloning established the cell line RBL-2H3; cells are described to release histamine as an IgE-mediated reaction
References:	Eccleston et al., Nature New Biol. 244: 73-75 (1973); Kulczycki et al., J. Exp. Med. 139: 600-616 (1974); Barsumian et al., Eur. J. Immunol. 11: 317-323 (1981); Hoffmann et al., J. Allergy Clin. Immunol. 99: 227-232 (1997)
Depositor:	Dr. A. Hoffmann, Paul-Ehrlich-Institut, Langen, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, fibroblast-like cells forming monolayers
Medium:	70% MEM (with Earle's salts) + 20% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:5 every 3 days using trypsin/EDTA; seed out at ca. 1-2 x 10 ⁶ cells/80 cm ²
Incubation:	at 37 °C without CO ₂ (keep lid of flask closed)
Doubling Time:	doubling time of ca. 50-60 hours
Harvest:	cell harvest of ca. 15-20 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1.5-3 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	histamine release confirmed by depositor
Species:	confirmed as rat with IEF of AST, LDH, NP
Cytogenetics:	rat hypertriploid karyotype; 68(62-72)<3n>; carries large metacentric marker
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RCH-ACV
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 548
Origin:	established from bone marrow cells taken at relapse of common acute lymphoblastic leukemia (cALL), seven months after diagnosis, from an 8-year-old girl treated with combination chemotherapy
References:	Jack et al., Cancer Genet. Cytogenet. 19: 261-269 (1986); Kim et al., Blood 88: 785-794 (1996)
Depositor:	Dr. T. Rabbitts, Cambridge, UK
DSMZ Cell Culture Data:	
Morphology:	small round cells growing singly or in small clumps in suspension; occasional adherent cells present
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	split saturated culture 1:2 to 1:10 every 2-6 days; seed out at ca. 1×10^6 cells/ml; maintain at ca. $0.5\text{-}2.5 \times 10^6$ cells/ml; maximum density at ca. 4×10^6 cells/ml; easy to culture, tolerates low cell densities
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-50 hours
Harvest:	cell harvest of ca. 4×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 8×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD10(+), CD13-, CD19+, CD20-, CD34(+), CD37-, CD79a-, cyCD79a+, CD80-, CD138-, HLA-DR+, sm/cyIgG-, smIgM-, cyIgM+, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hyperdiploid karyotype with 2% polyploidy; 43-50<2n>XX, +8, t(1;19)(q23;p13.3); sideline with idem, +16, +der(19)t(1;19); carries t(1;19) effecting E2A-PBX1 fusion; matches published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RD-ES
Cell Type:	human Ewing's sarcoma
DSMZ No:	ACC 260
Origin:	established from the primary tumor of a 19-year-old Caucasian man with Ewing's sarcoma of the humerus in 1984
References:	ATCC HTB 166
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent cells growing in monolayers, sometimes growing loosely adherent in clusters/clumps
Medium:	85% RPMI 1640 + 15% FBS
Subculture:	split confluent culture 1:4 every 5-6 days using trypsin/EDTA (occasional loosely adherent clumps can be detached by tapping the flask); seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 60 hours
Harvest:	cell harvest of about 10×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human flat-moded, near-triploid karyotype; 69(63-72)<3n>XXY/XXYY, +1, -4, -8, +9, +12, +15, -17, +18, -22, der(9)t(9;11)(p13;q11)t(11;22)(q24;q12)x2, add(14)(p11); carries two masked copies of 11;22 translocation associated with Ewing's sarcoma; presence of type-2 EWS-FLI1 hybrid transcripts (394 bp) confirmed by RT-PCR
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RED-1
Cell Type:	mouse erythroid leukemia
DSMZ No:	ACC 407
Origin:	established from a Rauscher murine leukemia virus-induced erythroid leukemia of a female DBA/2 mouse; cells were described to be responsive to erythropoietin (EPO) and to form benzidine-positive colonies upon stimulation with EPO or DMSO; cells were received at passage #60, frozen at DSMZ at passage #77
References:	Hagemeijer et al., J. Natl. Cancer Inst. 69: 945-951 (1982)
Depositor:	Dr. N. J. de Both, Erasmus University, Rotterdam, The Netherlands
DSMZ Cell Culture Data:	
Morphology:	round, single cells growing in suspension
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	maintain at 0.5×10^6 cells/ml; split saturated culture at 1:4 to 1:5 every second day; seed out at ca. $0.5\text{--}1.0 \times 10^6$ cells/ml; viability immediately after thawing may be rather low (<30%), but cells recover quickly and viability will raise over the next few passages; maximal density at ca. $2\text{--}3 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	cell harvest of ca. $1\text{--}2 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, NP
Cytogenetics:	murine hypodiploid karyotype with 10% polyploidy; 39(38-40)<2n>, t(X;1)(q10;q10); two centric-fusion markers present; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RED-4
Cell Type:	mouse erythroid leukemia
DSMZ No:	ACC 425
Origin:	established from a Rauscher murine leukemia virus-induced erythroid leukemia of a DBA/2 mouse; in contrast to RED-1 (DSM ACC 407), cells were described to be unresponsive to erythropoietin (EPO) or DMSO; cells were received at passage #11
References:	Hagemeijer et al., J. Natl. Cancer Inst. 69: 945-951 (1982)
Depositor:	Dr. N. J. de Both, Erasmus University, Rotterdam, The Netherlands
DSMZ Cell Culture Data:	
Morphology:	round, single cells growing in suspension
Medium:	85-90% Dulbecco's MEM + 10-15% FBS
Subculture:	maintain at 0.5×10^6 cells/ml; split saturated culture at 1:2 to 1:3 every day; seed out at ca. 1.0×10^6 cells/ml; viability immediately after thawing may be rather low (<30%), but cells recover subsequently over the next week and viability will raise over the next few passages (spent medium should be exchanged without discarding cells); cells grow better in 24-well-plates; maximal density at ca. 1.0×10^6 cells/ml
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	cell harvest of ca. 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of LDH, NP
Cytogenetics:	murine near diploid karyotype with 12.5% polyploidy; 39-42<2n>XX, Df(2E2F1); 1-2 centric fusion markers present; resembles published karyotype
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RED-5
Cell Type:	mouse erythroid leukemia
DSMZ No:	ACC 451
Origin:	established from a Rauscher murine leukemia virus-induced erythroid leukemia of a female DBA/2 mouse; cells were described as being responsive to erythropoietin (EPO) and to form benzidine-positive colonies upon stimulation with EPO or DMSO
References:	Hagemeijer et al., J. Natl. Cancer Inst. 69: 945-951 (1982)
Depositor:	Dr. N. J. de Both, Erasmus University, Rotterdam, The Netherlands
DSMZ Cell Culture Data:	
Morphology:	round, single cells growing in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture at 1:2 to 1:3 every two days; seed out at ca. $0.5-1.0 \times 10^6$ cells/ml; viability immediately after thawing may be rather low (<30%), but cells recover subsequently over the next week and viability will raise over the next few passages (spent medium should be exchanged without discarding cells); cells grow initially better in 24-well-plates; maximal density at ca. 2.0×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. 2.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 8×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Cytogenetics:	murine karyotype

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RED-6
Cell Type:	mouse erythroid leukemia
DSMZ No:	ACC 466
Origin:	established from the liver of a Rauscher murine leukemia virus-induced erythroid leukemia of a female DBA/2 mouse; cells were described as being differentiation-responsive to erythropoietin (EPO) and unresponsive to DSMO; cells were received at passage #24
References:	Hagemeijer et al., J. Natl. Cancer Inst. 69: 945-951 (1982)
Depositor:	Dr. N. J. de Both, Erasmus University, Rotterdam, The Netherlands
DSMZ Cell Culture Data:	
Morphology:	round single cells growing in suspension, some cells are slightly adherent
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture at 1:2 to 1:3 every day; seed out at ca. $0.2-0.5 \times 10^6$ cells/ml; viability immediately after thawing may be rather low (<30%), but cells recover subsequently over the next week and viability will raise over the next few passages; maximal density at ca. 3.0×10^6 cells/ml; maintain at ca. $0.5-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 18 hours
Harvest:	cell harvest of ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 7×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse by PCR using beta-globin primers
Cytogenetics:	murine near-diploid karyotype with 10% polyploidy

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INFORMATION SHEET FOR CELL LINES

Cell Line:	REH
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 22
Origin:	established from the peripheral blood of a 15-year-old North African girl with acute lymphoblastic leukemia (ALL at first relapse) in 1973
References:	Rosenfeld et al., Nature 267: 841-843 (1977); Matsuo & Drexler, Leukemia Res. 22: 567-579 (1998)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	small, round, single cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	initially after thawing cells may be difficult to culture! maintain at high density ($>2 \times 10^6$ cells/ml); cells grow slowly; minimum density at ca. 0.5×10^6 cells/ml; seed out at ca. 1×10^6 cells/ml with initially 20% FBS
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-70 hours
Harvest:	maximum density at ca. $3-5 \times 10^6$ cells/ml or higher
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5-8 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD5-, CD10+, CD13-, CD19+, CD20+, CD21-, CD34-, CD37-, CD38+, CD71+, CD79a-, cyCD79a+, CD80-, CD86-, CD138+, HLA-DR+, sm/cyIgG-, sm/cyIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human pseudodiploid karyotype; 46(44-47)<2n>X, -X, +16, del(3)(p22), t(4;12;21;16)(q32;p13;q22;q24.3)-inv(12)(p13q22), t(5;12)(q31-q32;p12), der(16)t(16;21)(q24.3;q22); sideline with inv(5)der(5)(p15q31),+18; carries t(12;21) and del(12) producing respective TEL-AML1 fusion and deletion of residual TEL; expression of the TEL-AML1 fusion gene was detected by RT-PCR analysis
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RGE
Cell Type:	rat glomerular endothelial cells
DSMZ No:	ACC 262
Origin:	established from freshly isolated glomeruli from kidneys of 6-week-old female Sprague-Dawley rats
References:	Laulajainen et al., Lab. Invest. 69: 183-192 (1993)
Depositor:	Dr. Holthöfer, University of Helsinki, Helsinki, Finland
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells growing as monolayer (after confluency cells tend to grow in multilayers)
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:2 to 1:3 twice a week using trypsin/EDTA (cells are only lightly adherent and can be detached by tapping the flask; however, to prepare single cell suspension, trypsin/EDTA treatment is necessary); seed out at ca. 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 17 hours
Harvest:	cell harvest of about 2.5×10^6 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as rat with IEF of AST, PEP B
Cytogenetics:	rat hypertriploid karyotype with 6% polyploidy; 67(67-72)<3n>; corresponds to published karyotype
Viruses:	ELISA: reverse transcriptase positive

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RH-1
Cell Type:	human rhabdomyosarcoma
DSMZ No:	ACC 493
Origin:	derived from the bone marrow metastasis of a malignant embryonal rhabdomyosarcoma in a male patient
References:	Roberts et al., Cancer Res. 49: 5407-5413 (1989); Morton & Potter, J. Pharmacol. Exp. Therapeut. 286: 1066-1073 (1998)
Depositor:	Dr. P. J. Houghton, St. Jude Children's Research Hospital, Memphis, TN, USA
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:4 to 1:20 every 3-10 days using trypsin/EDTA (short exposure only); seed out at ca. $0.5-1.0 \times 10^6$ cells/80 cm ² ; maximal density at ca. 4×10^5 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	cell harvest of ca. $1-2 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypotriploid karyotype with 12% polyploidy; 67(61-67)<3n>XX, -Y, +2, -3, +7, -8, -10, +12, -14, -15, -16, +19, -22, add(X)(p22), del(1)(p3?5p3?6), del(2)(p14)x1-2, del(2)(q23), del(9)(p11p13), ?add(11)(q25)x2, der(11)t(2;11)(q?3?q?3?)t(2;3)(q3?;p21), add(12)(q13)/dup(12)(q13q21), del(13)(?q21), der(14)t(Y;14)(q11.2;p11)x2, der(16)t(16;22)(q22;q22)x2, del(18)(q21), add(19)(q13), del(22)(q13); sideline with +7, +21, add(2)(q11), add(13)(p11); absence of t(2;13) confirmed by FISH
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RH-18
Cell Type:	human rhabdomyosarcoma
DSMZ No:	ACC 491
Origin:	cell line was derived from the tumor material xenografted in immuno-deficient mice; tumor material: primary pediatric embryonal rhabdomyosarcoma exhibiting moderate histological differentiation
References:	Hazelton et al., Cancer Res. 47: 4501-4507 (1987)
Depositor:	Dr. P. J. Houghton, St. Jude Children's Research Hospital, Memphis, TN, USA
DSMZ Cell Culture Data:	
Morphology:	large epitheloid cells growing as monolayers
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split confluent culture 1:2 to 1:3 every 4-8 days using trypsin/EDTA; seed out at ca. $1-2 \times 10^4$ cells/cm ² ; maintain at $0.5-0.2 \times 10^4$ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60-70 hours
Harvest:	maximum cell harvest of ca. 4×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $0.5-1.0 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin-, desmin+, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat moded hypotetraploid karyotype with 20% polyploidy; 78-87<4n>XX, -3, -4, -4, +5, +7, +7, -10, -10, -12, -12, -15, -19, +20, +20, +20, -21, +1-2mar, +0-4 dmin, add(1)(p11), del(5)(q13), add(16)(p13), del(16)(p13.2); sideline with i(3q)
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RH-30
Cell Type:	human rhabdomyosarcoma
DSMZ No:	ACC 489
Origin:	established from the bone marrow metastasis of a 17-year-old man with undifferentiated (standard classification) or unclassified (Palmer classification) alveolar rhabdomyosarcoma; cells were described
References:	as carrying a p53 mutation, expressing high levels of both myogenin and myoD, and expressing the Pax3/FKHR fusion protein secondary to the t(2;13)(q35;q14) translocation Douglass et al., Cytogenet. Cell Genet. 45: 148-155 (1987); Gibson et al., Clin. Cancer Res. 4: 145-152 (1998); Taylor et al., Med. Ped. Oncol. 35: 96-103 (2000)
Depositor:	Dr. P. J. Houghton, St. Jude Children's Research Hospital, Memphis, TN, USA
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing as monolayer
Medium:	90-95% RPMI 1640 + 5-10% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 3-4 days using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ² ; maintain at $0.05-0.1 \times 10^6$ cells/cm ² ; maximal density at ca. 0.2×10^6 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	cell harvest of ca. 0.1×10^6 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin+, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RLC-18
Cell Type:	rat embryonal liver
DSMZ No:	ACC 332
Origin:	established from the normal liver of a 17-day rat embryo (strain JAR-2) in 1975
References:	Takaoka et al., J. Exp. Med. 45: 317-326 (1975)
Depositor:	obtained from RIKEN Cell Bank (RCB0070), Tsubuka Science City, Ibaraki, Japan
DSMZ Cell Culture Data:	
Morphology:	epithelial-like growth in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:4 every 3 days using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ² + 7-10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35-40 hours
Harvest:	harvest of ca. 20×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as rat with IEF of AST, MDH, NP
Cytogenetics:	rat hypertriploid karyotype with 13% polyploidy; 64(58-66)<3n>; large submetacentric marker present in most cells
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RLD-1
Cell Type:	mouse T cell leukemia
DSMZ No:	ACC 415
Origin:	established from transplantable tumors induced in syngeneic mice inoculated with cell suspensions prepared from lymph nodes infiltrated by lymphatic leukemia induced in 6-week-old DBA/2 mice by Rauscher virus (present in supernate of JLSV-9 cells); described as expressing T cell surface markers (Thy-1, T-200, Lyt-1), high constitutive TdT activity, and as differentiating along T-helper lineage in response to treatment with DMSO or phorbol ester TPA
References:	de Both et al., Int. J. Cancer 32: 501-506 (1983)
Depositor:	Dr. N. J. de Both, Erasmus University, Rotterdam, The Netherlands
DSMZ Cell Culture Data:	
Morphology:	spheroidal cells growing singly in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.2-0.7 \times 10^6$ cells/ml; optimal split ratio of 1:2 to 1:6 every 2-4 days; seed out at ca. $0.3-0.5 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 20-30 hours
Harvest:	saturation density of ca. $1.5-2.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, NP
Cytogenetics:	murine hyperdiploid karyotype; $42-2n > XX, +14, +15$; no centric fusion markers present; matches published karyotype
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RMB-1
Cell Type:	mouse myeloid leukemia
DSMZ No:	ACC 391
Origin:	tumor induced in nude mouse injected with blast cells from ascites fluid of newborn Balb/c mouse injected intraperitoneally with Rauscher murine leukemia virus (supernate of producer cell line JLS-V9); descibed to be heterotransplantable, to form colonies in soft agar, to display myeloid cytochemical (myeloperoxidase) and functional (phagocytosis) features, and to differentiate terminally in response to DMSO
References:	de Both et al., Cell Differentiation 10: 13-21 (1981)
Depositor:	Dr. N. J. de Both, Erasmus University Rotterdam, The Netherlands
DSMZ Cell Culture Data:	
Morphology:	spheroidal cells growing singly in suspension
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	maintain at $1-2 \times 10^6$ cells/ml; split saturated culture 1:4 to 1:10 every 2-4 days; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-35 hours
Harvest:	maximal density at ca. 3×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 7×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as murine with IEF of AST, LDH, MDH, NP
Cytogenetics:	murine hyperdiploid karyotype; $41(38-41)<2n>XY$, structurally and numerically rearranged
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RO
Cell Type:	human B lymphoblastoid cells (from patient with severe combined immunodeficiency, SCID)
DSMZ No:	ACC 452
Origin:	established from the peripheral blood of a 3-year-old boy with severe combined immunodeficiency (SCID) by EBV-transformation using the cell line B95-8 as source for EBV; cells were described as lacking MHC class II molecules
References:	Hadam et al., in Progress in Immunodeficiency Research and Therapy I (p. 43) and II (p89), Amsterdam: Elsevier (1984 + 1986); De Preval et al., New Engl. J. Med. 318: 1295-1300 (1988); Steimle et al., Genes Development 9: 1021-1032 (1995)
Depositor:	Prof. Dr. M. Hadam, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing in suspension, singly or in small clusters
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	split saturated culture at 1:2 every two days; seed out at ca. 1.0×10^6 cells/ml; it is advisable to culture the cells initially in 24-well-plates; cell growth may be slow for the first week maximal density at ca. 2.0×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	cell harvest of ca. $1.0-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a-, cyCD79a+, CD80+, CD138-, HLA-DR-, sm/cyIgG (5-10%)+, sm/cyIgM (20-40%)+, sm/cykappa (5-10%)+, sm/cylambda (10-20%)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Cytogenetics:	human near-diploid karyotype with 8% polpoidy; 46(43-46)<2n>XY, no consistent abnormality present
Viruses:	PCR: EBV+, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RPMI-2650
Cell Type:	human nasal septum squamous cell carcinoma
DSMZ No:	ACC 287
Origin:	from the pleural effusion of a 52-year-old man with anaplastic squamous cell carcinoma of the nasal septum in 1962
References:	Moore & Sandberg, Cancer 17: 170-175 (1964); Moorhead, Exp. Cell Res. 39: 190-196 (1965)
Depositor:	obtained from DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, epitheloid, very small cells growing in clusters
Medium:	90% MEM (with Earle's salts) + 10% FBS + 1x non-essential amino acids
Subculture:	split confluent culture 1:4 to 1:8 every 4-5 days using trypsin/EDTA; seed out at about 0.5-1.0 x 10 ⁶ cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 40-50 hours
Harvest:	cell harvest of about 0.6 x 10 ⁶ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, cytokeratin-8+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human near-diploid karyotype with 10% polyploidy; 45(43-49)<2n>XY, -4, -16, -18, +2mar, del(1)(p11), add(4)(p16), del(8)(q22), add(14)(p11), add(14)(q32), del(19)(p13); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RPMI-7951
Cell Type:	human malignant melanoma
DSMZ No:	ACC 66
Origin:	established from an involved lymph node of an 18-year-old Caucasian woman with malignant melanoma in 1971
References:	Quinn et al., J. Natl. Cancer Inst. 59: 301-307 (1977)
Depositor:	Dr. H. Kirchner, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial-like cells growing as monolayers
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:4 to 1:10 every 4-6 days using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 60 hours
Harvest:	about 10-12 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1.0-2.0 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human flat-moded hyperdiploid karyotype with 22% polyploidy; 54(50-60)<2n>X, -X, +1, +2, +6, +7, +8, +20, +20, +3mar, der(8)t(7;8)(p11;p12), der(9)t(9;11)(p21;q11), add(18)(q23;HSR); shares t(9;11) with IPC-298 defining a new recurrent translocation in malignant melanoma
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RPMI-8226
Cell Type:	human multiple myeloma
DSMZ No:	ACC 402
Origin:	established from the peripheral blood of a 61-year-old man with multiple myeloma (IgG lambda-type) at diagnosis in 1966; described to produce and secrete only lambda light chains (but not heavy chains)
References:	Matsuoka et al., Proc. Soc. Exp. Biol. Med. 125: 1246-1250 (1967); review article: Drexler & Matsuo, Leukemia Res 24: 681-703 (2000)
Depositor:	Prof. Dr. H. W. L. Ziegler-Heitbrock, Institute of Immunology, University of Munich, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	cells grow mostly as single cells in suspension, some cells may grow slightly adherent (occasionally up to 50%)
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:2 every 3 days; cells may grow slightly adherent, but can be detached by pipetting or with a rubber policeman or with TNE (40 mM TrisCl pH 7.5, 1 mM EDTA pH 7.5, 150 mM NaCl) (trypsin is not necessary); seed out at ca. 0.4-0.5 x 10 ⁶ cells/ml; maximal density at ca. 1.0 x 10 ⁶ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60-70 hours
Harvest:	cell harvest of ca. 25 x 10 ⁶ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4-5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD5-, CD10-, CD13-, CD14(+), CD19-, CD20-, CD21(+), CD34+, CD37+, HLA-DR+, sm/cyIgM-, sm/cyIgG-, sm/cykappa-, smlambda-, cylambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human flat-modulated hypodiploid karyotype with 7.5% polyploidy; 62-67<3n>XXY, -1, +5, +6, +7, -8, -9, -10, +11, -12, -13, -14, +16, +18, -19, +21, -22, der(1)t(1;1)(p36.3;q11-12), add(3)(q27-29)x2, del(4)(p14p15.4), der(5;6)(q10;p10)x1-2, del(5)(q32), del(6)(q15), add(9)(p24), add(11)(p15), del(11)(q25), der(14)t(1;14)(q11-12;q32)x2, del(15)(q11.2q14)x2, add(16)(q23-24), der(17)t(?8;17)(q21.2;q25), del(22)(q13.2); der(1), 6q-, 14q+ and 22q- associated with multiple myeloma; identified by disomic 14q+
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RPMI-8402
Cell Type:	human T cell acute lymphoblastic leukemia
DSMZ No:	ACC 290
Origin:	established from the peripheral blood of a 16-year-old woman with acute lymphoblastic leukemia (ALL) in 1972
References:	Moore et al., In Vitro 8: 434 (1973); Huang et al., J. Natl. Cancer Inst. 53: 655-660 (1974)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round single cells in suspension, growing partly in clumps
Medium:	90% RPMI 1640 + 10% FBS + 2 mM L-glutamine
Subculture:	optimal split ratio of 1:3 to 1:7 once a week; seed out at about 0.25×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 47 hours
Harvest:	saturation density at about $1-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contaminaton was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2+, CD3-, CD4 (clone S3.5)+, CD4 (clone Leu-3a)-, CD5+, CD7+, CD8-, CD13-, CD19-, CD34-, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypotetraploid karyotype with 2% polyploidy; 90(79-91)<4n>XXX, -X, +3, +3, -10, -13, -14, +15, -18, -20, +2mar, dup(4)(q13q23)x2, del(6)(q14q22)x2, t(11;14)(p15;q11)x2, add(15)(p13); sideline with der(1)t(1;9)(p35/36;q11), add(13)(q34); carries t(11;14) associated with TAL1-TCRD rearrangement in T-ALL, resembles published karyotypes
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RS4;11
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 508
Origin:	established from the bone marrow of a 32-year-old woman with acute lymphoblastic leukemia (ALL L2) in first relapse
References:	Stong et al., Blood 65: 21-31 (1985)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single, relatively small, round cells growing in suspension
Medium:	90% alpha-MEM + 10% FBS
Subculture:	split saturated culture 1:3 to 1:4 every 3-4 days; seed out at ca. 0.5×10^6 cells/ml; maintain at about $0.5\text{-}1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	maximal density of about 4.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5\text{-}10 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20-, CD34(+), CD37-, CD79a-, cyCD79a+, CD80-, CD138-, HLA-DR+, sm/cyIgG-, sm/cyIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype; 47/48<2n>X/XX, +8, +18, t(4;11)(q21;q23), i(7q); sideline without +8/18; carries MLL-AF4 fusion gene; resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RT-112
Cell Type:	human urinary bladder transitional cell carcinoma
DSMZ No:	ACC 418
Origin:	established from the transitional cell carcinoma (histological grade G2) excised from a woman (age unknown) with untreated primary urinary bladder carcinoma in 1973; described to form tumors in nude mice
References:	Marshall et al., J. Natl. Cancer Inst. 58: 1743-1747 (1977); Masters et al., Cancer Res. 46: 3630-3636 (1986)
Depositor:	Dr. J. R. W. Masters, University College London, London, UK
DSMZ Cell Culture Data:	
Morphology:	adherent epithelioid cells growing as monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split culture prior to confluence 1:2 to 1:4 every 2-5 days using extended trypsin/EDTA treatment (5-10 min), rinsing first with PBS; seed out at ca. 2×10^6 cells/80 cm ² in 10 ml medium with initially 20% FBS; maximal density at ca. 3×10^5 cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	cell harvest of ca. $1-2 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule (frozen at passage #59)
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, NP
Cytogenetics:	human hyperdiploid karyotype with 4% polyploidy; 47(44-47)<2n>XX/XXX, +X, +10, +11, +14, -17, +mar, add(2)(q3?2), del(3)(p14p24.1), i(4p), add(6)(q2?), i(8q), del(9)(p21p22), add(10)(p12), der(11)t(3;11)(q11-13;p15), add(17)(p1?1), add(18)(q11-22); sideline with: idem, del(9)(p11), del(11)(q23), add(12)(q24.32), del(13)(q11q14), add(21)(p11); resembles published karyotypes
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RT-4
Cell Type:	human urinary bladder transitional cell carcinoma
DSMZ No:	ACC 412
Origin:	established from a recurrent well-differentiated transitional papillary tumor of the urinary bladder (clinical stage T2, histological grade G1, treated with gold grains 2 years earlier and diathermy after recurrence 10 months later) of a 63-year-old man in 1968
References:	Rigby & Franks, Brit. J. Cancer 24: 746-757 (1970); Marshall et al., J. Natl. Cancer Inst. 58: 1743-1747 (1977); Hastings & Franks, Int. J. Cancer 27: 15-21 (1981)
Depositor:	Dr. J. R. W. Masters, University College London, London, UK
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing as monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:2 to 1:3 twice a week using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/25 cm ² in 10 ml medium; maximal density at ca. $5-7 \times 10^6$ cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 80 hours
Harvest:	cell harvest of ca. $3-4 \times 10^6$ cells/25 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, NP
Cytogenetics:	human flat-moded hypotetraploid karyotype with 4% polyploidy; 82-88<4n>XXYY, -1, -4, -6, -9, -10, +12, -13, -18, +mar, der(17)t(9;17)(q11-12;q25); apparent tetraploid derivate of original karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RTG-2
Cell Type:	fish rainbow trout gonadal cells
DSMZ No:	ACC 352
Origin:	established from the gonadal cells of a rainbow trout (<i>salmo gairdneri</i>)
References:	Wolf & Quimby, Science 135: 1065-1066 (1962)
Depositor:	Dr. M. Kohlpoth, Akademie für Tierschutz, Neubiberg, Germany
DSMZ Cell Culture Data:	
Morphology:	long spindle-like and fibroblast-like cells growing adherently in monolayers
Medium:	90% MEM (with Earle's salts) + 10% FBS + 2 mM L-glutamine + 50 mg/L neomycin sulfate
Subculture:	split confluent culture 1:4 every 5-7 days using trypsin/EDTA; seed out at ca. 1×10^6 cells/80 cm ²
Incubation:	maximum temperature at 26 °C, optimal at 20 °C, grown without CO ₂ (keep lid of flask closed)
Doubling Time:	doubling time of ca. 72 hours
Harvest:	cell harvest of ca. 4×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	apparently fish; different in IEF of AST, LDH, MDH from primate and murine IEF patterns

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RV-C2
Cell Type:	mouse hybridoma (anti-rat fiber/troponin)
DSMZ No:	ACC 289
Origin:	described as secreting a mouse monoclonal antibody (IgG{2b) reacting with fetal, neonatal and adult cardiac fibers in immunofluorescence and with rat cardiac fetal and adult troponin T in Western blotting; in immunofluorescence this antibody stains some rat fetal and neonatal skeletal muscle fibers which express fetal cardiac troponin T as also shown by the Western blotting of rat fetal and neonatal skeletal muscle; this cell line was produced by fusing NSO myeloma cells with spleen cells from a Balb/c mouse that had been immunized with a myofibrillar preparation from hypothyroid rabbit ventricle
References:	Saggin et al., J. Biol. Chem 263: 18488; Saggin et al., Development 110: 547 (1990)
Depositor:	Prof. S. Schiaffino, University of Padova, Padova, Italy
DSMZ Cell Culture Data:	
Morphology:	round single cells in suspension
Medium:	80% Dulbecco's MEM (4.5 g/L glucose) + 20% horse serum + 2 mM L-glutamine
Subculture:	optimal split ratio of 1:2 to 1:3 twice a week; seed out at about $0.5-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 32 hours
Harvest:	maximal density at about 2.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% horse serum, 10% DMSO at about 10×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgG ₂ b monoclonal antibody
Species:	confirmed as murine with IEF of AST, LDH, MDH, NP
Cytogenetics:	murine hypertetraploid/pentaploid karyotype; 88-99<4n>; 2 centric fusion markers and 1-2 dmin present in most cells
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	RVH-421
Cell Type:	human melanoma
DSMZ No:	ACC 127
Origin:	established from the tumor biopsy of an intracranial secondary melanoma in a 28-year-old European man in 1979; it was described that increasing the tyrosine concentration enhances synthesis of melanin
References:	McCormick et al., Brit. J. Exp. Pathol. 64: 103-115 (1983)
Depositor:	Dr. H. Harris & Dr. D. McCormick, Dunn School of Pathology, University of Oxford, Oxford, UK
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial-like cells growing in monolayers
Medium:	90% RPMI 1640 (or Leibovitz's L-15) + 10% FBS
Subculture:	split confluent culture up to 1:10 one to two times per week using trypsin/EDTA; seed out 1-2 x 10 ⁶ cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Harvest:	about 15 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, HMB-45+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human near-triploid karyotype with 30% polyploidy; 69(60-70)<3n>XXY, -10, +13, -15, +16, -20, +mar, del(6)(q15), add(8)(q24), del(9)(p22), der(9)t(7;9)(q12;q34)x2, add(11)(q24-25), ins(12;abr)(?q13), der(13)t(6;13)(p13;q33)x2-3, add(19)(p13)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	S-117
Cell Type:	human thyroid sarcoma
DSMZ No:	ACC 266
Origin:	established from the sarcoma of the thyroid gland from a 47-year-old woman in 1985
References:	Hilbert et al., J. Cancer Res. Clin. Oncol. 115 (suppl.), 1989
Depositor:	obtained from the DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently as monolayer
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:2 about once a week using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 37 hours
Harvest:	cell harvest of about 0.5-1.0 x 10 ⁵ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP+, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human flat-moded hexaploid/octaploid karyotype with 14% polyploidy; 156(140-188)<n>XXXX, -X, -X; multiple copies of: i(1p), i(1q), i(3p), i(6p), i(9q), add(5)(p15), add(16)(q23), i(mar); 1-3 dmin/rings in 40% cells; isochromosome formation previously reported in thyroid tumors
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SAOS-2
Cell Type:	human osteogenic sarcoma
DSMZ No:	ACC 243
Origin:	established from the primary osteogenic sarcoma of an 11-year-old Caucasian woman in 1973 (ATCC HTB 85)
References:	Fogh & Trempe, in "Human Tumor Cells in Vitro", 1975, Plenum Press, pp. 115-159; Fogh et al., J. Natl Cancer Inst. 58: 209-214 (1977) and 59: 221-226 (1977)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial-like cells growing in monolayers
Medium:	85% McCoy's 5A + 15% FBS
Subculture:	split confluent culture 1:2 to 1:3 every 2-3 days using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/80 cm ² in 10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 43 hours
Harvest:	cell harvest of about 10×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human highly rearranged hyperdiploid karyotype with 16% polyploidy; 56(52-58)<2n>X, -X, -2, -2, +3, -6, -8, +10, -12, +14, +18, -19, +20, -21, -21, -22, +15mar; add(1)(p22), add(3)(q27)x2, add(4)(p15), add(6)(p21), add(7)(p22), add(8)(q24), add(9)(q35), add(11)(p15), add(12)(p13), add(13)(q14), add(14)(p12)x2; 13q14 breakpoint is associated with osteosarcomas; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SBC-2
Cell Type:	human cervix carcinoma (derivative of HELA)
DSMZ No:	ACC 228
Origin:	these cells were thought to be "established from the malignant urinary bladder carcinoma of a 82-year-old Caucasian woman (stage pTa G1-G2) in 1992"; same donor as for cell line SBC-7 (DSMZ ACC 229); however, DNA fingerprinting and cytogenetic analysis at the DSMZ established unequivocally that this cell line must be considered de facto a subclone of HELA
References:	Thanhäuser et al., Immunology 80: 151-156 (1993); Böhle et al., J. Urology 150: 1932-1937 (1993)
Depositor:	Prof. A. J. Ulmer, Forschungsinstitut Borstel, Borstel, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing adherently in monolayers, often triangle-formed cells
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:10 to 1:20 every second day using trypsin/EDTA; seed out at about 1 x 10 ⁶ cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 40 hours
Harvest:	cell harvest of about 20-25 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament(+), vimentin+
Fingerprinting:	same DNA profile as HELA using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human hypertriploid karyotype with 2% polyploidy; 75(66-80)<3n>XX, -X, +1, +3, -4, +5, +6, +8, -9, +10, +11, +11, +12, -13, -14, -15, +16, +17, -18, -22, +3-5mar, del(3)(q11), der(3)t(1;3)(q11;q13), der(3)t(3;5)(q13;q11), i(5p), i(7p)x1-2, der(8)t(8;11)(q24;q11), der(9)t(9;11)(p21;q11), i(11p)x3, add(12)(?q21), add(15)(p11), del(17)(p12); karyotype closely related to SBC-7; HELA markers present
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SBC-7
Cell Type:	human cervix carcinoma (derivative of HELA)
DSMZ No:	ACC 229
Origin:	these cells were thought to be "established from the malignant urinary bladder carcinoma of a 82-year-old Caucasian woman (stage pTa G1-G2) in 1992"; same donor as for cell line SBC-2 (DSM ACC 228); however, DNA fingerprinting and cytogenetic analysis at the DSMZ established unequivocally that this cell line must be considered de facto a subclone of HELA
References:	Thanhäuser et al., Immunology 80: 151-156 (1993); Böhle et al., J. Urology 150: 1932-1937 (1993)
Depositor:	Prof. A. J. Ulmer, Forschungsinstitut Borstel, Borstel, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epitheloid cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:5 to 1:10 about twice a week using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 15-17 hours
Harvest:	cell harvest of about 1-3 x 10 ⁵ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, cytokeratin-7(+), cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament+, vimentin+
Fingerprinting:	same DNA profile as HELA using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypertriploid karyotype with 15% polyploidy; 73/74(70-77)<3n>XX/XXX, +1, -4, +5, +6, -9, -9, +11, -13, -15, +16, +16, +17, +17, -22, -22, +4mar, der(1)t(1;14)(p11;q11), i(3q), del(3)(q11), add(5)(p11), i(7q), der(8)t(8;?11)(q24;q11), i(11p)x2, add(15)(p11), del(17)(p12)x2; HELA markers present
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SC-71
Cell Type:	mouse hybridoma (anti-rat fiber/myosin)
DSMZ No:	ACC 217
Origin:	described as secreting a mouse monoclonal antibody (IgG ₁) reacting with adult rat skeletal muscle type 2A (but not 2B) fibers in immunofluorescence and with undigested type 2A MHCs (myosin heavy chain) as well as both rod and light meromyosin fragments in Western blotting; rat neonatal and embryonic MHCs were not studied by Western blotting; established in 1985 by fusion of NSO myeloma cells with spleen cells from a BALB/c mouse immunized with purified myosin from bovine subcutaneous muscle
References:	Schiaffino et al., J. Muscle Res. Cell Motil. 10: 197-205 (1989)
Depositor:	Prof. S. Schiaffino, University of Padova, Padova, Italy
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS + 2 mM L-glutamine
Subculture:	maintain at $0.1-0.5 \times 10^6$ cells/ml; split saturated culture at 1:5 to 1:10 every 2-5 days; seed out at ca. 0.5×10^6 cells/ml; daily exchange of medium is recommended
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 20 hours
Harvest:	maximal density at about 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgG ₁ monoclonal antibody
Species:	confirmed as mouse with IEF of NP, PEP B
Cytogenetics:	murine karyotype
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SC-75
Cell Type:	mouse hybridoma (anti-rat fiber/myosin)
DSMZ No:	ACC 232
Origin:	described as secreting a mouse monoclonal antibody (IgM) reacting with all adult rat type 2 fibers in immunofluorescence and with all adult rat type 2 MHCs (myosin heavy chain) in Western blotting; recognizing also rat neonatal fibers in immunofluorescence and rat neonatal MHCs in Western blotting; established in 1985 by fusion of NSO myeloma cells with spleen cells from a BALB/c mouse immunized with purified myosin from bovine subcutaneous muscle
References:	Schiaffino et al., J. Muscle Res. Cell Motil. 10: 197-205 (1989)
Depositor:	Prof. S. Schiaffino, University of Padova, Padova, Italy
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension; some cells are loosely adherent (detachable by shaking the flask)
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% horse serum + 2 mM L-glutamine
Subculture:	maintain at $0.7-1.0 \times 10^6$ cells/ml; split saturated culture at 1:2 to 1:3 every 2 day (maximal split ratio of 1:5)
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 23 hours
Harvest:	maximal density at about 2.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% horse serum, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgM monoclonal antibody
Species:	confirmed as mouse with IEF of AST, NP, PEP B
Cytogenetics:	murine hypertetraploid karyotype with 5% polyploidy; $86-96 < 4n >$; 1-3 centric fusion markers and 1-2 dmin present in most cells
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SCHNEIDER-2
Cell Type:	insect - fruitfly (<i>Drosophila melanogaster</i>)
DSMZ No:	ACC 130
Origin:	established from the late embryo of a <i>Drosophila melanogaster</i> (fruitfly) in 1972; originally the cells were diploid with 5-10% XY, currently the cell line has only XX cells that are now 60-80% tetraploid
References:	Schneider, J. Embryol. Exp. Morph. 27: 353-365 (1972)
Depositor:	Dr. W. J. Gehring, Biocenter of the University of Basel, Basel, Switzerland
DSMZ Cell Culture Data:	
Morphology:	small adherent cells growing in monolayers; a small number of cells is also in suspension
Medium:	85-90% Schneider's <i>Drosophila</i> medium + 10-15% FBS
Subculture:	maintain at room temperature without gas exchange (keep lid of flask closed); cells are resistant to trypsinization, for propagation use cells in suspension or remove adherent cells with a rubber policeman about twice a week; after thawing cells may have a long lag period (3-7 days) before they start to grow; split confluent culture 1:2 to 1:4 every 2-4 days; seed out at ca. $1-2 \times 10^6$ in 80 cm ² flask
Incubation:	at room temperature without CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	about $15-20 \times 10^6$ cells/25 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	a mix-up with human or murine cells was excluded with IEF of AST, MDH, PEP B

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SCLC-21H
Cell Type:	human small cell lung carcinoma
DSMZ No:	ACC 372
Origin:	established from the pleural effusion of a 46-year-old Caucasian man with small cell lung carcinoma; sister cell line of cell line SCLC-22H (DSM ACC 373)
References:	Bepler et al., J. Cancer Res. Clin. Oncol. 113: 31-40 (1987); Bepler et al., Differentiation 37: 158-171 (1988)
Depositor:	Dr. G. Bepler, Duke University Medical Center, Durham, North Carolina, USA
DSMZ Cell Culture Data:	
Morphology:	growing in suspension in small or very big clusters (sheets) - cells are not adherent
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS
Subculture:	maintain at ca. 0.5×10^6 cells/ml; split 1:2 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml; cells grow in clumps and are difficult to disperse, thus cell counting may be difficult
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 2 days
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin(+), desmin-, endothel-, GFAP-, neurofilament-, vimentin(+)
Fingerprinting:	identical DNA profiles of SCLC-16HC, SCLC-21H, SCLC-22H and SCLC-24H using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hypodiploid karyotype with 12% polyploidy; 42(40-42)<2n>XY, +1, -2, -5, +6, -10, +11, -17, -18, -19, -21, +2mar, del(1)(p22p32), add(5)(p15), del(6)(q14), add(8)(p21), add(11)(q24), del(11)(p11), t(12;17)(q10;q10), del(13)(q12q14), add(15)(q25); resembles published karyotype (and that of SCLC-16HC)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SCLC-22H
Cell Type:	human small cell lung carcinoma
DSMZ No:	ACC 373
Origin:	established from the pericardial effusion of a 46-year-old Caucasian man with small cell lung carcinoma, oat cell type; patient had received chemotherapy; sister cell line of SCLC-21H (DSM ACC 372)
References:	Bepler et al., J. Cancer Res. Clin. Oncol. 113: 31-40 (1987); Bepler et al., Differentiation 37: 158-171 (1988); Bepler et al., Anticancer Res. 9: 1525-1530 (1989)
Depositor:	Dr. G. Bepler, Duke University Medical Center, Durham, North Carolina, USA
DSMZ Cell Culture Data:	
Morphology:	growing in suspension in small or very big clusters (sheets) - cells are not adherent
Medium:	80-90% Dulbecco's MEM + 10-20% FBS
Subculture:	maintain at ca. 1×10^6 cells/ml; split 1:2 every 2-3 days; seed out at ca. 1.0×10^6 cells/ml; cells grow in clumps and are difficult to disperse; thus cell counting may be difficult; dispersing the cells may also decrease the viability
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-6 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	identical DNA profiles of SCLC-16HC, SCLC-21H, SCLC-22H and SCLC-24H using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, LHD, NP
Cytogenetics:	human hypodiploid karyotype with 8% polyploidy; 43(40-44)<2n>XY, -8, -10, +11, +12, -16, -17, del(4)(q32), add(5)(p15), del(11)(p11), t(12;17)(q10;q10), add(12)(q23), del(13)(q12q14), add(21)(p11), add(22)(q13); sideline with del(3)(p14); closely resembles karyotype of sister cell line SCLC-21H
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SD-1
Cell Type:	human peripheral blood B lymphoblastoid cells (from patient with acute lymphoblastic leukemia)
DSMZ No:	ACC 366
Origin:	established from the peripheral blood of a woman with acute lymphoblastic leukemia (pre B-ALL) at diagnosis; cells were immortalized with EBV; shown at DSMZ by RT-PCR to carry minor breakpoint bcr-abl fusion (e1-a2 and e1-a3) for which it is a suitable positive control for RT-PCR
References:	Dhut et al., Leukemia 5: 49-55 (1991); Drexler et al., Leukemia Res. 23: 207-215 (1999)
Depositor:	Prof. B. D. Young, ICRF, St. Bartholomew's Hospital, London, UK
DSMZ Cell Culture Data:	
Morphology:	ovoid to round cells growing in suspension, singly or in clumps
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.3\text{--}1.0 \times 10^6$ cells/ml; feed cells twice-weekly, split of ca. 1:3 to 1:5; seed out at ca. 0.3×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 28-32 hours
Harvest:	reaches maximum density of ca. $1.0\text{--}1.3 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19+, CD20+, CD34+, CD37+, CD79a+, CD80+, CD138-, HLA-DR+, sm/cyIgG+, sm/cyIgM-, sm/cykappa+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human near-tetraploid karyotype; 92(88-92)<4n>XXXX, t(9;22)(q34;q11)x2; carries two balanced Ph translocations; tetraploid derivative of original diploid karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SEM
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 546
Origin:	established from the peripheral blood of a 5-year-old girl at relapse of acute lymphoblastic leukemia (ALL) in 1990; reported to carry the t(4;11) MLL-AF4 which was confirmed at DSMZ
References:	Greil et al., Brit. J. Haematol. 86: 275-283; Reichel et al., Oncogene 17: 3035-3044 (1998); review article: Drexler et al., Leukemia (submitted)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round to polygonal single rather small cells in suspension
Medium:	90% Iscove's MDM + 10% FBS
Subculture:	split saturated culture 1:2 to 1:6 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. $0.2\text{--}2.0 \times 10^6$ cells/ml; maximum density at ca. $3\text{--}4 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	cell harvest of ca. $3\text{--}4 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 8×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD4-, CD13+, CD14-, CD15+, CD19+, CD33+, CD34-, HLA-DR+, sm/cyIgG-, sm/cyIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypodiploid karyotype with 1.5% polyploidy; 45(40-46)<2n>XX, -13, t(4;11)(q21;q23), del(7)(p14); carries t(4;11) with breakpoints at AF4 and MLL; matches published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SER-W3
Cell Type:	rat Sertoli cells
DSMZ No:	ACC 420
Origin:	Sertoli cells dissected from normal testis from 17-day-old Wistar rat and immortalized with SV40 large-T antigen; cells were described as displaying morphological and functional features of Sertoli cells (phagocytosis, cisplatin sensitivity, transferrin secretion)
References:	Pognan et al., Cell Biol. Toxicol. 13: 453-463 (1997)
Depositor:	Dr. F. Pognan, Zeneca, Macclesfield, UK
DSMZ Cell Culture Data:	
Morphology:	epitheloid cells growing in monolayers
Medium:	90% Dulbecco's MEM + 10% FBS + 2 mM L-glutamine
Subculture:	split 90% confluent culture 1:2 to 1:5 every 2-6 days using trypsin/EDTA; seed out at ca. $0.3-0.5 \times 10^5$ cells/cm ² ; maintain at ca. $0.1-0.3 \times 10^5$ cells/cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	cell harvest of ca. 0.8×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1.5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as rat with IEF of AST, LDH, NP
Cytogenetics:	rat flat-moded hypotetraploid karyotype; 70-78<4n>XXYY/XXY; supernumerary t/acf present in most cells
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SEWA
Cell Type:	mouse osteogenic sarcoma
DSMZ No:	ACC 523
Origin:	established from a polyoma virus-induced osteogenic sarcoma in an A.SW mouse; cells were converted into ascitic form growing in suspension by serial intraperitoneal inoculation into genetically compatible mice; described to have lost double minute chromosomes which were still present in the original tumor cell line, but reoccur when retransplanted into mice
References:	Sjogren et al., Cancer Res. 21: 329-337 (1961); Levan et al., Cytogenet. Cell Genet. 20: 12-23 (1978)
Depositor:	Prof. Dr. H. Hameister, University of Ulm, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly or in clumps in suspension
Medium:	90-95% RPMI 1640 + 5-10% FBS
Subculture:	split saturated culture 1:4 to 1:10 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. $0.3\text{--}1.5 \times 10^6$ cells/ml; maximal density of ca. 2.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 20 hours
Harvest:	cell harvest of ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Species:	confirmed as mouse by cytogenetics
Cytogenetics:	murine flat-modulated near-diploid karyotype; 36-44<2n>; dmin rare or absent

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SF-158
Cell Type:	insect - fall armyworm (<i>Spodoptera frugiperda</i>)
DSMZ No:	ACC 292
Origin:	cloned from the parent line, IPLB-SF-21-AE, derived from pupal ovarian tissue of the fall armyworm (species <i>Spodoptera frugiperda</i> , family <i>Sciaridae</i> , order <i>Lepidoptera</i>); cells are susceptible to infection with MNPV and other Baculoviruses; cells can be used with Baculovirus expression vectors
References:	Fiebich et al., Eur. J. Biochem. 211: 19-26 (1993)
Depositor:	Dr. H. Weich, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	polymorphic round cells, mostly adherent
Medium:	95% TC 100 medium + 5% FBS
Subculture:	split culture 1:4 to 1:5 every 5-6 day (cells grow adherent, but trypsin is not necessary as cells can be detached by tapping the flask or with a rubber scraper); seed out about $1-2 \times 10^6$ cells/25 cm ²
Incubation:	maintain at room temperature (20-27 °C) without CO ₂ (keep lid of flask closed)
Doubling Time:	doubling time of about 70 hours (at room temperature)
Harvest:	cell harvest of about 0.2×10^6 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SF-21
Cell Type:	insect - fall armyworm (<i>Spodoptera frugiperda</i>)
DSMZ No:	ACC 119
Origin:	this cell line (full name: IPLB-SF-21-AE) was derived from immature ovaries of fall armyworm <i>Spodoptera frugiperda</i> pupae (species <i>Spodoptera frugiperda</i> , family <i>Sciaridae</i> , order <i>Lepidoptera</i>); cells are susceptible to Baculovirus infection; see also SF-9 (DSM ACC 125) which was derived from SF-21
References:	Vaughn et al., <i>In Vitro</i> 13: 213-217 (1977)
Depositor:	Dr. V. Jäger, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	90% round cells; 10% spindle-shaped, adherent
Medium:	90% TC 100 medium + 10% FBS
Subculture:	split culture 1:3 to 1:5 every 4-5 days; cells grow partly adherent, but the use of trypsin is not necessary, detach cells with rubber scraper; seed out at ca. 1×10^6 cells/25 cm ²
Incubation:	at about 20 °C (room temperature) without CO ₂ (keep lid of flask closed)
Doubling Time:	doubling time of ca. 3 days
Harvest:	about 25×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed with IEF of AST, PEP B as not primate or murine

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SF-9
Cell Type:	insect - fall armyworm (<i>Spodoptera frugiperda</i>)
DSMZ No:	ACC 125
Origin:	cloned from the parent cell line, IPLB-SF-21-AE, derived from pupal ovarian tissue of the fall armyworm (species <i>Spodoptera frugiperda</i> , family <i>Sciaridae</i> , order <i>Lepidoptera</i>); cells are susceptible to infection with MNPV and other Baculoviruses; cells can be used with Baculovirus expression vectors
References:	Vaughn et al., <i>In Vitro</i> 13: 213-217 (1977); Smith et al., <i>Proc. Natl. Acad. Sci. USA</i> 82: 8404-8408 (1985)
Depositor:	Dr. Asta Cramer, Institute for Virology, Marburg, Germany
DSMZ Cell Culture Data:	
Morphology:	polymorphic round cells, adherent
Medium:	90% TC 100 medium + 10% FBS
Subculture:	split culture 1:2 to 1:3 every 3-5 days (cells grow adherent, but trypsin is not necessary as cells can be detached by tapping the flask or with a rubber policeman); seed out at ca. 2×10^6 cells/80 cm ² ; immediately after thawing, cell viability may be low
Incubation:	at room temperature (about 20 °C) without CO ₂ (keep lid of flask closed)
Doubling Time:	doubling time of ca. 50 hours
Harvest:	about $5-10 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed with IEF of AST, PEP B as not primate or murine

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SGE-1
Cell Type:	rat kidney glomerular epithelial
DSMZ No:	ACC 343
Origin:	established from isolated renal glomeruli of a 3-week-old normal male Wistar rat in 1988
References:	Yamada et al., Cell Structure Function 13: 495-513 (1988); Yamada et al., Eur. J. Cell Biol. 49: 252-258 (1989)
Depositor:	Dr. M. Kawaguchi, Hyogo College of Medicine, Nishinomiya, Japan
DSMZ Cell Culture Data:	
Morphology:	adherent cells growing as monolayer with polygonal shapes
Medium:	45% Dulbecco's MEM + 45% Ham's F12 + 10% FBS + 1x insulin-transferrin-sodium selenite + 10 ng/ml EGF
Subculture:	split culture prior to confluency, 1:10 to 1:20 twice a week using trypsin/EDTA; seed out at ca. $2-2.5 \times 10^4$ cells/cm ² ; cells grow also without coating of the flask with collagen
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 33 hours
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as rat with IEF of AST, MDH, NP, PEP B
Cytogenetics:	rat flat-moded hyperdiploid karyotype; 43-48<2n>
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SH-SY5Y
Cell Type:	human neuroblastoma
DSMZ No:	ACC 209
Origin:	clonal subline of the neuroepithelioma cell line SK-N-SH that had been established in 1970 from the bone marrow biopsy of a 4-year-old girl with metastatic neuroblastoma
References:	Biedler et al., Cancer Res. 33: 2643-2652 (1973); Jalava et al., Cancer Res. 50: 3422-3428 (1990)
Depositor:	Dr. J. Heikkilä, University of Turku, Turku, Finland
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing as monolayer and in cell clusters (when cultured at high density); cells do not grow to complete confluency
Medium:	80-85% Dulbecco's MEM + 15-20% FBS
Subculture:	split culture 1-2 times a week using trypsin/EDTA; seed out with 2×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of >55 hours
Harvest:	cell harvest of about 15×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, neurofilament+, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human near-diploid karyotype with 1.8% polyploidy; 46/47(42-48)<2n>X/XX, +7, ins(1)(q32q11q43), add(9)(q34), der(22)t(?17;22)(q22;q13); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SIG-M5
Cell Type:	human acute monocytic leukemia
DSMZ No:	ACC 468
Origin:	established from the bone marrow of a 63-year-old man with acute myeloid leukemia of monocytic origin (AML FAB M5a) at diagnosis
References:	Walter et al., Brit. J. Haematol. 109: 396-404 (2000)
Depositor:	Dr. Gabriele Schoedon, University Hospital Zurich, Zurich, Switzerland
DSMZ Cell Culture Data:	
Morphology:	large round cells growing singly in suspension
Medium:	80% Iscove's MDM + 20% FBS
Subculture:	split saturated culture 1:2 every 3 days; seed out at ca. 1×10^4 cells/ml; after thawing, it is advisable to initially grow the cells in 12- or 24-well plates; cell viability remains continuously at or below 70%
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 72 hours
Harvest:	maximum density of about 2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4-, CD13+, CD14+, CD15+, CD19-, CD33+, CD34-, CD68+, HLA-DR-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypertriploid/hypotetraploid karyotype with 4% polyploidy; 78(77-81)<3n>XX/XXY, +1, +2, +3, +4, +5, +8, -9, -10, +12, +15, +16, +17, +18, +21, +22, +0-2mar; i(1q), add(17)(p1?1); resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SIMA
Cell Type:	human neuroblastoma
DSMZ No:	ACC 164
Origin:	established from the adrenal tumor tissue resected after treatment from a 20-month-old boy of European origin with neuroblastoma (stage III) in 1991
References:	Marini et al., Cancer Genet. Cytogenet. 112: 161-164 (1999)
Depositor:	Dr. P. Marini, Neuklinikum Schnarrenberg, Tübingen, Germany
DSMZ Cell Culture Data:	
Morphology:	cell shape is bipolar, cells growing as monolayer or forming foci
Medium:	90% RPMI 1640 + 10% FBS + 2 mM L-glutamine
Subculture:	split confluent culture 1:3 to 1:5 every 3-6 days using trypsin/EDTA; seed out at ca. 5×10^6 cells/80 cm ² ; cells may grow slowly initially forming non-adherent large cell aggregates
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	highly variable doubling time of about 34-100 hours
Harvest:	cell harvest of about $0.3-0.6 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, neurofilament+, vimentin+
Fingerprinting:	unique DNA profile using multiplex PCR at D1S80, D2S44, D17S5 and ApoB
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypotetraploid karyotype with 3.5% polypoidy; 88/89<4n>XXY/XXYY, -Y, -11, -17, -17, der(1qh+)(1;17)(p34-35;q12)x2, t(6;9)(p21;q33)x2, ?inv(12)(q21q24)x2, der(22)t(17;22)(q21;p12)x2; ca. 50-100 NMYC-containing dmin present in each cell; t(1;17) together with NMYC amplification is restricted to neuroblastoma; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SISO
Cell Type:	human cervix adenocarcinoma
DSMZ No:	ACC 327
Origin:	established from the surgically removed, chemotherapy-resistant cervix adenocarcinoma of a 67-year-old Japanese woman (clinical stage FIGO Ib)
References:	Sonoda et al., Int. J. Oncol. 6: 1099-1104 (1995)
Depositor:	Dr. T. Watanabe, Kyushu University, Fukuoka, Japan
DSMZ Cell Culture Data:	
Morphology:	epithelial growth in compact colonies; cells tend to pile up after confluency
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 5-7 days using trypsin/EDTA; seed out at ca. 2×10^6 cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	cell harvest of ca. 40×10^6 cells/175 cm ² (ca. 2.3×10^5 cells/cm ²)
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 3×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypotriploid karyotype with 12% polyploidy; 61<3n>XXX, -1, -2, -3, -4, -6, -8, -10, -13, -14, -16, -17, -19, -22, +4mar, add(X)(q28), add(5)(p15), add(6)(p23), del(9)(q34), add(11)(q24), del(12)(q15), der(12)t(12;17)(p12;q11), add(13)(p12), add(18)(q23)x1-2; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SK-ES-1
Cell Type:	human Ewing's sarcoma
DSMZ No:	ACC 518
Origin:	established from the Ewing's sarcoma of an 18-year-old man in 1971; described to lack expression of HLA-ABC and E, but HLA-ABC can be reconstituted by TNF-alpha and IFN-gamma; cells are sensitive to adenovirus infection and express high levels of growth/differentiation factor midkine
References:	Bloom, Cancer Res. 32: 960-967 (1972); Borowski et al., Immunobiol. 200: 1-20 (1999); Adachi et al., Cancer Res. 61: 7882-7888 (2001)
Depositor:	Dr. K. Scotlandi, Istituti Ortopedici Rizzoli, Bologna, Italy
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblastoid cells growing as monolayer, cells do not become confluent (max. ca 25% confluent), formation of dense clumps which become larger
Medium:	80-90% McCoy's 5A + 10-20% FBS
Subculture:	split culture 1:3 to 1:5 twice a week using trypsin/EDTA; seed out at ca. 2×10^6 cells/80 cm ² flask; maximal density at ca. 1.5×10^5 cells/cm ² ; use new flask after splitting; clumps can be dispersed with trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	cell harvest of about 7×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, neurofilament+, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hyperdiploid karyotype; 49(46-51)<2n>XY, +7, +2-4mar, add(Y)(q12), add(3)(q27), del(3)(q13), add(8)(p11), ?idic(9)(q3?4.2)del(9)(q1?q2?)add(9)(p11), t(11;22)(q24;q12), add(12)(p13), del(16)(q21); carries t(11;22) effecting EWS-FLI1 fusion yielding longer type II RT-PCR transcript
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SK-HEP-1
Cell Type:	human liver adenocarcinoma
DSMZ No:	ACC 141
Origin:	established from the ascites of a 52-year-old Caucasian man in 1971
References:	Fogh et al., J. Natl. Cancer Inst. 59: 221-226 (1977); Heffelfinger et al., In Vitro Cell Dev. Biol. 28A: 136-142 (1992)
Depositor:	Dr. H. Kirchner, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like cells growing as monolayer
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split confluent culture 1:10 to 1:20 about 1-2 times a week using trypsin/EDTA; cells grow fast; seed out at ca. $3-4 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	about $15-18 \times 10^6$ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, PEP B
Cytogenetics:	human hypotriploid karyotype with 9% polyploidy; 62(60-63)<3n>X, -X, -X, -4, +6, +7, -9, +12, -13, -14, -15, -16, -18, der(1;7)(p10;q10), ?der(2;20)(p10;q10), der(6;13)(p10;q10)x2-3, add(12)(p1?), i(14q), add(19)(p13), add(21)(p11); sideline with add(15)(p11); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SK-MEL-1
Cell Type:	human melanoma
DSMZ No:	ACC 303
Origin:	established from the lymph node metastasis (thoracic duct) of a 29-year-old Caucasian man with primary melanoma in 1966; cells were described to produce pigment and to lead to pigment-producing tumors upon heterotransplantation into hamster
References:	Oettgen et al., J. Natl. Cancer Inst. 41: 827-843 (1968)
Depositor:	Dr. J. Atzpodien, Medical School Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	mostly round cells growing singly or in clumps (occasional giant, polynucleated cells) in suspension
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	maintain between $0.1-0.3 \times 10^6$ cells/ml; cultures should be fed by removal of cell-free spent medium; a cell density of 0.5×10^6 cells/ml should not be exceeded
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	cells grow very slowly with a doubling time of several days, but with rapid medium acidification
Harvest:	ca. $0.5-1.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, cytokeratin-8-, desmin-, endothel-, GFAP-, HMB-45+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypertriploid karyotype with 4% polyploidy; 78(72-79)<3n>XXYY, +X/Y, +2, +4, -5, +6, +7, +11, -12, -13, -13, +15, -17, +19, -21, +6mar; add(2)(p24), add(3)(q11), i(4q)x2, del(6)(q22), add(6)(q27), add(7)(q35)x2, del(9)(p23)x2, add(10)(p13), del(11)(p15), del(12)(q24), add(14)(p11), add(14)(q32), i(17q), add(17)(p12), add(19)(q13), del(?22)(q11); carries pair of large submetacentric markers; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative: PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SK-MEL-3
Cell Type:	human melanoma
DSMZ No:	ACC 321
Origin:	established from the lymph node metastasis of a malignant melanoma from a 42-year-old Caucasian woman
References:	Fogh et al., in Human Tumor Cells in Vitro, edited J. Fogh, Plenum Press, New York 1975, p. 115-159
Depositor:	Dr. J. Atzpodien, Medical School Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial cells growing as monolayer
Medium:	90% McCoy's 5A + 10% FBS
Subculture:	split confluent culture 1:3 to 1:4 every 3-4 days using trypsin/EDTA; seed out ca. 2×10^6 cells/80 cm ² in 8-10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	cell harvest of ca. 15×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, HMB-45+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human near-tetraploid karyotype with 3% polyploidy; 92(80-93)<4n>XX, -X, -X, -6, +7, +7, +7, -10, -10, -16, -16, +20, +20, +20, +20, +22, dup(7)(p15p22)x2, der(9)t(7;9)(p11;p21)dup(7)(p15p22)x2; sideline with add(2)(p23) marker; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SK-MEL-30
Cell Type:	human melanoma
DSMZ No:	ACC 151
Origin:	derived from tumor tissue (subcutis metastasis) of a 67-year-old Caucasian man with malignant melanoma in 1975
References:	Carey et al., Proc. Natl. Acad. Sci. USA 73: 3278-3282 (1976)
Depositor:	Dr. J. Atzpodien & Dr. H. Kirchner, Medical School Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, epithelial-like cells growing as monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 2-3 times per week using trypsin/EDTA; seed out about 5×10^6 cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	about 6×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, HMB-45+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypertriploid karyotype with 3.5% polyploidy; 80(69-83)<3n>XXY-Y/XXYY, +1, +1, +2, +2, +3, +7, +10, -11, +12, +17, +18, +20, -21, +2mar, inv(6)(p11;q25), der(7)t(7;14)(p22;q22), der(9)t(9;?21)(p11;q11)x2, 14p+, der(15)t(15;?)(p12;?)x2, der(16)t(16;?)(q23;?), der(18)t(18;?)(p11;?), mar: ?der(19)t(1;19)(q12;q13); additional subclonal rearrangements of ch : der(1)t(1;?)(p12;?) were present in a few cells
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SK-MES-1
Cell Type:	human lung squamous cell carcinoma
DSMZ No:	ACC 353
Origin:	established from the pleural effusion of a 65-year-old Caucasian man with lung squamous cell carcinoma in 1970
References:	Fogh & Trempe, Human Tumor Cells in Vitro, Plenum Press, New York, 1975, 115-159
Depositor:	obtained from DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells growing as monolayers
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture about 1:3 to 1:5 every 3-4 days using trypsin/EDTA; seed out at ca. 1-2 x 10 ⁶ cells/80 cm ² + 10 ml medium
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 50 hours
Harvest:	cell harvest of ca. 15 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1-2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human flat-moded hyperdiploid karyotype with 44% polyploidy; 54-60<2n>X, -Y, +1, +1, -2, +4, +5, +7, +7, +11, +13, +19, +22, +5mar, add(X)(p21), i(1q), del(1)(q12)x1-2, add(2)(q37), del(3)(p22p26), add(4)(q32), der(5)t(5;6)(p15;p11), del(5)(q13), add(7)(q11;abr), add(11)(q2)x2, add(13)(p12)x1-2, add(13)(p12), add(15)(p12), der(16)t(8;16)(q12;q23), del(17)(p11), add(19)(p13)x2-3; sideline with additional der(7;8)(q10;p10), add(22)(q13), der(11)dup(11)(q13q22); carries 3p-, 5q-, +7, 17p- associated with NSCLC; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-1-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SK-MM-2
Cell Type:	human plasma cell leukemia
DSMZ No:	ACC 430
Origin:	established from the peripheral blood of a 54-year-old man with plasma cell leukemia (Igkappa) (refractory, terminal state) in 1982; plasma cell leukemia is related to multiple myeloma
References:	Eton et al., Leukemia 3: 729-735 (1989); O'Boyle et al., Leukemia Lymphoma 21: 255-266 (1996); Drexler & Matsuo, Leukemia Res. 24: 681-703 (2000)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round single cells growing in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:2 every 3-4 days; seed out at ca. 1.0×10^6 cells/ml; do not culture below 0.6×10^6 cells/ml; maximum density of ca. 1.3×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 100 hours
Harvest:	cell harvest of ca. 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10-, CD13-, CD19-, CD20-, CD34-, CD37-, CD38+, CD79a-, cyCD79a+, CD80-, CD138(+), HLA-DR-, sm/cyIgG-, sm/cyIgM-, smIgkappa-, cyIgkappa+, sm/cyIglambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH, NP
Cytogenetics:	human bimodal hypodiploid/hypotetraploid karyotype with 9% polyploidy (including a subpopulation with 32-64N); 42(38-43)<2n>XX, -13, -18, -22, del(1)(p32p36), der(2)t(1;2)(p32;q3?1)t(1;13)(p36q1?4), der(4;14)(4pter->4p15::4p11->4p10::14q10->14q22::4q11-12->4qter), del(6)(q15), i(8q), der(14)t(11;14)(q13;q32) (46%); 84(78-84)<4n>idemx2 (45%); carries non-reciprocal t(11;14) causing rearrangement of BCL1 associated with chronic lymphoproliferative disorders; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SK-N-MC
Cell Type:	human neuroblastoma
DSMZ No:	ACC 203
Origin:	established from the supraorbital metastasis of a neuroblastoma of a 14-year-old girl in 1971 (see ATCC HTB 10); although described in the original paper as neuroblastoma, SK-N-MC is now widely regarded as having originated from the morphologically similar Askin's tumor related to Ewing's sarcoma; FLI1-EWS gene fusion, the molecular pathological change in Ewing's sarcoma and Askin's tumor, was shown in SK-N-MC by Dunn et al. (Cancer Genet. Cytogenet. 76: 19-22, 1994)
References:	Biedler et al., Cancer Res. 33: 2643-2652 (1973); Spengler et al., In Vitro 8: 410 (1973)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent, fibroblast-like cells growing as monolayers
Medium:	90% Dulbecco's MEM + 10% FBS + 2 mM L-glutamine + 1x MEM non-essential amino acids
Subculture:	split confluent culture 1:3 to 1:6 every 3-5 days using trypsin/EDTA; seed out at ca. $1-4 \times 10^6$ cells/80 cm ² ; cells may grow slowly initially
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of about 15×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human pseudodiploid karyotype with 6% polyploidy; 46(40-47)<2n>XX, +8, -16, t(1;4)(p36;p13), t(2;3)(q24;q29), der(3)t(3;3)(p13;q22), del(10)(q25), t(11;22)(q24;q12), t(15;22)(q21;q12), add(17)(p12), add(21)(p12); dmin in 9% of cells; resembles published karyotype; carries masked 11;22 translocation recurrent in Ewing's and Askin's tumor; presence of type-2 EWS-FLI1 hybrid transcripts (328 bp) confirmed by RT-PCR
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SKM-1
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 547
Origin:	established from the peripheral blood of a 76-year-old Japanese man with acute monoblastic leukemia (AML M5) in 1989 following myelodysplastic syndromes (MDS)
References:	Nakagawa et al., Brit. J. Haematol. 77: 250-252 (1991); Nakagawa et al., Brit. J. Haematol. 85: 469-476 (1993); Nakagawa & Matozaki, Leukemia Lymphoma 17: 335-339 (1995)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. $0.5\text{--}2.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of ca. 2.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD3-, CD4(+), CD11b+, CD13+, CD14(+), CD15+, CD19-, CD33+, CD34+, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hypodiploid karyotype with 8% polyploidy; 43(38-43)<2n>XY, +1, -12, -14, -20, -21, t(1;19)(q21;q13), del(2)(p11), del(9)(q12), add(17)(p1?); sideline with idem, der(10)t(10;?21)(p11;q11), -21; consistent with published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SKW-3
Cell Type:	human T cell leukemia
DSMZ No:	ACC 53
Origin:	established from the peripheral blood of a 61-year-old man with T cell chronic lymphocytic leukemia (CLL) in 1977
References:	Hirono et al., J. Immunol. 123: 1133-1140 (1979)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.3\text{--}1.0 \times 10^6$ cells/ml; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	saturation density at about 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, PCR assays
Immunology:	CD2+, CD3-, CD4+, CD5+, CD6+, CD7+, CD8+, CD13-, CD19-, CD34-, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human near diploid karyotype with 4% polyploidy; 46(43-48)<2n>XY, +8, -14, t(3;3)(q11;q27), der(8)t(8;14)(q24;q11)x2, t(8;11)(p21;p12), der(12)t(12;?)(q24;?), der(14)t(8;14)(q24;q11)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SNB-19
Cell Type:	human glioblastoma
DSMZ No:	ACC 325
Origin:	established from the surgical resection of a left parieto-occipital glioblastoma from a 47-year-old man in 1980; cells were described to secrete plasminogen activator, to be clonogenic in soft agar and to be tumorigenic in nude mice
References:	Gross et al., Cancer Res. 48: 291-296 (1985)
Depositor:	Dr. H. Weich, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblastic cells growing as monolayer with contact inhibition, occasional giant cells
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:10 every 3-5 days using trypsin/EDTA; seed out ca. $2-4 \times 10^6$ cells/80 cm ² in 8-10 ml medium
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	cell harvest of ca. 15×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP+, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypotriploid karyotype with 15% polyploidy; 63(58-63)<3n>XXY, +1, +7, -8, -10, -12, -13, -14, -15, -16, -18, -21, -22, +2mar; der(1)del(1)(q23)ins(1;4)(p32;q?23q27), del(1)(q13), del(4)(q23q27), del(4)(q28q35), add(8)(q24), add(11)(p15), der(19)add(19)(p13)add(19)(q13); large submetacentric, der(19) and der(1) markers; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SOM-4D10
Cell Type:	mouse hybridoma (CD71)
DSMZ No:	ACC 480
Origin:	described as secreting a mouse monoclonal antibody (IgG ₂ b) CD71; the cell line was established in 1986 by fusion of PX63AG8.653 cells with spleen cells of a mouse immunized with an EBV-transformed B cell line from a patient with MHC class II deficiency
References:	Hadam, Leucocyte Typing VI: 524-527 (1997)
Depositor:	Dr. M. R. Hadam, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent cells growing as monolayers, some cells grow in suspension
Medium:	80% Dulbecco's MEM (4.5 g/L glucose) + 20% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement" for the first 1-2 weeks post-thawing)
Subculture:	split saturated cultures 1:3 every 2 days; seed out at 0.2×10^6 cell/ml; maintain at $0.2\text{-}0.4 \times 10^6$ cells/ml
Incubation:	at 37°C with 5-10% CO ₂
Doubling Time:	doubling time of about 20-30 hours
Harvest:	maximal density of about 5×10^6 cells/25 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgG ₂ b monoclonal antibody
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SP2/0-AG14
Cell Type:	mouse myeloma
DSMZ No:	ACC 146
Origin:	SP2/0-AG14 is a hybrid between BALB/c spleen cells and the myeloma cell line P3X63AG8; cells were described as not synthesizing or secreting immunoglobulin chains, being resistant to azaguanine at 20 : g/ml, and not growing in HAT medium; cell line is used as a fusion partner for the production of hybridomas secreting specific monoclonal antibodies
References:	Shulman et al., Nature 276: 269-270 (1978); Ozato et al., J. Immunol. 126: 317-321 (1981)
Depositor:	Dr. G. Köhler, Max-Planck-Institute, Freiburg, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension
Medium:	92.5% Iscove's MDM + 7.5% FBS + 2 mM L-glutamine
Subculture:	maintain at 0.1-0.5 x 10 ⁶ cells/ml; split 1:3 to 1:10 every 2-4 days; seed out at ca. 0.5 x 10 ⁶ cells/ml
Incubation:	at 37 /C with 7.5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	maximal density at 0.5-1.0 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of LDH, NP, PEP B
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SPC-BM-36
Cell Type:	insect - silkworm (<i>Bombyx mori</i>)
DSMZ No:	ACC 186
Origin:	derived from larvae of a silkworm (species <i>Bombyx mori</i> , family Bombycidae, order Lepidoptera) (Seidenspinner)
References:	no reference available
Depositor:	Dr. H. S. Conradt, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	large, mostly adherent cells; 90% round cells (singly or aggregates), 10% spindled-shaped cells with long processes
Medium:	80% TC 100 medium + 20% FBS
Subculture:	cells are difficult to culture and grow slowly! split confluent culture 1:3 to 1:6 every 3-4 days using a (plastic) cell scraper
Incubation:	cells grow without CO ₂ (keep lid of flask closed) at room temperature (20-25 /C)
Doubling Time:	doubling time of ca. 50-100 hours
Harvest:	cell harvest of up to 12 x 10 ⁶ cells/80 cm ² flask
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as not of primate or murine origin with IEF of AST, PEP B

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SPI-801
Cell Type:	human chronic myeloid leukemia in blast crisis (derivative of K-562)
DSMZ No:	ACC 86
Origin:	allegedly derived "from the peripheral blood of a 10-year-old girl with T-ALL in relapse in 1980"; but DNA fingerprinting showed cross-contamination with K-562; K-562 was established from the pleural effusion of a 53-year-old woman with chronic myeloid leukemia (CML) in blast crisis in 1970; SPI-801 cells express and produce hemoglobins (quantitatively and qualitatively different from those of K-562 cells)
References:	Gignac et al., Leukemia Lymphoma 10: 359-368 (1993); for K-562: Lozzio et al., J. Natl. Cancer Inst. 50: 535-538 (1973); Lozzio et al., Blood 45: 321-334 (1975)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round, large, single cells in suspension
Medium:	90-95% RPMI 1640 + 5-10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximal density at about 1.2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD3-, CD13+, CD19-, GlyA+
Fingerprinting:	same DNA profile as K-562 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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Dept. Human and Animal Cell Cultures

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SPI-802
Cell Type:	human chronic myeloid leukemia in blast crisis (derivative of K-562)
DSMZ No:	ACC 92
Origin:	allegedly derived "from the peripheral blood of a 10-year-old girl with T-ALL in relapse in 1980"; but DNA fingerprinting showed cross-contamination with K-562; K-562 was established from the pleural effusion of a 53-year-old woman with chronic myeloid leukemia (CML) in blast crisis in 1970; SPI-802 cells express and produce hemoglobins (quantitatively and qualitatively different from those of K-562 cells)
References:	Gignac et al., Leukemia Lymphoma 10: 359-368 (1993); for K-562: Lozzio et al., J. Natl. Cancer Inst. 50: 535-538 (1973); Lozzio et al., Blood 45: 321-334 (1975)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round, large, single cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximal density at about 1.2×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	CD3-, CD13+, CD19-, GlyA+
Fingerprinting:	same DNA profile as K-562 using multiplex PCR of minisatellite markers
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypotriploid karyotype without sharp mode; 60-66<3n>XX, -X, -4, -5, -8, -10, -13, -16, -20, -21, -22, +5mar, der(3)t(3;?)(q22;?), del(4)(q22), del(6)(q15)x2, der(2q11q)t(2;11)(2p11p;2q11q), der(15)t(15;18)(q21;q11/12), der(17)t(17;?)(p11/13;?), der(18)t(18;?)(p11;?), del(20)(q11/12), del(X)(p22); results of FISH analysis show that one or two marker chromosomes probably derive from Ph
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SR-4987
Cell Type:	mouse stromal cells
DSMZ No:	ACC 323
Origin:	established from the bone marrow of a female BDF/1 mouse in 1987; described to produce M-CSF, but not G-CSF, GM-CSF, or IL-3
References:	Pessina et al., Cytotechnology 8: 93-102 (1992)
Depositor:	Dr. A. Pessina, University of Milan, Milano, Italia
DSMZ Cell Culture Data:	
Morphology:	fibroblast-like cells growing adherently in monolayers
Medium:	90% McCoy's 5A + 10% FBS
Subculture:	split confluent culture 1:2 to 1:5 twice a week using trypsin/EDTA; seed out ca. $1.0\text{--}1.5 \times 10^6$ cells/25 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24-36 hours
Harvest:	cell harvest of ca. $3.0\text{--}3.5 \times 10^5$ /cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as murine with IEF of AST, MDH, NP
Cytogenetics:	murine hypotetraploid karyotype; 70(59-70)<4n>; 1-3 centric fusion, and a small ring, markers present
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SR-786
Cell Type:	human anaplastic large T cell lymphoma
DSMZ No:	ACC 369
Origin:	established from the pleural effusion an 11-year-old young boy with CD30+ (Ki-1) large T cell lymphoma in 1983 (also described as "SR"); described to express the NPM-ALK fusion gene
References:	Su et al., Am. J. Pathol. 132: 192-198 (1988); Su et al., Am. J. Pathol. 135: 439-445 (1989); Beckwith et al. J. Natl. Cancer Inst. 82: 501-509 (1990); Su et al., Int. J. Oncol. 4: 1089-1095 (1994); Dirks et al., Leukemia 10: 142-149 (1996)
Depositor:	Dr. M. J. S. Dyer, Institute of Cancer Research, Sutton, Surrey, UK
DSMZ Cell Culture Data:	
Morphology:	ovoid-to-round, sometimes irregular, occasionally with apical processes, growing in suspension, singly or in small clumps
Medium:	85% RPMI 1640 + 15% FBS
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; split 1:2 to 1:2.5 every 3-4 days; seed out at ca. 0.5×10^6 cells/ml; to maximize viability, it may be necessary to replenish medium daily; cultures show a temporary loss of viability from one to five days after recovery of frozen ampoules
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	maximum density of ca. $0.9-1.3 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3-6 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4+, CD5-, CD7-, CD8-, CD10-, CD11b-, CD13-, CD14-, CD15+, CD19-, CD20-, CD21-, CD25+, CD30+, CD33-, CD34-, CD68+, CD71+, HLA-DR+, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hypertriploid, flat-moded karyotype; 70-84<3n>XX, +1, +2, -4, +5, +6, +7, +8, -13, -13, +14, -18, +19, +22, +6-9mar, add(1)(q11), del(1)(p11)/der(?)t(1;?)(q11;?), der(2)t(2;5)(p23;q35)inv(2)(p23q14)x2, del(4)(q22), der(5)t(2;5)(p23;q35)x2, del(7)(q21), der(9)t(1;9)(q11;p24)x2, der(12)t(12;13)(q24.32;q11)x1-2, del(13)(q13q31), add(14)(p11)/der(?)t(14;?)(q11;?)x2, del(21)(q22); carries masked t(2;5) associated with ALCL
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	ST-2
Cell Type:	mouse stromal cells
DSMZ No:	ACC 333
Origin:	established from Whitlock-Witte type long-term bone marrow culture of BC8 mice
References:	Ogawa et al., EMBO J. 7: 1337-1343 (1988)
Depositor:	obtained from RIKEN Cell Bank (RCB0224), Tsubuka Science City, Ibaraki, Japan
DSMZ Cell Culture Data:	
Morphology:	adherent, fibroblast-like cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split nearly confluent culture 1:4 to 1:5 every 3-5 days using trypsin/EDTA (cell layer might detach spontaneously when cells grow too densely); seed out $0.5-1.0 \times 10^6$ cells/75 cm ² in 7-10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-50 hours
Harvest:	cell harvest of ca. 5×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, MDH, NP
Cytogenetics:	murine flat-moded hypotetraploid karyotype; 68-77<4n>; centric fusion marker present in most cells
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SU-DHL-1
Cell Type:	human anaplastic large cell lymphoma
DSMZ No:	ACC 356
Origin:	established from the pleural effusion of a 10-year-old boy in 1973; original diagnosis was diffuse histiocytic lymphoma, corrected to diffuse large cell lymphoma (1984) and malignant histiocytosis (1989); according to current REAL classification considered to represent an anaplastic large cell lymphoma (ALCL); described to carry the t(2;5)(p23;q35) NPM-ALK fusion gene
References:	Epstein et al., Cancer 34: 1851-1872 (1974) and 42: 2379-2391 (1978); Mason et al., Brit. J. Haematol. 74: 161-168 (1990); Dirks et al., Leukemia 10: 142-149 (1996)
Depositor:	Dr. J. Minowada, Fujiaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	polymorphic cells growing in suspension, singly or in small clusters
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.3-1.0 \times 10^6$ cells/ml; seed out at ca. 0.3×10^6 cells/ml; split 1:2 to 1:3 every 2-3 days
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	maximal density at ca. 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5-8 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD5+, CD7-, CD10-, CD11b-, CD13-, CD14-, CD15-, CD19-, CD20-, CD25+, CD30+, CD33-, CD34-, CD37-, CD71+, CD79a-, cyCD79a+, CD80-, CD138-, HLA-DR-, sm/cyIgG-, sm/cyIgM-, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human flat-moded hypertriploid karyotype with 16% polyploidy; 74(67-75)<3n>XX, -Y, +1, +2, +3, +5, -7, +12, -16, -18, +19, -20, +21, +3mar, del(1)(p21), t(2;5)(p23;q35)x2, del(6)(q23)x1-2, add(8)(p12), add(9)(p21), del(10)(p14), add(12)(q24), add(14)(p12), add(16)(q24), dup(19)(q13.1qter); sideline with der(12)t(12;19)(q13;q13); carries two copies of balanced t(2;5) associated with ALCL; matches published karyotype; expression of NPM-ALK fusion gene was detected by RT-PCR
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SU-DHL-4
Cell Type:	human B cell lymphoma
DSMZ No:	ACC 495
Origin:	established from the peritoneal effusion of a 38-year-old man with B-NHL (diffuse large cell, cleaved cell type; originally described as "diffuse histiocytic lymphoma") in 1975
References:	Epstein et al., Cancer 42: 2379-2391 (1978), Cancer 37: 2158-2186 (1976), Cancer Res. 39: 1748-1759 (1979); Kaiser-McCaw Hecht et al., Cancer Genet. Cytogenet. 14: 205-218 (1985); Winter et al., Blood 63: 140-146 (1984)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single round cells growing in suspension
Medium:	80-90% RPMI 1640 + 10-20% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 2 days; seed out at ca. 0.5×10^6 cells/ml; after thawing, during the first week cells should be kept with 20% FBS initially; culture in 24-well-plate is of advantage; maximum cell density of 1.8×10^6 cells/ml; maintain at about 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	cell harvest of about $1.0\text{-}1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a+, CD80(+), CD138-, HLA-DR+, cy/smlgG+, cy/smlgM-, cy/smkappa+, cy/smlambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human hyperdiploid karyotype with 4% polyploidy; 50(47-51)<2n>XY/XXY, -3, +7, +8, +12, +13, -16, +18, der(3)t(3;3)(p2?4;p1?2), der(8)t(3;8)(p1?2;q12)?inv(3)(p1?p2?), der(12)t(3;12)(p1?q24)add(3)(p2?4), add(13)(p11), t(14;18)(q32.3;q21.1), add(16)(p13), der(18)t(14;18)(q32.3;q21.1); carries t(14;18) effecting BCL2(MBR)/IGH fusion; matches published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SUP-B15
Cell Type:	human B cell precursor leukemia
DSMZ No:	ACC 389
Origin:	established from the bone marrow of a 9-year-old boy with acute lymphoblastic leukemia (B cell precursor ALL) in second relapse in 1984; described to carry the ALL-variant (m-bcr) of the BCR-ABL fusion gene (e1-a2)
References:	Naumovski et al., Cancer Res. 48: 2876-2879 (1988); review article: Drexler et al., Leukemia Res. 24: 109-115 (2000)
Depositor:	Dr. S. D. Smith, University of Chicago, Chicago, Illinois, USA
DSMZ Cell Culture Data:	
Morphology:	small round cells growing singly in suspension
Medium:	80% McCoy's 5A + 20% FBS
Subculture:	maintain at ca. 1×10^6 cells/ml; split 1:2 to 1:3 about once a week; seed out at ca. 5×10^6 cells in 5 ml medium; it is advisable to culture the cells initially in 24-well-plates, cell growth may be slow during the first week after thawing
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60 hours
Harvest:	cell harvest of ca. 1.5×10^6 cells/ml; maximum density of ca. 3×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13+, CD19+, CD20-, CD34+, CD37-, CD38+, CD79a+, cyCD79a+, CD80-, CD138-, HLA-DR+, sm/cyIgG-, smlgM-, cyIgM+, sm/cykappa-, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human pseudodiploid karyotype; 46<2n>XY, der(1)t(1;1)(p11;q31), add(3)(q2?7), der(4)t(1;4)(p11;q35), t(9;22)(q34;q11), add(10)(q25), ?del(14)(q23q31), der(16)t(9;16)(q11;p13); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SUP-M2
Cell Type:	human anaplastic large cell lymphoma
DSMZ No:	ACC 509
Origin:	derived from the cerebrospinal fluid of a 5-year-old girl with refractory malignant histiocytosis; described to carry the typical t(2;5)(p23;q35) leading to the NPM-ALK fusion gene and to express CD30, defining the cells as anaplastic large cell lymphoma (ALCL)
References:	Morgan et al., Blood 73: 2155-2164 (1989); Mason et al., Brit. J. Haematol. 74: 161-168 (1990); Bischof et al., Mol. Cell. Biol. 17: 2312-2325 (1997)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round cells growing singly or in clumps in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:3 to 1:5 every second day; seed out at ca. 0.3-0.5 x 10 ⁶ cells/ml; maintain at ca. 0.5-1.0 x 10 ⁶ cells/ml; maximal density of ca. 1.5 x 10 ⁶ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35 hours
Harvest:	cell harvest of about 1.0 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	CD2+, CD3-, CD4+, CD13-, CD14-, CD15(+), CD19-, CD25+, CD30+, CD33-, CD34-, CD68-, CD71+, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-modulated near-diploid karyotype with 3.5% polyploidy; 44-50<2n>XX/XXX, +1, +7, +8, add(X)(q26), der(X;7)(q10;p10), add(1)(q21), der(1)(1pter->q43::1q43->q21::9q31->qter), t(2;5)(p23;q35), der(6)t(6;6)(q27;p11), add(9)(q31), add(11)(q24)/der(11)t(4;11)(?q34;q24), del(19)(q13.2); extensive subclonal changes incl. dicentric t(1;1); carries t(2;5) effecting NPM-ALK fusion; resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SUP-T1
Cell Type:	human T cell lymphoma
DSMZ No:	ACC 140
Origin:	established from the pleural effusion of an 8-year-old boy with T-lymphoblastic lymphoma in relapse
References:	Smith et al., Blood 67: 650-656 (1986)
Depositor:	Dr. M. Gramatzki, University Hospital Erlangen, Erlangen, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.3\text{--}1.0 \times 10^6$ cells/ml; split ratio of 1:2 to 1:3 every 2-3 days; after thawing the viability may drop to less than 50%, but the cells recover during the following 4-5 days; seed out at ca. $0.5\text{--}1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30 hours
Harvest:	maximal density of $1.5\text{--}2.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4\text{--}5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3-, CD4+, CD5+, CD6+, CD7+, CD8+, CD13(+), CD19-, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human hypotetraploid karyotype with 1.8% polyploidy; 85<4n>XXX/XXX?Y, -8, -8, -9, -12, inv(2)(p22q11)x2, t(2;?20)(p13;p11), del(4)(q31q35), del(6)(q25)x2, add(7)(q32), add(9)(q34)x2, inv(14)(q11q32)x2; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SW-1710
Cell Type:	human urinary bladder transitional cell carcinoma
DSMZ No:	ACC 426
Origin:	established from the bladder tumor of an 84-year-old Caucasian woman following transurethral tumor resection in 1977
References:	Kyriazis et al., Cancer Res. 44: 3997-4005 (1984)
Depositor:	Dr. W. A. Schulz, Urologische Klinik, Düsseldorf, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like, elongated cells growing adherently as monolayers
Medium:	85% Dulbecco's MEM + 15% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 3-6 days using trypsin/EDTA; seed out at ca. 1.5×10^6 cells/80 cm ² in 10 ml medium
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 25-32 hours
Harvest:	cell harvest of ca. $6-8 \times 10^6$ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-1.5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, NP
Cytogenetics:	human flat-modulated hypotetraploid karyotype with 8% polyploidy; 77-84<4n>X, -X, -X, -X, -4, -4, -5, -5, +6, +6, -8, -9, -10, -11, -13, -14, -14, -16, -19, +20, +20, +4mar, der(X)t(X;6)(q11-13;q25-q27), del(2)(p12)x2, add(3)(p14-21), der(6)t(X;6)(q11-13;q25-27)x2, der(6)del(6)(p11p12.2)add(6)(q25-27)x2, add(13)(q33-34); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SW-403
Cell Type:	human colon adenocarcinoma
DSMZ No:	ACC 294
Origin:	established from the adenocarcinoma (grade III) of a 51-year-old Caucasian woman; cells were described to produce carcinoembryonic antigen (CEA) and to be tumorigenic in nude mice
References:	Leibovitz et al., Cancer Res. 36: 4562-4569 (1976)
Depositor:	Dr. H. Weich, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent round cells, sometimes aggregated in complexes of 20-100 cells
Medium:	90% Dulbecco's MEM + 10% FBS
Subculture:	split confluent culture 1:3 to 1:4 every 3-4 days using trypsin/EDTA; seed out at about 2.5-3.0 x 10 ⁶ cells/25 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of about 40-50 hours
Harvest:	cell harvest of about 40 x 10 ⁶ cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2.5-3.0 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH
Cytogenetics:	human hypotriploid karyotype with 12% polyploidy; 62(58-70)<3n>XX/XXX, +2, +2, -8, -9, -10, -10, -12, -13, -14, -18, -19, -22, +mar, der(1)t(1;1)(p21;q22), del(4)(q21), add(5)(p15), del(6)(q24-q25), i(8q), del(9)(p21), i(9q), add(13)(p13), add(14)(p13); carries very large isochromosome-marker, does not correspond to published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SW-480
Cell Type:	human colon adenocarcinoma
DSMZ No:	ACC 313
Origin:	established from the tumor of a 50-year-old Caucasian man with colon adenocarcinoma (grade 4, Duke class B)
References:	Leibovitz et al., Cancer Res. 36: 4562-4569 (1976)
Depositor:	Dr. H. Kirchner, Medical School Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	endothelial-like adherent cells, mostly bipolar with microvilli growing in monolayers
Medium:	90-95% RPMI 1640 + 5-10% FBS
Subculture:	split confluent culture 1:5 to 1:10 once a week using trypsin/EDTA; seed out at about 1.0×10^6 cells/80 cm ² ; an 80 cm ² culture flask becomes confluent within 1 week
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of 25-30 hours
Harvest:	cell harvest of about $10-15 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human hypotriploid karyotype with 12% polyploidy; 58(51-58)<3n>XX, -Y, -1, -3, -4, -5, -6, +7, -8, -10, -12, -15, -18, -19, +20, -22, der(1)t(1;9)(q22;q12), t(2;12)(q35;q13.1), del(5)(q13.2q32), add(7)(q35), dup(7)(q11.22q21.3), der(8)t(1;8)(p11;p11)t(1;9)(q22;q12), add(10)(p14), inv(14)(q13q32), del(18)(q12.2), der(19)t(6;19)(p11;q13), der(20)t(14;20)(q13;p13)x2; identified by large submetacentric der(2)t(2;12) marker
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	SW-948
Cell Type:	human colon adenocarcinoma
DSMZ No:	ACC 326
Origin:	established from the colon tumor of a 81-year-old Caucasian woman with colon adenocarcinoma (Duke C)
References:	Leibovitz et al., Cancer Res. 36: 4562-4569 (1976)
Depositor:	obtained from ATCC (CCL 237), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	cells grow adherent in monolayers in tight clusters or islands of epithelium-like cells
Medium:	90% Leibovitz's L-15 + 10% FBS
Subculture:	split confluent culture optimally at 1:2 to 1:7 once or twice a week using trypsin/EDTA; seed out at ca. 6×10^4 cells/cm ² or 2×10^6 cells/25 cm ² flask
Incubation:	at 37 °C without CO ₂
Doubling Time:	doubling time of 2-4 days
Harvest:	cell harvest of $2.0\text{--}2.5 \times 10^5$ cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2\text{--}4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, desmin-, endothel-, GFAP-, neurofilament-, vimentin(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH
Cytogenetics:	human hypotriploid karyotype with 12% polyploidy; 68(65-70)X ₃ , +7, -8, -10, +13, +13, -16, +2mar, del(X)(q22.2), del(1)(p12), i(9q), add(17)(p12), add(19)(q13)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	T-24
Cell Type:	human urinary bladder carcinoma
DSMZ No:	ACC 376
Origin:	established from the primary tumor of an 81-year-old Caucasian woman with urinary bladder carcinoma (transitional cell carcinoma, grade III) in 1970; described to produce a variety of cytokines (e.g. G-CSF, IL-6 and SCF) and to carry a p53 mutation
References:	Bubenik et al., Int. J. Cancer 5: 310-319 (1970); Bubenik et al., Int. J. Cancer 11: 765-773 (1973); Grimm et al., Urol. Res. 23: 293-300 (1995)
Depositor:	Dr. W. A. Schulz & Dr. B. J. Schmitz-Dräger, Urologische Klinik, Düsseldorf, Germany
DSMZ Cell Culture Data:	
Morphology:	epitheloid adherent cells growing in monolayers
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS
Subculture:	split confluent culture 1:5 twice a week using trypsin/EDTA; seed out at ca. $1-2 \times 10^5$ cells/80 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of ca. 10×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hypertriploid karyotype with 8% polyploidy; 70-75<3n>XX, -X, +7, -10, -16, -17, -18, +20, +22, +4mar, add(1)(q11), add(1)(p34), del(6)(q21), del(8)(p21), del(9)(p21), del(10)(p11), add(11)(p13), del(13)(q13q14), i(14q), der(15q21q), add(18)(p11); matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	TALL-1
Cell Type:	human T cell leukemia
DSMZ No:	ACC 521
Origin:	established from the bone marrow of a 28-year-old man who developed the terminal leukemic phase of lymphomarcoma in 1976
References:	Miyoshi et al., Nature 267: 843-844 (1977)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	single cells growing in suspension
Medium:	85% RPMI 1640 + 15% FBS
Subculture:	split saturated culture 1:2 to 1:3 every 3 days; seed out at ca. 0.5×10^6 cells/ml; maintain at ca. $0.5\text{--}2.0 \times 10^6$ cells/ml; maximal density of ca. 2.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 50-70 hours
Harvest:	cell harvest of about 2.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5\text{--}6 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in microbiological culture, PCR assays
Immunology:	CD2+, CD3+, CD4+, CD5+, CD6+, CD7+, CD8+, CD13-, CD19-, TCRalpha/beta(+), TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hypertetraploid karyotype; $90\text{--}102\text{--}4n\text{--}XXY/XXYY$, -3, -9, -12, +13, +14, +14, -15, +20, +20, +21, +21, +2-4mar, der(X;1)(p10;p10)/add(1)(q11), add(1)(p11), add(5)(p1?5), del(2)(p22p24), del(12)(q24.1), dup(21)(q11qter)x3-4; resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	TANOUE
Cell Type:	human B cell leukemia
DSMZ No:	ACC 399
Origin:	established from the peripheral blood of an 11-year-old boy with acute lymphoblastic leukemia (ALL FAB L2) in 1990
References:	El-Sonbaty et al., Leukemia Res. 19: 249-256 (1995)
Depositor:	obtained from RIKEN Cell Bank (RCB1180), Tsubuka Science City, Ibaraki, Japan
DSMZ Cell Culture Data:	
Morphology:	single, small, round cells growing in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $1-2 \times 10^6$ cells/ml; split ratio of 1:3 to 1:4 once a week; seed out at ca. 2.0×10^6 cells/ml (it might be preferable to start culture in a 24-well-plate and then to switch to plastic flasks)
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	maximal density at ca. $3-5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD34-, CD37(+), HLA-DR(+), sm/cyIgG-, sm/cyIgM+, sm/cykappa(+), sm/cylambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LHD, NP
Cytogenetics:	human hyperdiploid karyotype with 12% polyploidy; 47/48<2n>X, -Y/XYqh+, +7, +14, dup(1)(q21.1/21.2q23.1/23.2), t(2;4)(q2?2;q2?5), del(6)(q27), t(8;14)(q24;q32), del(13)(q34), add(22)(q13); extensive subclonal variation; carries t(8;14), +7, and dup(1q) typical of B-ALL/Burkitt's lymphoma; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	TC-71
Cell Type:	human Ewing's sarcoma
DSMZ No:	ACC 516
Origin:	established from the tumor of a 22-year-old man with metastatic Ewing's sarcoma that arose in the humerus; cell line was derived in 1981 from a biopsy of recurrent tumor at the primary site
References:	Whang-Peng et al., Cancer Genet. Cytogenet. 21: 185-208 (1986); Scotlandi et al., Cancer Res. 56: 4570-4574 (1996)
Depositor:	Dr. K. Scotlandi, Istituti Ortopedici Rizzoli, Bologna, Italy
DSMZ Cell Culture Data:	
Morphology:	loosely adherent cells growing in monolayers
Medium:	90% Iscove's MDM + 10% FBS
Subculture:	split confluent culture 1:2 to 1:6 every 2-4 days, trypsin/EDTA is not necessary, cells can be detached by tapping the flask; seed out at ca. 2×10^6 cells/80 cm ² flask
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 24 hours
Harvest:	cell harvest of about 15×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, neurofilament+, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-moded hypertriploid karyotype with 4% polyploidy; 70-80<3n>XX, -Y, +1, +2, -3, +6, +8, -9, +11, -17, +21, +2-11 mar, add(3)(q29), add(6)(q25)x2-3, del(8)(p22), add(8)(p22), del(9)(p24), t(11;22)(q24;q12)x2, i(13q)x1-2, add(14)(p11); sideline with der(2)t(2;3)(p23;p11), add(11)(p15)x2; carries two copies of t(11;22) effecting EWS-FLI-1 fusion, yielding type 1 (shorter) product; broadly resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	TCC-SUP
Cell Type:	human urinary bladder transitional cell carcinoma
DSMZ No:	ACC 377
Origin:	established from the tumor specimen resected from the urinary bladder transitional cell carcinoma (undifferentiated, grade IV) of a 67-year-old woman in 1974 (without therapy)
References:	Nayak et al., Brit. J. Cancer 35: 142-151 (1977); Grimm et al., Urol. Res. 23: 293-300 (1995)
Depositor:	Dr. W. A. Schulz & Dr. B. J. Schmitz-Dräger, Urologische Klinik, Düsseldorf, Germany
DSMZ Cell Culture Data:	
Morphology:	epithelial-like adherent cells with smooth borders growing as monolayer
Medium:	80-85% Dulbecco's MEM (4.5 g/L glucose) + 10-15% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 3-4 days using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/80 cm ² in 10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	cell harvest of ca. 6×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1.5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Baytril (enrofloxacin), then negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human hypotriploid karyotype with 8% polyploidy; 67(58-67)<3n>XX, -X, +1, -2, -4, +5, -9, -12, -18, -21, +1-2mar, der(1;21)(q10;q10), del(1)(q13), i(5p)x3, del(6)(q14), add(6)(p23)/der(6)t(3;6)(p22;p22), add(6)(p22/23), del(10)(p11), add(20)(q12), add(21)(p11/12); i(5p) and rearrangements of chs 1 and 3p recurrent in bladder TCC; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	TE-671
Cell Type:	human rhabdomyosarcoma
DSMZ No:	ACC 263
Origin:	in the original publication and in the ATCC catalogue this cell line is described as "medulloblastoma"; recent studies (Stratton et al., Nature 337: 311-312, 1989; Chen et al., Nature 340: 106, 1989) and the characterization carried out at the DSMZ indicated that TE-671 are muscle cells, most likely identical to the rhabdomyosarcoma cell line RD (also known as TE32 or 130T).
References:	McAllister et al., Int. J. Cancer 20: 206-212 (1977)
Depositor:	obtained from DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	round and elongated cells with spindle-shape extensions; the round cells contain almost no cytoplasm; cells grow adherent as monolayer
Medium:	80-85% Dulbecco's MEM + 15-20% FBS
Subculture:	split before cells are entirely confluent, about every 5-10 days (1:3 to 1:5 once or twice a week), using trypsin/EDTA; seed out about 1×10^6 cells/80 cm ²
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of 70-80 hours
Harvest:	cell harvest of about 4×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2.5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, cytokeratin-7-, cytokeratin-8-, cytokeratin-17-, cytokeratin-18-, desmin+, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human bimodal hyperdiploid/tetraploid karyotype with 9.7% polyploidy; 43-50/86-98<4n>XX, -X, -X, +1, -2, -4, +7, +8, -9, -9, -13, -14, -14, -15, -15, -16, -16, -17, -18, -18, -19, -19, -20, -20, -21, -21, +8-11mar, del(1)(p31), add(1)(p31), i(2p), add(3)(p11), add(5)(q35)x2, del(6)(q22)x2, add(9)(q34)x2, add(12)(q24)x2, der(13)t(6;13)(p12.1;p13)x2, add(14)(p11), add(?14)(q24), i(17q), der(19)t(17;19)(q12;p/q13)x2; ploidy resembles RD embryonal rhabdomyosarcoma of which TE-671 is a subline
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	TF-1
Cell Type:	human erythroleukemia
DSMZ No:	ACC 334
Origin:	established from the bone marrow of a 35-year-old man with erythroleukemia (AML M6) in 1987; cell line was first called MFD-1; cells are described to be proliferatively responsive to several hematopoietic growth factors including EPO, GM-CSF, IFN-gamma, IL-3, IL-4, IL-5, IL-6, IL-13, LIF, NGF, OSM, SCF, TNF-alpha and TPO (see Mire-Sluis et al., J. Immunol. Methods 187: 191-199, 1995); we have noted that the cell line can become very quickly cytokine-independent (within 3-4 weeks), presumably due to outgrowth of independent cells
References:	Kitamura et al., Blood 73: 375-380 (1989); Kitamura et al., J. Cell Physiol. 140: 323-334 (1989); Drexler et al., Leukemia 11: 701-708 (1997)
Depositor:	Dr. T. Kitamura, University of Tokyo, Tokyo, Japan
DSMZ Cell Culture Data:	
Morphology:	large, single, round cells in suspension; cells shed large amounts of cytoplasmatic particles that accumulate in the culture (not to be confused with bacteria)
Medium:	70% RPMI 1640 + 20% FBS + 10-20% vol conditioned medium of cell line 5637 (DSM ACC 35) (or 1-5 ng/ml recombinant GM-CSF or IL-3)
Subculture:	cells are difficult to culture; split saturated culture 1:2 every 3 days; seed out at ca. 1×10^6 cells/ml; optimal density at ca. $0.5-1.0 \times 10^6$ cells/ml. CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 70 hours
Harvest:	maximal density at ca. $1.3-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with BM-Cyclin (tiamulin & minocycline), then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD13+, CD14-, CD15-, CD19-, CD33+, CD34(+), CD41+, CD42(+), CD61+, CD68(+), CD71+, GlyA-, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human highly rearranged hyperdiploid karyotype with 12% polyploidy; 52-57<2n>XY/XXY, +3, +5, +6, -8, +12, +15, +19, +19, +20, +20, +3mar, der(1)?dup(1)(p21p31)t(1;8)(p36;q11), t(2;12)(q32;q14), t(3;12)(p13-14;p12-13), add(3)(q21), add(5)(q11-13), der(8)t(1;8)(p36;q11), der(12)t(3;12)(p13-14;p12-13)t(1;12)q31-32;q24, add(14)(p12), iso(17)(q10)add(17)(q21), add(19)(q13), trp(19)(q12;q13.3), der(21)t(19;21)(q13.1;q22)dup(19)(q13.1q13.3)t(11;19)(q13;q13.3), der(22)t(19;22)(q11;p11); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	TFK-1
Cell Type:	human bile duct carcinoma
DSMZ No:	ACC 344
Origin:	established from the surgical specimen extracted from a 63-year-old Japanese man with extrahepatic bile duct carcinoma; described to express c-erbB-2 protein
References:	Saijyo et al., Tohoku J. Exp. Med. 177: 61-71 (1995)
Depositor:	Dr. T. Kudo & Dr. S. Saijyo, Tohoku University, Sendai, Japan
DSMZ Cell Culture Data:	
Morphology:	epithelial cells growing as monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:3 to 1:5 every 2-3 days using trypsin/EDTA (for 10-15 min); seed out at ca. $1-2 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40 hours
Harvest:	cell harvest of ca. $10-30 \times 10^6$ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17+, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human flat-moded hypertriploid karyotype with 20% polyploidy; 72(64-81)<3n>XY/XYY, +1, +3, +5, +5, +6, -9, +11, -13, -16, -18, +20, +2mar, del(1)(q21), del(1)(p11), i(1q), del(3)(p12p21)x2, add(5)(q35), i(5p)x2, inv(7)(q11q35), del(11)(p11)x1-2, i(15p), i(17p); matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	THB-5
Cell Type:	mouse hybridoma (anti-C3d receptor and anti-EBV receptor)
DSMZ No:	ACC 88
Origin:	established by fusion of the P3X63AG8.653 mouse myeloma cell line with spleen cells of a BALB/c mouse immunized with human B cell line SB; cells were reported to produce a monoclonal antibody (IgG ₂ akappa) reacting with the human C3d receptor (CD21) and an EBV receptor of 145 kDa
References:	Weis et al., Proc. Natl. Acad. Sci. USA 81: 881-885 (1984)
Depositor:	obtained from ATCC (HB 135), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	single, round cells in suspension
Medium:	85% RPMI 1640 + 15% FBS
Subculture:	optimal density at 0.5×10^6 cells/ml, minimal density at 0.1×10^6 cells/ml; split 1:3 to 1:5; after thawing, it is advised to use feeder cells for the first 7-10 days; seed out at ca. $0.5-1.0 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 50 hours
Harvest:	maximal density at 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgG ₂ a monoclonal antibody
Fingerprinting:	unique Hinf I-(gtg) ₅ DNA profile
Species:	confirmed as mouse with IEF of AST, MDH, PEP B
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	THP-1
Cell Type:	human acute monocytic leukemia
DSMZ No:	ACC 16
Origin:	established from the peripheral blood of a 1-year-old boy with acute monocytic leukemia (AML) at relapse in 1978; the cells can be used for induction of differentiation studies; the cells were described to produce lysozyme and to be phagocytic
References:	Tsuchiya et al., Int. J. Cancer 26: 171-176 (1980); Tsuchiya et al., Cancer Res. 42: 1530-1536 (1982)
Depositor:	Dr. J. Minowada, Fujisaki Cell Center, Okayama, Japan
DSMZ Cell Culture Data:	
Morphology:	round, single cells in suspension, partly in clusters
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.1-1.0 \times 10^6$ cells/ml; split 1:2 to 1:3 every 3-4 days; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 35-50 hours
Harvest:	saturation density at about 1.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ml
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, PCR assays
Immunology:	CD3-, CD13+, CD14(+), CD15+, CD19-, CD33+, CD34-, CD68(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human near-tetraploid karyotype; 94(88-96)<4n>XY/XXY, -Y, +1, +3, +6, +6, -8, -13, -19, -22, -22, +2mar, add(1)(p11), del(1)(q42.2), i(2q), del(6)(p21)x2-4, i(7p), der(9)t(9;11)(p22;q23)i(9)(p10)x2, der(11)t(9;11)(p22;q23)x2, add(12)(q24)x1-2, der(13)t(8;13)(p11;p12), add(?18)(q21); carries t(9;11) associated with AML M5
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	TI-1
Cell Type:	mouse hybridoma (anti-rat fiber/troponin I)
DSMZ No:	ACC 285
Origin:	described as secreting a mouse monoclonal antibody (IgG ₂ b) reacting with rat adult cardiac fibers in immunofluorescence and with rat adult cardiac troponin I in Western blotting; this cell line was produced by fusing NSO myeloma cells with spleen cells from a BALB/c mouse immunized with myofibrillar preparation from hypothyroid rabbit ventricle
References:	Saggin et al., J. Biol. Chem. 264: 16299-16303 (1989)
Depositor:	Prof. S. Schiaffino, University of Padova, Padova, Italy
DSMZ Cell Culture Data:	
Morphology:	single round cells in suspension; culture contains almost always some cell debris and dead cells
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% horse serum + 2 mM L-glutamine
Subculture:	split saturated culture at 1:2 to 1:3 every 1-2 days; optimal density at about 0.5-0.8 x 10 ⁶ cells/ml (not to be cultured below 0.5 x 10 ⁶ cells/ml)
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 50 hours
Harvest:	maximal density at about 1.5 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% horse serum, 10% DMSO at about 3-4 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgG ₂ b monoclonal antibody
Species:	confirmed as mouse with IEF of AST, LDH, MDH, NP
Cytogenetics:	murine hypertetraploid/hypopentaploid karyotype; 92(87-92)<4n>, 1-3 dm present in most cells
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	TI-4
Cell Type:	mouse hybridoma (anti-rat fiber/troponin I)
DSMZ No:	ACC 286
Origin:	described as secreting a mouse monoclonal antibody (IgG ₁) reacting with rat cardiac fibers in immunofluorescence and with rat adult and fetal cardiac troponin I in Western blotting; this cell line was produced by fusing NSO myeloma cells with spleen cells from a BALB/c mouse immunized with purified troponin complex from bovine adult heart
References:	Saggin et al., J. Biol. Chem. 264: 16299-16303 (1989)
Depositor:	Prof. S. Schiaffino, University of Padova, Padova, Italy
DSMZ Cell Culture Data:	
Morphology:	cells grow in suspension as single cells or in small clumps
Medium:	80% Dulbecco's MEM (4.5 g/L glucose) + 10-20% horse serum + 2 mM L-glutamine
Subculture:	split saturated culture at 1:2 to 1:8 every 3-6 days; after thawing seed out at about 0.8×10^6 cells/ml and culture initially at 20% horse serum in a 24-well-plate
Incubation:	at 37 °C with 5% CO ₂
Harvest:	maximal density at about 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% horse serum, 10% DMSO at about $4-6 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	secretion of a IgG ₁ monoclonal antibody
Species:	confirmed as mouse with IEF of MDH, NP
Cytogenetics:	murine karyotype
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	TMM
Cell Type:	human B lymphoblastoid cells (from patient with chronic myeloid leukemia in blast crisis)
DSMZ No:	ACC 95
Origin:	established from the peripheral blood of a 62-year-old man with Philadelphia-chromosome negative chronic myeloid leukemia (CML) in blast crisis in 1985; cells were described to be Ph-negative, bcr-abl-negative; cells appear to be rather a "normal" B-lymphoblastoid cell line (B-LCL)
References:	McCarty et al., Blood 70: 1665-1672 (1987)
Depositor:	Dr. E. B. Thompson, University of Texas, Galveston, Texas, USA
DSMZ Cell Culture Data:	
Morphology:	polygonal cells, single or in clusters in suspension
Medium:	90% Iscove's MDM + 10% FBS
Subculture:	cells are difficult to culture! maintain at $0.2-0.8 \times 10^6$ cells/ml; split 1:1.5 to 1:2 every 2 to 3 days
Incubation:	at 37 °C with 5% CO ₂
Harvest:	saturation density at about $2.5-4.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture assays
Immunology:	CD3-, CD10-, CD13-, CD14-, CD15-, CD19+, CD20+, CD33-, CD37+, HLA-DR+, anti-EBNA2+, sm/cyIgG-, sm/cyIgM-, sm/cykappa+, sm/cylambda-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human near-diploid karyotype with 8% polyploidy; 46(42-46)<2n>XY; small ring present in 6% cells; no consistent numerical or structural rearrangement; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	TN-368
Cell Type:	insect - Tiger moth (<i>Trichoplusia ni</i>)
DSMZ No:	ACC 177
Origin:	established from the larvae of a tiger moth (species <i>Trichoplusia ni</i> = <i>Autographa brassicae</i> = <i>Plusia ni</i> , family Noctuidae, subfamily Plusiinae, order Lepidoptera)
References:	no reference available
Depositor:	Dr. H. S. Conradt, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	spindle-shaped cells growing in suspension (90%); cells tend to cluster in aggregates
Medium:	80% TC 100 medium + 20% FBS
Subculture:	split saturated culture 1:10 every 4-5 days
Incubation:	without exposure to CO ₂ (keep lid of flask closed) at room temperature (20-25 °C)
Harvest:	cell harvest of about 0.6 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as not murine and not primate with IEF of AST, PEP B

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INFORMATION SHEET FOR CELL LINES

Cell Line:	TT2609-C02
Cell Type:	human follicular thyroid carcinoma
DSMZ No:	ACC 510
Origin:	established in 1996 from the tumor tissue of a 57-year-old Caucasian man with local recurrence of a previously iodine-irradiated follicular thyroid cancer (well-differentiated with dispersed poorly differentiated areas)
References:	Geldof et al., Thyroid 11: 909-917 (2001)
Depositor:	Dr. A. A. Geldof, University Hospital, Amsterdam, The Netherlands
DSMZ Cell Culture Data:	
Morphology:	small cells growing adherent but irregularly in monolayers
Medium:	80-85% RPMI 1640 + 15-20% FBS + 1 mM sodium pyruvate + insulin-transferrin-selenium medium supplement
Subculture:	split confluent culture 1:3 to 1:4 every week using trypsin/EDTA (for 10 min); seed out at ca. 1×10^6 cells/25 cm ² ; maintain at ca. 8×10^6 cells/175 cm ² ; maximal density at about 14×10^6 cells/175 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60-80 hours
Harvest:	cell harvest of about 10×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, PCR assays
Immunology:	cytokeratin+, cytokeratin-7-, cytokeratin-8+, cytokeratin-17-, cytokeratin-18+, desmin-, endothel-, GFAP-, neurofilament-, thyroglobulin-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human by cytogenetics
Cytogenetics:	human flat-modulated hypertriploid karyotype with 4% polyploidy; 69-75<3n>XXYY, -2, +8, +8, +9, +10, +14, -15, -16, +17, -22, +1-2mar, der(1)t(1;2)(p32;q11-13), add(4)(p15.2), ?inv(8)(q12q21.2)x1-2, add(9)(p24)x2, i(15q), del(17)(p11)x1-2, add(22)(p11)x2; resembles published karyotype
Viruses:	PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	U-138-MG
Cell Type:	human glioblastoma
DSMZ No:	ACC 291
Origin:	established from the glioblastoma of a 47-year-old Caucasian man
References:	Ponten et al., Acta Pathol. Microbiol. Scand. 14: 465-486 (1968); Hum. Hered. 21, 238 (1971)
Depositor:	obtained from DKFZ Tumorbank, Heidelberg, Germany
DSMZ Cell Culture Data:	
Morphology:	fibroblastoid fusiform cells growing as mono- or multilayers
Medium:	90% Ham's F10 + 10% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 4-8 days using trypsin/EDTA; seed out at about 0.1-0.2 x 10 ⁵ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 70 hours
Harvest:	cell harvest of ca. 0.5-0.7 x 10 ⁵ /cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1-2 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, MDH, NP
Cytogenetics:	human flat-moded hypotriploid karyotype with 20% polyploidy; 60(57-63)<3n>XYY, -3, -5, -6, -8, -9, -10, -11, -13, -14, -17, -19, +2mar, add(4)(p15), del(6)(q15), dup(7)(q35q21), add(11)(p15), add(12)(p11), add(14)(q32)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	U-2197
Cell Type:	human malignant fibrous histiocytoma
DSMZ No:	ACC 406
Origin:	established from the subcutaneous malignant fibrous histiocytoma of a 79-year-old Caucasian man in 1984 (third recurrence after repeated surgery, radially invading the thumb extensor); described to express high levels of PDGF-A which is not secreted
References:	Genberg et al., Am. J. Pathol. 135: 1185-1196 (1989); Afrakhte et al., Int. J. Cancer 68: 802-809 (1996)
Depositor:	Dr. M. Nister, University of Uppsala, Uppsala, Sweden
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblastoid large polygonal cells with extensive lamellae
Medium:	80% MEM (with Earle's salts) + 20% FBS + 0.135% Na-bicarbonate
Subculture:	split confluent culture 1:2 to 1:3 once a week using trypsin/EDTA; maintain at 6×10^6 cells/175 cm ² ; seed out at ca. 1×10^6 cells/80 cm ² in 10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 100-120 hours
Harvest:	cell harvest of ca. 8×10^6 cells/175 cm ² ; maximal density of 10×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 1×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP(+), neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human hypertetraploid karyotype with hypotriploid sideline and 20% polyploidy; 100(92-101)<4n>XXXX, -2, -4, +6, +6, +7, -8, -9, -9, +14, +14, -15, -15, -16, -16, +19, -20, -20, -21, -21, +13mar, i(Xq)x2, add(X)(p11)x2, del(1)(q11)x2, del(4)(p11), add(6)(q15)x2, i(6p)x2-4, add(7)(q35)x2, add(7)(p14)x2, del(7)(p11), add(12)(q24)x2, add(13)(p11)x2, add(13)(p11)x2, add(19)(q13), del(19)(q13)x2; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	U-266
Cell Type:	human multiple myeloma
DSMZ No:	ACC 9
Origin:	established from the peripheral blood of a 53-year-old man with IgE-secreting myeloma (refractory, terminal) in 1968; cells were described to produce IgE lambda; possible fusion partner for hybridoma production; cells express mRNA for bcl-2 gene
References:	Nilsson et al., Clin. Exp. Immunol. 7: 477-489 (1970); review article: Drexler & Matsuo, Leukemia Res 24: 681-703 (2000)
Depositor:	Dr. K. Nilsson, University Hospital, Uppsala, Sweden
DSMZ Cell Culture Data:	
Morphology:	round to polygonal, single or clustered cells in suspension, some loosely adherent
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split saturated culture 1:2 to 1:4 every 2-3 days; maintain at $0.2-1.0 \times 10^6$ cells/ml; cells grow partially (loosely) adherent and can be removed with a bent Pasteur pipette (trypsin is not necessary); seed out at ca. $0.5-0.7 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 55 hours
Harvest:	saturation density at $1.0-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Ciprobay (ciprofloxacin), then negative in DAPI, microbiological culture, PCR assays
Immunology:	CD3-, CD4-, CD5-, CD10-, CD13-, CD14-, CD15-, CD19-, CD21-, CD22-, CD33+, CD34-, CD37-, CD38(+), CD138+, HLA-DR+, sm/cyIgG-, sm/cyIgM-, sm/cykappa-, smlambda-, cylambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human complex hypodiploid karyotype with 6.5% polyploidy; 44(40-46)<2n>XY, -8, -10, -13, -15, +2mar, t(1;11)(p33;q13), add(3)(q27), t(4;11)(q?21;q23), add(7)(q32), add(8)(q24), add(9)(q34), add(10)(p14), add(14)(p11), add(17)(p11), add(18)(p12), der(22)t(15;22)(q21;q13); 11q13 breakpoint recurrent in multiple myeloma
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	U-698-M
Cell Type:	human B cell lymphoma
DSMZ No:	ACC 4
Origin:	established from the involved tonsil of a 7-year-old boy with lymphoblastic lymphoma at diagnosis prior to therapy in 1972; cells were described to be EBV-negative
References:	Nilsson et al., Int. J. Cancer 13: 808-823 (1974); Nilsson et al., Int. J. Cancer 19: 337-344 (1977)
Depositor:	Dr. K. Nilsson, University Hospital, Uppsala, Sweden
DSMZ Cell Culture Data:	
Morphology:	single or clumped cells in suspension; some cells adhere to plastic
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.2-0.7 \times 10^6$ cells/ml; split 1:2 to 1:5 every 2-3 days; some cells adhere to plastic, but can be detached mechanically (trypsin not necessary); seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	maximal density at 1.5×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture assays
Immunology:	CD3-, CD13-, CD19+, CD37+, HLA-DR+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human hyperdiploid karyotype with 5.5% polyploidy; 49(44-50)<2n>XY, +3, +7, -14, +mar, dup(1)(q43q21.2), der(2)t(2;3)(p16;p11), add(3)(p11), del(6)(q15q22), del(9)(p22), dup(11)(q23q13), add(13)(p12), add(16)(q24); carries large submetacentric dup(1) marker
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	U-937
Cell Type:	human histiocytic lymphoma
DSMZ No:	ACC 5
Origin:	established from the pleural effusion of a 37-year-old man with generalized diffuse histiocytic lymphoma in 1974; cells were described to express markers and properties of monocytes; patented cell line
References:	Sundström et al., Int. J. Cancer 17: 565-577 (1976)
Depositor:	Dr. K. Nilsson, University Hospital, Uppsala, Sweden
DSMZ Cell Culture Data:	
Morphology:	round to polygonal, single cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.1-1.0 \times 10^6$ cells/ml; split 1:2 to 1:5 every 2-3 days; seed out at ca. 1×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30-40 hours
Harvest:	saturation density at about 2.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, PCR assays
Immunology:	CD3-, CD4+, CD13(+), CD14-, CD15+, CD19-, CD33+, CD34+, CD54+, CD68+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human flat-moded hypotriploid karyotype; 63(58-69)XYY, -2, -4, -6, +7, -9, -20, -21, +3mar, t(1;12)(q21;p13), der(5)t(1;5)(p22;q35), add(9)(p22), t(10;11)(p14;q23), i(11q), i(12p), add(16)(q22), add(19)(q13); carries t(10;11) (seen in AML M5); t(1;5) resembles variant of t(2;5) (histiocytic lymphoma)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	UT-7
Cell Type:	human acute myeloid leukemia
DSMZ No:	ACC 137
Origin:	established from the bone marrow of a 64-year-old man with acute myeloid leukemia (AML M7) at diagnosis in 1988; cells are constitutively cytokine-dependent and responsive to various cytokines
References:	Komatsu et al., Cancer Res. 51: 341-348 (1991); Drexler et al., Leukemia 11: 701-709 (1997)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round (relatively large) cells growing singly in suspension (ca. 1-2% cells are slightly adherent)
Medium:	80% alpha-MEM + 20% FBS + 5 ng/ml GM-CSF
Subculture:	maintain at $0.5-1.0 \times 10^6$ cells/ml; cultivate in 12- or 24-well-plate for optimal growth; split saturated culture 1:2 every 2-3 days; seed out at ca. 0.5×10^6 cells/ml. CAVEAT: by favoring selective growth, suboptimal culture of cytokine-dependent cell lines may promote outgrowth of factor-independent subclones. Selective conditions include cytokine insufficiency and cell density. Thus the DSMZ cannot guarantee indefinite stability of factor-dependence of cell lines.
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60-70 hours
Harvest:	saturation density at ca. $1.0-1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $4-5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD3-, CD4-, CD13+, CD14-, CD15-, CD19-, CD33+, CD34-, CD41(+), CD42-, CD68+, GlyA(+), HLA-DR(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of LDH, NP
Cytogenetics:	human flat-modulated hypotetraploid karyotype; 72-88<4n>XXYY, +1, -8, -9, -9, -9, -11, -11, -12, -13, -13, -14, -18, -21, -22, +6mar, del(1)(q42)x2, t(2;4)(p16-23;q27-31)x1-2, der(2)t(2;5)(p16-23;q12-14)x1-2, add(5)(q11), del(5)(q13), der(6;13)(p10;q10), i(7q)x1-2, add(8)(q24)x1-3, del(9)(q22), add(12)(p12-13), der(13)t(13;13)(p11;q14), add(14)(p11), add(15)(p11), add(16)(p1?), add(17)(p11), add(19)(p13)x2, add(19)(q13), i(21q)x1-2; extensive subclonal variation present; resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	V-79
Cell Type:	Chinese hamster lung fibroblasts
DSMZ No:	ACC 335
Origin:	established from the lung tissue of a Chinese hamster in 1958
References:	Ford & Yerganian, J. Natl. Cancer Inst. 21: 393-425 (1958); Elkind, Nature 184: 1293-1295 (1959)
Depositor:	obtained from RIKEN Cell Bank (RCB0008), Tsubuka Science City, Ibaraki, Japan
DSMZ Cell Culture Data:	
Morphology:	fibroblast-like adherent cells growing in clumps
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 2-3 days; cells can be detached by pipetting with PBS or by mild treatment with trypsin/EDTA; seed out at ca. 3×10^6 cells/80 cm ² ; after splitting, it is recommend to seed out the cells in fresh culture flasks as possibly the cells may not attach any longer in the old vessels
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 16 hours
Harvest:	cell harvest of ca. 25×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-3 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Fingerprinting:	unique DNA profile with (gtg) ₅ multilocus probe
Species:	confirmed as Chinese hamster with IEF of MDH, NP
Cytogenetics:	hamster highly rearranged near-diploid karyotype with 20% polyploidy; 22(20-24); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	VA-ES-BJ
Cell Type:	human epithelioid sarcoma
DSMZ No:	ACC 328
Origin:	established from the bone marrow metastasis of an epithelioid sarcoma (epidural tumor) of a 41-year-old man following unsuccessful chemotherapy in 1991; cells were described to grow rapidly after heterotransplantation into nude mice, to secrete functional GM-CSF and to be positive for AE-1, AE-3, vimentin and epithelial membrane antigen
References:	Helson et al., Int. J. Oncol. 7: 51-56 (1995)
Depositor:	Dr. C. Helson, New York Medical College, Valhalla, New York, USA
DSMZ Cell Culture Data:	
Morphology:	adherent cells growing in monolayers - epithelioid, partly fibroblastoid large cells
Medium:	80% Dulbecco's MEM + 20% FBS + 1% non-essential amino acids + 4 mM L-glutamine
Subculture:	split confluent culture 1:2 to 1:5 once or twice a week; seed out at ca. $3-5 \times 10^4$ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 60-90 hours
Harvest:	cell harvest of max. 1.5×10^5 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $3-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH
Cytogenetics:	human hypotriploid karyotype with 13% polyploidy; 68(65-70)X _n XX, -Y, -3, -4, +6, +7, +7, -11, -12, +13, -14, -15, +17, +20, -22, add(1)(p32-34), add(1)(q43)x2, add(9)(q34)x3, del(22)(q13.2); matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	VERO-B4
Cell Type:	African green monkey kidney
DSMZ No:	ACC 33
Origin:	established from the kidney of a normal adult African green monkey in Japan in 1962; cells are susceptible to a number of viruses; used in virus replication studies and plaque assays, and as indicator cell line for mycoplasma testing
References:	Yasumura et al., Nippon Rinsho 21: 1209 (1963)
Depositor:	Dr. H. W. L. Ziegler-Heitbrock, Institute of Immunology, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent-elongated, fibroblast-like cells in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent cultures 1:3 to 1:10 using trypsin/EDTA; seed out at ca. $0.5-1.0 \times 10^6$ cells/80 cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 25 hours
Harvest:	cell harvest of ca. 10×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Fingerprinting:	unique Hinf I-(gtg) ₅ DNA profile
Species:	confirmed as monkey with IEF of AST, LDH, NP
Cytogenetics:	simian rearranged flat-moded hypodiploid karyotype with 12% polypoidy; 54(48-61)<2n>X; VERO-B4 markers present
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	VIP-VIIIC8
Cell Type:	mouse hybridoma (CD2)
DSMZ No:	ACC 479
Origin:	described as secreting a mouse monoclonal antibody (IgG ₂ a) CD2; the cell line was established in 1987 by fusion of PX63AG8.653 cells with spleen cells of a mouse immunized with fresh peripheral blood lymphocytes from patients with expansions of gamma-delta Tcells
References:	not published
Depositor:	Dr. M. R. Hadam, Medical School of Hannover, Hannover, Germany
DSMZ Cell Culture Data:	
Morphology:	single round cells growing in suspension
Medium:	80% Dulbecco's MEM (4.5 g/L glucose) + 20% FBS + HFCS ("Hybridoma Fusion and Cloning Supplement")
Subculture:	split saturated cultures 1:2 to 1:3 every 3 days; seed out at 0.8×10^6 cell/ml, best in 12-well-plate with HFCS; maintain at $0.3\text{--}0.5 \times 10^6$ cells/ml; some cells attach to the surface of the culture flask, but can be detached by tapping the flask; for the first 2 weeks, HFCS should be used which thereafter can be reduced to 50% or 25% of the recommended concentration; cells grow very slowly in the complete absence of HFCS
Incubation:	at 37°C with 5-10% CO ₂
Doubling Time:	doubling time of about 40-50 hours
Harvest:	maximal density of about 1×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	secretion of a IgG ₂ a monoclonal antibody
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	VLM
Cell Type:	mouse SV40-transformed embryo cells
DSMZ No:	ACC 429
Origin:	established from SV40-transformed BALB/c mouse embryo cells; cells are described as possessing SV40 T-antigen and tumor-specific transplantation antigens and as not releasing spontaneously SV40; tumorigenic in mice
References:	Zarling & Tevethia, J. Natl. Cancer Inst. 50: 137-147 (1973)
Depositor:	Prof. G. Brandner, Institute for Med. Microbiology and Hygiene, Freiburg, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent fibroblastoid cells growing in monolayers
Medium:	80-90% Dulbecco's MEM + 10-20% FBS
Subculture:	split confluent culture 1:3 to 1:6 every 3-6 days using trypsin/EDTA; seed out at ca. $2-3 \times 10^5$ cells/ml in 10 ml medium;
Incubation:	at 37 °C with 5-10% CO ₂
Doubling Time:	doubling time of ca. 25 hours
Harvest:	cell harvest of maximal 13×10^6 cells/25 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, NP
Cytogenetics:	murine hypertriploid karyotype
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	VM-CUB1
Cell Type:	human urinary bladder transitional cell carcinoma
DSMZ No:	ACC 400
Origin:	established from the primary transitional cell carcinoma of the urinary bladder of a male patient; described to carry a p53 point mutation
References:	Williams, Invest. Urol. 17: 359-363 (1980); Grimm et al., Urol. Res. 23: 293-300 (1995)
Depositor:	Dr. W. A. Schulz & Dr. B. J. Schmitz-Dräger, Urologische Klinik, Düsseldorf, Germany
DSMZ Cell Culture Data:	
Morphology:	adherent epithelial-like cells growing as monolayers
Medium:	90% Dulbecco's MEM (4.5 g/L glucose) + 10% FBS
Subculture:	split confluent culture 1:5 to 1:10 every 3-5 days using trypsin/EDTA; seed out at ca. $1-2 \times 10^6$ cells/80 cm ² in 10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-45 hours
Harvest:	cell harvest of ca. 40×10^6 cells/175 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-2 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human hypertriploid karyotype with 4% polyploidy; 80<3n>XX/XX?Y, +1, +1, +5, +6, +7, +9, -11, +15, -18, +19, +20, +22, +22, +7mar, del(1)(q21), del(1)(q11), der(1;6)(p10;q10)x2, add(3)(p21.3)x2, i(5p), der(6)t(1;6)(q11;q22), ?inv(7)(p15q35), i(9q), del(10)(p11), del(10)(q?22?24), add(12)(p11)x2, add(15)(p11), add(21)(p11); sideline with i(5q)
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

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INFORMATION SHEET FOR CELL LINES

Cell Line:	WEHI-164S
Cell Type:	mouse fibrosarcoma
DSMZ No:	ACC 25
Origin:	established from a fibrosarcoma chemically induced in a BALB/c mouse; cells were described to be sensitive to human cytotoxic monocytes, to tumor necrosis factor (TNF-alpha) and lymphotoxin (TNF-beta)
References:	Rollinghoff et al., Proc. Soc. Exp. Biol. Med. 144: 813-818 (1973); Ziegler-Heitbrock et al., Blood 65: 8-14 (1985)
Depositor:	Dr. H. W. L. Ziegler-Heitbrock, Institute of Immunology, University of Munich, Munich, Germany
DSMZ Cell Culture Data:	
Morphology:	fibroblast-like cells growing as monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split confluent cultures at 1:3 to 1:6; cells are adherent, slapping of the back of the flask by hand will detach the cells; seed out at ca. 4×10^6 cells/80 cm ² in 10 ml medium
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-50 hours
Harvest:	ca. 15×10^6 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture assays
Immunology:	see available graph: lysis of MTT-labelled cells exposed to TNF
Species:	confirmed as mouse with IEF of AST, MDH, NP, PEP B
Cytogenetics:	murine hypertriploid karyotype with 20% polyploidy; 60-62<3n>; supernumerary ring/acf but no centric fusion marker present in most cells
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	WEHI-3B
Cell Type:	mouse myelomonocytic leukemia
DSMZ No:	ACC 26
Origin:	macrophage-like cells derived from a BALB/c mouse treated for tumor induction; cells produce IL-3
References:	Warner et al., J. Natl. Cancer Inst. 43: 963 (1969); J. Immunol. 119: 950-954 (1977); Cancer Res. 37: 546-550 (1977)
Depositor:	Dr. E. Schmitt, Institute of Immunology, University of Mainz, Mainz, Germany
DSMZ Cell Culture Data:	
Morphology:	small, round, single cells growing adherently and partly in suspension
Medium:	90-95% Iscove's MDM + 5-10% FBS + 2 mM L-glutamine + 2.5×10^{-5} M mercaptoethanol
Subculture:	split confluent culture 1:2 to 1:5 twice to three times weekly; cells are trypsin-sensitive, detach mechanically after loosening by incubating in PBS (without Ca^{2+} or Mg^{2+}) for 10-15 min; seed out at ca. $0.2-0.5 \times 10^5$ cells/cm ²
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 20-30 hours
Harvest:	maximum cell harvest of about 1×10^6 cells/cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $1-1.5 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, PCR assays
Species:	confirmed as mouse with IEF of LDH, NP
Cytogenetics:	murine hypotetraploid karyotype with 19% polyploidy; 73/74(70-79)<4n>; 2-3 centric fusion markers present
Viruses:	ELISA: reverse transcriptase negative

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INFORMATION SHEET FOR CELL LINES

Cell Line:	WERI-RB-1
Cell Type:	human retinoblastoma
DSMZ No:	ACC 90
Origin:	established from retinoblastoma tissue of a 1-year-old Caucasian girl in 1974; corresponds to ATCC HTB 169
References:	McFall et al., Cancer Res. 37: 1003-1010 (1977); McFall et al., Cancer Res. 38: 2827-2835 (1978)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	growing as grape-like clusters of round cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	optimal split ratio of 1:2 to 1:3 once a week; maximal cell density of about 1.5×10^6 cells/ml; seed out at ca. $0.3\text{--}0.5 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 49 hours
Harvest:	cell harvest of about $0.5\text{--}1.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS and 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, neurofilament-, vimentin-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, MDH, NP
Cytogenetics:	human pseudodiploid karyotype with 3.9% polyploidy; 46(41-48)<2n>XX, +6, -10, -10, -14, -22, +3mar, add(3)(q25), add(3)(q25), add(4)(p15), add(5)(q35), i(6q), del(7)(p21), add(9)(q33), der(13)x2, add(16)(q23), add(16)(q23), i(17q), add(19)(q13); apparently (uniparental?) disomic rearrangement of ch 13; corresponds to reported karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	WMP-2
Cell Type:	mouse lymphoblasts
DSMZ No:	ACC 453
Origin:	established from spleen and lymph nodes of an adult Maloney murine leukemia virus (MoMuLV)-infected female WMP/WMP wild mouse; cells are described to carry nine pairs of metacentric Robertsonian translocation chromosomes containing 1 to 18; cells can be used for physical gene mapping by in situ hybridization in mice
References:	Zörnig et al., Cytogenet. Cell Genet. 71: 37-40 (1995)
Depositor:	Dr. H. Hameister, University of Ulm, Ulm, Germany
DSMZ Cell Culture Data:	
Morphology:	single round cells growing in suspension
Medium:	80% RPMI 1640 + 20% FBS + 50 : M mercaptoethanol
Subculture:	split saturated culture at 1:3 to 1:5 every 2-4 days; seed out at ca. 2.5 x 10 ⁵ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 30-40 hours
Harvest:	cell harvest of ca. 2.0-2.5 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in microbiological culture, RNA hybridization, PCR assays
Cytogenetics:	murine karyotype with a preponderance of centric fusion markers; resembles published karyotype

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INFORMATION SHEET FOR CELL LINES

Cell Line:	WSU-NHL
Cell Type:	human B cell lymphoma
DSMZ No:	ACC 58
Origin:	established from the pleural effusion of a 46-year-old woman with nodular histiocytic lymphoma (refractory, progressive) in 1986; reported to carry the 14;18 translocation and to express mRNA for the bcl-2 oncogene
References:	Mohamed et al., Leukemia Res. 12: 833-843 (1988)
Depositor:	Dr. A. Al-Katib, Wayne State University, Detroit, Michigan, USA
DSMZ Cell Culture Data:	
Morphology:	round to elongated cells in suspension, single or in small clusters
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at 0.5-2.0 x 10 ⁶ cells; split 1:2 to 1:4 every 2-4 days; seed out at ca. 0.5-0.8 x 10 ⁶ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 45 hours
Harvest:	maximum density at about 2.5 x 10 ⁶ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, PCR assays
Immunology:	CD3-, CD10+, CD13-, CD19+, CD20+, CD34-, CD37+, CD79a+, CD80+, CD138+, HLA-DR+, sm/cyIgG+, sm/cyIgM- (it appears that about 3-5% are positive), sm/cykappa-, sm/cylambda+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP, PEP B
Cytogenetics:	human hypodiploid karyotype with 6% polyploidy; 45(42-45)<2n>XX, -4, -15, +mar, der(1)t(1;4)(p36;q11), der(1)t(1;7)(p34;q31.2)t(1;16)(q22;q22), del(5)(q32q35), der(6)t(1;6)(q22;p21), t(9;13)(p22;q13-14), der(13)t(13;?16)(p11;p11), t(14;18)(q32;q21), der(16)t(16;?)(q22;?)(?;5)(?;q31), add(17)(p1?), add(19)(q13.2); matches published karyotype; carries t(14;18) associated with NHL
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	X63AG8.653
Cell Type:	mouse myeloma
DSMZ No:	ACC 43
Origin:	X63AG8.653 is a non-secreting clone derived from P3X63AG8 (a Balb/c myeloma) and is recommended as fusion parent for production of hybridomas
References:	Kearny et al., J. Immunol. 123: 1548-1550 (1979)
Depositor:	Dr. J. Wehland, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	single, round, large cells in suspension
Medium:	90% Dulbecco's MEM or RPMI 1640 + 10% FBS
Subculture:	maintain at $0.1-1.0 \times 10^6$ cells/ml; optimal split ratio of 1:10 to 1:20 every 48 hours; seed out at about 1×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 30-50 hours
Harvest:	maximal density of ca. $1.5-2.0 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Fingerprinting:	unique Hinf I-(gtg) ₅ DNA profile
Species:	confirmed as mouse with IEF of LDH, MDH, NP
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	XC
Cell Type:	rat sarcoma
DSMZ No:	ACC 118
Origin:	the original tumor was induced in a Wistar rat by intramuscular injection of Rous sarcoma virus; cells have been used as indicator cells for detection of murine leukemia viruses
References:	Svoboda, Nature 186: 980-981 (1960); Klement, Proc. Natl. Acad. Sci. USA 63: 753-758 (1969); Roew et al., Virology 42: 1136-1139 (1970)
Depositor:	obtained from ATCC (CCL 165), Rockville, Maryland, USA
DSMZ Cell Culture Data:	
Morphology:	epithelial-like, small cells growing as monolayers
Medium:	90% MEM (with Earle's salts) + 10% FBS + non-essential amino acids
Subculture:	split confluent culture 1:4 every 3 days using trypsin/EDTA
Incubation:	at 37 °C with 5% CO ₂
Harvest:	about 1-2 x 10 ⁷ cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 2-3 x 10 ⁶ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Fingerprinting:	unique Hinf I-(gtg) ₅ DNA profile
Species:	confirmed as rat with IEF of AST, NP
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	XTH-2
Cell Type:	South African clawed toad (<i>Xenopus laevis</i>)
DSMZ No:	ACC 190
Origin:	established from the embryonal (stage 52/53) normal heart endothelium (tadpole heart) of a South African clawed toad (<i>Xenopus laevis</i> Daudin) in 1979; cells were described to synthesize factor-VIII-related antigen (AHG-a.p.), fibronectin and laminin and to be tetraploid (72 ± 5 , $2N = 36$)
References:	Schlage et al., Eur. J. Cell Biol. 24: 21A (1981); Schlage & Bereiter-Hahn, Micr. Acta 87: 19-34 (1983); Schlage, Cell Tissue Res. 245: 445-451 (1986)
Depositor:	Dr. W. K. Schlage, Institute for Biological Research, Köln, Germany
DSMZ Cell Culture Data:	
Morphology:	large epitheloid (low density monolayer), cobblestone with occasional piling-up (high density monolayer)
Medium:	90% Dulbecco's MEM (diluted with 20% distilled water) + 10% FBS + 20-30 mM HEPES
Subculture:	split confluent culture 1:2 every 10-12 days using trypsin/EDTA; avoid centrifugation, just dilute the harvested cells with medium and seed into a new flask; cells grow extremely slowly with a doubling time of about 10-12 days; thaw in cold water (10°C), dilute with 10-fold volume of fresh culture medium, centrifuge gently (about $80 \times g$ for 5 min)
Incubation:	keep at room temperature in closed flasks (without exposure to CO_2)
Doubling Time:	doubling time of ca. 10-12 days
Harvest:	cell harvest of about 2.5×10^6 cells/80 cm ²
Storage:	frozen with 90% complete medium + 10% glycerol (not DMSO) at about 0.5×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization assays
Species:	confirmed as <i>Xenopus</i> by a species-specific PCR

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INFORMATION SHEET FOR CELL LINES

Cell Line:	Y-79
Cell Type:	human retinoblastoma
DSMZ No:	ACC 246
Origin:	established from the primary tumor (right eye) of a 2-year-old Caucasian girl in 1971
References:	Reid et al., J. Natl. Cancer Inst. 53: 347-360 (1974)
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	cells grow in clusters in suspension
Medium:	80% RPMI 1640 + 20% FBS
Subculture:	optimal split ratio at 1:2 two to three times per week; seed out at ca. 0.5×10^6 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of about 40 hours
Harvest:	cell harvest of about $1.0\text{--}1.5 \times 10^6$ cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 8×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin-, desmin-, endothel-, GFAP-, neurofilament-, vimentin+
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of MDH, NP
Cytogenetics:	human hyperdiploid karyotype with 10% polyploidy; 47(45-48)<2n>XX, +mar, der(4)t(4;?6)(p15;p21.2), add(12)(q24.32), add(15)(q23), add(20)(q13); large submetacentric marker with q-arm comprising HSR
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

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INFORMATION SHEET FOR CELL LINES

Cell Line:	YAC-1
Cell Type:	mouse lymphoma
DSMZ No:	ACC 96
Origin:	established by inoculation of the Moloney leukemia virus (MLV) into a newborn A/Sn mouse; cells were described to be sensitive to the cytotoxic activity of natural killer (NK) cells and are therefore often used as target cells in NK assays
References:	Kiesling et al., Eur. J. Immunol. 5: 112-117 (1975)
Depositor:	Prof. P. F. Mührladt, GBF, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round to oval, single or clustered cells in suspension
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	maintain at $0.3-2.0 \times 10^6$ cells/ml; cells grow very fast; split ratio of 1:5 to 1:10; seed out at ca. 5×10^5 cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 20 hours
Harvest:	maximal density of about 3.0×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $5-10 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	contamination was eliminated with Mycoplasma Removal Agent, then negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Species:	confirmed as mouse with IEF of AST, MDH, PEP B
Cytogenetics:	hyperdiploid with 7% polyploidy; $41(38-42) \times 2n$; mouse chromosome morphology confirmed
Viruses:	ELISA: reverse transcriptase positive

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INFORMATION SHEET FOR CELL LINES

Cell Line:	YAPC
Cell Type:	human pancreas carcinoma
DSMZ No:	ACC 382
Origin:	established from ascites obtained during exploratory laparotomy of a 43-year-old Japanese man with pancreas carcinoma (terminal, no therapy) in December 1993; cells were described as being dependent on autocrinely secreted IL-1alpha
References:	Yamada et al., Int. J. Cancer 76: 141-147 (1998)
Depositor:	Dr. T. Yamada, Gunma University, Japan
DSMZ Cell Culture Data:	
Morphology:	cobblestone-like adherent cells growing in monolayers
Medium:	90% RPMI 1640 + 10% FBS
Subculture:	split culture 1:10 every 4 to 6 days prior to confluence (as the monolayer detaches upon reaching confluency) using trypsin/EDTA; seed out at ca. $1-2 \times 10^5$ cells/cm ² ; optimal density is at about 5×10^6 cells/80 cm ² ; it is recommended to add 10% of the spent medium to any fresh culture; after thawing cells may have a lag time of 1-2 days before reassuming growth
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 48 hours
Harvest:	cell harvest of ca. 2.0×10^7 cells/80 cm ²
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about $2-4 \times 10^6$ cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	cytokeratin+, cytokeratin-7+, cytokeratin-8+, cytokeratin-17(+), cytokeratin-18+, desmin-, endothel-, GFAP-, HMB-45-, neurofilament-, vimentin(+)
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, NP
Cytogenetics:	human near-triploid karyotype with 8% polyploidy; 66-71<3n>XXYY, +3, +7, +8, +9, +10, +14, -17, -18, -19, -20, -22, -22, -22, +3mar, der(1)del(1)(q23-25)inv(1)(p11q21), der(1;5)(q10;q10), add(3)(p11)/del(3)(p11p21)x2, del(5)(p11), add(6)(p21), del(7)(q32), add(9)(p1?), der(9;15)(q10;q10), der(10)t(10;11)(p11;q13)x2, del(12)(p12), add(13)(p11)x2, add(13)(p11), inv(16)(p13q11.2), del(18)(q21), add(20)(p11)/der(20)t(1;20)(p34;p11); loss of 1p, 3p, 9p, 18q associated with pancreas carcinoma; matches published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV-, HBV-, HCV-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzti.htm for details) .

Deutsche Sammlung von Mikroorganismen und Zellkulturen
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INFORMATION SHEET FOR CELL LINES

Cell Line:	YT
Cell Type:	human T/NK cell leukemia
DSMZ No:	ACC 434
Origin:	established from the pericardial fluid of a 15-year-old man with acute lymphoblastic leukemia (ALL) at relapse (with accompanying thymoma) in 1983; cells were described as: (i) showing NK activity and ADCC (antigen-dependent cell-mediated cytotoxicity), (ii) having their T cell receptor (TCR) genes in germline configuration, and (iii) expressing no TCR proteins (also CD3-, CD16-, CD56+, CD57-)
References:	Yodoi et al., J. Immunol. 134: 1623-1630 (1985); Yoneda et al., Leukemia 6: 136-141 (1992); Kanegane et al., Leukemia Lymphoma 29: 491-498 (1998); review article: Drexler & Matsuo, Leukemia 14: 777-782 (2000).
Depositor:	Dr. H. G. Drexler, DSMZ, Braunschweig, Germany
DSMZ Cell Culture Data:	
Morphology:	round to polygonal cells growing singly or in clumps in suspension
Medium:	80% Iscove's MDM + 20% FBS
Subculture:	split saturated culture 1:2 every second day; start culture in a 24-well-plate and seed out at ca. $0.8-1.5 \times 10^6$ cells/ml, initially after thawing the viability may be reduced down to ca. 20-50% during the first 1-2 weeks (during that initial period, cells should not be diluted, medium may be exchanged without centrifugation); in their vigorous growth phase, maintain at $0.2-0.5 \times 10^6$ cells/ml; maximal density at ca. $0.6-0.8 \times 10^6$ cells/ml
Incubation:	at 37 °C with 5% CO ₂
Doubling Time:	doubling time of ca. 40-50 hours
Harvest:	cell harvest of ca. 0.6×10^6 cells/ml
Storage:	frozen with 70% medium, 20% FBS, 10% DMSO at about 4×10^6 cells/ampoule
DSMZ Scientific Data:	
Mycoplasma:	negative in DAPI, microbiological culture, RNA hybridization, PCR assays
Immunology:	CD2-, CD3-, CD4-, CD5-, CD6-, CD7+, CD8-, CD13-, CD19-, CD25+, CD30+, CD34-, TCRalpha/beta-, TCRgamma/delta-
Fingerprinting:	multiplex PCR of minisatellite markers revealed a unique DNA profile
Species:	confirmed as human with IEF of AST, LDH, NP
Cytogenetics:	human flat-modulated near-tetraploid karyotype with 12% polyploidy; 84-98<4n>XXYY, -6, -9, -15, -19, +6-8mar, der(X)t(X;7)(q25;q21), der(1)dup(1)(q12.2q22)t(1;17)(q3?1;q1?2), der(4)t(1;4)(q32.2;q35)x2, der(5)t(X;5)(p1?3;q21)x2, del(6)(p11), der(10;17)(q10;q10), dup(14)(q1?2?), add(16)(q24), add(17)(q11); resembles published karyotype
Viruses:	ELISA: reverse transcriptase negative; PCR: EBV+, HBV-, HCV-, HHV-8-, HIV-, HTLV-I/II-

According to the DSMZ-Protocols of Identity-, Quality and Characterization Programs (see DSMZ Catalogue of Human and Animal Cell Lines or on www.dsmz.de/mutz/mutzi.htm for details) .